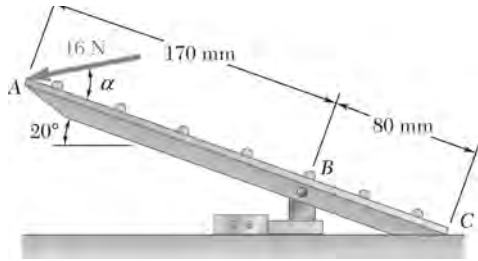


CHAPTER 3



PROBLEM 3.1

A foot valve for a pneumatic system is hinged at B . Knowing that $\alpha = 28^\circ$, determine the moment of the 16-N force about Point B by resolving the force into horizontal and vertical components.

SOLUTION

Note that

$$\theta = \alpha - 20^\circ = 28^\circ - 20^\circ = 8^\circ$$

and

$$F_x = (16 \text{ N}) \cos 8^\circ = 15.8443 \text{ N}$$

$$F_y = (16 \text{ N}) \sin 8^\circ = 2.2268 \text{ N}$$

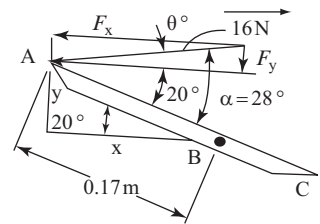
Also

$$x = (0.17 \text{ m}) \cos 20^\circ = 0.159748 \text{ m}$$

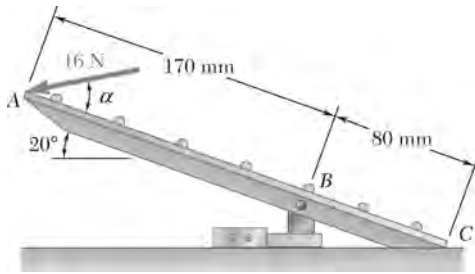
$$y = (0.17 \text{ m}) \sin 20^\circ = 0.058143 \text{ m}$$

Noting that the direction of the moment of each force component about B is counterclockwise,

$$\begin{aligned} M_B &= xF_y + yF_x \\ &= (0.159748 \text{ m})(2.2268 \text{ N}) \\ &\quad + (0.058143 \text{ m})(15.8443 \text{ N}) \\ &= 1.277 \text{ N} \cdot \text{m} \end{aligned}$$



$$\text{or } \mathbf{M}_B = 1.277 \text{ N} \cdot \text{m} \curvearrowright$$



PROBLEM 3.2

A foot valve for a pneumatic system is hinged at B . Knowing that $\alpha = 28^\circ$, determine the moment of the 16-N force about Point B by resolving the force into components along ABC and in a direction perpendicular to ABC .

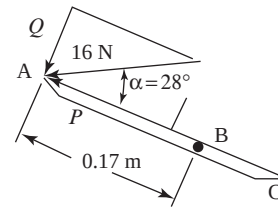
SOLUTION

First resolve the 4-lb force into components $\mathbf{P}$ and $\mathbf{Q}$, where

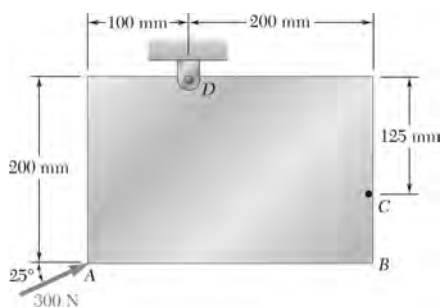
$$\begin{aligned} Q &= (16 \text{ N}) \sin 28^\circ \\ &= 7.5115 \text{ N} \end{aligned}$$

Then

$$\begin{aligned} M_B &= r_{A/B} Q \\ &= (0.17 \text{ m})(7.5115 \text{ N}) \\ &= 1.277 \text{ N} \cdot \text{m} \end{aligned}$$



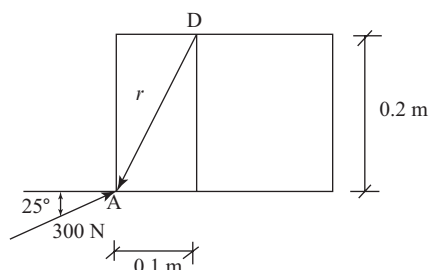
$$\text{or } \mathbf{M}_B = 1.277 \text{ N} \cdot \text{m} \curvearrowleft$$



PROBLEM 3.3

A 300-N force is applied at A as shown. Determine (a) the moment of the 300-N force about D, (b) the smallest force applied at B that creates the same moment about D.

SOLUTION



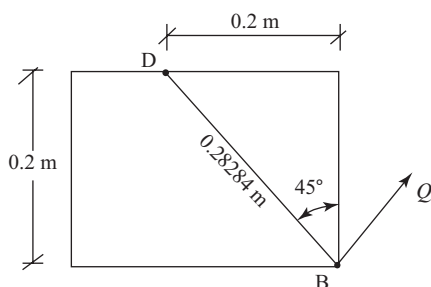
$$\begin{aligned}
 (a) \quad F_x &= (300 \text{ N}) \cos 25^\circ \\
 &= 271.89 \text{ N} \\
 F_y &= (300 \text{ N}) \sin 25^\circ \\
 &= 126.785 \text{ N} \\
 \mathbf{F} &= (271.89 \text{ N})\mathbf{i} + (126.785 \text{ N})\mathbf{j}
 \end{aligned}$$

$$\mathbf{r} = \overline{DA} = -(0.1 \text{ m})\mathbf{i} - (0.2 \text{ m})\mathbf{j}$$

$$\mathbf{M}_D = \mathbf{r} \times \mathbf{F}$$

$$\begin{aligned}
 \mathbf{M}_D &= [-(0.1 \text{ m})\mathbf{i} - (0.2 \text{ m})\mathbf{j}] \times [(271.89 \text{ N})\mathbf{i} + (126.785 \text{ N})\mathbf{j}] \\
 &= -(12.6785 \text{ N} \cdot \text{m})\mathbf{k} + (54.378 \text{ N} \cdot \text{m})\mathbf{k} \\
 &= (41.700 \text{ N} \cdot \text{m})\mathbf{k}
 \end{aligned}$$

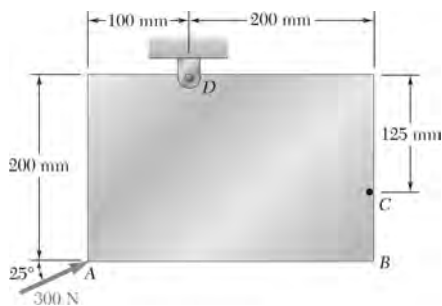
$$\mathbf{M}_D = 41.7 \text{ N} \cdot \text{m} \curvearrowright \blacktriangleleft$$



(b) The smallest force Q at B must be perpendicular to $\overline{DB}$ at 45°

$$\mathbf{M}_D = Q(\overline{DB})$$

$$41.700 \text{ N} \cdot \text{m} = Q(0.28284 \text{ m}) \quad Q = 147.4 \text{ N} \nearrow 45^\circ \blacktriangleleft$$



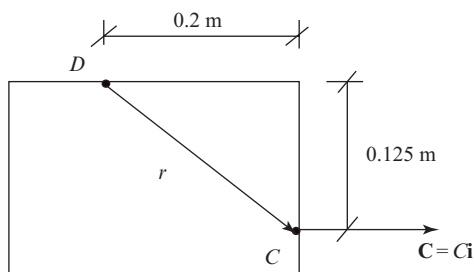
PROBLEM 3.4

A 300-N force is applied at A as shown. Determine (a) the moment of the 300-N force about D, (b) the magnitude and sense of the horizontal force applied at C that creates the same moment about D, (c) the smallest force applied at C that creates the same moment about D.

SOLUTION

- (a) See Problem 3.3 for the figure and analysis leading to the determination of $\mathbf{M}_D$

$$\mathbf{M}_D = 41.7 \text{ N} \cdot \text{m} \curvearrowleft$$



- (b) Since $\mathbf{C}$ is horizontal $\mathbf{C} = C\mathbf{i}$

$$\mathbf{r} = \overline{DC} = (0.2 \text{ m})\mathbf{i} - (0.125 \text{ m})\mathbf{j}$$

$$\mathbf{M}_D = \mathbf{r} \times C\mathbf{i} = C(0.125 \text{ m})\mathbf{k}$$

$$41.7 \text{ N} \cdot \text{m} = (0.125 \text{ m})(C)$$

$$C = 333.60 \text{ N}$$

$$C = 334 \text{ N} \quad \blacktriangleleft$$

- (c) The smallest force C must be perpendicular to DC ; thus, it forms α with the vertical

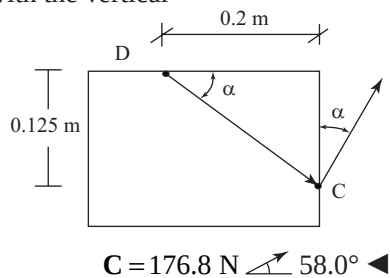
$$\tan \alpha = \frac{0.125 \text{ m}}{0.2 \text{ m}}$$

$$\alpha = 32.0^\circ$$

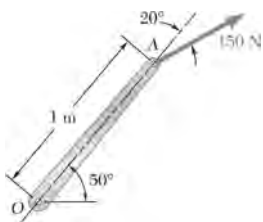
$$\mathbf{M}_D = C(DC); \quad DC = \sqrt{(0.2 \text{ m})^2 + (0.125 \text{ m})^2}$$

$$= 0.23585 \text{ m}$$

$$41.70 \text{ N} \cdot \text{m} = C(0.23585 \text{ m})$$



$$C = 176.8 \text{ N} \nearrow 58.0^\circ \quad \blacktriangleleft$$



PROBLEM 3.5

A 150-N force acts on the end of the 1 m lever as shown. Determine the moment of the force about O.

SOLUTION

First note

$$P_x = 150 \cos 30^\circ$$

$$= 129.9 \text{ N}$$

$$P_y = 150 \sin 30^\circ$$

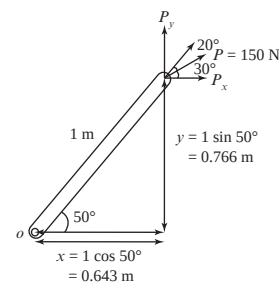
$$= 75 \text{ N}$$

Noting that the direction of the moment of P_x about O is clockwise and P_y about O is counterclockwise,

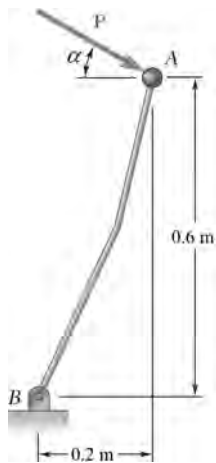
$$M_B = xP_y - yP_x$$

$$= (0.643)(75) - (0.766)(129.9)$$

$$= -51.3 \text{ N} \cdot \text{m}$$



$$\text{or } \mathbf{M}_B = 51.3 \text{ N} \cdot \text{m} \quad \curvearrowright \blacktriangleleft$$



PROBLEM 3.6

For the shift lever shown, determine the magnitude and the direction of the smallest force $\mathbf{P}$ that has a $25 \text{ N} \cdot \text{m}$ clockwise moment about B .

SOLUTION

For $\mathbf{P}$ to be minimum, it must be perpendicular to the line joining Points A and B . Thus,

$$\begin{aligned}\alpha &= \theta \\ &= \tan^{-1} \frac{0.2 \text{ m}}{0.6 \text{ m}} \\ &= 18.43^\circ\end{aligned}$$

and

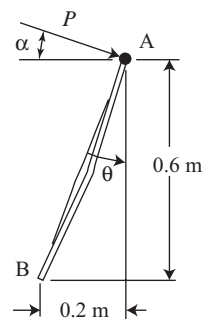
$$M_B = dP_{\min}$$

where

$$\begin{aligned}d &= r_{A/B} \\ &= \sqrt{(0.2 \text{ m})^2 + (0.6 \text{ m})^2} \\ &= 0.632 \text{ m}\end{aligned}$$

Then

$$\begin{aligned}P_{\min} &= \frac{25 \text{ N} \cdot \text{m}}{0.632 \text{ m}} \\ &= 39.6 \text{ N}\end{aligned}$$



$$\mathbf{P}_{\min} = 39.6 \text{ N} \searrow 18.43^\circ \blacktriangleleft$$



PROBLEM 3.7

A 55-N force $\mathbf{P}$ is applied to a shift lever. The moment of $\mathbf{P}$ about B is clockwise and has a magnitude of $34 \text{ N} \cdot \text{m}$. Determine the value of α .

SOLUTION

By definition

$$M_B = r_{A/B} P \sin \theta$$

where

$$\theta = \alpha + (90^\circ - \phi)$$

and

$$\phi = \tan^{-1} \frac{0.2 \text{ m}}{0.6 \text{ m}} = 18.43^\circ$$

also

$$r_{A/B} = \sqrt{(0.2 \text{ m})^2 + (0.6 \text{ m})^2} \\ = 0.632 \text{ m}$$

Then

$$34 \text{ N} \cdot \text{m} = (0.632 \text{ m})(55 \text{ N}) \sin(\alpha + 90^\circ - 18.43^\circ)$$

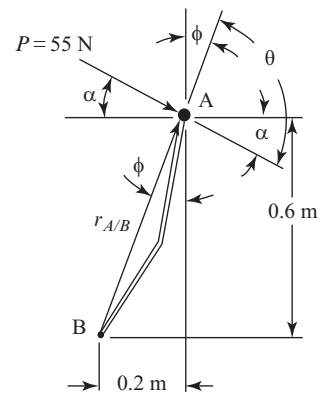
or

$$\sin(\alpha + 71.57^\circ) = 0.978$$

$$\alpha + 71.57^\circ = 77.96^\circ$$

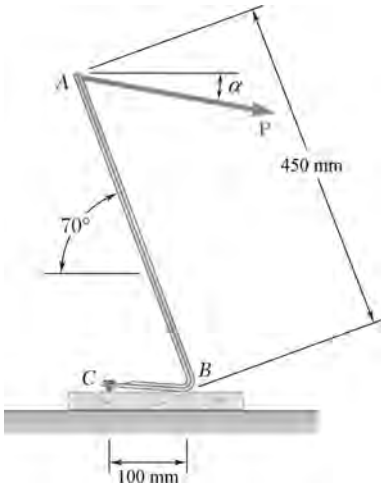
or

$$\alpha + 71.57^\circ = 102.04^\circ$$



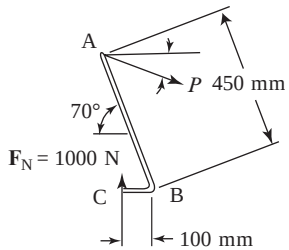
$$\alpha = 6.39^\circ, 30.5^\circ \quad \blacktriangleleft$$

PROBLEM 3.8



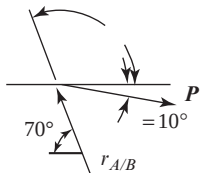
It is known that a vertical force of 1000 N is required to remove the nail at C from the board. As the nail first starts moving, determine (a) the moment about B of the force exerted on the nail, (b) the magnitude of the force **P** that creates the same moment about B if $\alpha = 10^\circ$, (c) the smallest force **P** that creates the same moment about B.

SOLUTION



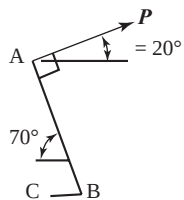
(a) We have $M_B = r_{C/B} F_N$
 $= (100 \text{ mm})(1000 \text{ N}) = 10^5 \text{ N} \cdot \text{mm}$
 $= 100 \text{ N} \cdot \text{m}$

or $M_B = 100 \text{ N} \cdot \text{m} \curvearrowright$



(b) By definition $M_B = r_{A/B} P \sin \theta$
 $\theta = 10^\circ + (180^\circ - 70^\circ)$
 $= 120^\circ$

Then $100 \text{ N} \cdot \text{m} = (0.45 \text{ m}) \times P \sin 120^\circ$ or $P = 257 \text{ N} \blacktriangleleft$



(c) For **P** to be minimum, it must be perpendicular to the line joining Points A and B. Thus, **P** must be directed as shown.

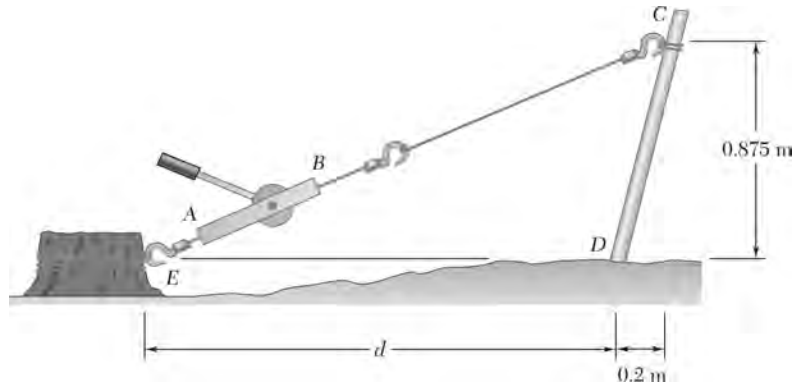
Thus $M_B = d P_{\min}$
 $d = r_{A/B}$

or $100 \text{ N} \cdot \text{m} = (0.45 \text{ m}) P_{\min}$

or $P_{\min} = 222 \text{ N}$ $P_{\min} = 222 \text{ N} \nearrow 20^\circ \blacktriangleleft$

PROBLEM 3.9

A winch puller AB is used to straighten a fence post. Knowing that the tension in cable BC is 1040 N and length d is 1.90 m , determine the moment about D of the force exerted by the cable at C by resolving that force into horizontal and vertical components applied (a) at Point C , (b) at Point E .



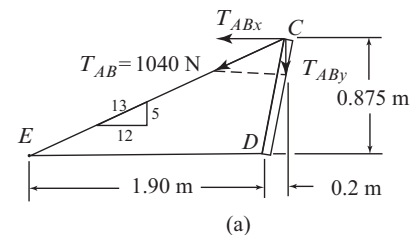
SOLUTION

(a) Slope of line $EC = \frac{0.875\text{ m}}{1.90\text{ m} + 0.2\text{ m}} = \frac{5}{12}$

Then $T_{ABx} = \frac{12}{13}(T_{AB})$
 $= \frac{12}{13}(1040\text{ N})$
 $= 960\text{ N}$

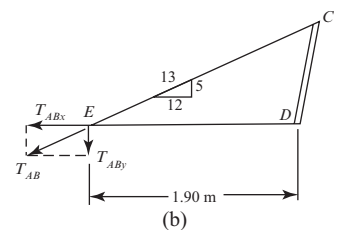
and $T_{ABy} = \frac{5}{13}(1040\text{ N})$
 $= 400\text{ N}$

Then $M_D = T_{ABx}(0.875\text{ m}) - T_{ABy}(0.2\text{ m})$
 $= (960\text{ N})(0.875\text{ m}) - (400\text{ N})(0.2\text{ m})$
 $= 760\text{ N} \cdot \text{m}$



or $M_D = 760\text{ N} \cdot \text{m} \curvearrowright$

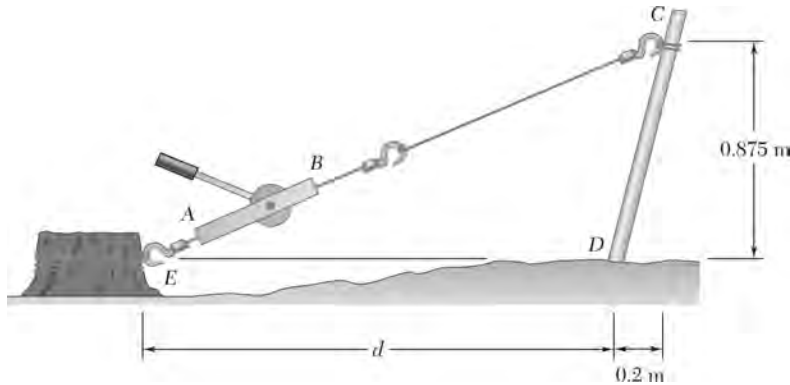
(b) We have $M_D = T_{ABx}(y) + T_{ABy}(x)$
 $= (960\text{ N})(0) + (400\text{ N})(1.90\text{ m})$
 $= 760\text{ N} \cdot \text{m}$



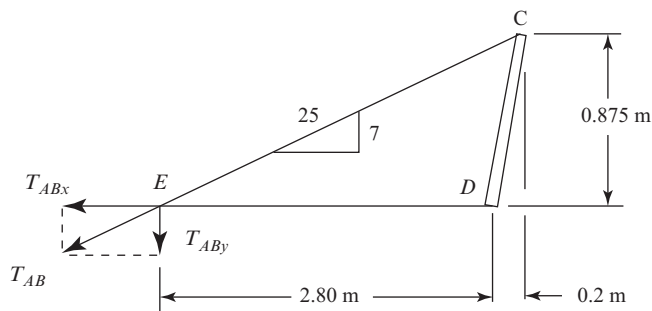
or $M_D = 760\text{ N} \cdot \text{m} \curvearrowright$

PROBLEM 3.10

It is known that a force with a moment of $960 \text{ N} \cdot \text{m}$ about D is required to straighten the fence post CD . If $d = 2.80 \text{ m}$, determine the tension that must be developed in the cable of winch puller AB to create the required moment about Point D .



SOLUTION



Slope of line $EC = \frac{0.875 \text{ m}}{2.80 \text{ m} + 0.2 \text{ m}} = \frac{7}{24}$

Then $T_{ABx} = \frac{24}{25} T_{AB}$

and $T_{ABy} = \frac{7}{25} T_{AB}$

We have $M_D = T_{ABx}(y) + T_{ABy}(x)$

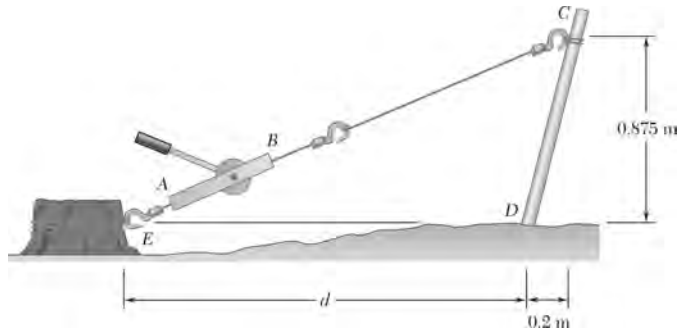
$$960 \text{ N} \cdot \text{m} = \frac{24}{25} T_{AB}(0) + \frac{7}{25} T_{AB}(2.80 \text{ m})$$

$$T_{AB} = 1224 \text{ N}$$

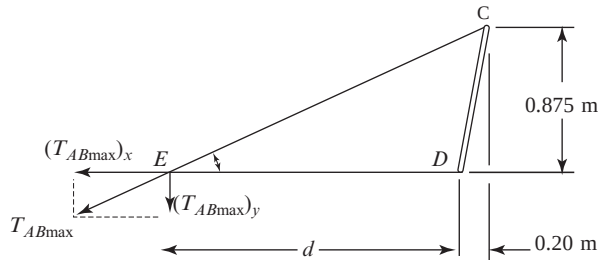
$$\text{or } T_{AB} = 1224 \text{ N} \quad \blacktriangleleft$$

PROBLEM 3.11

It is known that a force with a moment of $960 \text{ N} \cdot \text{m}$ about D is required to straighten the fence post CD . If the capacity of winch puller AB is 2400 N , determine the minimum value of distance d to create the specified moment about Point D .



SOLUTION



The minimum value of d can be found based on the equation relating the moment of the force $\mathbf{T}_{AB}$ about D :

$$M_D = (T_{AB \max})_y (d)$$

where

$$M_D = 960 \text{ N} \cdot \text{m}$$

$$(T_{AB \max})_y = T_{AB \max} \sin \theta = (2400 \text{ N}) \sin \theta$$

Now

$$\sin \theta = \frac{0.875 \text{ m}}{\sqrt{(d + 0.20)^2 + (0.875)^2} \text{ m}}$$

$$960 \text{ N} \cdot \text{m} = 2400 \text{ N} \left[\frac{0.875}{\sqrt{(d + 0.20)^2 + (0.875)^2}} \right] (d)$$

$$\text{or} \quad \sqrt{(d + 0.20)^2 + (0.875)^2} = 2.1875d$$

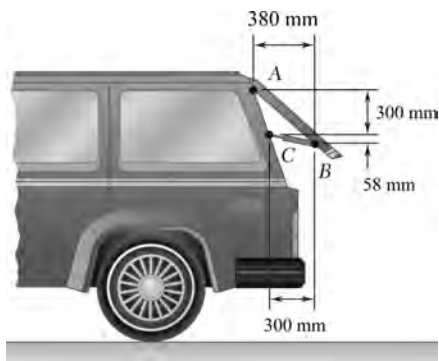
$$\text{or} \quad (d + 0.20)^2 + (0.875)^2 = 4.7852d^2$$

$$\text{or} \quad 3.7852d^2 - 0.40d - .8056 = 0$$

Using the quadratic equation, the minimum values of d are 0.51719 m and $-.41151 \text{ m}$.

Since only the positive value applies here, $d = 0.51719 \text{ m}$

or $d = 517 \text{ mm} \quad \blacktriangleleft$



PROBLEM 3.12

The tailgate of a car is supported by the hydraulic lift BC . If the lift exerts a 625-N force directed along its centerline on the ball and socket at B , determine the moment of the force about A .

SOLUTION

First note

$$d_{CB} = \sqrt{(300 \text{ mm})^2 + (58 \text{ mm})^2} \\ = 305.6 \text{ mm}$$

Then

$$\cos \theta = \frac{300 \text{ mm}}{305.6 \text{ mm}}$$

$$\sin \theta = \frac{58 \text{ mm}}{305.6 \text{ mm}}$$

and

$$\mathbf{F}_{CB} = F_{CB} \cos \theta \mathbf{i} - F_{CB} \sin \theta \mathbf{j} \\ = \frac{625 \text{ N}}{305.6 \text{ mm}} [(300 \text{ mm})\mathbf{i} - (58 \text{ mm})\mathbf{j}]$$

Now

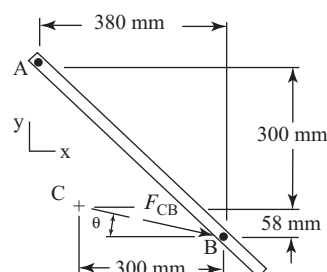
$$\mathbf{M}_A = \mathbf{r}_{B/A} \times \mathbf{F}_{CB}$$

where

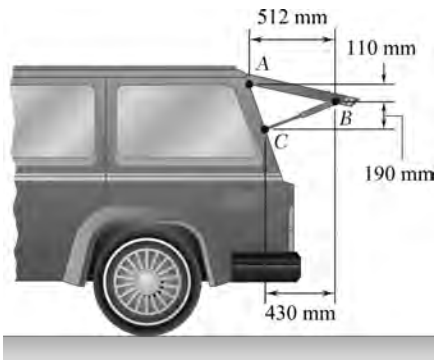
$$\mathbf{r}_{B/A} = (380 \text{ mm})\mathbf{i} - (300 \text{ mm} + 58 \text{ mm})\mathbf{j} \\ = (380 \text{ mm})\mathbf{i} - (358 \text{ mm})\mathbf{j}$$

Then

$$\mathbf{M}_A = [(380 \text{ mm})\mathbf{i} - (358 \text{ mm})\mathbf{j}] \times \frac{625 \text{ N}}{305.6 \text{ mm}} (300 \text{ mm}\mathbf{i} - 58 \text{ mm}\mathbf{j}) \\ = -45075 \text{ N} \cdot \text{mm}\mathbf{i} + 219650 \text{ N} \cdot \text{mm}\mathbf{k} \\ = 174575 \text{ N} \cdot \text{mm}\mathbf{k} \\ = 175 \text{ N} \cdot \text{m}\mathbf{k}$$



$$\text{or } \mathbf{M}_A = 175 \text{ N} \cdot \text{m} \curvearrowright$$



PROBLEM 3.13

The tailgate of a car is supported by the hydraulic lift BC . If the lift exerts a 625-N force directed along its centerline on the ball and socket at B , determine the moment of the force about A .

SOLUTION

First note

$$d_{CB} = \sqrt{(430 \text{ mm})^2 + (190 \text{ mm})^2}$$

$$= 470.1 \text{ mm}$$

Then

$$\cos \theta = \frac{430 \text{ mm}}{470.1 \text{ mm}}$$

$$\sin \theta = \frac{190 \text{ mm}}{470.1 \text{ mm}}$$

and

$$\mathbf{F}_{CB} = (F_{CB} \cos \theta)\mathbf{i} - (F_{CB} \sin \theta)\mathbf{j}$$

$$= \frac{625 \text{ N}}{470.1 \text{ mm}}[(430 \text{ mm})\mathbf{i} + (190 \text{ mm})\mathbf{j}]$$

Now

$$\mathbf{M}_A = \mathbf{r}_{B/A} \times \mathbf{F}_{CB}$$

where

$$\mathbf{r}_{B/A} = (512 \text{ mm})\mathbf{i} - (110 \text{ mm})\mathbf{j}$$

Then

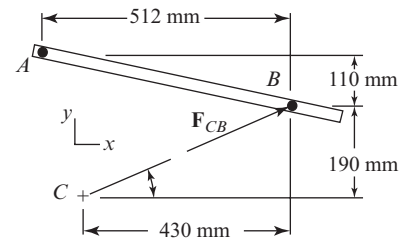
$$\mathbf{M}_A = [(512 \text{ mm})\mathbf{i} - (110 \text{ mm})\mathbf{j}] \times \frac{625 \text{ N}}{(470.1 \text{ mm})} (430 \text{ mm } \mathbf{i} + 190 \text{ mm } \mathbf{j})$$

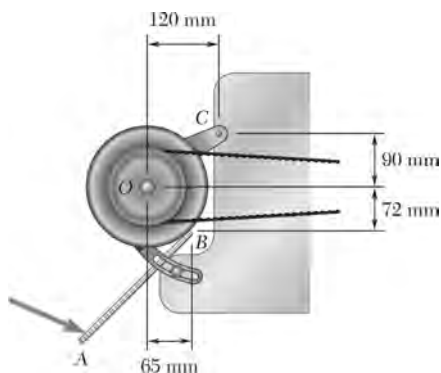
$$= 129334 \text{ N} \cdot \text{mm } \mathbf{k} + 62886 \text{ N} \cdot \text{mm } \mathbf{k}$$

$$= 192220 \text{ N} \cdot \text{mm } \mathbf{k}$$

$$= 192 \text{ N} \cdot \text{m } \mathbf{k}$$

$$\text{or } \mathbf{M}_A = 192 \text{ N} \cdot \text{m } \curvearrowright$$





PROBLEM 3.14

A mechanic uses a piece of pipe AB as a lever when tightening an alternator belt. When he pushes down at A , a force of 485 N is exerted on the alternator at B . Determine the moment of that force about bolt C if its line of action passes through O .

SOLUTION

We have

$$\mathbf{M}_C = \mathbf{r}_{B/C} \times \mathbf{F}_B$$

Noting the direction of the moment of each force component about C is clockwise.

$$M_C = xF_{By} + yF_{Bx}$$

where

$$x = 120 \text{ mm} - 65 \text{ mm} = 55 \text{ mm}$$

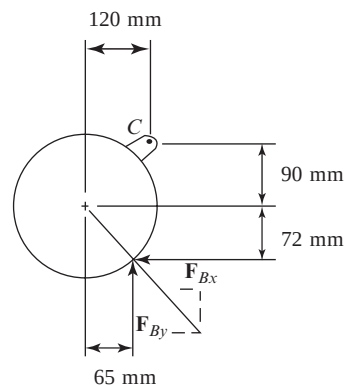
$$y = 72 \text{ mm} + 90 \text{ mm} = 162 \text{ mm}$$

and

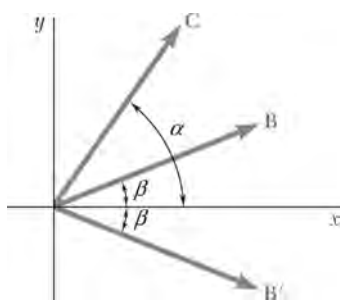
$$F_{Bx} = \frac{65}{\sqrt{(65)^2 + (72)^2}} (485 \text{ N}) = 325 \text{ N}$$

$$F_{By} = \frac{72}{\sqrt{(65)^2 + (72)^2}} (485 \text{ N}) = 360 \text{ N}$$

$$\begin{aligned} M_C &= (55 \text{ mm})(360 \text{ N}) + (162)(325 \text{ N}) \\ &= 72450 \text{ N} \cdot \text{mm} \\ &= 72.450 \text{ N} \cdot \text{m} \end{aligned}$$



$$\text{or } \mathbf{M}_C = 72.5 \text{ N} \cdot \text{m} \curvearrowright \blacktriangleleft$$



PROBLEM 3.15

Form the vector products $\mathbf{B} \times \mathbf{C}$ and $\mathbf{B}' \times \mathbf{C}$, where $B = B'$, and use the results obtained to prove the identity

$$\sin \alpha \cos \beta = \frac{1}{2} \sin(\alpha + \beta) + \frac{1}{2} \sin(\alpha - \beta).$$

SOLUTION

Note:

$$\mathbf{B} = B(\cos \beta \mathbf{i} + \sin \beta \mathbf{j})$$

$$\mathbf{B}' = B(\cos \beta \mathbf{i} - \sin \beta \mathbf{j})$$

$$\mathbf{C} = C(\cos \alpha \mathbf{i} + \sin \alpha \mathbf{j})$$

By definition:

$$|\mathbf{B} \times \mathbf{C}| = BC \sin(\alpha - \beta) \quad (1)$$

$$|\mathbf{B}' \times \mathbf{C}| = BC \sin(\alpha + \beta) \quad (2)$$

Now

$$\begin{aligned} \mathbf{B} \times \mathbf{C} &= B(\cos \beta \mathbf{i} + \sin \beta \mathbf{j}) \times C(\cos \alpha \mathbf{i} + \sin \alpha \mathbf{j}) \\ &= BC(\cos \beta \sin \alpha - \sin \beta \cos \alpha) \mathbf{k} \end{aligned} \quad (3)$$

and

$$\begin{aligned} \mathbf{B}' \times \mathbf{C} &= B(\cos \beta \mathbf{i} - \sin \beta \mathbf{j}) \times C(\cos \alpha \mathbf{i} + \sin \alpha \mathbf{j}) \\ &= BC(\cos \beta \sin \alpha + \sin \beta \cos \alpha) \mathbf{k} \end{aligned} \quad (4)$$

Equating the magnitudes of $\mathbf{B} \times \mathbf{C}$ from Equations (1) and (3) yields:

$$BC \sin(\alpha - \beta) = BC(\cos \beta \sin \alpha - \sin \beta \cos \alpha) \quad (5)$$

Similarly, equating the magnitudes of $\mathbf{B}' \times \mathbf{C}$ from Equations (2) and (4) yields:

$$BC \sin(\alpha + \beta) = BC(\cos \beta \sin \alpha + \sin \beta \cos \alpha) \quad (6)$$

Adding Equations (5) and (6) gives:

$$\sin(\alpha - \beta) + \sin(\alpha + \beta) = 2 \cos \beta \sin \alpha$$

$$\text{or } \sin \alpha \cos \beta = \frac{1}{2} \sin(\alpha + \beta) + \frac{1}{2} \sin(\alpha - \beta) \quad \blacktriangleleft$$

PROBLEM 3.16

A line passes through the Points (20 m, 16 m) and (−1 m, −4 m). Determine the perpendicular distance d from the line to the origin O of the system of coordinates.

SOLUTION

$$d_{AB} = \sqrt{[20 \text{ m} - (-1 \text{ m})]^2 + [16 \text{ m} - (-4 \text{ m})]^2}$$

$$= 29 \text{ m}$$

Assume that a force $\mathbf{F}$, or magnitude $F(\text{N})$, acts at Point A and is directed from A to B .

Then,

$$\mathbf{F} = F\lambda_{AB}$$

where

$$\lambda_{AB} = \frac{\mathbf{r}_B - \mathbf{r}_A}{d_{AB}}$$

$$= \frac{1}{29}(21\mathbf{i} + 20\mathbf{j})$$

By definition

$$\mathbf{M}_O = |\mathbf{r}_A \times \mathbf{F}| = dF$$

where

$$\mathbf{r}_A = -(1 \text{ m})\mathbf{i} - (4 \text{ m})\mathbf{j}$$

Then

$$\mathbf{M}_O = [-(1 \text{ m})\mathbf{i} - (4 \text{ m})\mathbf{j}] \times \frac{F}{29 \text{ m}}[(21 \text{ m})\mathbf{i} + (20 \text{ m})\mathbf{j}]$$

$$= \frac{F}{29}[-(20)\mathbf{k} + (84)\mathbf{k}]$$

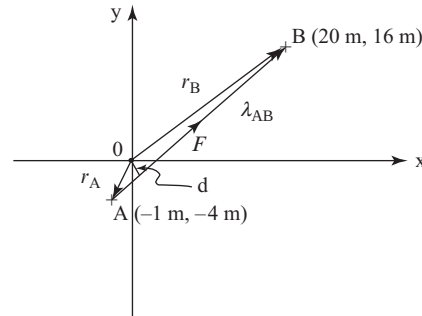
$$= \left(\frac{64}{29}F\right)\mathbf{k} \text{ N} \cdot \text{m}$$

Finally

$$\left(\frac{64}{29}F\right) = d(F)$$

$$d = \frac{64}{29} \text{ m}$$

$$d = 2.21 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.17

The vectors $\mathbf{P}$ and $\mathbf{Q}$ are two adjacent sides of a parallelogram. Determine the area of the parallelogram when (a) $\mathbf{P} = -7\mathbf{i} + 3\mathbf{j} - 3\mathbf{k}$ and $\mathbf{Q} = 2\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$, (b) $\mathbf{P} = 6\mathbf{i} - 5\mathbf{j} - 2\mathbf{k}$ and $\mathbf{Q} = -2\mathbf{i} + 5\mathbf{j} - \mathbf{k}$.

SOLUTION

(a) We have

$$A = |\mathbf{P} \times \mathbf{Q}|$$

where

$$\mathbf{P} = -7\mathbf{i} + 3\mathbf{j} - 3\mathbf{k}$$

$$\mathbf{Q} = 2\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$$

Then

$$\begin{aligned}\mathbf{P} \times \mathbf{Q} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -7 & 3 & -3 \\ 2 & 2 & 5 \end{vmatrix} \\ &= [(15 + 6)\mathbf{i} + (-6 + 35)\mathbf{j} + (-14 - 6)\mathbf{k}] \\ &= (21)\mathbf{i} + (29)\mathbf{j} - (20)\mathbf{k}\end{aligned}$$

$$A = \sqrt{(20)^2 + (29)^2 + (-20)^2} \quad \text{or } A = 41.0 \quad \blacktriangleleft$$

(b) We have

$$A = |\mathbf{P} \times \mathbf{Q}|$$

where

$$\mathbf{P} = 6\mathbf{i} - 5\mathbf{j} - 2\mathbf{k}$$

$$\mathbf{Q} = -2\mathbf{i} + 5\mathbf{j} - \mathbf{k}$$

Then

$$\begin{aligned}\mathbf{P} \times \mathbf{Q} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 6 & -5 & -2 \\ -2 & 5 & -1 \end{vmatrix} \\ &= [(5 + 10)\mathbf{i} + (4 + 6)\mathbf{j} + (30 - 10)\mathbf{k}] \\ &= (15)\mathbf{i} + (10)\mathbf{j} + (20)\mathbf{k}\end{aligned}$$

$$A = \sqrt{(15)^2 + (10)^2 + (20)^2} \quad \text{or } A = 26.9 \quad \blacktriangleleft$$

PROBLEM 3.18

A plane contains the vectors **A** and **B**. Determine the unit vector normal to the plane when **A** and **B** are equal to, respectively, (a) $\mathbf{i} + 2\mathbf{j} - 5\mathbf{k}$ and $4\mathbf{i} - 7\mathbf{j} - 5\mathbf{k}$, (b) $3\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$ and $-2\mathbf{i} + 6\mathbf{j} - 4\mathbf{k}$.

SOLUTION

(a) We have $\lambda = \frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|}$

where $\mathbf{A} = 1\mathbf{i} + 2\mathbf{j} - 5\mathbf{k}$

$$\mathbf{B} = 4\mathbf{i} - 7\mathbf{j} - 5\mathbf{k}$$

Then
$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & +2 & -5 \\ 4 & -7 & -5 \end{vmatrix}$$
$$= (-10 - 35)\mathbf{i} + (20 + 5)\mathbf{j} + (-7 - 8)\mathbf{k}$$
$$= 15(3\mathbf{i} - \mathbf{j} - \mathbf{k})$$

and $|\mathbf{A} \times \mathbf{B}| = 15\sqrt{(-3)^2 + (-1)^2 + (-1)^2} = 15\sqrt{11}$

$$\lambda = \frac{15(-3\mathbf{i} - \mathbf{j} - \mathbf{k})}{15\sqrt{11}} \quad \text{or } \lambda = \frac{1}{\sqrt{11}}(-3\mathbf{i} - \mathbf{j} - \mathbf{k}) \quad \blacktriangleleft$$

(b) We have $\lambda = \frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|}$

where $\mathbf{A} = 3\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$

$$\mathbf{B} = -2\mathbf{i} + 6\mathbf{j} - 4\mathbf{k}$$

Then
$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -3 & 2 \\ -2 & 6 & -4 \end{vmatrix}$$
$$= (12 - 12)\mathbf{i} + (-4 + 12)\mathbf{j} + (18 - 6)\mathbf{k}$$
$$= (8\mathbf{j} + 12\mathbf{k})$$

and $|\mathbf{A} \times \mathbf{B}| = 4\sqrt{(2)^2 + (3)^2} = 4\sqrt{13}$

$$\lambda = \frac{4(2\mathbf{j} + 3\mathbf{k})}{4\sqrt{13}} \quad \text{or } \lambda = \frac{1}{\sqrt{13}}(2\mathbf{j} + 3\mathbf{k}) \quad \blacktriangleleft$$

PROBLEM 3.19

Determine the moment about the origin O of the force $\mathbf{F} = 4\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$ that acts at a Point A . Assume that the position vector of A is (a) $\mathbf{r} = 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$, (b) $\mathbf{r} = 2\mathbf{i} + 2.5\mathbf{j} - 1.5\mathbf{k}$, (c) $\mathbf{r} = 2\mathbf{i} + 5\mathbf{j} + 6\mathbf{k}$. The components of $\mathbf{F}$ and $\mathbf{r}$ are in N and m respectively.

SOLUTION

$$\begin{aligned} (a) \quad \mathbf{M}_O &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -3 & 4 \\ 4 & 5 & -3 \end{vmatrix} \\ &= (9 - 20)\mathbf{i} + (16 + 6)\mathbf{j} + (10 + 12)\mathbf{k} \quad \mathbf{M}_O = -11\mathbf{i} + 22\mathbf{j} + 22\mathbf{k} \text{ N} \cdot \text{m} \quad \blacktriangleleft \end{aligned}$$

$$\begin{aligned} (b) \quad \mathbf{M}_O &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2.5 & -1.5 \\ 4 & 5 & -3 \end{vmatrix} \\ &= (-7.5 + 7.5)\mathbf{i} + (-6 + 6)\mathbf{j} + (10 - 10)\mathbf{k} \quad \mathbf{M}_O = 0 \quad \blacktriangleleft \end{aligned}$$

$$\begin{aligned} (c) \quad \mathbf{M}_O &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 5 & 6 \\ 4 & 5 & -3 \end{vmatrix} \\ &= (-15 - 30)\mathbf{i} + (24 + 6)\mathbf{j} + (10 - 20)\mathbf{k} \quad \mathbf{M}_O = -45\mathbf{i} + 30\mathbf{j} - 10\mathbf{k} \text{ N} \cdot \text{m} \quad \blacktriangleleft \end{aligned}$$

Note: The answer to Part b could have been anticipated since the elements of the last two rows of the determinant are proportional.

PROBLEM 3.20

Determine the moment about the origin O of the force $\mathbf{F} = -2\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$ that acts at a Point A . Assume that the position vector of A is (a) $\mathbf{r} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, (b) $\mathbf{r} = 2\mathbf{i} + 3\mathbf{j} - 5\mathbf{k}$, (c) $\mathbf{r} = -4\mathbf{i} + 6\mathbf{j} + 10\mathbf{k}$. The components of $\mathbf{F}$ and $\mathbf{r}$ are in N and m respectively.

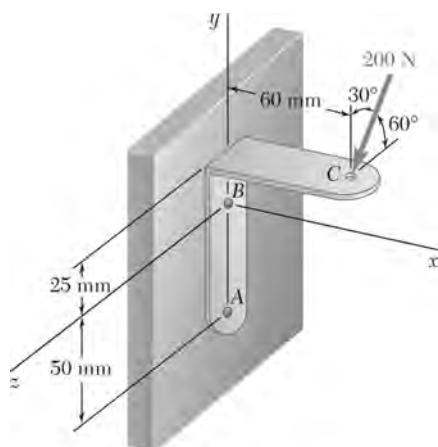
SOLUTION

$$\begin{aligned} (a) \quad \mathbf{M}_O &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ -2 & 3 & 5 \end{vmatrix} \\ &= (5-3)\mathbf{i} + (-2-5)\mathbf{j} + (3+2)\mathbf{k} & \mathbf{M}_O = 2\mathbf{i} - 7\mathbf{j} + 5\mathbf{k} \text{ N} \cdot \text{m} \quad \blacktriangleleft \end{aligned}$$

$$\begin{aligned} (b) \quad \mathbf{M}_O &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & -5 \\ -2 & 3 & 5 \end{vmatrix} \\ &= (15+15)\mathbf{i} + (10-10)\mathbf{j} + (6+6)\mathbf{k} & \mathbf{M}_O = 30\mathbf{i} + 12\mathbf{k} \text{ N} \cdot \text{m} \quad \blacktriangleleft \end{aligned}$$

$$\begin{aligned} (c) \quad \mathbf{M}_O &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -4 & 6 & 10 \\ -2 & 3 & 5 \end{vmatrix} \\ &= (30-30)\mathbf{i} + (-20+20)\mathbf{j} + (-12+12)\mathbf{k} & \mathbf{M}_O = 0 \quad \blacktriangleleft \end{aligned}$$

Note: The answer to Part c could have been anticipated since the elements of the last two rows of the determinant are proportional.



PROBLEM 3.21

A 200-N force is applied as shown to the bracket ABC . Determine the moment of the force about A .

SOLUTION

We have

$$\mathbf{M}_A = \mathbf{r}_{C/A} \times \mathbf{F}_C$$

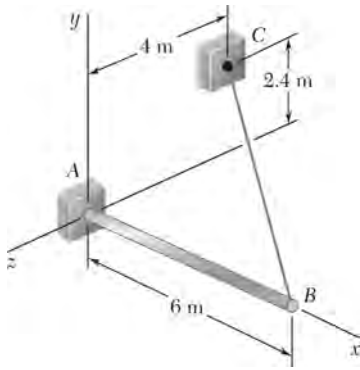
where

$$\mathbf{r}_{C/A} = (0.06 \text{ m})\mathbf{i} + (0.075 \text{ m})\mathbf{j}$$

$$\mathbf{F}_C = -(200 \text{ N})\cos 30^\circ \mathbf{j} + (200 \text{ N})\sin 30^\circ \mathbf{k}$$

Then

$$\begin{aligned} \mathbf{M}_A &= 200 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.06 & 0.075 & 0 \\ 0 & -\cos 30^\circ & \sin 30^\circ \end{vmatrix} \\ &= 200[(0.075 \sin 30^\circ)\mathbf{i} - (0.06 \sin 30^\circ)\mathbf{j} - (0.06 \cos 30^\circ)\mathbf{k}] \\ &\text{or } \mathbf{M}_A = (7.50 \text{ N} \cdot \text{m})\mathbf{i} - (6.00 \text{ N} \cdot \text{m})\mathbf{j} - (10.39 \text{ N} \cdot \text{m})\mathbf{k} \quad \blacktriangleleft \end{aligned}$$



PROBLEM 3.22

The 6-m boom AB has a fixed end A . A steel cable is stretched from the free end B of the boom to a Point C located on the vertical wall. If the tension in the cable is 2.5 kN, determine the moment about A of the force exerted by the cable at B .

SOLUTION

First note

$$d_{BC} = \sqrt{(-6)^2 + (2.4)^2 + (-4)^2} \\ = 7.6 \text{ m}$$

Then

$$\mathbf{T}_{BC} = \frac{2.5 \text{ kN}}{7.6} (-6\mathbf{i} + 2.4\mathbf{j} - 4\mathbf{k})$$

We have

$$\mathbf{M}_A = \mathbf{r}_{B/A} \times \mathbf{T}_{BC}$$

where

$$\mathbf{r}_{B/A} = (6 \text{ m})\mathbf{i}$$

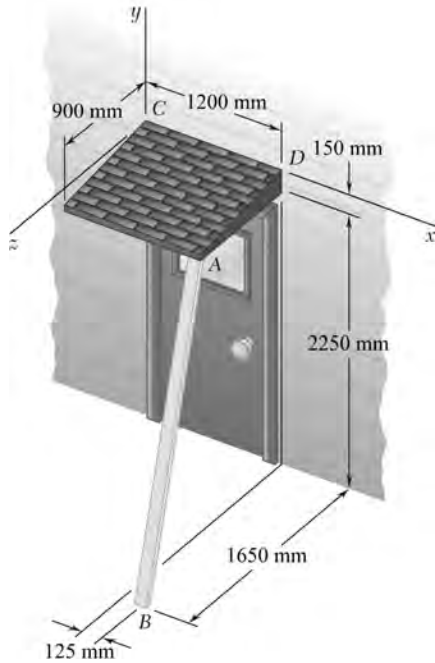
Then

$$\mathbf{M}_A = (6 \text{ m})\mathbf{i} \times \frac{2.5 \text{ kN}}{7.6} (-6\mathbf{i} + 2.4\mathbf{j} - 4\mathbf{k})$$

$$\text{or } \mathbf{M}_A = (7.89 \text{ kN} \cdot \text{m})\mathbf{j} + (4.74 \text{ kN} \cdot \text{m})\mathbf{k} \quad \blacktriangleleft$$

PROBLEM 3.23

A wooden board AB , which is used as a temporary prop to support a small roof, exerts at Point A of the roof a 300-N force directed along BA . Determine the moment about C of that force.



SOLUTION

We have $\mathbf{M}_C = \mathbf{r}_{A/C} \times \mathbf{F}_{BA}$

where $\mathbf{r}_{A/C} = (1200 \text{ mm})\mathbf{i} - (150 \text{ mm})\mathbf{j} + (900 \text{ mm})\mathbf{k}$

and $\mathbf{F}_{BA} = \lambda_{BA} F_{BA}$

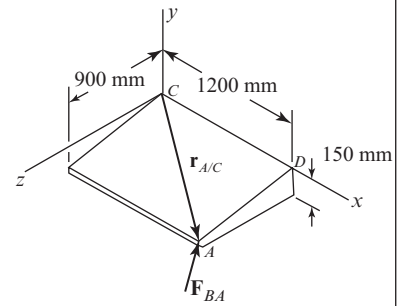
$$= \left[\frac{-(125 \text{ mm})\mathbf{i} + (2250 \text{ mm})\mathbf{j} - (750 \text{ mm})\mathbf{k}}{\sqrt{(125)^2 + (2250)^2 + (30)^2}} \right] (300 \text{ N})$$

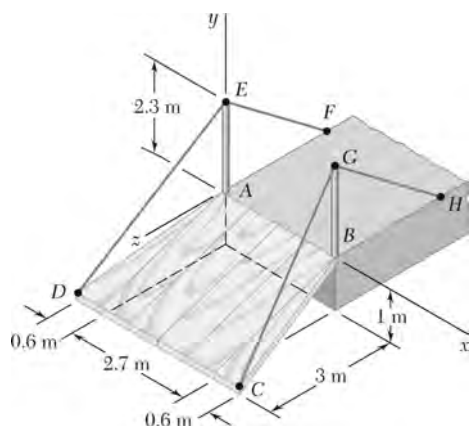
$$= -(16.64 \text{ N})\mathbf{i} + (299.5 \text{ N})\mathbf{j} - (99.84 \text{ N})\mathbf{k}$$

$$\mathbf{M}_C = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1200 & -150 & 900 \\ -16.64 & 299.5 & -99.84 \end{vmatrix} \text{ N} \cdot \text{m}$$

$$= -(254574 \text{ N} \cdot \text{mm})\mathbf{i} + (104832 \text{ N} \cdot \text{mm})\mathbf{j} + (356904 \text{ N} \cdot \text{mm})\mathbf{k}$$

$$\text{or } \mathbf{M}_C = -(255 \text{ N} \cdot \text{m})\mathbf{i} + (104.8 \text{ N} \cdot \text{m})\mathbf{j} + (357 \text{ N} \cdot \text{m})\mathbf{k} \quad \blacktriangleleft$$





PROBLEM 3.24

The ramp $ABCD$ is supported by cables at corners C and D . The tension in each of the cables is 810 N . Determine the moment about A of the force exerted by (a) the cable at D , (b) the cable at C .

SOLUTION

(a) We have

$$\mathbf{M}_A = \mathbf{r}_{E/A} \times \mathbf{T}_{DE}$$

where

$$\mathbf{r}_{E/A} = (2.3 \text{ m})\mathbf{j}$$

$$\mathbf{T}_{DE} = \lambda_{DE} T_{DE}$$

$$= \frac{(0.6 \text{ m})\mathbf{i} + (3.3 \text{ m})\mathbf{j} - (3 \text{ m})\mathbf{k}}{\sqrt{(0.6)^2 + (3.3)^2 + (3)^2} \text{ m}} (810 \text{ N})$$

$$= (108 \text{ N})\mathbf{i} + (594 \text{ N})\mathbf{j} - (540 \text{ N})\mathbf{k}$$

$$\mathbf{M}_A = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2.3 & 0 \\ 108 & 594 & -540 \end{vmatrix} \text{ N} \cdot \text{m}$$

$$= -(1242 \text{ N} \cdot \text{m})\mathbf{i} - (248.4 \text{ N} \cdot \text{m})\mathbf{k}$$

$$\text{or } \mathbf{M}_A = -(1242 \text{ N} \cdot \text{m})\mathbf{i} - (248 \text{ N} \cdot \text{m})\mathbf{k} \quad \blacktriangleleft$$

(b) We have

$$\mathbf{M}_A = \mathbf{r}_{G/A} \times \mathbf{T}_{CG}$$

where

$$\mathbf{r}_{G/A} = (2.7 \text{ m})\mathbf{i} + (2.3 \text{ m})\mathbf{j}$$

$$\mathbf{T}_{CG} = \lambda_{CG} T_{CG}$$

$$= \frac{-(0.6 \text{ m})\mathbf{i} + (3.3 \text{ m})\mathbf{j} - (3 \text{ m})\mathbf{k}}{\sqrt{(0.6)^2 + (3.3)^2 + (3)^2} \text{ m}} (810 \text{ N})$$

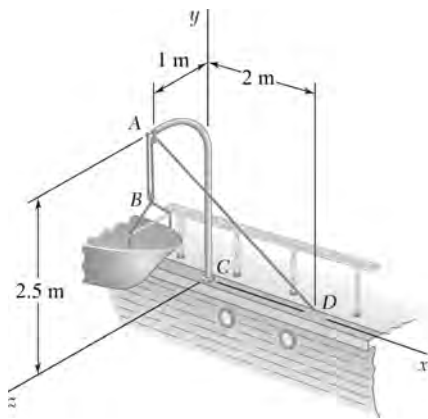
$$= -(108 \text{ N})\mathbf{i} + (594 \text{ N})\mathbf{j} - (540 \text{ N})\mathbf{k}$$

$$\mathbf{M}_A = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2.7 & 2.3 & 0 \\ -108 & 594 & -540 \end{vmatrix} \text{ N} \cdot \text{m}$$

$$= -(1242 \text{ N} \cdot \text{m})\mathbf{i} + (1458 \text{ N} \cdot \text{m})\mathbf{j} + (1852 \text{ N} \cdot \text{m})\mathbf{k}$$

$$\text{or } \mathbf{M}_A = -(1242 \text{ N} \cdot \text{m})\mathbf{i} + (1458 \text{ N} \cdot \text{m})\mathbf{j} + (1852 \text{ N} \cdot \text{m})\mathbf{k} \quad \blacktriangleleft$$

PROPRIETARY MATERIAL. © 2010 The McGraw-Hill Companies, Inc. All rights reserved. No part of this Manual may be displayed, reproduced or distributed in any form or by any means, without the prior written permission of the publisher, or used beyond the limited distribution to teachers and educators permitted by McGraw-Hill for their individual course preparation. If you are a student using this Manual, you are using it without permission.



PROBLEM 3.25

A small boat hangs from two davits, one of which is shown in the figure. The tension in line AD is 410 N. Determine the moment about C of the resultant force $\mathbf{R}_A$ exerted on the davit at A .

SOLUTION

We have

$$\mathbf{R}_A = 2\mathbf{F}_{AB} + \mathbf{F}_{AD}$$

where

$$\mathbf{F}_{AB} = -(410 \text{ N})\mathbf{j}$$

and

$$\mathbf{F}_{AD} = F_{AD} \frac{\overrightarrow{AD}}{AD} = (410 \text{ N}) \frac{(2\mathbf{i} - 2.5\mathbf{j} - \mathbf{k})}{\sqrt{(2)^2 + (2.25)^2 + (1)^2}}$$

$$\mathbf{F}_{AD} = (258.5\text{N})\mathbf{i} - (323.1\text{N})\mathbf{j} - (129.25\text{N})\mathbf{k}$$

Thus

$$\mathbf{R}_A = 2\mathbf{F}_{AB} + \mathbf{F}_{AD} = (258.5\text{N})\mathbf{i} - (1143.1\text{N})\mathbf{j} - (129.25\text{N})\mathbf{k}$$

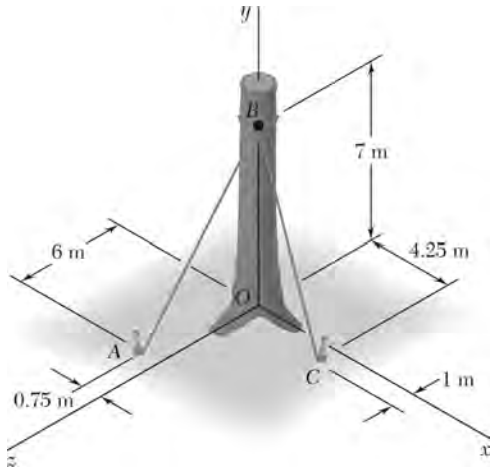
Also

$$\mathbf{r}_{A/C} = (2.5\text{m})\mathbf{j} + (1\text{m})\mathbf{k}$$

Using Eq. (3.21):

$$\begin{aligned} \mathbf{M}_C &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2.5 & 1 \\ 258.5 & -1143.1 & -129.25 \end{vmatrix} \\ &= (820 \text{ N} \cdot \text{m})\mathbf{i} + (259 \text{ N} \cdot \text{m})\mathbf{j} - (646 \text{ N} \cdot \text{m})\mathbf{k} \end{aligned}$$

$$\mathbf{M}_C = (820 \text{ N} \cdot \text{m})\mathbf{i} + (259 \text{ N} \cdot \text{m})\mathbf{j} - (646 \text{ N} \cdot \text{m})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 3.26

In Sample Problem 3.5, determine the perpendicular distance from Point O to cable AB .

SAMPLE PROBLEM 3.5 Before the trunk of a large tree is felled, cables AB and BC are attached as shown. Knowing that the tensions in cables AB and BC are 555 N and 660 N, respectively, determine the moment about O of the resultant force exerted on the tree by the cables at B .

SOLUTION

We have

$$|\mathbf{M}_O| = T_{BA}d$$

where

d = perpendicular distance from O to line AB .

Now

$$\mathbf{M}_O = \mathbf{r}_{B/O} \times \mathbf{T}_{BA}$$

and

$$\mathbf{r}_{B/O} = (7 \text{ m})\mathbf{j}$$

$$\mathbf{T}_{BA} = \lambda_{BA}T_{AB}$$

$$= \frac{-(0.75 \text{ m})\mathbf{i} - (7 \text{ m})\mathbf{j} + (6 \text{ m})\mathbf{k}}{\sqrt{(0.75)^2 + (7)^2 + (6)^2} \text{ m}} (555 \text{ N})$$

$$= -(45.0 \text{ N})\mathbf{i} - (420 \text{ N})\mathbf{j} + (360 \text{ N})\mathbf{k}$$

$$\mathbf{M}_O = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 7 & 0 \\ -45 & -420 & 360 \end{vmatrix} \text{ N} \cdot \text{m}$$

$$= (2520.0 \text{ N} \cdot \text{m})\mathbf{i} + (315.00 \text{ N} \cdot \text{m})\mathbf{k}$$

and

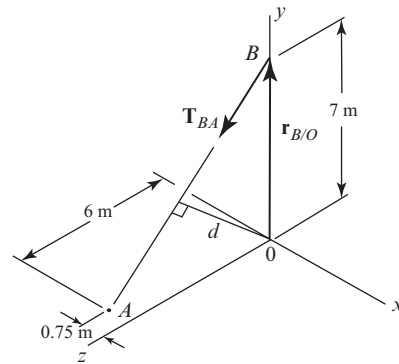
$$|\mathbf{M}_O| = \sqrt{(2520.0)^2 + (315.00)^2}$$

$$= 2539.6 \text{ N} \cdot \text{m}$$

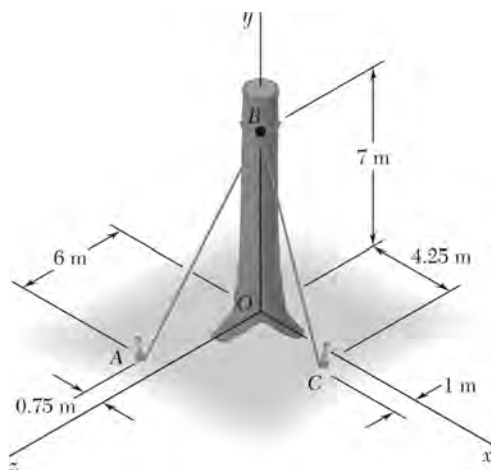
$$2539.6 \text{ N} \cdot \text{m} = (555 \text{ N})d$$

or

$$d = 4.5759 \text{ m}$$



$$\text{or } d = 4.58 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.27

In Sample Problem 3.5, determine the perpendicular distance from Point O to cable BC .

SAMPLE PROBLEM 3.5 Before the trunk of a large tree is felled, cables AB and BC are attached as shown. Knowing that the tensions in cables AB and BC are 555 N and 660 N, respectively, determine the moment about O of the resultant force exerted on the tree by the cables at B .

SOLUTION

We have

$$|\mathbf{M}_O| = T_{BC}d$$

where

d = perpendicular distance from O to line BC .

$$\mathbf{M}_O = \mathbf{r}_{B/O} \times \mathbf{T}_{BC}$$

$$\mathbf{r}_{B/O} = 7 \text{ m} \mathbf{j}$$

$$\mathbf{T}_{BC} = \lambda_{BC} T_{BC}$$

$$= \frac{(4.25 \text{ m})\mathbf{i} - (7 \text{ m})\mathbf{j} + (1 \text{ m})\mathbf{k}}{\sqrt{(4.25)^2 + (7)^2 + (1)^2} \text{ m}} (660 \text{ N})$$

$$= (340 \text{ N})\mathbf{i} - (560 \text{ N})\mathbf{j} + (80 \text{ N})\mathbf{k}$$

$$\mathbf{M}_O = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 7 & 0 \\ 340 & -560 & 80 \end{vmatrix}$$

$$= (560 \text{ N} \cdot \text{m})\mathbf{i} - (2380 \text{ N} \cdot \text{m})\mathbf{k}$$

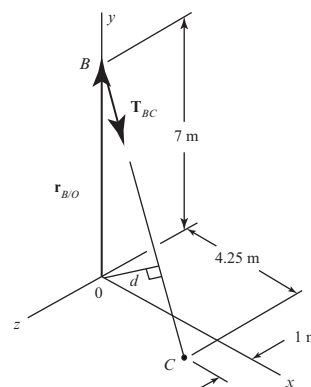
and

$$|\mathbf{M}_O| = \sqrt{(560)^2 + (2380)^2}$$

$$= 2445.0 \text{ N} \cdot \text{m}$$

$$2445.0 \text{ N} \cdot \text{m} = (660 \text{ N})d$$

$$d = 3.7045 \text{ m}$$

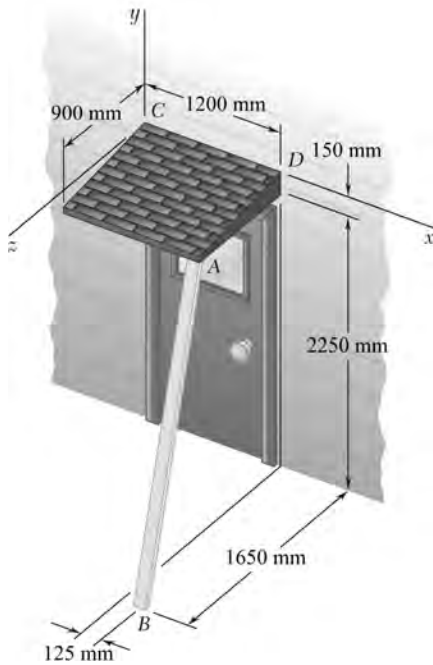


$$\text{or } d = 3.70 \text{ m} \blacktriangleleft$$

PROBLEM 3.28

In Problem 3.23, determine the perpendicular distance from Point D to a line drawn through Points A and B .

PROBLEM 3.23 A wooden board AB , which is used as a temporary prop to support a small roof, exerts at Point A of the roof a 300 N force directed along BA . Determine the moment about C of that force.



SOLUTION

We have

$$|\mathbf{M}_D| = F_{BA}d$$

where

d = perpendicular distance from D to line AB .

$$\mathbf{M}_D = \mathbf{r}_{A/D} \times \mathbf{F}_{BA}$$

$$\mathbf{r}_{A/D} = -(150 \text{ mm})\mathbf{j} + (900 \text{ mm})\mathbf{k}$$

$$\mathbf{F}_{BA} = \lambda_{BA} F_{BA}$$

$$= \left[\frac{-(125 \text{ mm})\mathbf{i} + (2250 \text{ mm})\mathbf{j} - (750 \text{ mm})\mathbf{k}}{\sqrt{(125)^2 + (2250)^2 + (30)^2}} \right] (300 \text{ N})$$

$$= -(16.64 \text{ N})\mathbf{i} + (299.5 \text{ N})\mathbf{j} - (99.84 \text{ N})\mathbf{k}$$

$$\mathbf{M}_D = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -150 & 900 \\ -16.64 & 299.5 & -99.84 \end{vmatrix} \text{ N} \cdot \text{m}$$

$$= -(254574 \text{ N} \cdot \text{mm})\mathbf{i} - (14976 \text{ N} \cdot \text{mm})\mathbf{j} - (2496 \text{ N} \cdot \text{mm})\mathbf{k}$$

and

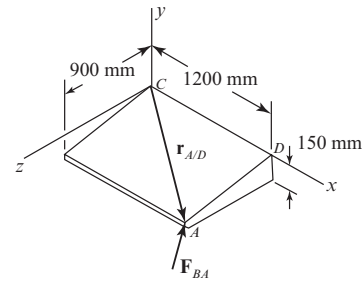
$$|\overline{\mathbf{M}}_D| = \sqrt{(254574)^2 + (14976)^2 + (2496)^2}$$

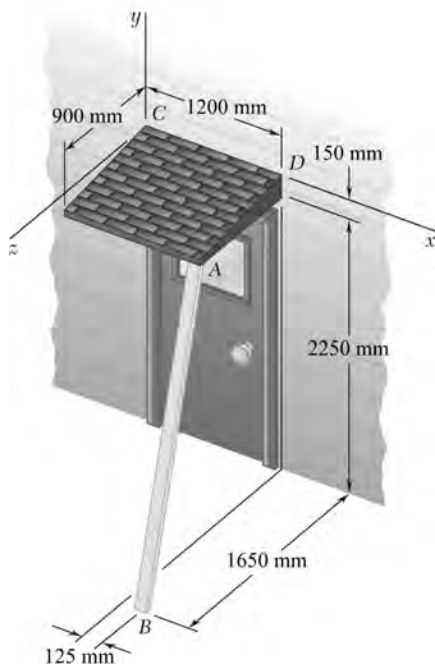
$$= 255026 \text{ N} \cdot \text{mm}$$

$$255026 \text{ N} \cdot \text{mm} = (300 \text{ N})d$$

$$d = 850 \text{ mm}$$

$$\text{or } d = 850 \text{ mm} \quad \blacktriangleleft$$





PROBLEM 3.29

In Problem 3.23, determine the perpendicular distance from Point C to a line drawn through Points A and B.

PROBLEM 3.23 A wooden board AB, which is used as a temporary prop to support a small roof, exerts at Point A of the roof a 300 N force directed along BA. Determine the moment about C of that force.

SOLUTION

We have

$$|\mathbf{M}_C| = F_{BA}d$$

where

d = perpendicular distance from C to line AB.

$$\mathbf{M}_C = \mathbf{r}_{A/C} \times \mathbf{F}_{BA}$$

$$\mathbf{r}_{A/C} = (1200 \text{ mm})\mathbf{i} - (150 \text{ mm})\mathbf{j} + (900 \text{ mm})\mathbf{k}$$

$$\mathbf{F}_{BA} = \lambda_{BA} F_{BA}$$

$$= \left[\frac{-(125 \text{ mm})\mathbf{i} + (2250 \text{ mm})\mathbf{j} - (750 \text{ mm})\mathbf{k}}{\sqrt{(125)^2 + (2250)^2 + (30)^2}} \right] (300 \text{ N})$$

$$= -(16.64 \text{ N})\mathbf{i} + (299.5 \text{ N})\mathbf{j} - (99.84 \text{ N})\mathbf{k}$$

$$\mathbf{M}_C = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1200 & -150 & 900 \\ -16.64 & 299.5 & -99.84 \end{vmatrix} \text{ N} \cdot \text{m}$$

$$= -(254574 \text{ N} \cdot \text{mm})\mathbf{i} + (104832 \text{ N} \cdot \text{mm})\mathbf{j} + (356904 \text{ N} \cdot \text{mm})\mathbf{k}$$

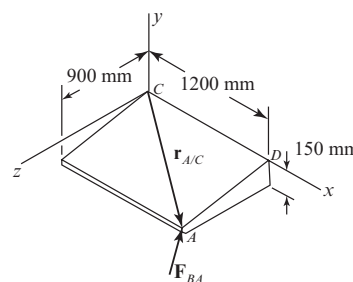
and

$$|\overline{\mathbf{M}}_C| = \sqrt{(254574)^2 + (104832)^2 + (356904)^2}$$

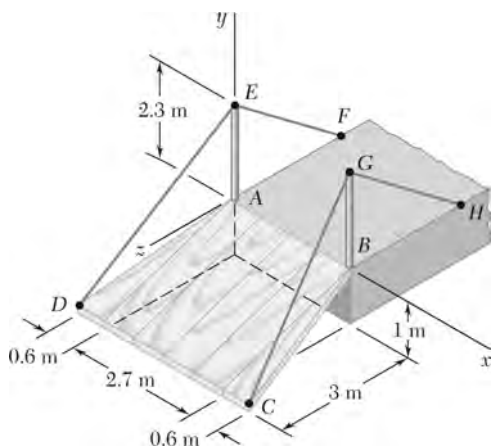
$$= 450753 \text{ N} \cdot \text{mm}$$

$$450753 = (300 \text{ N})d$$

$$d = 1503 \text{ mm}$$



$$\text{or } d = 1.503 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.30

In Problem 3.24, determine the perpendicular distance from Point A to portion DE of cable DEF.

PROBLEM 3.24 The ramp ABCD is supported by cables at corners C and D. The tension in each of the cables is 810 N. Determine the moment about A of the force exerted by (a) the cable at D, (b) the cable at C.

SOLUTION

We have

$$|\mathbf{M}_A| = T_{DE}d$$

where

d = perpendicular distance from A to line DE.

$$\mathbf{M}_A = \mathbf{r}_{E/A} \times \mathbf{T}_{DE}$$

$$\mathbf{r}_{E/A} = (2.3 \text{ m})\mathbf{j}$$

$$\mathbf{T}_{DE} = \lambda_{DE} \mathbf{T}_{DE}$$

$$= \frac{(0.6 \text{ m})\mathbf{i} + (3.3 \text{ m})\mathbf{j} - (3 \text{ m})\mathbf{k}}{\sqrt{(0.6)^2 + (3.3)^2 + (3)^2} \text{ m}} (810 \text{ N})$$

$$= (108 \text{ N})\mathbf{i} + (594 \text{ N})\mathbf{j} - (540 \text{ N})\mathbf{k}$$

$$\mathbf{M}_A = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2.3 & 0 \\ 108 & 594 & 540 \end{vmatrix} \text{ N} \cdot \text{m}$$

$$= -(1242.00 \text{ N} \cdot \text{m})\mathbf{i} - (248.00 \text{ N} \cdot \text{m})\mathbf{k}$$

and

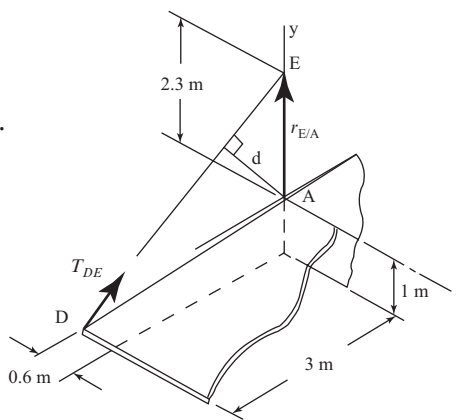
$$|\mathbf{M}_A| = \sqrt{(1242.00)^2 + (248.00)^2}$$

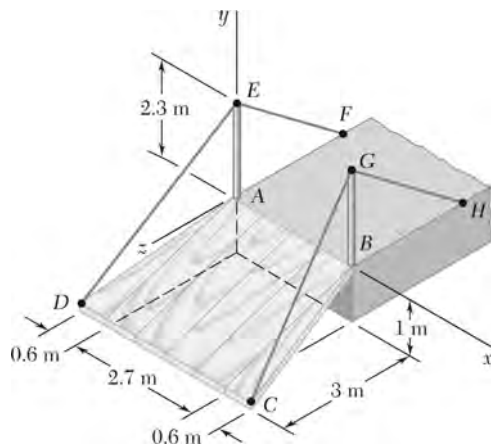
$$= 1266.52 \text{ N} \cdot \text{m}$$

$$1266.52 \text{ N} \cdot \text{m} = (810 \text{ N})d$$

$$d = 1.56360 \text{ m}$$

$$\text{or } d = 1.564 \text{ m} \quad \blacktriangleleft$$





PROBLEM 3.31

In Problem 3.24, determine the perpendicular distance from Point A to a line drawn through Points C and G.

PROBLEM 3.24 The ramp ABCD is supported by cables at corners C and D. The tension in each of the cables is 810 N. Determine the moment about A of the force exerted by (a) the cable at D, (b) the cable at C.

SOLUTION

We have

$$|\mathbf{M}_A| = T_{CG}d$$

where

d = perpendicular distance from A to line CG.

$$\mathbf{M}_A = \mathbf{r}_{G/A} \times \mathbf{T}_{CG}$$

$$\mathbf{r}_{G/A} = (2.7 \text{ m})\mathbf{i} + (2.3 \text{ m})\mathbf{j}$$

$$\begin{aligned} \mathbf{T}_{CG} &= \lambda_{CG} T_{CG} \\ &= \frac{-(0.6 \text{ m})\mathbf{i} + (3.3 \text{ m})\mathbf{j} - (3 \text{ m})\mathbf{k}}{\sqrt{(0.6)^2 + (3.3)^2 + (3)^2}} (810 \text{ N}) \\ &= -(108 \text{ N})\mathbf{i} + (594 \text{ N})\mathbf{j} - (540 \text{ N})\mathbf{k} \end{aligned}$$

$$\begin{aligned} \mathbf{M}_A &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2.7 & 2.3 & 0 \\ -108 & 594 & -540 \end{vmatrix} \text{ N} \cdot \text{m} \\ &= -(1242.00 \text{ N} \cdot \text{m})\mathbf{i} + (1458.00 \text{ N} \cdot \text{m})\mathbf{j} + (1852.00 \text{ N} \cdot \text{m})\mathbf{k} \end{aligned}$$

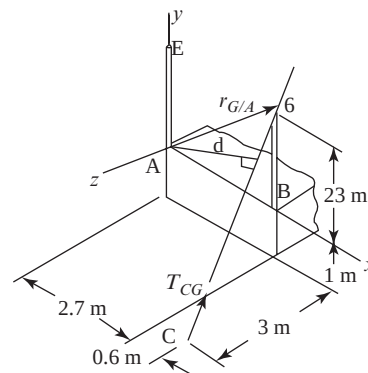
and

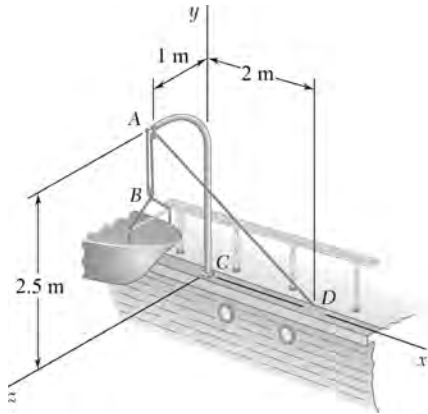
$$\begin{aligned} |\mathbf{M}_A| &= \sqrt{(1242.00)^2 + (1458.00)^2 + (1852.00)^2} \\ &= 2664.3 \text{ N} \cdot \text{m} \end{aligned}$$

$$2664.3 \text{ N} \cdot \text{m} = (810 \text{ N})d$$

$$d = 3.2893 \text{ m}$$

$$\text{or } d = 3.29 \text{ m} \quad \blacktriangleleft$$





PROBLEM 3.32

In Problem 3.25, determine the perpendicular distance from Point C to portion AD of the line $ABAD$.

PROBLEM 3.25 A small boat hangs from two davits, one of which is shown in the figure. The tension in line $ABAD$ is 410 N. Determine the moment about C of the resultant force $\mathbf{R}_A$ exerted on the davit at A .

SOLUTION

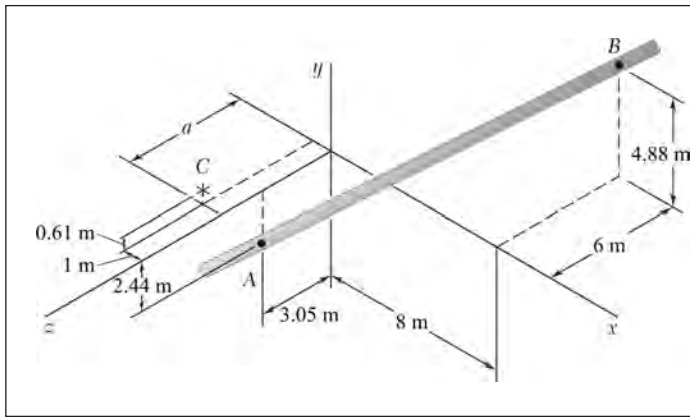
First compute the moment about C of the force $\mathbf{F}_{DA}$ exerted by the line on D :

$$\begin{aligned}\text{From Problem 3.25: } \mathbf{F}_{DA} &= -\mathbf{F}_{AD} \\ &= -(258.5 \text{ N})\mathbf{i} + (323.1 \text{ N})\mathbf{j} + (129.25 \text{ N})\mathbf{k} \\ \mathbf{M}_C &= \mathbf{r}_{D/C} \times \mathbf{F}_{DA} \\ &= +(2 \text{ m})\mathbf{i} \times [-(258.5 \text{ N})\mathbf{i} + (323.1 \text{ N})\mathbf{j} + (129.25 \text{ N})\mathbf{k}] \\ &= -(258.5 \text{ N} \cdot \text{m})\mathbf{j} + (646.2 \text{ N} \cdot \text{m})\mathbf{k} \\ \mathbf{M}_C &= \sqrt{(258.5)^2 + (646.2)^2} \\ &= 695.99 \text{ N} \cdot \text{m}\end{aligned}$$

$$\text{Then } \mathbf{M}_C = F_{DA} d$$

$$\begin{aligned}\text{Since } F_{DA} &= 410 \text{ N} \\ d &= \frac{M_C}{F_{DA}} \\ &= \frac{695.99 \text{ N} \cdot \text{m}}{410 \text{ N}}\end{aligned}$$

$$d = 1.698 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.33

Determine the value of a that minimises the perpendicular distance from Point C to a section of pipeline that passes through Points A and B.

SOLUTION

Assuming a force $\mathbf{F}$ acts along AB,

$$|\mathbf{M}_C| = |\mathbf{r}_{A/C} \times \mathbf{F}| = F(d)$$

where

d = perpendicular distance from C to line AB

$$\mathbf{F} = \lambda_{AB} \mathbf{F}$$

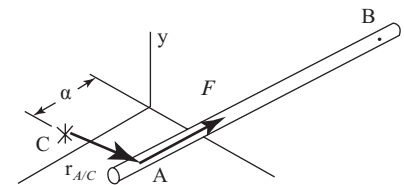
$$= \frac{(8\text{ m})\mathbf{i} + (7.32\text{ m})\mathbf{j} - (9.05\text{ m})\mathbf{k}}{\sqrt{(8)^2 + (7.32)^2 + (9.05)^2}}$$

$$= F(0.5664\mathbf{i} + 0.5183\mathbf{j} - 0.6407\mathbf{k})$$

$$\mathbf{r}_{A/C} = (1\text{ m})\mathbf{i} - (3.05\text{ m})\mathbf{j} - (a - 3.05\text{ m})\mathbf{k}$$

$$\mathbf{M}_C = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -3.05 & 3.05 - a \\ 0.5664 & 0.5183 & -0.6407 \end{vmatrix} F$$

$$= [(0.37332 + 0.5183a)\mathbf{i} + (2.36822 - 0.5664a)\mathbf{j} + (2.24582)\mathbf{k}] F$$



Since

$$|\mathbf{M}_C| = \sqrt{|\mathbf{r}_{A/C} \times \mathbf{F}|^2} \quad \text{or} \quad |\mathbf{r}_{A/C} \times \mathbf{F}| = (dF)^2$$

$$(0.3733 + 0.5183a)^2 + (2.368 - 0.5664a)^2 + (2.246)^2 = d^2$$

Setting $\frac{d}{da}(d^2) = 0$ to find a to minimize d

$$[2(0.5183)(0.3733 + 0.5183a) + 2(-0.5664)(2.368 - 0.5664a)] = 0$$

Solving

$$a = 1.947 \text{ m}$$

$$\text{or } a = 1.947 \text{ m} \quad \blacktriangleleft$$

PROBLEM 3.34

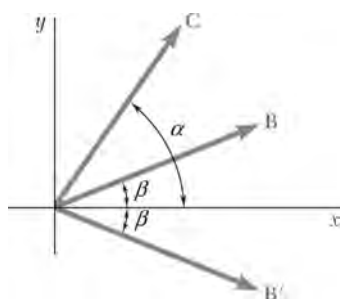
Given the vectors $\mathbf{P} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $\mathbf{Q} = 4\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$, and $\mathbf{S} = -2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$, compute the scalar products $\mathbf{P} \cdot \mathbf{Q}$, $\mathbf{P} \cdot \mathbf{S}$, and $\mathbf{Q} \cdot \mathbf{S}$.

SOLUTION

$$\begin{aligned}\mathbf{P} \cdot \mathbf{Q} &= (3\mathbf{i} - \mathbf{j} + 2\mathbf{k}) \cdot (4\mathbf{i} - 5\mathbf{j} - 3\mathbf{k}) \\ &= (3)(4) + (-1)(-5) + (2)(-3) \\ &= 1\end{aligned}\qquad \text{or} \qquad \mathbf{P} \cdot \mathbf{Q} = 1 \quad \blacktriangleleft$$

$$\begin{aligned}\mathbf{P} \cdot \mathbf{S} &= (3\mathbf{i} - \mathbf{j} + 2\mathbf{k}) \cdot (-2\mathbf{i} + 3\mathbf{j} - \mathbf{k}) \\ &= (3)(-2) + (-1)(3) + (2)(-1) \\ &= -11\end{aligned}\qquad \text{or} \qquad \mathbf{P} \cdot \mathbf{S} = -11 \quad \blacktriangleleft$$

$$\begin{aligned}\mathbf{Q} \cdot \mathbf{S} &= (4\mathbf{i} - 5\mathbf{j} - 3\mathbf{k}) \cdot (-2\mathbf{i} + 3\mathbf{j} - \mathbf{k}) \\ &= (4)(-2) + (5)(3) + (-3)(-1) \\ &= 10\end{aligned}\qquad \text{or} \qquad \mathbf{Q} \cdot \mathbf{S} = 10 \quad \blacktriangleleft$$

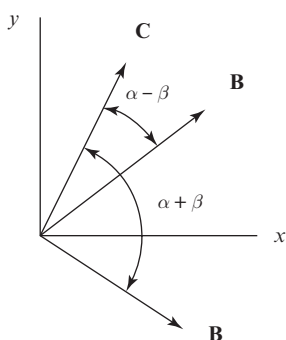


PROBLEM 3.35

Form the scalar products $\mathbf{B} \cdot \mathbf{C}$ and $\mathbf{B}' \cdot \mathbf{C}$, where $B = B'$, and use the results obtained to prove the identity

$$\cos \alpha \cos \beta = \frac{1}{2} \cos(\alpha + \beta) + \frac{1}{2} \cos(\alpha - \beta).$$

SOLUTION



By definition $\mathbf{B} \cdot \mathbf{C} = BC \cos(\alpha - \beta)$

where

$$\mathbf{B} = B[(\cos \beta)\mathbf{i} + (\sin \beta)\mathbf{j}]$$

$$\mathbf{C} = C[(\cos \alpha)\mathbf{i} + (\sin \alpha)\mathbf{j}]$$

$$(B \cos \beta)(C \cos \alpha) + (B \sin \beta)(C \sin \alpha) = BC \cos(\alpha - \beta)$$

or

$$\cos \beta \cos \alpha + \sin \beta \sin \alpha = \cos(\alpha - \beta) \quad (1)$$

By definition $\mathbf{B}' \cdot \mathbf{C} = BC \cos(\alpha + \beta)$

where

$$\mathbf{B}' = [(\cos \beta)\mathbf{i} - (\sin \beta)\mathbf{j}]$$

$$(B \cos \beta)(C \cos \alpha) + (-B \sin \beta)(C \sin \alpha) = BC \cos(\alpha + \beta)$$

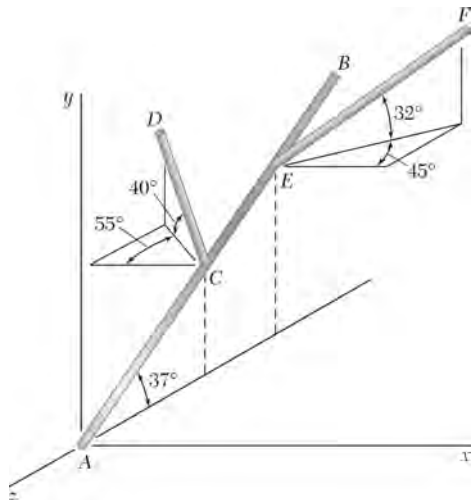
or

$$\cos \beta \cos \alpha - \sin \beta \sin \alpha = \cos(\alpha + \beta) \quad (2)$$

Adding Equations (1) and (2),

$$2 \cos \beta \cos \alpha = \cos(\alpha - \beta) + \cos(\alpha + \beta)$$

$$\text{or } \cos \alpha \cos \beta = \frac{1}{2} \cos(\alpha + \beta) + \frac{1}{2} \cos(\alpha - \beta) \quad \blacktriangleleft$$



PROBLEM 3.36

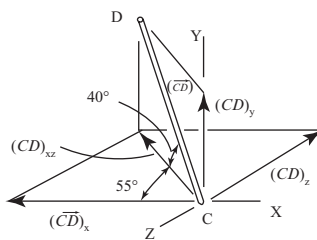
Section AB of a pipeline lies in the yz plane and forms an angle of 37° with the z axis. Branch lines CD and EF join AB as shown. Determine the angle formed by pipes AB and CD .

SOLUTION

First note

$$\overline{AB} = AB(\sin 37^\circ \mathbf{j} - \cos 37^\circ \mathbf{k})$$

$$\overline{CD} = CD(-\cos 40^\circ \cos 55^\circ \mathbf{j} + \sin 40^\circ \mathbf{j} - \cos 40^\circ \sin 55^\circ \mathbf{k})$$



Now

$$\overline{AB} \cdot \overline{CD} = (AB)(CD) \cos \theta$$

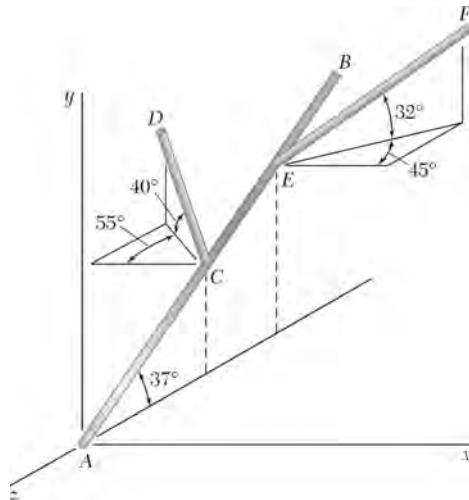
or

$$\begin{aligned} AB(\sin 37^\circ \mathbf{j} - \cos 37^\circ \mathbf{k}) \cdot CD(-\cos 40^\circ \cos 55^\circ \mathbf{j} + \sin 40^\circ \mathbf{j} - \cos 40^\circ \sin 55^\circ \mathbf{k}) \\ = (AB)(CD) \cos \theta \end{aligned}$$

or

$$\begin{aligned} \cos \theta &= (\sin 37^\circ)(\sin 40^\circ) + (-\cos 37^\circ)(-\cos 40^\circ \sin 55^\circ) \\ &= 0.88799 \end{aligned}$$

$$\text{or } \theta = 27.4^\circ \blacktriangleleft$$



PROBLEM 3.37

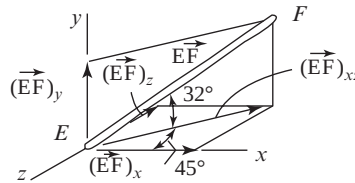
Section AB of a pipeline lies in the yz plane and forms an angle of 37° with the z axis. Branch lines CD and EF join AB as shown. Determine the angle formed by pipes AB and EF .

SOLUTION

First note

$$\overrightarrow{AB} = AB(\sin 37^\circ \mathbf{j} - \cos 37^\circ \mathbf{k})$$

$$\overrightarrow{EF} = EF(\cos 32^\circ \cos 45^\circ \mathbf{i} + \sin 32^\circ \mathbf{j} - \cos 32^\circ \sin 45^\circ \mathbf{k})$$



Now

$$\overrightarrow{AB} \cdot \overrightarrow{EF} = (AB)(EF) \cos \theta$$

or

$$AB(\sin 37^\circ \mathbf{j} - \cos 37^\circ \mathbf{k}) \cdot EF(\cos 32^\circ \cos 45^\circ \mathbf{j} + \sin 32^\circ \mathbf{j} - \cos 32^\circ \sin 45^\circ \mathbf{k})$$

$$= (AB)(EF) \cos \theta$$

or

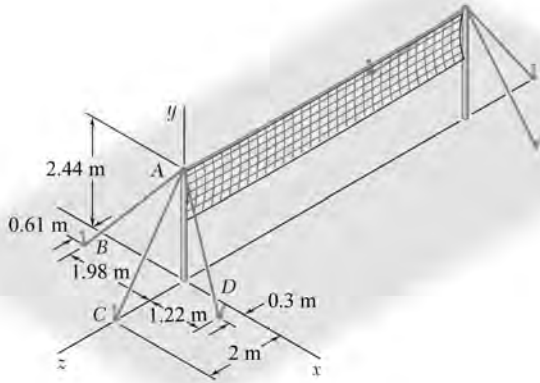
$$\cos \theta = (\sin 37^\circ)(\sin 32^\circ) + (-\cos 37^\circ)(-\cos 32^\circ \sin 45^\circ)$$

$$= 0.79782$$

$$\text{or } \theta = 37.1^\circ \blacktriangleleft$$

PROBLEM 3.38

Consider the volleyball net shown. Determine the angle formed by guy wires AB and AC.



SOLUTION

First note

$$AB = \sqrt{(-1.98)^2 + (-2.44)^2 + (0.61)^2}$$

$$= 3.2 \text{ m}$$

$$AC = \sqrt{(0)^2 + (-2.44)^2 + (2)^2}$$

$$= 3.155 \text{ m}$$

and

$$\overrightarrow{AB} = -(1.98 \text{ m})\mathbf{i} - (2.44 \text{ m})\mathbf{j} + (0.61 \text{ m})\mathbf{k}$$

$$\overrightarrow{AC} = -(2.44 \text{ m})\mathbf{j} + (2 \text{ m})\mathbf{k}$$

By definition

$$\overrightarrow{AB} \cdot \overrightarrow{AC} = (AB)(AC) \cos \theta$$

or

$$[-(1.98)\mathbf{i} - (2.44)\mathbf{j} + (0.61)\mathbf{k}] \cdot [-(2.44)\mathbf{j} + (2)\mathbf{k}] = (3.2)(3.155) \cos \theta$$

$$[(-2.44) \times (-2.44)] + [(0.61)(2)] = 10.096 \cos \theta$$

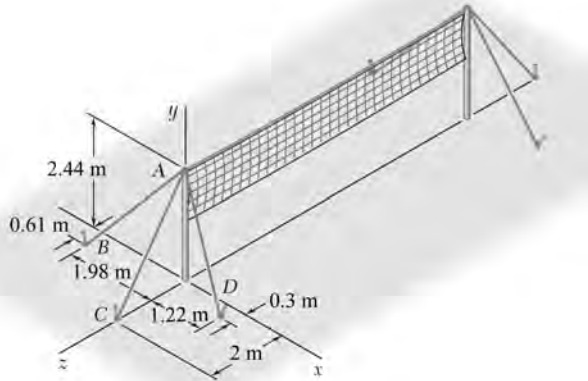
or

$$\cos \theta = 0.711$$

$$\text{or } \theta = 44.7^\circ \blacktriangleleft$$

PROBLEM 3.39

Consider the volleyball net shown. Determine the angle formed by guy wires AC and AD.



SOLUTION

First note

$$AC = \sqrt{(0)^2 + (-2.44)^2 + (2)^2} \\ = 3.155 \text{ m}$$

$$AD = \sqrt{(1.22)^2 + (-2.44)^2 + (0.3)^2} \\ = 2.744 \text{ m}$$

and

$$\overrightarrow{AC} = (-2.44 \text{ m})\mathbf{j} + (2 \text{ m})\mathbf{k}$$

$$\overrightarrow{AD} = (1.22 \text{ m})\mathbf{i} - (2.44 \text{ m})\mathbf{j} + (0.3 \text{ m})\mathbf{k}$$

By definition

$$\overrightarrow{AC} \cdot \overrightarrow{AD} = (AC)(AD) \cos \theta$$

or

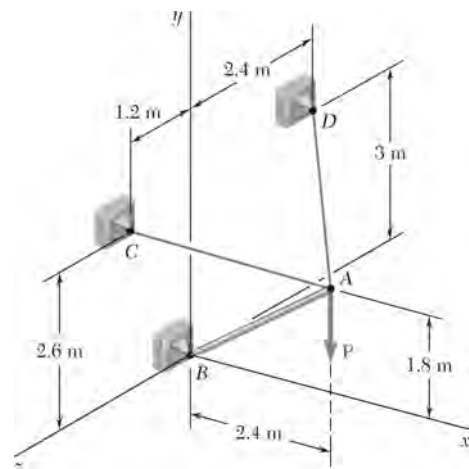
$$(-2.44\mathbf{j} + 2\mathbf{k}) \cdot (1.22\mathbf{i} - 2.44\mathbf{j} + 0.3\mathbf{k}) = (3.155)(2.744) \cos \theta$$

$$(-2.44)(-2.44) + 2(0.3) = 8.6573 \cos \theta$$

or

$$\cos \theta = 0.757$$

$$\text{or } \theta = 40.8^\circ \quad \blacktriangleleft$$



PROBLEM 3.40

Knowing that the tension in cable AC is 1260 N, determine (a) the angle between cable AC and the boom AB, (b) the projection on AB of the force exerted by cable AC at Point A.

SOLUTION

(a) First note

$$AC = \sqrt{(-2.4)^2 + (0.8)^2 + (1.2)^2} \\ = 2.8 \text{ m}$$

$$AB = \sqrt{(-2.4)^2 + (-1.8)^2 + (0)^2} \\ = 3.0 \text{ m}$$

and

$$\overrightarrow{AC} = -(2.4 \text{ m})\mathbf{i} + (0.8 \text{ m})\mathbf{j} + (1.2 \text{ m})\mathbf{k}$$

$$\overrightarrow{AB} = -(2.4 \text{ m})\mathbf{i} - (1.8 \text{ m})\mathbf{j}$$

By definition $\overrightarrow{AC} \cdot \overrightarrow{AB} = (AC)(AB)\cos\theta$

$$\text{or } (-2.4\mathbf{i} + 0.8\mathbf{j} + 1.2\mathbf{k}) \cdot (-2.4\mathbf{i} - 1.8\mathbf{j}) = (2.8)(3.0) \times \cos\theta$$

$$\text{or } (-2.4)(-2.4) + (0.8)(-1.8) + (1.2)(0) = 8.4 \cos\theta$$

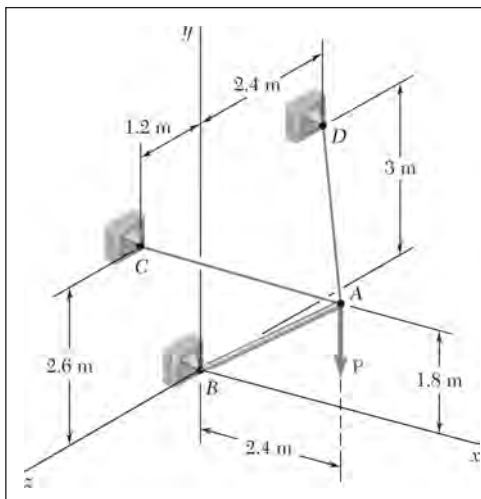
$$\text{or } \cos\theta = 0.51429$$

$$\text{or } \theta = 59.0^\circ \blacktriangleleft$$

(b) We have

$$(T_{AC})_{AB} = T_{AC} \cdot \lambda_{AB} \\ = T_{AC} \cos\theta \\ = (1260 \text{ N})(0.51429)$$

$$\text{or } (T_{AC})_{AB} = 648 \text{ N} \blacktriangleleft$$



PROBLEM 3.41

Knowing that the tension in cable AD is 405 N, determine (a) the angle between cable AD and the boom AB , (b) the projection on AB of the force exerted by cable AD at Point A .

SOLUTION

(a) First note

$$AD = \sqrt{(-2.4)^2 + (1.2)^2 + (-2.4)^2} \\ = 3.6 \text{ m}$$

$$AB = \sqrt{(-2.4)^2 + (-1.8)^2 + (0)^2} \\ = 3.0 \text{ m}$$

and

$$\mathbf{AD} = -(2.4 \text{ m})\mathbf{i} + (1.2 \text{ m})\mathbf{j} - (2.4 \text{ m})\mathbf{k}$$

$$\mathbf{AB} = -(2.4 \text{ m})\mathbf{i} - (1.8 \text{ m})\mathbf{j}$$

By definition,

$$\mathbf{AD} \cdot \mathbf{AB} = (AD)(AB) \cos \theta$$

$$(-2.4\mathbf{i} + 1.2\mathbf{j} - 2.4\mathbf{k}) \cdot (-2.4\mathbf{i} - 1.8\mathbf{j}) = (3.6)(3.0) \cos \theta$$

$$(-2.4)(-2.4) + (1.2)(-1.8) + (-2.4)(0) = 10.8 \cos \theta$$

$$\cos \theta = \frac{1}{3}$$

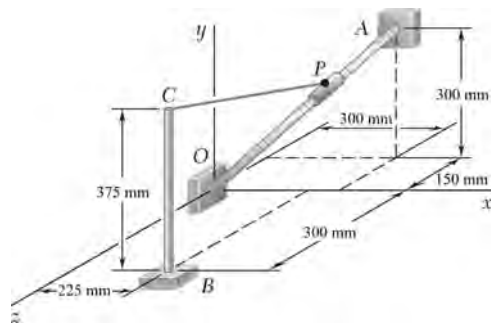
$$\theta = 70.5^\circ \quad \blacktriangleleft$$

(b)

$$(T_{AD})_{AB} = T_{AD} \cdot \lambda_{AB} \\ = T_{AD} \cos \theta$$

$$= (405 \text{ N}) \left(\frac{1}{3} \right)$$

$$(T_{AD})_{AB} = 135.0 \text{ N} \quad \blacktriangleleft$$



PROBLEM 3.42

Slider P can move along rod OA . An elastic cord PC is attached to the slider and to the vertical member BC . Knowing that the distance from O to P is 150 mm and that the tension in the cord is 15 N, determine (a) the angle between the elastic cord and the rod OA , (b) the projection on OA of the force exerted by cord PC at Point P .

SOLUTION

First note $OA = \sqrt{(300)^2 + (300)^2 + (-150)^2}$
 $= 450 \text{ mm}$

Then $\lambda_{OA} = \frac{\overline{OA}}{OA} = \frac{1}{450}(300\mathbf{i} + 300\mathbf{j} - 150\mathbf{k})$
 $= \frac{1}{3}(2\mathbf{i} + 2\mathbf{j} - \mathbf{k})$

Now $OP = 150 \text{ mm} \Rightarrow OP = \frac{1}{3}(OA)$

The coordinates of Point P are (100 mm, 100 mm, - 50 mm)

so that $\overline{PC} = (125 \text{ mm})\mathbf{i} + (275 \text{ mm})\mathbf{j} + (350 \text{ mm})\mathbf{k}$

and $PC = \sqrt{(125)^2 + (275)^2 + (350)^2} = 462.33 \text{ mm}$

(a) We have $\overline{PC} \cdot \lambda_{OA} = (PC) \cos \theta$

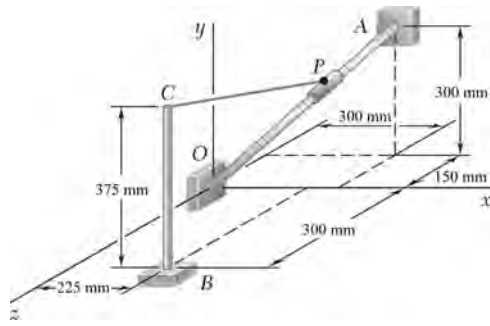
or $(125\mathbf{i} + 275\mathbf{j} + 350\mathbf{k}) \cdot \frac{1}{3}(2\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = 462.33 \cos \theta$

or $\cos \theta = 0.3244$

or $\theta = 71.1^\circ \blacktriangleleft$

(b) We have $(T_{PC})_{OA} = T_{PC} \cdot \lambda_{OA}$
 $= (T_{PC} \lambda_{PC}) \cdot \lambda_{OA}$
 $= T_{PC} \frac{\overline{PC}}{PC} \cdot \lambda_{OA}$
 $= T_{PC} \cos \theta$
 $= (15 \text{ N})(0.3244)$

or $(T_{PC})_{OA} = 4.87 \text{ N} \blacktriangleleft$



PROBLEM 3.43

Slider P can move along rod OA . An elastic cord PC is attached to the slider and to the vertical member BC . Determine the distance from O to P for which cord PC and rod OA are perpendicular.

SOLUTION

First note

$$OA = \sqrt{(300)^2 + (300)^2 + (-150)^2} \\ = 450 \text{ mm}$$

Then

$$\lambda_{OA} = \frac{\overline{OA}}{OA} = \frac{1}{450}(300\mathbf{i} + 300\mathbf{j} - 150\mathbf{k}) \\ = \frac{1}{3}(2\mathbf{i} + 2\mathbf{j} - \mathbf{k})$$

Let the coordinates of Point P be $(x \text{ mm}, y \text{ mm}, z \text{ mm})$. Then

$$\overline{PC} = [(225 - x)\mathbf{i} + (375 - y)\mathbf{j} + (300 - z)\mathbf{k}]$$

Also,

$$\overline{OP} = d_{OP}\lambda_{OA} = \frac{d_{OP}}{3}(2\mathbf{i} + 2\mathbf{j} - \mathbf{k})$$

and

$$\overline{OP} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

$$x = \frac{2}{3}d_{OP} \quad y = \frac{2}{3}d_{OP} \quad z = \frac{1}{3}d_{OP}$$

The requirement that OA and PC be perpendicular implies that

$$\lambda_{OA} \cdot \overline{PC} = 0$$

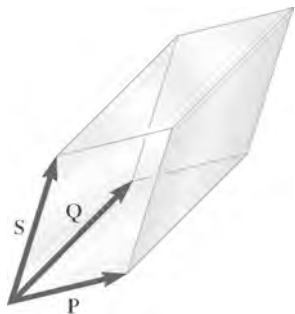
or

$$\frac{1}{3}(2\mathbf{i} + 2\mathbf{j} - \mathbf{k}) \cdot [(225 - x)\mathbf{i} + (375 - y)\mathbf{j} + (300 - z)\mathbf{k}] = 0$$

or

$$(2)\left(225 - \frac{2}{3}d_{OP}\right) + (2)\left(375 - \frac{2}{3}d_{OP}\right) + (-1)\left[300 - \left(-\frac{1}{3}d_{OP}\right)\right] = 0$$

$$\text{or } d_{OP} = 300 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 3.44

Determine the volume of the parallelepiped of Fig. 3.25 when (a) $\mathbf{P} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$, $\mathbf{Q} = -2\mathbf{i} - 5\mathbf{j} + \mathbf{k}$, and $\mathbf{S} = 7\mathbf{i} + \mathbf{j} - \mathbf{k}$, (b) $\mathbf{P} = 5\mathbf{i} - \mathbf{j} + 6\mathbf{k}$, $\mathbf{Q} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$, and $\mathbf{S} = -3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$.

SOLUTION

Volume of a parallelepiped is found using the mixed triple product.

$$\begin{aligned}
 (a) \quad \text{Vol} &= \mathbf{P} \cdot (\mathbf{Q} \times \mathbf{S}) \\
 &= \begin{vmatrix} 4 & -3 & 2 \\ -2 & -5 & 1 \\ 7 & 1 & -1 \end{vmatrix} \\
 &= (20 - 21 - 4 + 70 + 6 - 4) \\
 &= 67
 \end{aligned}$$

or Volume = 67.0 ◀

$$\begin{aligned}
 (b) \quad \text{Vol} &= \mathbf{P} \cdot (\mathbf{Q} \times \mathbf{S}) \\
 &= \begin{vmatrix} 5 & -1 & 6 \\ 2 & 3 & 1 \\ -3 & -2 & 4 \end{vmatrix} \\
 &= (60 + 3 - 24 + 54 + 8 + 10) \\
 &= 111
 \end{aligned}$$

or Volume = 111.0 ◀

PROBLEM 3.45

Given the vectors $\mathbf{P} = 4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$, $\mathbf{Q} = 2\mathbf{i} + 4\mathbf{j} - 5\mathbf{k}$, and $\mathbf{S} = S_x\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, determine the value of S_x for which the three vectors are coplanar.

SOLUTION

If $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{S}$ are coplanar, then $\mathbf{P}$ must be perpendicular to $(\mathbf{Q} \times \mathbf{S})$.

$$\mathbf{P} \cdot (\mathbf{Q} \times \mathbf{S}) = 0$$

(or, the volume of a parallelepiped defined by $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{S}$ is zero).

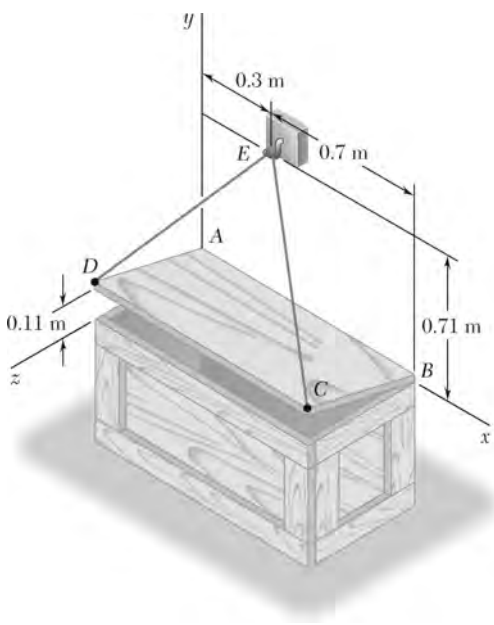
Then

$$\begin{vmatrix} 4 & -2 & 3 \\ 2 & 4 & -5 \\ S_x & -1 & 2 \end{vmatrix} = 0$$

or

$$32 + 10S_x - 6 - 20 + 8 - 12S_x = 0$$

$$S_x = 7 \quad \blacktriangleleft$$



PROBLEM 3.46

The 0.61×1.00 -m lid $ABCD$ of a storage bin is hinged along side AB and is held open by looping cord DEC over a frictionless hook at E . If the tension in the cord is 66 N, determine the moment about each of the coordinate axes of the force exerted by the cord at D .

SOLUTION

First note

$$z = \sqrt{(0.61)^2 - (0.11)^2} \\ = 0.60 \text{ m}$$

Then

$$d_{DE} = \sqrt{(0.3)^2 + (0.6)^2 + (-0.6)^2} \\ = 0.9 \text{ m}$$

and

$$\mathbf{T}_{DE} = \frac{66 \text{ N}}{0.9} (0.3\mathbf{i} + 0.6\mathbf{j} - 0.6\mathbf{k}) \\ = 22[(1 \text{ N})\mathbf{i} + (2 \text{ N})\mathbf{j} - (2 \text{ N})\mathbf{k}]$$

Now

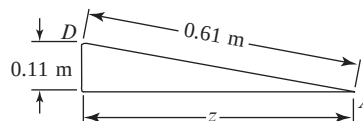
$$\mathbf{M}_A = \mathbf{r}_{D/A} \times \mathbf{T}_{DE}$$

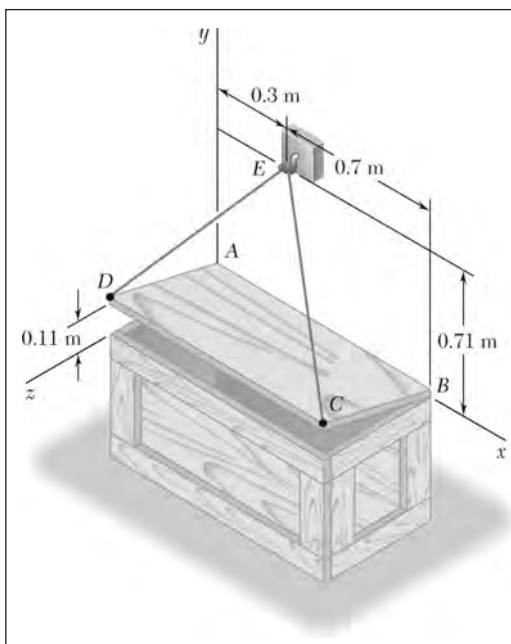
where

$$\mathbf{r}_{D/A} = (0.11 \text{ m})\mathbf{j} + (0.60 \text{ m})\mathbf{k}$$

Then

$$\mathbf{M}_A = 22 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0.11 & 0.60 \\ 1 & 2 & -2 \end{vmatrix} \\ = 22[(-0.22 - 1.20)\mathbf{i} + 0.60\mathbf{j} - 0.11\mathbf{k}] \\ = -(31.24 \text{ N} \cdot \text{m})\mathbf{i} + (13.20 \text{ N} \cdot \text{m})\mathbf{j} - (2.42 \text{ N} \cdot \text{m})\mathbf{k} \\ M_x = -31.2 \text{ N} \cdot \text{m}, \quad M_y = 13.20 \text{ N} \cdot \text{m}, \quad M_z = -2.42 \text{ N} \cdot \text{m} \blacktriangleleft$$





PROBLEM 3.47

The 0.61×1.00 -m lid $ABCD$ of a storage bin is hinged along-side AB and is held open by looping cord DEC over a frictionless hook at E . If the tension in the cord is 66 N, determine the moment about each of the coordinate axes of the force exerted by the cord at C .

SOLUTION

First note

$$z = \sqrt{(0.61)^2 - (0.11)^2} \\ = 0.60 \text{ m}$$

Then

$$d_{CE} = \sqrt{(-0.7)^2 + (0.6)^2 + (-0.6)^2} \\ = 1.1 \text{ m}$$

and

$$\mathbf{T}_{CE} = \frac{66 \text{ N}}{1.1} (-0.7\mathbf{i} + 0.6\mathbf{j} - 0.6\mathbf{k}) \\ = 6[-(7 \text{ N})\mathbf{i} + (6 \text{ N})\mathbf{j} - (6 \text{ N})\mathbf{k}]$$

Now

$$\mathbf{M}_A = \mathbf{r}_{E/A} \times \mathbf{T}_{CE}$$

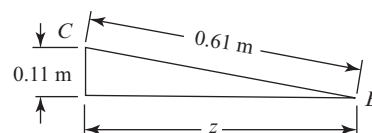
where

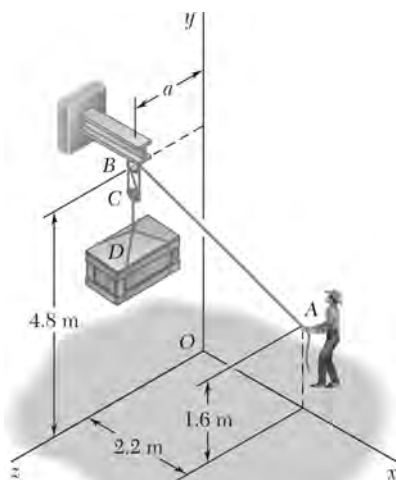
$$\mathbf{r}_{E/A} = (0.3 \text{ m})\mathbf{i} + (0.71 \text{ m})\mathbf{j}$$

Then

$$\mathbf{M}_A = 6 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.3 & 0.71 & 0 \\ -7 & 6 & -6 \end{vmatrix} \\ = 6[-4.26\mathbf{i} + 1.8\mathbf{j} + (1.8 + 4.97)\mathbf{k}] \\ = -(25.56 \text{ N} \cdot \text{m})\mathbf{i} + (10.80 \text{ N} \cdot \text{m})\mathbf{j} + (40.62 \text{ N} \cdot \text{m})\mathbf{k}$$

$$M_x = -25.6 \text{ N} \cdot \text{m}, \quad M_y = 10.80 \text{ N} \cdot \text{m}, \quad M_z = 40.6 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$





PROBLEM 3.48

To lift a heavy crate, a man uses a block and tackle attached to the bottom of an I-beam at hook B . Knowing that the moments about the y and the z axes of the force exerted at B by portion AB of the rope are, respectively, $120 \text{ N} \cdot \text{m}$ and $-460 \text{ N} \cdot \text{m}$, determine the distance a .

SOLUTION

First note

$$\overline{BA} = (2.2 \text{ m})\mathbf{i} - (3.2 \text{ m})\mathbf{j} - (a \text{ m})\mathbf{k}$$

Now

$$\mathbf{M}_D = \mathbf{r}_{A/D} \times \mathbf{T}_{BA}$$

where

$$\mathbf{r}_{A/D} = (2.2 \text{ m})\mathbf{i} + (1.6 \text{ m})\mathbf{j}$$

$$\mathbf{T}_{BA} = \frac{T_{BA}}{d_{BA}} (2.2\mathbf{i} - 3.2\mathbf{j} - a\mathbf{k}) \text{ (N)}$$

Then

$$\begin{aligned} \mathbf{M}_D &= \frac{T_{BA}}{d_{BA}} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2.2 & 1.6 & 0 \\ 2.2 & -3.2 & -a \end{vmatrix} \\ &= \frac{T_{BA}}{d_{BA}} \{-1.6a\mathbf{i} + 2.2a\mathbf{j} + [(2.2)(-3.2) - (1.6)(2.2)]\mathbf{k}\} \end{aligned}$$

Thus

$$M_y = 2.2 \frac{T_{BA}}{d_{BA}} a \quad (\text{N} \cdot \text{m})$$

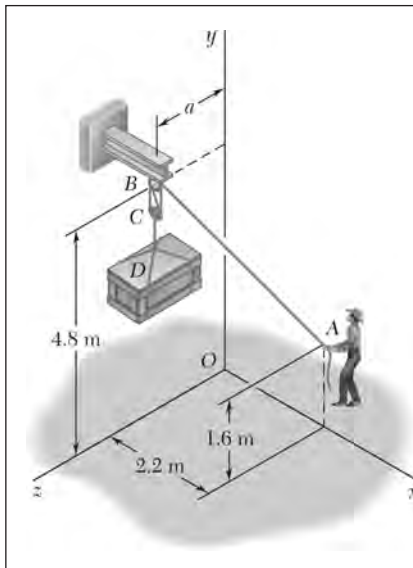
$$M_z = -10.56 \frac{T_{BA}}{d_{BA}} \quad (\text{N} \cdot \text{m})$$

Then forming the ratio

$$\frac{M_y}{M_z}$$

$$\frac{120 \text{ N} \cdot \text{m}}{-460 \text{ N} \cdot \text{m}} = \frac{2.2 \frac{T_{BA}}{d_{BA}} (\text{N} \cdot \text{m})}{-10.56 \frac{T_{BA}}{d_{BA}} (\text{N} \cdot \text{m})}$$

$$\text{or } a = 1.252 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.49

To lift a heavy crate, a man uses a block and tackle attached to the bottom of an I-beam at hook B . Knowing that the man applies a 195-N force to end A of the rope and that the moment of that force about the y axis is $132 \text{ N} \cdot \text{m}$, determine the distance a .

SOLUTION

First note

$$d_{BA} = \sqrt{(2.2)^2 + (-3.2)^2 + (-a)^2}$$

$$= \sqrt{15.08 + a^2} \text{ m}$$

and

$$\mathbf{T}_{BA} = \frac{195 \text{ N}}{d_{BA}} (2.2\mathbf{i} - 3.2\mathbf{j} - a\mathbf{k})$$

Now

$$M_y = \mathbf{j} \cdot (\mathbf{r}_{A/D} \times \mathbf{T}_{BA})$$

where

$$\mathbf{r}_{A/O} = (2.2 \text{ m})\mathbf{i} + (1.6 \text{ m})\mathbf{j}$$

Then

$$M_y = \frac{195}{d_{BA}} \begin{vmatrix} 0 & 1 & 0 \\ 2.2 & 1.6 & 0 \\ 2.2 & -3.2 & -a \end{vmatrix}$$

$$= \frac{195}{d_{BA}} (2.2a) (\text{N} \cdot \text{m})$$

Substituting for M_y and d_{BA}

$$132 \text{ N} \cdot \text{m} = \frac{195}{\sqrt{15.08 + a^2}} (2.2a)$$

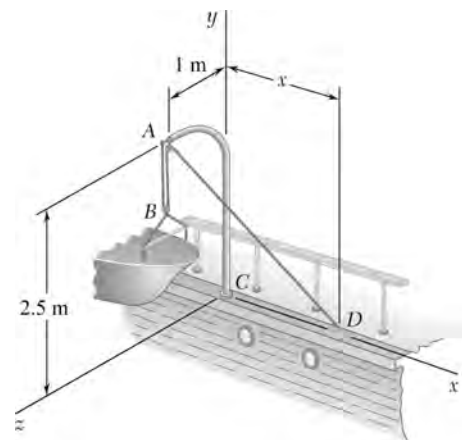
or

$$0.30769\sqrt{15.08 + a^2} = a$$

Squaring both sides of the equation

$$0.094675(15.08 + a^2) = a^2$$

$$\text{or } a = 1.256 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.50

A small boat hangs from two davits, one of which is shown in the figure. It is known that the moment about the z axis of the resultant force $\mathbf{R}_A$ exerted on the davit at A must not exceed $470 \text{ N}\cdot\text{m}$ in absolute value. Determine the largest allowable tension in line $ABAD$ when $x = 2 \text{ m}$.

SOLUTION

First note

$$\mathbf{R}_A = 2\mathbf{T}_{AB} + \mathbf{T}_{AD}$$

Also note that only $\mathbf{T}_{AD}$ will contribute to the moment about the z axis.

Now

$$AD = \sqrt{(2)^2 + (-2.5)^2 + (-1)^2} = 3.354 \text{ m}$$

Then,

$$\begin{aligned} \mathbf{T}_{AD} &= T \frac{\overline{AD}}{AD} \\ &= \frac{T}{3.354} (2\mathbf{i} - 2.5\mathbf{j} - \mathbf{k}) \end{aligned}$$

Now

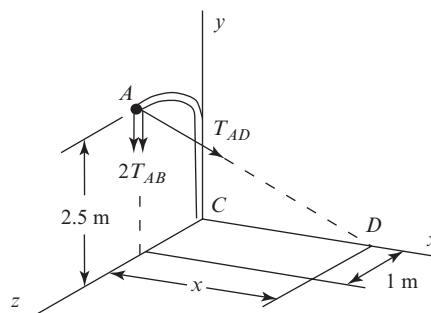
$$M_z = \mathbf{k} \cdot (\mathbf{r}_{A/C} \times \mathbf{T}_{AD})$$

where

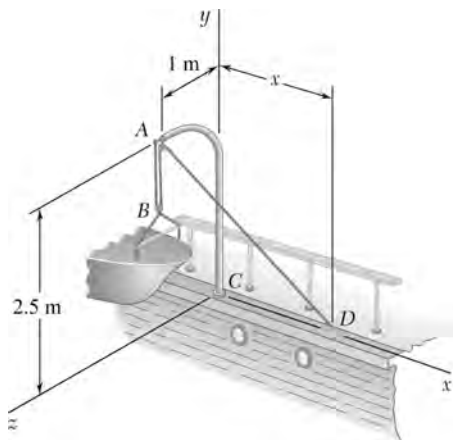
$$\mathbf{r}_{A/C} = (2.5\text{m})\mathbf{j} + (1\text{m})\mathbf{k}$$

Then for $T_{\max}$,

$$\begin{aligned} 470 &= \begin{vmatrix} T_{\max} & 0 & 0 & 1 \\ 3.354 & 0 & 2.5 & 1 \\ 2 & -2.5 & -1 & \end{vmatrix} \\ &= \frac{T_{\max}}{3.354} (2)(2.25) \end{aligned}$$



$$\text{or } T_{\max} = 350 \text{ N} \quad \blacktriangleleft$$



PROBLEM 3.51

For the davit of Problem 3.50, determine the largest allowable distance x when the tension in line $ABAD$ is 300 N.

SOLUTION

From the solution of Problem 3.50, $\mathbf{T}_{AD}$ is now

$$\begin{aligned} T_{AD} &= T \frac{AD}{AD} \\ &= \frac{300\text{N}(\mathbf{x}\mathbf{i} - 2.5\mathbf{j} - \mathbf{k})}{\sqrt{x^2 + (-2.5)^2 + (-1)^2}} \end{aligned}$$

Then $M_z = \mathbf{k} \cdot (\mathbf{r}_{A/C} \times \mathbf{T}_{AD})$ becomes

$$470 = \begin{vmatrix} 300 \\ \sqrt{x^2 + (-2.5)^2 + (-1)^2} \end{vmatrix} \begin{vmatrix} 0 & 0 & 1 \\ 0 & 2.5 & 1 \\ x & -2.5 & -1 \end{vmatrix}$$

$$470 = \frac{300}{\sqrt{x^2 + 7.25}} |-(2.5)(x)|$$

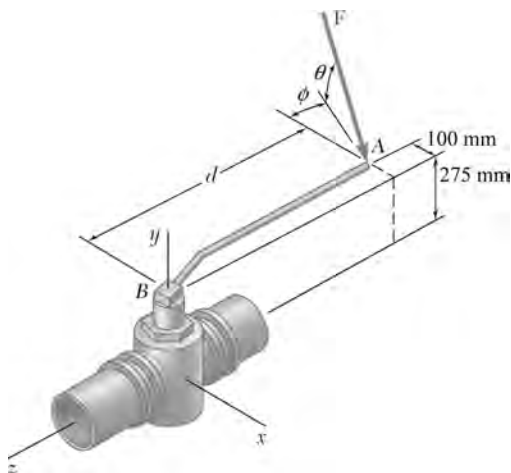
$$470\sqrt{x^2 + 7.25} = 750x$$

$$0.627\sqrt{x^2 + 7.25} = x$$

Squaring both sides: $0.393x^2 + 2.850 = x^2$

$$x^2 = 4.695$$

$$x = 2.17 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.52

To loosen a frozen valve, a force $\mathbf{F}$ of magnitude 350 N is applied to the handle of the valve. Knowing that $\theta = 25^\circ$, $M_x = -102 \text{ N} \cdot \text{m}$ and $M_z = -72 \text{ N} \cdot \text{m}$, determine ϕ and d .

SOLUTION

We have

$$\Sigma \mathbf{M}_O: \mathbf{r}_{A/O} \times \mathbf{F} = \mathbf{M}_O$$

where

$$\mathbf{r}_{A/O} = -(100 \text{ mm})\mathbf{i} + (275 \text{ mm})\mathbf{j} - (d)\mathbf{k}$$

$$\mathbf{F} = F(\cos \theta \cos \phi \mathbf{i} - \sin \theta \mathbf{j} + \cos \theta \sin \phi \mathbf{k})$$

For

$$F = 350 \text{ N}, \quad \theta = 25^\circ$$

$$\mathbf{F} = (350 \text{ N})[(0.90631 \cos \phi)\mathbf{i} - 0.42262\mathbf{j} + (0.90631 \sin \phi)\mathbf{k}]$$

$$\begin{aligned} \mathbf{M}_O &= (350 \text{ N}) \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -100 & 275 & -d \\ 0.90631 \cos \phi & -0.42262 & 0.90631 \sin \phi \end{vmatrix} \text{ mm} \\ &= (350 \text{ N})[(249.2 \sin \phi - 0.42262d)\mathbf{i} + (-0.90631d \cos \phi + 90.631 \sin \phi)\mathbf{j} \\ &\quad + (42.262 - 249.2 \cos \phi)\mathbf{k}] \text{ mm} \end{aligned}$$

and

$$M_x = (350 \text{ N})(249.2 \sin \phi - 0.4226d) \text{ mm} = -(102) \text{ N} \cdot \text{m} \times 10^3 \text{ mm/m} \quad (1)$$

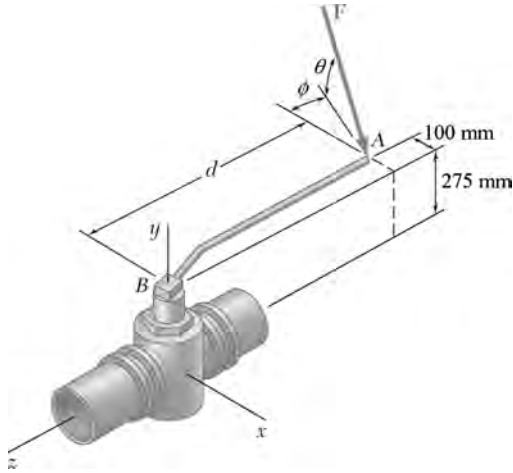
$$M_z = (350 \text{ N})(42.262 - 249.2 \cos \phi) \text{ mm} = -72 (\text{N} \cdot \text{m})(1000 \text{ mm/m}) \quad (2)$$

From Equation (2)

$$\phi = \cos^{-1} \left(\frac{247.98}{249.2} \right) = 5.68^\circ \quad \text{or} \quad \phi = 5.68^\circ \quad \blacktriangleleft$$

From Equation (1)

$$d = \left(\frac{316.09}{0.4226} \right) = 747.97 \text{ mm} \quad \text{or} \quad d = 748 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 3.53

When a force $\mathbf{F}$ is applied to the handle of the valve shown, its moments about the x and z axes are, respectively, $M_x = -130 \text{ N} \cdot \text{m}$ and $M_z = -135 \text{ N} \cdot \text{m}$. For $d = 675 \text{ mm}$, determine the moment M_y of $\mathbf{F}$ about the y axis.

SOLUTION

We have

$$\Sigma \mathbf{M}_O: \mathbf{r}_{A/O} \times \mathbf{F} = \mathbf{M}_O$$

Where

$$\mathbf{r}_{A/O} = -(100 \text{ mm})\mathbf{i} + (275 \text{ mm})\mathbf{j} - (675 \text{ mm})\mathbf{k}$$

$$\mathbf{F} = F(\cos \theta \cos \phi \mathbf{i} - \sin \theta \mathbf{j} + \cos \theta \sin \phi \mathbf{k})$$

$$\mathbf{M}_O = F \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -100 & 275 & -675 \\ \cos \theta \cos \phi & -\sin \theta & \cos \theta \sin \phi \end{vmatrix} \text{ N} \cdot \text{mm}$$

$$= F[(275 \cos \theta \sin \phi - 675 \sin \theta)\mathbf{i} + (-675 \cos \theta \cos \phi + 100 \cos \theta \sin \phi)\mathbf{j} + (100 \sin \theta - 275 \cos \theta \cos \phi)\mathbf{k}] \text{ N} \cdot \text{mm}$$

and

$$M_x = F(275 \cos \theta \sin \phi - 675 \sin \theta) (\text{N} \cdot \text{mm}) \quad (1)$$

$$M_y = F(-675 \cos \theta \cos \phi + 100 \cos \theta \sin \phi) (\text{N} \cdot \text{mm}) \quad (2)$$

$$M_z = F(100 \sin \theta - 275 \cos \theta \cos \phi) (\text{N} \cdot \text{mm}) \quad (3)$$

Now, Equation (1) $\cos \theta \sin \phi = \frac{1}{275} \left(\frac{M_x}{F} + 675 \sin \theta \right) \quad (4)$

and Equation (3) $\cos \theta \cos \phi = \frac{1}{275} \left(100 \sin \theta - \frac{M_z}{F} \right) \quad (5)$

Substituting Equations (4) and (5) into Equation (2),

$$M_y = F \left\{ -675 \left[\frac{1}{275} \left(100 \sin \theta - \frac{M_z}{F} \right) \right] + 100 \left[\frac{1}{275} \left(\frac{M_x}{F} + 675 \sin \theta \right) \right] \right\}$$

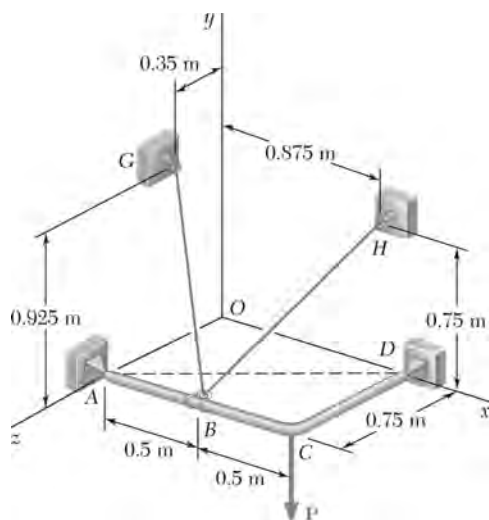
or

$$M_y = \frac{1}{275} (675 M_z + 100 M_x)$$

PROBLEM 3.53 (Continued)

Noting that the ratios $\frac{675}{275}$ and $\frac{100}{275}$ are the ratios of lengths, have

$$\begin{aligned}M_y &= \frac{675}{275}(-135 \text{ N} \cdot \text{m}) + \frac{100}{275}(-130 \text{ N} \cdot \text{m}) \\&= (331.36) - (47.273) \\&= -(378.63) \text{ N} \cdot \text{m} \qquad \text{or} \quad M_y = -379 \text{ N} \cdot \text{m} \quad \blacktriangleleft\end{aligned}$$



PROBLEM 3.54

The frame ACD is hinged at A and D and is supported by a cable that passes through a ring at B and is attached to hooks at G and H . Knowing that the tension in the cable is 450 N, determine the moment about the diagonal AD of the force exerted on the frame by portion BH of the cable.

SOLUTION

$$M_{AD} = \lambda_{AD} \cdot (\mathbf{r}_{B/A} \times \mathbf{T}_{BH})$$

where

$$\lambda_{AD} = \frac{1}{5}(4\mathbf{i} - 3\mathbf{k})$$

$$\mathbf{r}_{B/A} = (0.5 \text{ m})\mathbf{i}$$

and

$$\begin{aligned} d_{BH} &= \sqrt{(0.375)^2 + (0.75)^2 + (-0.75)^2} \\ &= 1.125 \text{ m} \end{aligned}$$

Then

$$\begin{aligned} \mathbf{T}_{BH} &= \frac{450 \text{ N}}{1.125} (0.375\mathbf{i} + 0.75\mathbf{j} - 0.75\mathbf{k}) \\ &= (150 \text{ N})\mathbf{i} + (300 \text{ N})\mathbf{j} - (300 \text{ N})\mathbf{k} \end{aligned}$$

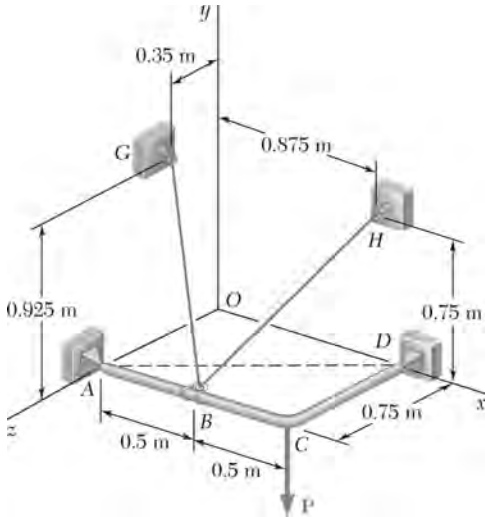
Finally

$$\begin{aligned} M_{AD} &= \frac{1}{5} \begin{vmatrix} 4 & 0 & -3 \\ 0.5 & 0 & 0 \\ 150 & 300 & -300 \end{vmatrix} \\ &= \frac{1}{5} [(-3)(0.5)(300)] \end{aligned}$$

$$\text{or } M_{AD} = -90.0 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$

PROBLEM 3.55

In Problem 3.54, determine the moment about the diagonal AD of the force exerted on the frame by portion BG of the cable.



SOLUTION

$$M_{AD} = \lambda_{AD} \cdot (\mathbf{r}_{B/A} \times \mathbf{T}_{BG})$$

where

$$\lambda_{AD} = \frac{1}{5}(4\mathbf{i} - 3\mathbf{k})$$

$$\mathbf{r}_{B/A} = (0.5 \text{ m})\mathbf{j}$$

and

$$\begin{aligned} BG &= \sqrt{(-0.5)^2 + (0.925)^2 + (-0.4)^2} \\ &= 1.125 \text{ m} \end{aligned}$$

Then

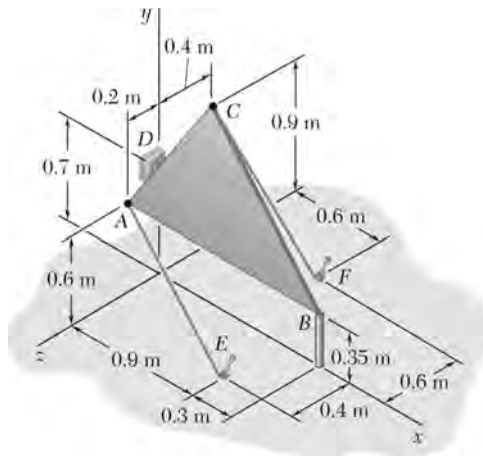
$$\begin{aligned} \bar{\mathbf{T}}_{BG} &= \frac{450 \text{ N}}{1.125}(-0.5\mathbf{i} + 0.925\mathbf{j} - 0.4\mathbf{k}) \\ &= -(200 \text{ N})\mathbf{i} + (370 \text{ N})\mathbf{j} - (160 \text{ N})\mathbf{k} \end{aligned}$$

Finally

$$M_{AD} = \frac{1}{5} \begin{vmatrix} 4 & 0 & -3 \\ 0.5 & 0 & 0 \\ -200 & 370 & -160 \end{vmatrix}$$

$$= \frac{1}{5}[(-3)(0.5)(370)]$$

$$M_{AD} = -111.0 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 3.56

The triangular plate ABC is supported by ball-and-socket joints at B and D and is held in the position shown by cables AE and CF . If the force exerted by cable AE at A is 55 N , determine the moment of that force about the line joining Points D and B .

SOLUTION

First note

$$d_{AE} = \sqrt{(0.9)^2 + (-0.6)^2 + (0.2)^2} = 1.1\text{ m}$$

Then

$$\begin{aligned}\mathbf{T}_{AE} &= \frac{55\text{ N}}{1.1} (0.9\mathbf{i} - 0.6\mathbf{j} + 0.2\mathbf{k}) \\ &= 5[(9\text{ N})\mathbf{i} - (6\text{ N})\mathbf{j} + (2\text{ N})\mathbf{k}]\end{aligned}$$

Also

$$\begin{aligned}DB &= \sqrt{(1.2)^2 + (-0.35)^2 + (0)^2} \\ &= 1.25\text{ m}\end{aligned}$$

Then

$$\begin{aligned}\lambda_{DB} &= \frac{\overline{DB}}{DB} \\ &= \frac{1}{1.25} (1.2\mathbf{i} - 0.35\mathbf{j}) \\ &= \frac{1}{25} (24\mathbf{i} - 7\mathbf{j})\end{aligned}$$

Now

$$M_{DB} = \lambda_{DB} \cdot (\mathbf{r}_{A/D} \times \mathbf{T}_{AE})$$

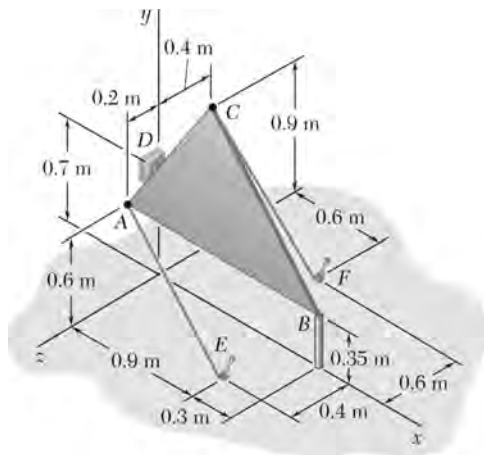
where

$$\mathbf{T}_{DA} = -(0.1\text{ m})\mathbf{j} + (0.2\text{ m})\mathbf{k}$$

Then

$$\begin{aligned}M_{DB} &= \frac{1}{25} (5) \begin{vmatrix} 24 & -7 & 0 \\ 0 & -0.1 & 0.2 \\ 9 & -6 & 2 \end{vmatrix} \\ &= \frac{1}{5} (-4.8 - 12.6 + 28.8)\end{aligned}$$

$$\text{or } M_{DB} = 2.28\text{ N} \cdot \text{m} \quad \blacktriangleleft$$



PROBLEM 3.57

The triangular plate ABC is supported by ball-and-socket joints at B and D and is held in the position shown by cables AE and CF . If the force exerted by cable CF at C is 33 N, determine the moment of that force about the line joining Points D and B .

SOLUTION

First note $d_{CF} = \sqrt{(0.6)^2 + (-0.9)^2 + (-0.2)^2} = 1.1 \text{ m}$

Then
$$\begin{aligned} \mathbf{T}_{CF} &= \frac{33 \text{ N}}{1.1} (0.6\mathbf{i} - 0.9\mathbf{j} + 0.2\mathbf{k}) \\ &= 3[(6 \text{ N})\mathbf{i} - (9 \text{ N})\mathbf{j} - (2 \text{ N})\mathbf{k}] \end{aligned}$$

Also
$$\begin{aligned} DB &= \sqrt{(1.2)^2 + (-0.35)^2 + (0)^2} \\ &= 1.25 \text{ m} \end{aligned}$$

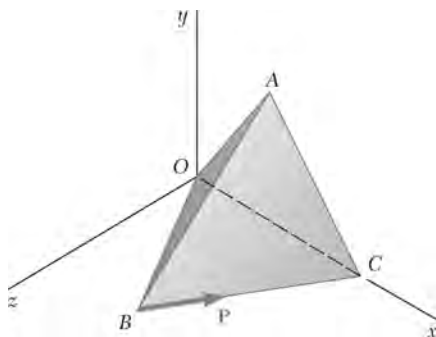
Then
$$\begin{aligned} \lambda_{DB} &= \frac{\overline{DB}}{DB} \\ &= \frac{1}{1.25} (1.2\mathbf{i} - 0.35\mathbf{j}) \\ &= \frac{1}{25} (24\mathbf{i} - 7\mathbf{j}) \end{aligned}$$

Now $M_{DB} = \lambda_{DB} \cdot (\mathbf{r}_{C/D} \times \mathbf{T}_{CF})$

where $\mathbf{r}_{C/D} = (0.2 \text{ m})\mathbf{j} - (0.4 \text{ m})\mathbf{k}$

Then
$$\begin{aligned} M_{DB} &= \frac{1}{25} (3) \begin{vmatrix} 24 & -7 & 0 \\ 0 & 0.2 & -0.4 \\ 6 & -9 & -2 \end{vmatrix} \\ &= \frac{3}{25} (-9.6 + 16.8 - 86.4) \end{aligned}$$

or $M_{DB} = -9.50 \text{ N} \cdot \text{m} \quad \blacktriangleleft$



PROBLEM 3.58

A regular tetrahedron has six edges of length a . A force $\mathbf{P}$ is directed as shown along edge BC . Determine the moment of $\mathbf{P}$ about edge OA .

SOLUTION

We have

$$M_{OA} = \lambda_{OA} \cdot (\mathbf{r}_{C/O} \times \mathbf{P})$$

where

From triangle OBC

$$(OA)_x = \frac{a}{2}$$

$$(OA)_z = (OA)_x \tan 30^\circ = \frac{a}{2} \left(\frac{1}{\sqrt{3}} \right) = \frac{a}{2\sqrt{3}}$$

Since

$$(OA)^2 = (OA)_x^2 + (OA)_y^2 + (OA)_z^2$$

or

$$a^2 = \left(\frac{a}{2} \right)^2 + (OA)_y^2 + \left(\frac{a}{2\sqrt{3}} \right)^2$$

$$(OA)_y = \sqrt{a^2 - \frac{a^2}{4} - \frac{a^2}{12}} = a\sqrt{\frac{2}{3}}$$

Then

$$\mathbf{r}_{A/O} = \frac{a}{2} \mathbf{i} + a\sqrt{\frac{2}{3}} \mathbf{j} + \frac{a}{2\sqrt{3}} \mathbf{k}$$

and

$$\lambda_{OA} = \frac{1}{2} \mathbf{i} + \sqrt{\frac{2}{3}} \mathbf{j} + \frac{1}{2\sqrt{3}} \mathbf{k}$$

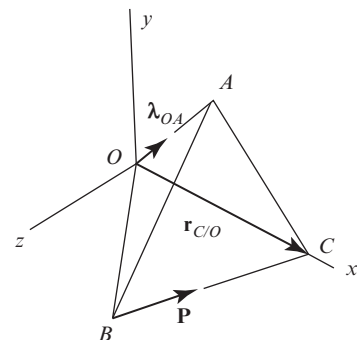
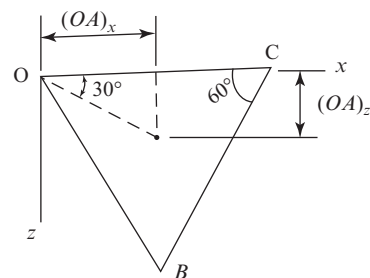
$$\mathbf{P} = \lambda_{BC} P = \frac{(a \sin 30^\circ) \mathbf{i} - (a \cos 30^\circ) \mathbf{k}}{a} (P) = \frac{P}{2} (\mathbf{i} - \sqrt{3} \mathbf{k})$$

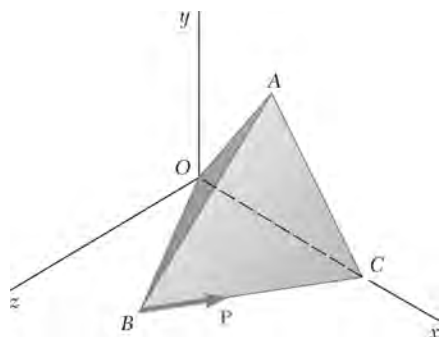
$$\mathbf{r}_{C/O} = a \mathbf{i}$$

$$M_{OA} = \begin{vmatrix} \frac{1}{2} & \sqrt{\frac{2}{3}} & \frac{1}{2\sqrt{3}} \\ 1 & 0 & 0 \\ 1 & 0 & -\sqrt{3} \end{vmatrix} (a) \left(\frac{P}{2} \right)$$

$$= \frac{aP}{2} \left(-\sqrt{\frac{2}{3}} \right) (1)(-\sqrt{3}) = \frac{aP}{\sqrt{2}}$$

$$M_{OA} = \frac{aP}{\sqrt{2}} \quad \blacktriangleleft$$





PROBLEM 3.59

A regular tetrahedron has six edges of length a . (a) Show that two opposite edges, such as OA and BC , are perpendicular to each other. (b) Use this property and the result obtained in Problem 3.59 to determine the perpendicular distance between edges OA and BC .

SOLUTION

- (a) For edge OA to be perpendicular to edge BC ,

$$\overline{OA} \cdot \overline{BC} = 0$$

where

From triangle OBC

$$(OA)_x = \frac{a}{2}$$

$$(OA)_z = (OA)_x \tan 30^\circ = \frac{a}{2} \left(\frac{1}{\sqrt{3}} \right) = \frac{a}{2\sqrt{3}}$$

$$\overline{OA} = \left(\frac{a}{2} \right) \mathbf{i} + (OA)_y \mathbf{j} + \left(\frac{a}{2\sqrt{3}} \right) \mathbf{k}$$

and

$$\begin{aligned} \overline{BC} &= (a \sin 30^\circ) \mathbf{i} - (a \cos 30^\circ) \mathbf{k} \\ &= \frac{a}{2} \mathbf{i} - \frac{a\sqrt{3}}{2} \mathbf{k} = \frac{a}{2} (\mathbf{i} - \sqrt{3} \mathbf{k}) \end{aligned}$$

Then

$$\left[\frac{a}{2} \mathbf{i} + (OA)_y \mathbf{j} + \left(\frac{a}{2\sqrt{3}} \right) \mathbf{k} \right] \cdot (\mathbf{i} - \sqrt{3} \mathbf{k}) \frac{a}{2} = 0$$

or

$$\frac{a^2}{4} + (OA)_y(0) - \frac{a^2}{4} = 0$$

$$\overline{OA} \cdot \overline{BC} = 0$$

so that

$\overline{OA}$ is perpendicular to $\overline{BC}$ ◀

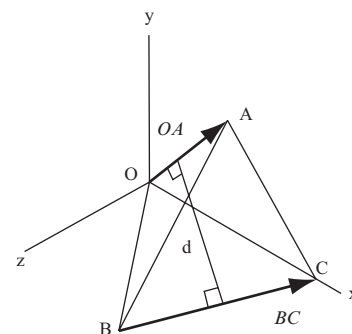
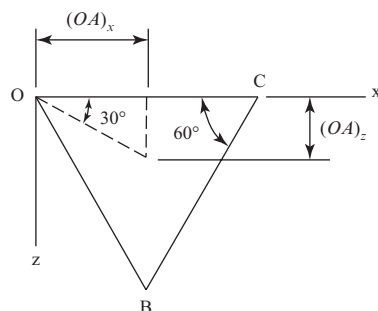
- (b) Have $M_{OA} = Pd$, with P acting along BC and d the perpendicular distance from $\overline{OA}$ to $\overline{BC}$.

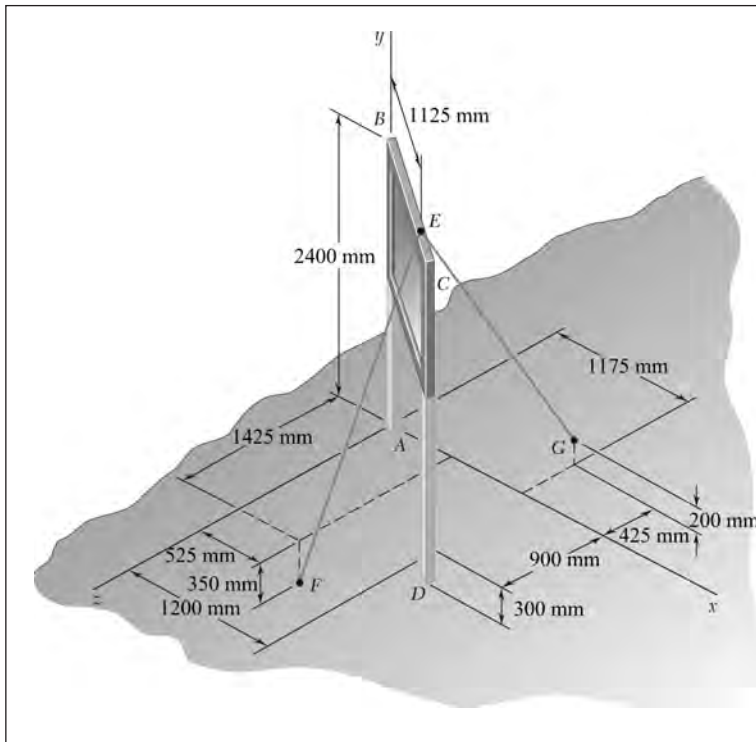
From the results of Problem 3.56

$$M_{OA} = \frac{Pa}{\sqrt{2}}$$

$$\frac{Pa}{\sqrt{2}} = Pd$$

$$\text{or } d = \frac{a}{\sqrt{2}} \quad \blacktriangleleft$$





PROBLEM 3.60

A sign erected on uneven ground is guyed by cables EF and EG . If the force exerted by cable EF at E is 230 N, determine the moment of that force about the line joining points A and D .

SOLUTION

First note that $BC = \sqrt{(1200)^2 + (900)^2} = 1500$ mm and that $\frac{BE}{BC} = \frac{1125}{1500} = \frac{3}{4}$. The coordinates of Point E are then $(\frac{3}{4} \times 1200, 2400, \frac{3}{4} \times 900)$ or $(900 \text{ mm}, 2400 \text{ mm}, 675 \text{ mm})$. Then

$$d_{EF} = \sqrt{(-15)^2 + (-110)^2 + (30)^2} \\ = 2875 \text{ mm}$$

Then

$$\mathbf{T}_{EF} = \sqrt{(-375)^2 + (-2750)^2 + (750)^2} \\ = \frac{230 \text{ N}}{2875} (-375\mathbf{i} - 2750\mathbf{j} + 750\mathbf{k}) \\ = 2[-(15 \text{ N})\mathbf{i} - (110 \text{ N})\mathbf{j} + (30 \text{ N})\mathbf{k}]$$

Also

$$AD = \sqrt{(1200)^2 + (-300)^2 + (900)^2} \\ = 1529.71 \text{ mm}$$

Then

$$\lambda_{AD} = \frac{\overrightarrow{AD}}{AD} \\ = \frac{1}{1529.71} (1200\mathbf{i} - 300\mathbf{j} + 900\mathbf{k}) \\ = (0.7845\mathbf{i} - 0.1961\mathbf{j} + 0.5883\mathbf{k})$$

PROBLEM 3.60 (Continued)

Now

$$M_{AD} = \lambda_{AD} \cdot (\mathbf{r}_{E/A} \times \mathbf{T}_{EF})$$

where

$$\mathbf{r}_{E/A} = (900 \text{ mm})\mathbf{i} + (2400 \text{ mm})\mathbf{j} + (675 \text{ mm})\mathbf{k}$$

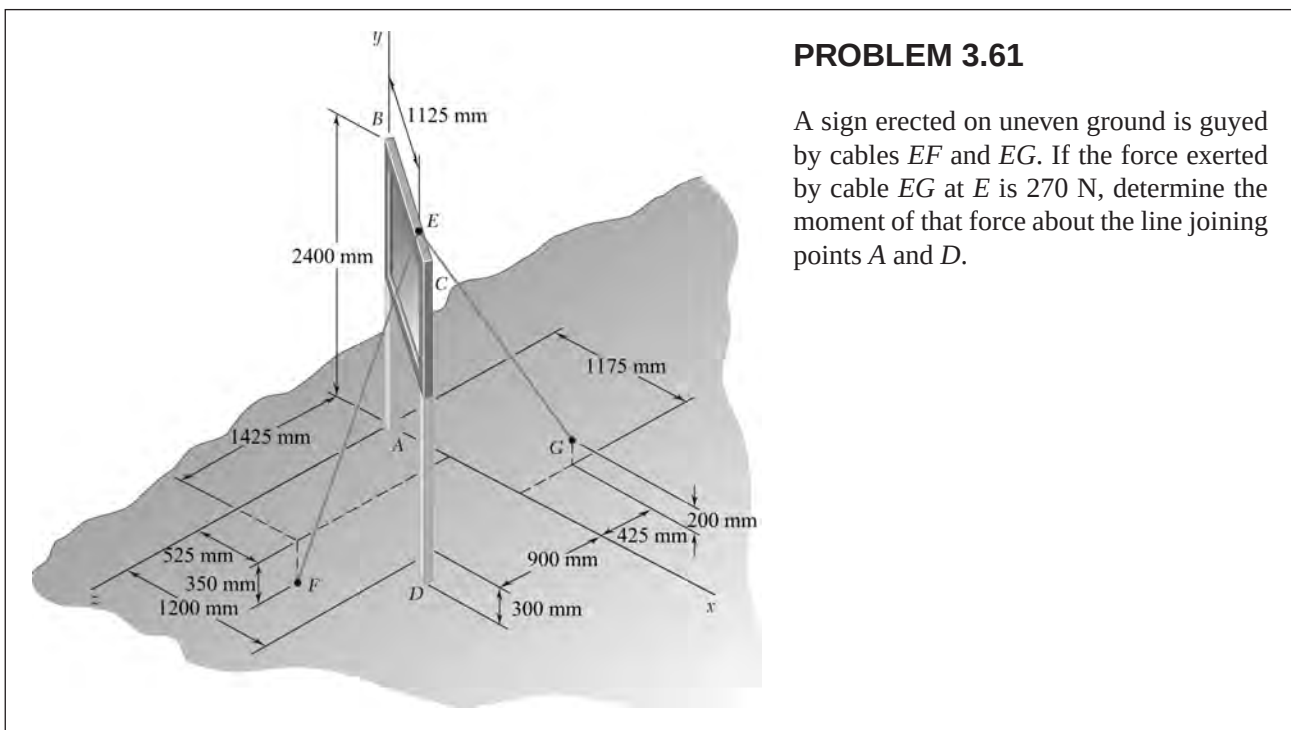
Then

$$M_{AD} = \begin{vmatrix} 0.7845 & -0.1961 & +0.5883 \\ 900 & 2400 & 675 \\ -30 & -220 & 60 \end{vmatrix}$$

$$= 229466 + 14560 - 74126$$

$$= 169900 \text{ N} \cdot \text{mm}$$

$$M_{AD} = 169.9 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$



SOLUTION

First note that $BC = \sqrt{(1200)^2 + (900)^2} = 1500 \text{ mm}$ and that $\frac{BE}{BC} = \frac{1125}{1500} = \frac{3}{4}$. The coordinates of Point E are then $(\frac{3}{4} \times 1200, 2400, \frac{3}{4} \times 900)$ or $(900 \text{ mm}, 2400 \text{ mm}, 675 \text{ mm})$. Then

$$d_{EG} = \sqrt{(275)^2 + (-2200)^2 + (-1100)^2} = 2475 \text{ mm}$$

Then

$$\begin{aligned} \mathbf{T}_{EG} &= \frac{270 \text{ N}}{2475} (275\mathbf{i} - 2200\mathbf{j} - 1100\mathbf{k}) \\ &= (30 \text{ N})\mathbf{i} - (240 \text{ N})\mathbf{j} - (120 \text{ N})\mathbf{k} \end{aligned}$$

Also

$$AD = \sqrt{(1200)^2 + (-300)^2 + (900)^2} = 1529.71 \text{ mm}$$

Then

$$\begin{aligned} \lambda_{AD} &= \frac{\overline{AD}}{AD} \\ &= \frac{1}{1529.71} (1200\mathbf{i} - 300\mathbf{j} + 900\mathbf{k}) \\ &= (0.7845\mathbf{i} - 0.1961\mathbf{j} + 0.5883\mathbf{k}) \end{aligned}$$

PROBLEM 3.61 (Continued)

Now $M_{AD} = \lambda_{AD} \cdot (\mathbf{r}_{E/A} \times \mathbf{T}_{EF})$

where $\mathbf{r}_{E/A} = (900 \text{ mm})\mathbf{i} + (2400 \text{ mm})\mathbf{j} + (675 \text{ mm})\mathbf{k}$

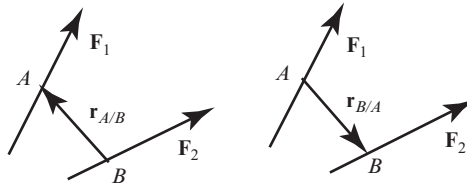
Then
$$M_{AD} = \begin{vmatrix} 0.7845 & -0.1961 & +0.5883 \\ 900 & 2400 & 675 \\ 30 & -240 & -120 \end{vmatrix}$$
$$= -98847 - 25150 - 169430 \text{ N} \cdot \text{mm}$$
$$= -293427 \text{ N} \cdot \text{mm}$$

or $M_{AD} = -293 \text{ N} \cdot \text{m} \blacktriangleleft$

PROBLEM 3.62

Two forces $\mathbf{F}_1$ and $\mathbf{F}_2$ in space have the same magnitude F . Prove that the moment of $\mathbf{F}_1$ about the line of action of $\mathbf{F}_2$ is equal to the moment of $\mathbf{F}_2$ about the line of action of $\mathbf{F}_1$.

SOLUTION



First note that

$$\mathbf{F}_1 = F_1 \lambda_1 \quad \text{and} \quad \mathbf{F}_2 = F_2 \lambda_2$$

Let M_1 = moment of $\mathbf{F}_2$ about the line of action of $\mathbf{F}_1$ and M_2 = moment of $\mathbf{F}_1$ about the line of action of $\mathbf{F}_2$

Now, by definition

$$\begin{aligned} M_1 &= \lambda_1 \cdot (\mathbf{r}_{B/A} \times \mathbf{F}_2) \\ &= \lambda_1 \cdot (\mathbf{r}_{B/A} \times \lambda_2) F_2 \\ M_2 &= \lambda_2 \cdot (\mathbf{r}_{A/B} \times \mathbf{F}_1) \\ &= \lambda_2 \cdot (\mathbf{r}_{A/B} \times \lambda_1) F_1 \end{aligned}$$

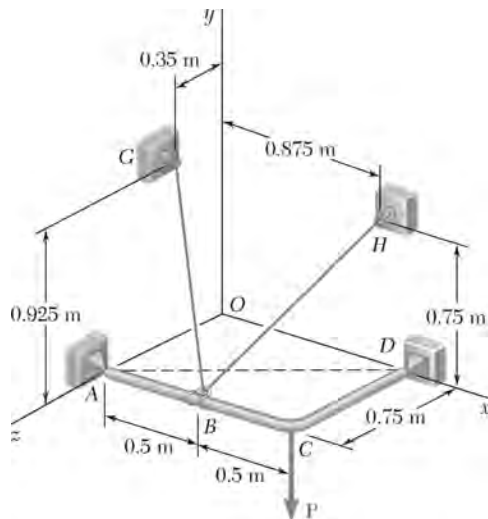
Since

$$\begin{aligned} F_1 &= F_2 = F \quad \text{and} \quad \mathbf{r}_{A/B} = -\mathbf{r}_{B/A} \\ M_1 &= \lambda_1 \cdot (\mathbf{r}_{B/A} \times \lambda_2) F \\ M_2 &= \lambda_2 \cdot (-\mathbf{r}_{B/A} \times \lambda_1) F \end{aligned}$$

Using Equation (3.38) $\lambda_1 \cdot (\mathbf{r}_{B/A} \times \lambda_2) = \lambda_2 \cdot (-\mathbf{r}_{B/A} \times \lambda_1)$

so that

$$M_2 = \lambda_1 \cdot (\mathbf{r}_{B/A} \times \lambda_2) F \qquad M_{12} = M_{21} \quad \blacktriangleleft$$



PROBLEM 3.63

In Problem 3.54, determine the perpendicular distance between portion BH of the cable and the diagonal AD .

PROBLEM 3.54 The frame ACD is hinged at A and D and is supported by a cable that passes through a ring at B and is attached to hooks at G and H . Knowing that the tension in the cable is 450 N, determine the moment about the diagonal AD of the force exerted on the frame by portion BH of the cable.

SOLUTION

From the solution to Problem 3.54:

$$T_{BH} = 450 \text{ N}$$

$$\mathbf{T}_{BH} = (150 \text{ N})\mathbf{i} + (300 \text{ N})\mathbf{j} - (300 \text{ N})\mathbf{k}$$

$$|M_{AD}| = 90.0 \text{ N} \cdot \text{m}$$

$$\lambda_{AD} = \frac{1}{5}(4\mathbf{i} - 3\mathbf{k})$$

Based on the discussion of Section 3.11, it follows that only the perpendicular component of $\mathbf{T}_{BH}$ will contribute to the moment of $\mathbf{T}_{BH}$ about line $\overline{AD}$.

Now

$$\begin{aligned} (T_{BH})_{\text{parallel}} &= \mathbf{T}_{BH} \cdot \lambda_{AD} \\ &= (150\mathbf{i} + 300\mathbf{j} - 300\mathbf{k}) \cdot \frac{1}{5}(4\mathbf{i} - 3\mathbf{k}) \\ &= \frac{1}{5}[(150)(4) + (-300)(-3)] \\ &= 300 \text{ N} \end{aligned}$$

Also

$$\mathbf{T}_{BH} = (\mathbf{T}_{BH})_{\text{parallel}} + (\mathbf{T}_{BH})_{\text{perpendicular}}$$

so that

$$(\mathbf{T}_{BH})_{\text{perpendicular}} = \sqrt{(450)^2 - (300)^2} = 335.41 \text{ N}$$

Since λ_{AD} and $(\mathbf{T}_{BH})_{\text{perpendicular}}$ are perpendicular, it follows that

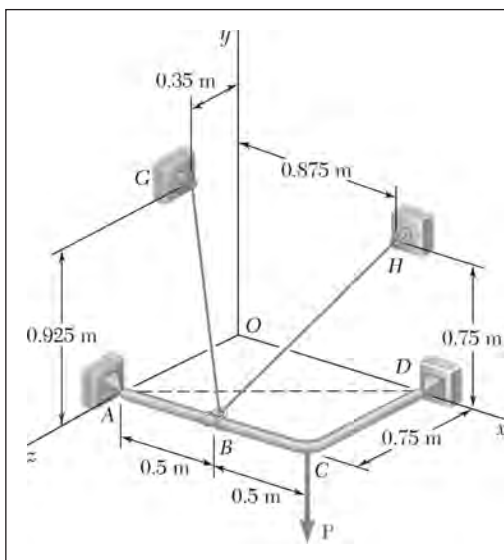
$$M_{AD} = d(\mathbf{T}_{BH})_{\text{perpendicular}}$$

or

$$90.0 \text{ N} \cdot \text{m} = d(335.41 \text{ N})$$

$$d = 0.26833 \text{ m}$$

$$d = 0.268 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.64

In Problem 3.55, determine the perpendicular distance between portion BG of the cable and the diagonal AD .

PROBLEM 3.55 In Problem 3.54, determine the moment about the diagonal AD of the force exerted on the frame by portion BG of the cable.

SOLUTION

From the solution to Problem 3.55:

$$\mathbf{T}_{BG} = 450 \text{ N}$$

$$\mathbf{T}_{BG} = -(200 \text{ N})\mathbf{i} + (370 \text{ N})\mathbf{j} - (160 \text{ N})\mathbf{k}$$

$$|M_{AD}| = 111 \text{ N} \cdot \text{m}$$

$$\lambda_{AD} = \frac{1}{5}(4\mathbf{i} - 3\mathbf{k})$$

Based on the discussion of Section 3.11, it follows that only the perpendicular component of $\mathbf{T}_{BG}$ will contribute to the moment of $\mathbf{T}_{BG}$ about line $\overline{AD}$.

Now

$$\begin{aligned} (\mathbf{T}_{BG})_{\text{parallel}} &= \mathbf{T}_{BG} \cdot \lambda_{AD} \\ &= (-200\mathbf{i} + 370\mathbf{j} - 160\mathbf{k}) \cdot \frac{1}{5}(4\mathbf{i} - 3\mathbf{k}) \\ &= \frac{1}{5}[(-200)(4) + (-160)(-3)] \\ &= -64 \text{ N} \end{aligned}$$

Also

$$\bar{\mathbf{T}}_{BG} = (\mathbf{T}_{BG})_{\text{parallel}} + (\mathbf{T}_{BG})_{\text{perpendicular}}$$

so that

$$(\mathbf{T}_{BG})_{\text{perpendicular}} = \sqrt{(450)^2 - (-64)^2} = 445.43 \text{ N}$$

Since λ_{AD} and $(\mathbf{T}_{BG})_{\text{perpendicular}}$ are perpendicular, it follows that

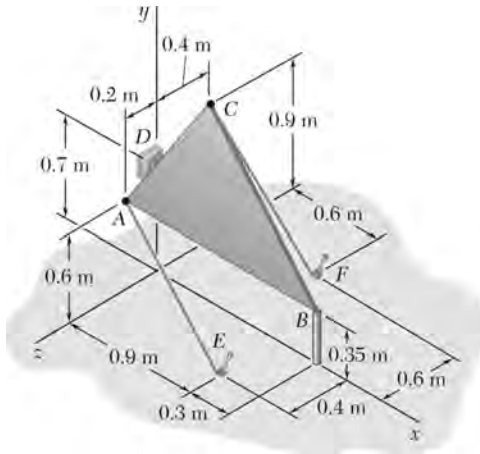
$$M_{AD} = d(\mathbf{T}_{BG})_{\text{perpendicular}}$$

or

$$111 \text{ N} \cdot \text{m} = d(445.43 \text{ N})$$

$$d = 0.24920 \text{ m}$$

$$d = 0.249 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.65

In Problem 3.56, determine the perpendicular distance between cable AE and the line joining Points D and B .

PROBLEM 3.56 The triangular plate ABC is supported by ball-and-socket joints at B and D and is held in the position shown by cables AE and CF . If the force exerted by cable AE at A is 55 N , determine the moment of that force about the line joining Points D and B .

SOLUTION

From the solution to Problem 3.56

$$\mathbf{T}_{AE} = 55\text{ N}$$

$$\mathbf{T}_{AE} = 5[(9\text{ N})\mathbf{i} - (6\text{ N})\mathbf{j} + (2\text{ N})\mathbf{k}]$$

$$|M_{DB}| = 2.28\text{ N} \cdot \text{m}$$

$$\lambda_{DB} = \frac{1}{25}(24\mathbf{i} - 7\mathbf{j})$$

Based on the discussion of Section 3.11, it follows that only the perpendicular component of $\mathbf{T}_{AE}$ will contribute to the moment of $\mathbf{T}_{AE}$ about line $\overline{DB}$.

Now

$$\begin{aligned} (\mathbf{T}_{AE})_{\text{parallel}} &= \mathbf{T}_{AE} \cdot \lambda_{DB} \\ &= 5(9\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}) \cdot \frac{1}{25}(24\mathbf{i} - 7\mathbf{j}) \\ &= \frac{1}{5}[(9)(24) + (-6)(-7)] \\ &= 51.6\text{ N} \end{aligned}$$

Also

$$\mathbf{T}_{AE} = (\mathbf{T}_{AE})_{\text{parallel}} + (\mathbf{T}_{AE})_{\text{perpendicular}}$$

so that

$$(\mathbf{T}_{AE})_{\text{perpendicular}} = \sqrt{(55)^2 + (51.6)^2} = 19.0379\text{ N}$$

Since λ_{DB} and $(\mathbf{T}_{AE})_{\text{perpendicular}}$ are perpendicular, it follows that

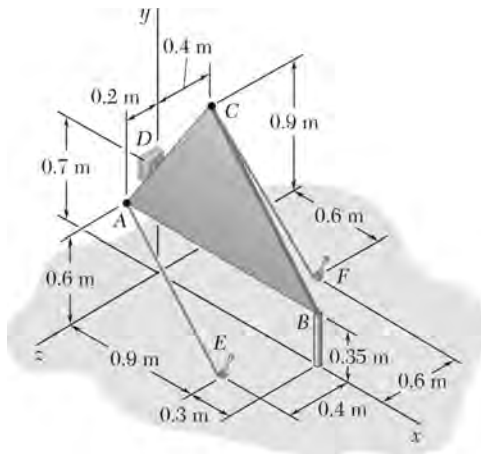
$$M_{DB} = d(\mathbf{T}_{AE})_{\text{perpendicular}}$$

or

$$2.28\text{ N} \cdot \text{m} = d(19.0379\text{ N})$$

$$d = 0.119761$$

$$d = 0.1198\text{ m} \quad \blacktriangleleft$$



PROBLEM 3.66

In Problem 3.57, determine the perpendicular distance between cable CF and the line joining Points D and B .

PROBLEM 3.57 The triangular plate ABC is supported by ball-and-socket joints at B and D and is held in the position shown by cables AE and CF . If the force exerted by cable CF at C is 33 N, determine the moment of that force about the line joining Points D and B .

SOLUTION

From the solution to Problem 3.57

$$\mathbf{T}_{CF} = 33 \text{ N}$$

$$\mathbf{T}_{CF} = 3[(6 \text{ N})\mathbf{i} - (9 \text{ N})\mathbf{j} - (2 \text{ N})\mathbf{k}]$$

$$|M_{DB}| = 9.50 \text{ N} \cdot \text{m}$$

$$\lambda_{DB} = \frac{1}{25}(24\mathbf{i} - 7\mathbf{j})$$

Based on the discussion of Section 3.11, it follows that only the perpendicular component of $\mathbf{T}_{CF}$ will contribute to the moment of $\mathbf{T}_{CF}$ about line DB .

Now

$$\begin{aligned} (\mathbf{T}_{CF})_{\text{parallel}} &= \mathbf{T}_{CF} \cdot \lambda_{DB} \\ &= 3(6\mathbf{i} - 9\mathbf{j} - 2\mathbf{k}) \cdot \frac{1}{25}(24\mathbf{i} - 7\mathbf{j}) \\ &= \frac{3}{25}[(6)(24) + (-9)(-7)] \\ &= 24.84 \text{ N} \end{aligned}$$

Also

$$\mathbf{T}_{CF} = (\mathbf{T}_{CF})_{\text{parallel}} + (\mathbf{T}_{CF})_{\text{perpendicular}}$$

so that

$$\begin{aligned} (\mathbf{T}_{CF})_{\text{perpendicular}} &= \sqrt{(33)^2 - (24.84)^2} \\ &= 21.725 \text{ N} \end{aligned}$$

Since λ_{DB} and $(\mathbf{T}_{CF})_{\text{perpendicular}}$ are perpendicular, it follows that

$$|M_{DB}| = d(\mathbf{T}_{CF})_{\text{perpendicular}}$$

or

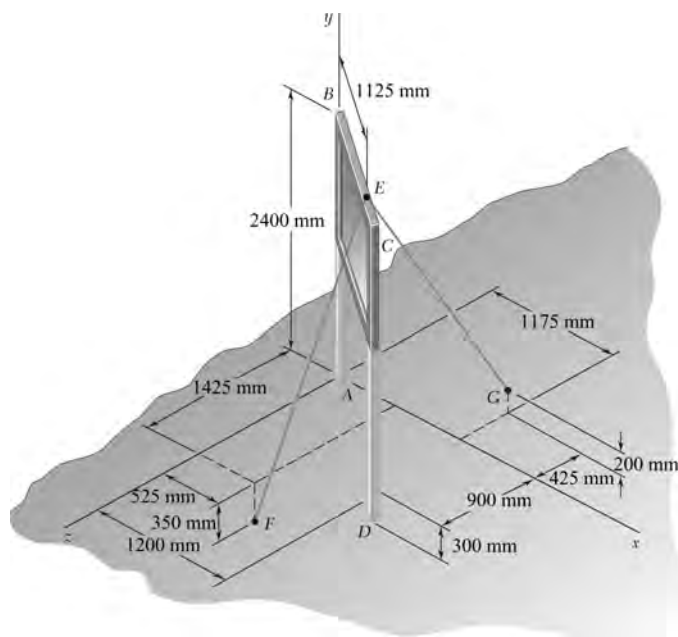
$$9.50 \text{ N} \cdot \text{m} = d \times 21.725 \text{ N}$$

$$\text{or } d = 0.437 \text{ m} \quad \blacktriangleleft$$

PROBLEM 3.67

In Problem 3.60, determine the perpendicular distance between cable EF and the line joining Points A and D .

PROBLEM 3.60 A sign erected on uneven ground is guyed by cables EF and EG . If the force exerted by cable EF at E is 230 N, determine the moment of that force about the line joining Points A and D .



SOLUTION

From the solution to Problem 3.61

$$\begin{aligned} T_{EF} &= 230 \text{ N} \\ \mathbf{T}_{EF} &= (-30\text{N})\mathbf{i} - (220\text{N})\mathbf{j} + (60\text{N})\mathbf{k} \\ |M_{AD}| &= 169.9 \text{ N} \cdot \text{m} \\ \lambda_{AD} &= (0.7845\mathbf{i} - 0.1961\mathbf{j} + 0.5883\mathbf{k}) \end{aligned}$$

Based on the discussion of Section 3.11, it follows that only the perpendicular component of $\mathbf{T}_{EF}$ will contribute to the moment of $\mathbf{T}_{EF}$ about line $\overline{AD}$.

$$\begin{aligned} \text{Now} \quad (T_{EF})_{\text{parallel}} &= \mathbf{T}_{EF} \cdot \lambda_{AD} \\ &= (-30\mathbf{i} - 220\mathbf{j} + 60\mathbf{k}) \cdot (0.7845\mathbf{i} - 0.1961\mathbf{j} + 0.5883\mathbf{k}) \\ &= 54.905 \text{ N} \end{aligned}$$

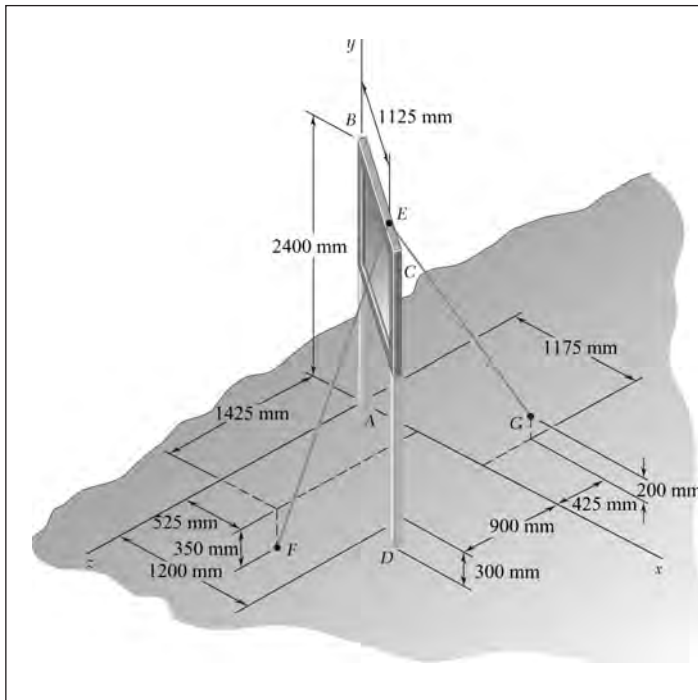
$$\text{Also} \quad \mathbf{T}_{EF} = (\mathbf{T}_{EF})_{\text{parallel}} + (\mathbf{T}_{EF})_{\text{perpendicular}}$$

$$\text{so that} \quad (\mathbf{T}_{EF})_{\text{perpendicular}} = \sqrt{(230)^2 - (54.905)^2} = 223.35 \text{ N}$$

Since λ_{AD} and $(\mathbf{T}_{EF})_{\text{perpendicular}}$ are perpendicular, it follows that

$$M_{AD} = d(T_{EF})_{\text{perpendicular}}$$

$$\text{or} \quad 169.9 \text{ N} \cdot \text{m} = d \times 223.35 \text{ N} \quad \text{or} \quad d = 0.761 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.68

In Problem 3.61, determine the perpendicular distance between cable EG and the line joining Points A and D .

PROBLEM 3.61 A sign erected on uneven ground is guyed by cables EF and EG . If the force exerted by cable EG at E is 270 N, determine the moment of that force about the line joining Points A and D .

SOLUTION

From the solution to Problem 3.62

$$T_{EG} = 270 \text{ N}$$

$$\mathbf{T}_{EG} = (30 \text{ N})\mathbf{i} - (240 \text{ N})\mathbf{j} - (120 \text{ N})\mathbf{k}$$

$$|M_{AD}| = 293 \text{ N} \cdot \text{m}$$

$$\lambda_{AD} = (0.7845\mathbf{i} - 0.1961\mathbf{j} + 0.5883\mathbf{k})$$

Based on the discussion of Section 3.11, it follows that only the perpendicular component of $\mathbf{T}_{EG}$ will contribute to the moment of $\mathbf{T}_{EG}$ about line $\overline{AD}$.

Now

$$\begin{aligned} (T_{EG})_{\text{parallel}} &= \mathbf{T}_{EG} \cdot \lambda_{AD} \\ &= (30\mathbf{i} - 240\mathbf{j} - 120\mathbf{k}) \cdot (0.7845\mathbf{i} - 0.1961\mathbf{j} + 0.5883\mathbf{k}) \\ &= 0 \end{aligned}$$

Thus,

$$(\mathbf{T}_{EG})_{\text{perpendicular}} = \mathbf{T}_{EG} = 270 \text{ N}$$

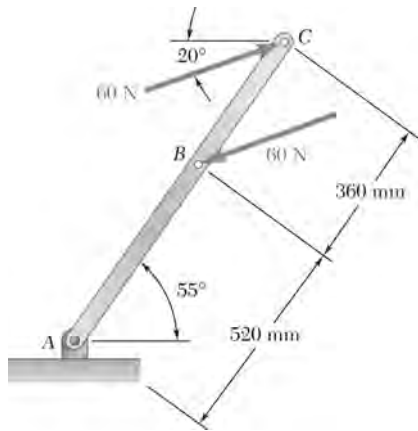
Since λ_{AD} and $(\mathbf{T}_{EG})_{\text{perpendicular}}$ are perpendicular, it follows that

$$|M_{AD}| = d(T_{EG})_{\text{perpendicular}}$$

or

$$293 \text{ N} \cdot \text{m} = d \times 270 \text{ N}$$

$$\text{or } d = 1.085 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.69

Two parallel 60-N forces are applied to a lever as shown. Determine the moment of the couple formed by the two forces (a) by resolving each force into horizontal and vertical components and adding the moments of the two resulting couples, (b) by using the perpendicular distance between the two forces, (c) by summing the moments of the two forces about Point A.

SOLUTION

(a) We have $\Sigma \mathbf{M}_B: -d_1 C_x + d_2 C_y = \mathbf{M}$

where $d_1 = (0.360 \text{ m}) \sin 55^\circ$
 $= 0.29489 \text{ m}$

$d_2 = (0.360 \text{ m}) \sin 55^\circ$
 $= 0.20649 \text{ m}$

$C_x = (60 \text{ N}) \cos 20^\circ$
 $= 56.382 \text{ N}$

$C_y = (60 \text{ N}) \sin 20^\circ$
 $= 20.521 \text{ N}$

$\mathbf{M} = -(0.29489 \text{ m})(56.382 \text{ N})\mathbf{k} + (0.20649 \text{ m})(20.521 \text{ N})\mathbf{k}$
 $= -(12.3893 \text{ N} \cdot \text{m})\mathbf{k}$

or $\mathbf{M} = 12.39 \text{ N} \cdot \text{m} \curvearrowleft$

(b) We have $\mathbf{M} = Fd(-\mathbf{k})$
 $= 60 \text{ N}[(0.360 \text{ m}) \sin(55^\circ - 20^\circ)](-\mathbf{k})$
 $= -(12.3893 \text{ N} \cdot \text{m})\mathbf{k}$

or $\mathbf{M} = 12.39 \text{ N} \cdot \text{m} \curvearrowleft$

(c) We have $\Sigma \mathbf{M}_A: \Sigma(\mathbf{r}_A \times \mathbf{F}) = \mathbf{r}_{B/A} \times \mathbf{F}_B + \mathbf{r}_{C/A} \times \mathbf{F}_C = \mathbf{M}$

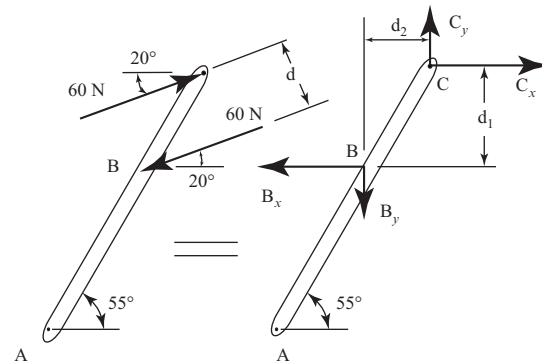
$$M = (0.520 \text{ m})(60 \text{ N}) \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos 55^\circ & \sin 55^\circ & 0 \\ -\cos 20^\circ & -\sin 20^\circ & 0 \end{vmatrix}$$

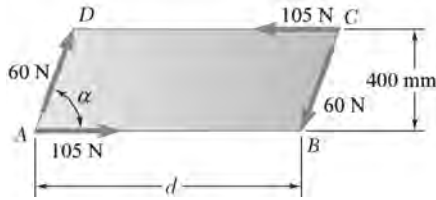
$$+ (0.800 \text{ m})(60 \text{ N}) \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos 55^\circ & \sin 55^\circ & 0 \\ \cos 20^\circ & \sin 20^\circ & 0 \end{vmatrix}$$

$= (17.8956 \text{ N} \cdot \text{m} - 30.285 \text{ N} \cdot \text{m})\mathbf{k}$

$= -(12.3892 \text{ N} \cdot \text{m})\mathbf{k}$

or $\mathbf{M} = 12.39 \text{ N} \cdot \text{m} \curvearrowleft$

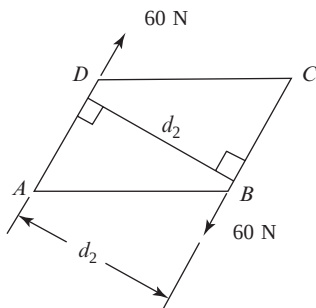




PROBLEM 3.70

A plate in the shape of a parallelogram is acted upon by two couples. Determine (a) the moment of the couple formed by the two 105 N forces, (b) the perpendicular distance between the 60 N forces if the resultant of the two couples is zero, (c) the value of α if the resultant couple is 9 N·m clockwise and d is 1050 mm.

SOLUTION



- (a) We have
where

$$M_1 = d_1 F_1$$

$$d_1 = 400 \text{ mm}$$

$$F_1 = 105 \text{ N}$$

$$M_1 = (400 \text{ mm})(105 \text{ N})$$

$$= 42000 \text{ N} \cdot \text{mm} \quad \text{or } \mathbf{M_1 = 42 \text{ N} \cdot \text{m} \curvearrowright}$$

- (b) We have

$$\mathbf{M_1 + M_2 = 0}$$

or

$$42 \text{ N} \cdot \text{m} - d_2(60 \text{ N}) = 0$$

$$d_2 = 0.7 \text{ m} = 700 \text{ mm} \quad \blacktriangleleft$$

- (c) We have

$$\mathbf{M_{total} = M_1 + M_2}$$

or

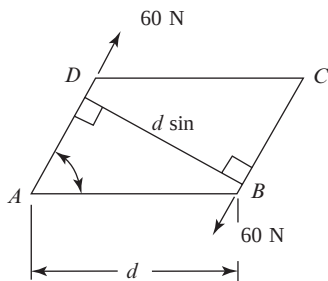
$$-9 \text{ N} \cdot \text{m} = 42 \text{ N} \cdot \text{m} - \left(\frac{1050}{1000} \text{ m} \right) (\sin \alpha) (60 \text{ N})$$

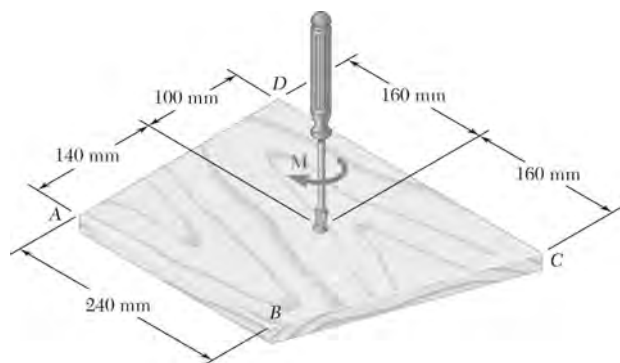
$$\sin \alpha = 0.80952$$

and

$$\alpha = 54.049^\circ$$

$$\text{or } \alpha = 54.0^\circ \quad \blacktriangleleft$$

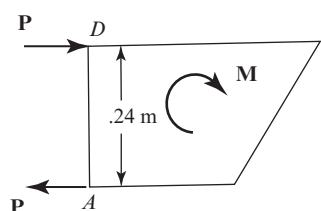




PROBLEM 3.71

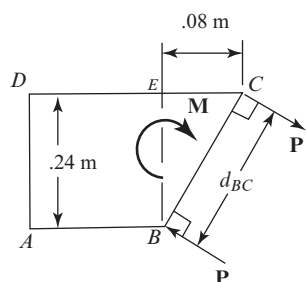
A couple $\mathbf{M}$ of magnitude $18 \text{ N}\cdot\text{m}$ is applied to the handle of a screwdriver to tighten a screw into a block of wood. Determine the magnitudes of the two smallest horizontal forces that are equivalent to $\mathbf{M}$ if they are applied (a) at corners A and D, (b) at corners B and C, (c) anywhere on the block.

SOLUTION



(a) We have $M = Pd$
 or $18 \text{ N}\cdot\text{m} = P(.24 \text{ m})$
 $P = 75.0 \text{ N}$

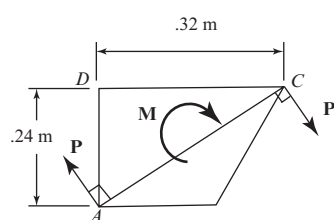
or $P_{\min} = 75.0 \text{ N} \blacktriangleleft$



(b) $d_{BC} = \sqrt{(BE)^2 + (EC)^2}$
 $= \sqrt{(.24 \text{ m})^2 + (.08 \text{ m})^2}$
 $= 0.25298 \text{ m}$

We have $M = Pd$
 $18 \text{ N}\cdot\text{m} = P(0.25298 \text{ m})$
 $P = 71.152 \text{ N}$

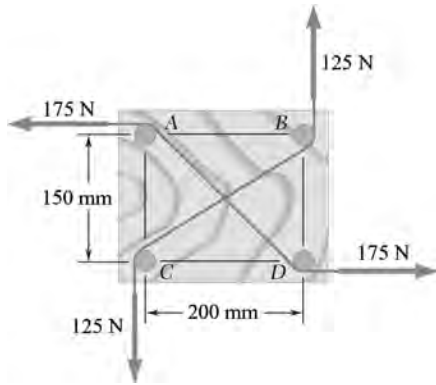
or $P = 71.2 \text{ N} \blacktriangleleft$



(c) $d_{AC} = \sqrt{(AD)^2 + (DC)^2}$
 $= \sqrt{(0.24 \text{ m})^2 + (0.32 \text{ m})^2}$
 $= 0.4 \text{ m}$

We have $M = Pd_{AC}$
 $18 \text{ N}\cdot\text{m} = P(0.4 \text{ m})$
 $P = 45.0 \text{ N}$

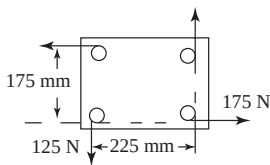
or $P = 45.0 \text{ N} \blacktriangleleft$



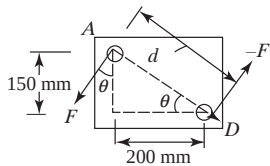
PROBLEM 3.72

Four 25 mm-diameter pegs are attached to a board as shown. Two strings are passed around the pegs and pulled with the forces indicated. (a) Determine the resultant couple acting on the board. (b) If only one string is used, around which pegs should it pass and in what directions should it be pulled to create the same couple with the minimum tension in the string? (c) What is the value of that minimum tension?

SOLUTION



$$\begin{aligned}
 (a) \quad +\curvearrowright M &= (175 \text{ N})(175 \text{ mm}) + (125 \text{ N})(225 \text{ mm}) \\
 &= 30625 \text{ N} \cdot \text{mm} + 28125 \text{ N} \cdot \text{mm} \\
 M &= 58750 \text{ N} \cdot \text{mm} \curvearrowright \blacktriangleleft
 \end{aligned}$$



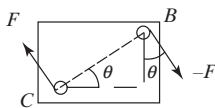
(b) With only one string, pegs A and D, or B and C should be used. We have

$$\tan \theta = \frac{150}{200} \quad \theta = 36.9^\circ \quad 90^\circ - \theta = 53.1^\circ$$

Direction of forces:

With pegs A and D: $\theta = 53.1^\circ \blacktriangleleft$

With pegs B and C: $\theta = 53.1^\circ \blacktriangleleft$



(c) The distance between the centers of the two pegs is

$$\sqrt{(200)^2 + (150)^2} = 250 \text{ mm}$$

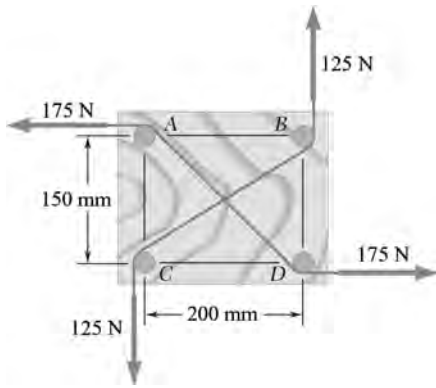
Therefore, the perpendicular distance d between the forces is

$$\begin{aligned}
 d &= 250 \text{ mm} + 2\left(\frac{25}{2} \text{ mm}\right) \\
 &= 275 \text{ mm}
 \end{aligned}$$

We must have $M = Fd = 58750 \text{ N} \cdot \text{mm}$

$$= F(275 \text{ mm}) \quad F = 213.6 \text{ N} = 214 \text{ N} \blacktriangleleft$$

PROBLEM 3.73



Four pegs of the same diameter are attached to a board as shown. Two strings are passed around the pegs and pulled with the forces indicated. Determine the diameter of the pegs knowing that the resultant couple applied to the board is $60.25 \text{ N} \cdot \text{m}$ counterclockwise.

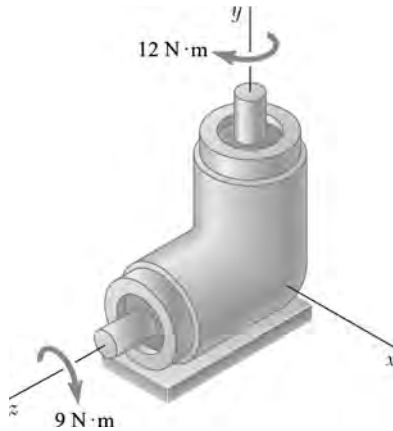
SOLUTION

$$M = d_{AD}F_{AD} + d_{BC}F_{BC}$$

$$M = 60.25 \text{ N} \cdot \text{m} = 60250 \text{ N} \cdot \text{mm}$$

$$60250 \text{ N} \cdot \text{mm} = [(150 + d)\text{mm}](175 \text{ N}) + [(200 + d)\text{mm}](125 \text{ N})$$

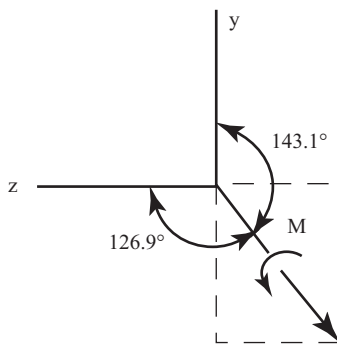
$$d = 30 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 3.74

The shafts of an angle drive are acted upon by the two couples shown. Replace the two couples with a single equivalent couple, specifying its magnitude and the direction of its axis.

SOLUTION



Based on $\mathbf{M} = \mathbf{M}_1 + \mathbf{M}_2$

where $\mathbf{M}_1 = -(12 \text{ N} \cdot \text{m})\mathbf{j}$

$\mathbf{M}_2 = -(9 \text{ N} \cdot \text{m})\mathbf{k}$

$\mathbf{M} = -(12 \text{ N} \cdot \text{m})\mathbf{j} - (9 \text{ N} \cdot \text{m})\mathbf{k}$

and

$|\mathbf{M}| = \sqrt{(12)^2 + (9)^2} = 15 \text{ N} \cdot \text{m}$

or $M = 15 \text{ N} \cdot \text{m} \quad \blacktriangleleft$

$\lambda = \frac{\mathbf{M}}{|\mathbf{M}|}$

$= \frac{-(12 \text{ N} \cdot \text{m})\mathbf{j} - (9 \text{ N} \cdot \text{m})\mathbf{k}}{15 \text{ N} \cdot \text{m}}$

$= -0.8\mathbf{j} - 0.6\mathbf{k}$

or

$\mathbf{M} = |\mathbf{M}|\lambda = (15 \text{ N} \cdot \text{m})(-0.8\mathbf{j} - 0.6\mathbf{k})$

$\cos \theta_x = 0 \quad \theta_x = 90^\circ$

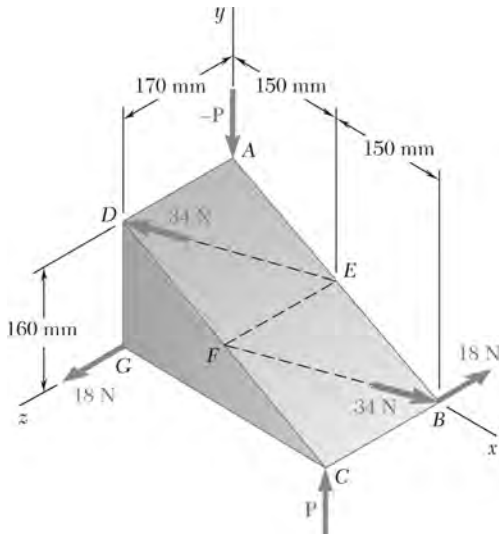
$\cos \theta_y = -0.8 \quad \theta_y = 143.130^\circ$

$\cos \theta_z = -0.6 \quad \theta_z = 126.870^\circ$

or $\theta_x = 90.0^\circ \quad \theta_y = 143.1^\circ \quad \theta_z = 126.9^\circ \quad \blacktriangleleft$

PROBLEM 3.75

If $P = 0$, replace the two remaining couples with a single equivalent couple, specifying its magnitude and the direction of its axis.



SOLUTION

We have $\mathbf{M} = \mathbf{M}_1 + \mathbf{M}_2$

where $\mathbf{M}_1 = \mathbf{r}_{G/C} \times \mathbf{F}_1$

$$\mathbf{r}_{G/C} = -(0.3 \text{ m})\mathbf{i}$$

$$\mathbf{F}_1 = (18 \text{ N})\mathbf{k}$$

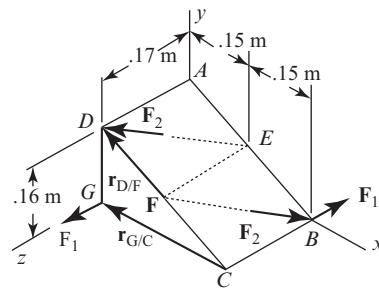
$$\begin{aligned}\mathbf{M}_1 &= -(0.3 \text{ m})\mathbf{i} \times (18 \text{ N})\mathbf{k} \\ &= (5.4 \text{ N} \cdot \text{m})\mathbf{j}\end{aligned}$$

Also, $\mathbf{M}_2 = \mathbf{r}_{D/F} \times \mathbf{F}_2$

$$\mathbf{r}_{D/F} = -(0.15 \text{ m})\mathbf{i} + (0.08 \text{ m})\mathbf{j}$$

$$\begin{aligned}\mathbf{F}_2 &= \lambda_{ED} \mathbf{F}_2 \\ &= \frac{(0.15 \text{ m})\mathbf{i} + (0.08 \text{ m})\mathbf{j} + (0.17 \text{ m})\mathbf{k}}{\sqrt{(0.15)^2 + (0.08)^2 + (0.17)^2} \text{ m}} (34 \text{ N}) \\ &= 141.421 \text{ N} \cdot \text{m} (.15\mathbf{i} + .08\mathbf{j} + .17\mathbf{k})\end{aligned}$$

$$\begin{aligned}\mathbf{M}_2 &= 141.421 \text{ N} \cdot \text{m} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -.15 & .08 & 0 \\ -.15 & .08 & .17 \end{vmatrix} \\ &= 141.421 (.0136\mathbf{i} + 0.0255\mathbf{j}) \text{ N} \cdot \text{m}\end{aligned}$$



PROBLEM 3.75 (Continued)

and

$$\begin{aligned}\mathbf{M} &= [(5.4 \text{ N} \cdot \text{m})\mathbf{j}] + [141.421(.0136\mathbf{i} + .0255\mathbf{j}) \text{ N} \cdot \text{m}] \\ &= (1.92333 \text{ N} \cdot \text{m})\mathbf{i} + (9.0062 \text{ N} \cdot \text{m})\mathbf{j}\end{aligned}$$

$$\begin{aligned}|\mathbf{M}| &= \sqrt{(M_x)^2 + (M_y)^2} \\ &= \sqrt{(1.92333)^2 + (9.0062)^2} \\ &= 9.2093 \text{ N} \cdot \text{m}\end{aligned}$$

$$\text{or } M = 9.21 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$

$$\begin{aligned}\lambda &= \frac{\mathbf{M}}{|\mathbf{M}|} = \frac{(1.92333 \text{ N} \cdot \text{m})\mathbf{i} + (9.0062 \text{ N} \cdot \text{m})\mathbf{j}}{9.2093 \text{ N} \cdot \text{m}} \\ &= 0.20885 + 0.97795\mathbf{j}\end{aligned}$$

$$\cos \theta_x = 0.20885$$

$$\theta_x = 77.945^\circ$$

$$\text{or } \theta_x = 77.9^\circ \quad \blacktriangleleft$$

$$\cos \theta_y = 0.97795$$

$$\theta_y = 12.054^\circ$$

$$\text{or } \theta_y = 12.05^\circ \quad \blacktriangleleft$$

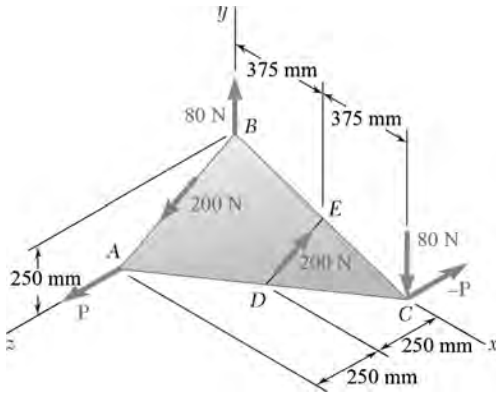
$$\cos \theta_z = 0.0$$

$$\theta_z = 90^\circ$$

$$\text{or } \theta_z = 90.0^\circ \quad \blacktriangleleft$$

PROBLEM 3.76

If $P = 0$, replace the two remaining couples with a single equivalent couple, specifying its magnitude and the direction of its axis.



SOLUTION

$$\mathbf{M} = \mathbf{M}_1 + \mathbf{M}_2; \quad F_1 = 80 \text{ N}, \quad F_2 = 200 \text{ N}$$

$$\mathbf{M}_1 = \mathbf{r}_C \times \mathbf{F}_1 = (750 \text{ mm})\mathbf{i} \times (-80 \text{ N})\mathbf{j} = -(60000 \text{ N}\cdot\text{mm})\mathbf{k}$$

$$\mathbf{M}_2 = \mathbf{r}_{E/B} \times \mathbf{F}_2; \quad \mathbf{r}_{E/B} = (375 \text{ mm})\mathbf{i} - (125 \text{ mm})\mathbf{j}$$

$$d_{DE} = \sqrt{(0)^2 + (125)^2 + (250)^2} = 125\sqrt{5} \text{ mm}$$

$$F_2 = \frac{200 \text{ N}}{125\sqrt{5}} (125\mathbf{j} - 250\mathbf{k})$$

$$= 40\sqrt{5} [(1\text{N})\mathbf{j} - (2\text{N})\mathbf{k}]$$

$$\mathbf{M}_2 = 40\sqrt{5} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 375 & -125 & 0 \\ 0 & 1 & -2 \end{vmatrix}$$

$$= 40\sqrt{5} [(250 \text{ N}\cdot\text{mm})\mathbf{i} + (750 \text{ N}\cdot\text{mm})\mathbf{j} + (375 \text{ N}\cdot\text{mm})\mathbf{k}]$$

$$\mathbf{M} = (22361\mathbf{i} + 67082\mathbf{j} + 33541\mathbf{k}) \text{ N}\cdot\text{mm}$$

$$= (22.361 \text{ N}\cdot\text{mm})\mathbf{i} + (67.082 \text{ N}\cdot\text{mm})\mathbf{j} - (26.459 \text{ N}\cdot\text{mm})\mathbf{k}$$

$$M = \sqrt{(22361)^2 + (67082)^2 + (-26459)^2}$$

$$= 75499 \text{ N}\cdot\text{mm}$$

$$M = 75.5 \text{ N}\cdot\text{m} \quad \blacktriangleleft$$

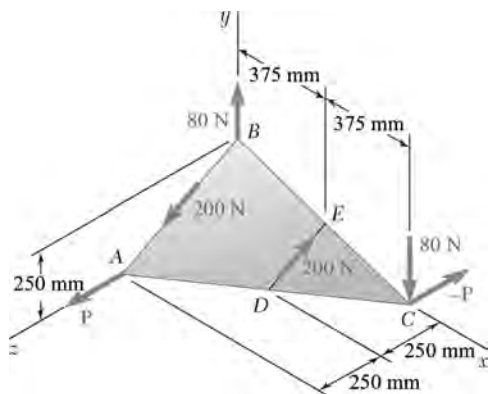
$$\lambda_{\text{axis}} = \frac{\mathbf{M}}{M} = 0.29617\mathbf{i} + 0.88852\mathbf{j} - 0.35045\mathbf{k}$$

$$\cos \theta_x = 0.29617$$

$$\cos \theta_y = 0.88852$$

$$\cos \theta_z = -0.35045$$

$$\theta_x = 72.8^\circ \quad \theta_y = 27.3^\circ \quad \theta_z = 110.5^\circ \quad \blacktriangleleft$$



PROBLEM 3.77

If $P = 100$ N, replace the three couples with a single equivalent couple, specifying its magnitude and the direction of its axis.

SOLUTION

From the solution to Problem. 3.76

$$80 \text{ N force: } M_1 = -60000 \text{ N} \cdot \text{mm} \mathbf{k}$$

$$200 \text{ N force: } M_2 = 8\sqrt{5}[(22361 \text{ N} \cdot \text{mm})\mathbf{i} + (67082 \text{ N} \cdot \text{mm})\mathbf{j} + (33541 \text{ N} \cdot \text{mm})\mathbf{k}]$$

$$\begin{aligned} P = 100 \text{ N} \quad M_3 &= \mathbf{r}_C \times P \\ &= (750 \text{ mm})\mathbf{i} \times (100 \text{ N})(-\mathbf{k}) \\ &= (75000 \text{ N} \cdot \text{mm})\mathbf{j} \end{aligned}$$

$$\begin{aligned} \mathbf{M} &= \mathbf{M}_1 + \mathbf{M}_2 + \mathbf{M}_3 \\ &= (22361 \text{ N} \cdot \text{mm})\mathbf{i} + (142082 \text{ N} \cdot \text{mm})\mathbf{j} - (26459 \text{ N} \cdot \text{mm})\mathbf{k} \end{aligned}$$

$$\begin{aligned} M &= \sqrt{(22361)^2 + (142082)^2 + (-26459)^2} \\ &= 146244 \text{ N} \cdot \text{mm} \end{aligned}$$

$$M = 146.2 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$

$$\lambda_{\text{axis}} = \frac{\mathbf{M}}{M} = 0.15290\mathbf{i} + 0.97154\mathbf{j} - 0.18092\mathbf{k}$$

$$\cos \theta_x = 0.15290$$

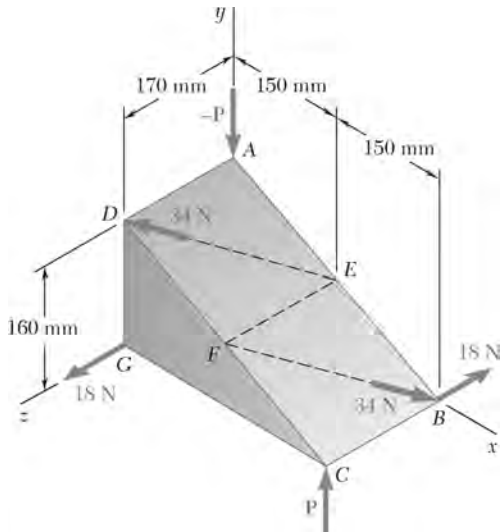
$$\cos \theta_y = 0.97154$$

$$\cos \theta_z = -0.180921$$

$$\theta_x = 81.2^\circ \quad \theta_y = 13.70^\circ \quad \theta_z = 100.4^\circ \quad \blacktriangleleft$$

PROBLEM 3.78

If $P = 20$ N, replace the three couples with a single equivalent couple, specifying its magnitude and the direction of its axis.



SOLUTION

We have

$$\mathbf{M} = \mathbf{M}_1 + \mathbf{M}_2 + \mathbf{M}_3$$

where

$$\mathbf{M}_1 = \mathbf{r}_{G/C} \times \mathbf{F}_1 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -0.3 & 0 & 0 \\ 0 & 0 & 18 \end{vmatrix} \text{ N} \cdot \text{m} = (5.4 \text{ N} \cdot \text{m})\mathbf{j}$$

$$\begin{aligned} \mathbf{M}_2 &= \mathbf{r}_{D/E} \times \mathbf{F}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -0.15 & 0.08 & 0 \\ -0.15 & 0.08 & 0.17 \end{vmatrix} 141.421 \text{ N} \cdot \text{m} \\ &= 141.421(0.0136\mathbf{i} + 0.0255\mathbf{j}) \text{ N} \cdot \text{m} \end{aligned}$$

(See Solution to Problem 3.75)

$$\begin{aligned} \mathbf{M}_3 &= \mathbf{r}_{C/A} \times \mathbf{F}_3 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.3 & 0 & 0.17 \\ 0 & 20 & 0 \end{vmatrix} \text{ N} \cdot \text{m} \\ &= -(3.4 \text{ N} \cdot \text{m})\mathbf{i} + (6 \text{ N} \cdot \text{m})\mathbf{k} \end{aligned}$$

$$\begin{aligned} \mathbf{M} &= [(1.92333 - 3.4)\mathbf{i} + (5.4 + 3.6062)\mathbf{j} + (6)\mathbf{k}] \text{ N} \cdot \text{m} \\ &= -(1.47667 \text{ N} \cdot \text{m})\mathbf{i} + (9.0062 \text{ N} \cdot \text{m})\mathbf{j} + (6 \text{ N} \cdot \text{m})\mathbf{k} \end{aligned}$$

$$\begin{aligned} |\mathbf{M}| &= \sqrt{M_x^2 + M_y^2 + M_z^2} \\ &= \sqrt{(1.47667)^2 + (9.0062)^2 + (6)^2} \\ &= 10.9221 \text{ N} \cdot \text{m} \end{aligned}$$

$$\text{or } M = 10.92 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$

PROBLEM 3.78 (Continued)

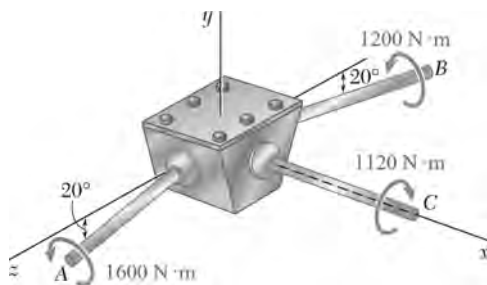
$$\lambda = \frac{\mathbf{M}}{|\mathbf{M}|} = \frac{-1.47667 + 9.0062 + 6}{10.9221}$$

$$= -0.135200\mathbf{i} + 0.82459\mathbf{j} + 0.54934\mathbf{k}$$

$$\cos \theta_x = -0.135200 \quad \theta_x = 97.770 \quad \text{or} \quad \theta_x = 97.8^\circ \quad \blacktriangleleft$$

$$\cos \theta_y = 0.82459 \quad \theta_y = 34.453 \quad \text{or} \quad \theta_y = 34.5^\circ \quad \blacktriangleleft$$

$$\cos \theta_z = 0.54934 \quad \theta_z = 56.678 \quad \text{or} \quad \theta_z = 56.7^\circ \quad \blacktriangleleft$$



PROBLEM 3.79

Shafts *A* and *B* connect the gear box to the wheel assemblies of a tractor, and shaft *C* connects it to the engine. Shafts *A* and *B* lie in the vertical *yz* plane, while shaft *C* is directed along the *x* axis. Replace the couples applied to the shafts with a single equivalent couple, specifying its magnitude and the direction of its axis.

SOLUTION

Represent the given couples by the following couple vectors:

$$\begin{aligned}\mathbf{M}_A &= -1600 \sin 20^\circ \mathbf{j} + 1600 \cos 20^\circ \mathbf{k} \\ &= -(547.232 \text{ N} \cdot \text{m}) \mathbf{j} + (1503.51 \text{ N} \cdot \text{m}) \mathbf{k}\end{aligned}$$

$$\begin{aligned}\mathbf{M}_B &= 1200 \sin 20^\circ \mathbf{j} + 1200 \cos 20^\circ \mathbf{k} \\ &= (410.424 \text{ N} \cdot \text{m}) \mathbf{j} + (1127.63 \text{ N} \cdot \text{m}) \mathbf{k}\end{aligned}$$

$$\mathbf{M}_C = -(1120 \text{ N} \cdot \text{m}) \mathbf{i}$$

The single equivalent couple is

$$\begin{aligned}\mathbf{M} &= \mathbf{M}_A + \mathbf{M}_B + \mathbf{M}_C \\ &= -(1120 \text{ N} \cdot \text{m}) \mathbf{i} - (136.808 \text{ N} \cdot \text{m}) \mathbf{j} + (2631.1 \text{ N} \cdot \text{m}) \mathbf{k}\end{aligned}$$

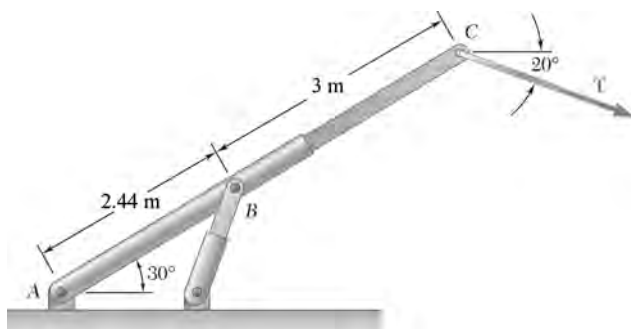
$$\begin{aligned}M &= \sqrt{(1120)^2 + (136.808)^2 + (2631.1)^2} \\ &= 2862.8 \text{ N} \cdot \text{m}\end{aligned}$$

$$\cos \theta_x = \frac{-1120}{2862.8}$$

$$\cos \theta_y = \frac{-136.808}{2862.8}$$

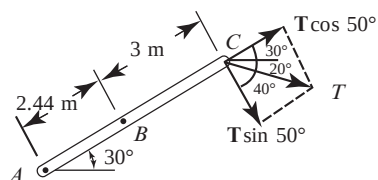
$$\cos \theta_z = \frac{2631.1}{2862.8}$$

$$M = 2860 \text{ N} \cdot \text{m} \quad \theta_x = 113.0^\circ \quad \theta_y = 92.7^\circ \quad \theta_z = 23.2^\circ \quad \blacktriangleleft$$



PROBLEM 3.80

The tension in the cable attached to the end C of an adjustable boom ABC is 2.8 kN. Replace the force exerted by the cable at C with an equivalent force-couple system (a) at A, (b) at B.



SOLUTION

(a) Based on

$$\Sigma F: F_A = T = 2800 \text{ N}$$

or

$$\mathbf{F}_A = 2800 \text{ N} \searrow 20^\circ \blacktriangleleft$$

$$\begin{aligned} \Sigma M_A: M_A &= (T \sin 50^\circ)(d_A) \\ &= (2800 \text{ N}) \sin 50^\circ (5.44 \text{ m}) \\ &= 11668 \text{ N} \cdot \text{m} \end{aligned}$$

or

$$\mathbf{M}_A = 11.67 \text{ kN} \cdot \text{m} \curvearrowright \blacktriangleleft$$

(b) Based on

$$\Sigma F: F_B = T = (2800) \text{ N}$$

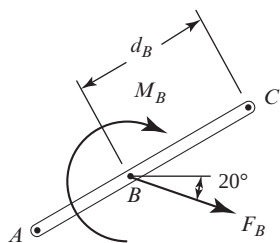
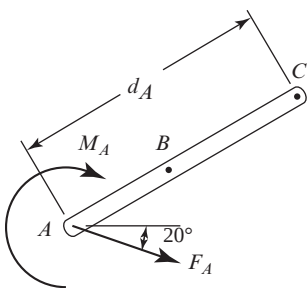
or

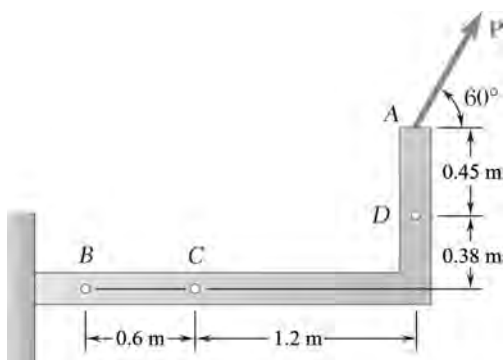
$$\mathbf{F}_B = 2800 \text{ N} \searrow 20^\circ \blacktriangleleft$$

$$\begin{aligned} \Sigma M_B: M_B &= (T \sin 50^\circ)(d_B) \\ &= (2800 \text{ N})(\sin 50^\circ)(3 \text{ m}) \\ &= 6435 \text{ N} \cdot \text{m} \end{aligned}$$

or

$$\mathbf{M}_B = 6.43 \text{ kN} \cdot \text{m} \curvearrowright \blacktriangleleft$$





PROBLEM 3.81

A 800 N force $\mathbf{P}$ is applied at Point A of a structural member. Replace $\mathbf{P}$ with (a) an equivalent force-couple system at C, (b) an equivalent system consisting of a vertical force at B and a second force at D.

SOLUTION

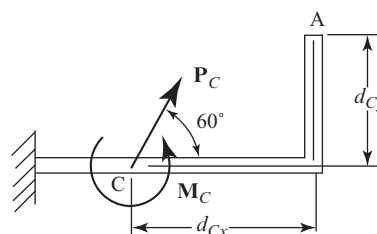
(a) Based on $\Sigma F: P_C = P = 800 \text{ N}$
 $\Sigma M_C: M_C = -P_x d_{Cy} + P_y d_{Cx}$

or $\mathbf{P}_C = 800 \text{ N } 60^\circ \swarrow$

where $P_x = (800 \text{ N}) \cos 60^\circ$
 $= 400 \text{ N}$
 $P_y = (800 \text{ N}) \sin 60^\circ$
 $= 692.82 \text{ N}$
 $d_{Cx} = 1.2 \text{ m}$
 $d_{Cy} = 0.83 \text{ m}$

$$M_C = (400 \text{ N})(0.83 \text{ m}) + (692.82 \text{ N})(1.2 \text{ m})$$

$$= 1163.4 \text{ N} \cdot \text{m}$$



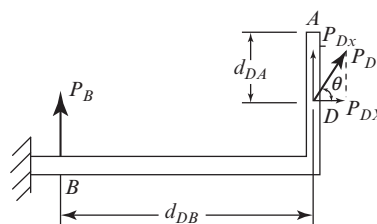
or $\mathbf{M}_C = 1163 \text{ N} \cdot \text{m } \curvearrowleft$

(b) Based on $\Sigma F_x: P_{Dx} = P \cos 60^\circ$
 $= (800 \text{ N}) \cos 60^\circ$
 $= 400 \text{ N}$

$$\Sigma M_D: (P \cos 60^\circ)(d_{DA}) = P_B(d_{DB})$$

$$[(800 \text{ N}) \cos 60^\circ](0.45 \text{ m}) = P_B(1.8 \text{ m})$$

$$P_B = 100 \text{ N}$$



or $\mathbf{P}_B = 100 \text{ N } \uparrow$

PROBLEM 3.81 (Continued)

$$\Sigma F_y: P \sin 60^\circ = P_B + P_{Dy}$$

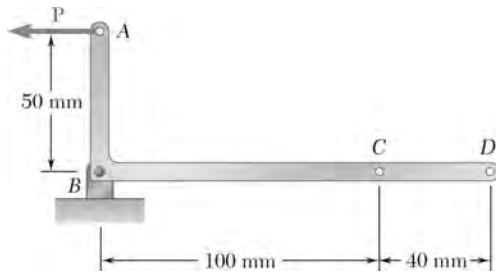
$$(800 \text{ N}) \sin 60^\circ = 100 \text{ N} + P_{Dy}$$

$$P_{Dy} = 592.82 \text{ N}$$

$$\begin{aligned} P_D &= \sqrt{(P_{Dx})^2 + (P_{Dy})^2} \\ &= \sqrt{(400)^2 + (592.82)^2} \\ &= 715.15 \text{ N} \end{aligned}$$

$$\begin{aligned} \theta &= \tan^{-1} \left(\frac{P_{Dy}}{P_{Dx}} \right) \\ &= \tan^{-1} \left(\frac{592.82}{400} \right) \\ &= 55.991^\circ \end{aligned}$$

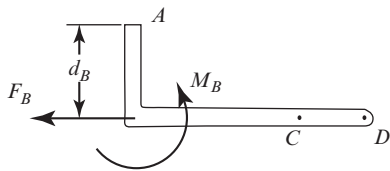
$$\text{or } P_D = 715 \text{ N } \angle 56.0^\circ \blacktriangleleft$$



PROBLEM 3.82

The 80-N horizontal force **P** acts on a bell crank as shown.
 (a) Replace **P** with an equivalent force-couple system at **B**.
 (b) Find the two vertical forces at **C** and **D** that are equivalent to the couple found in Part *a*.

SOLUTION

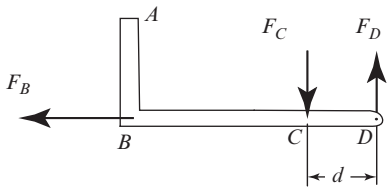


(a) Based on $\Sigma F: F_B = F = 80 \text{ N}$ or $\mathbf{F}_B = 80.0 \text{ N} \leftarrow \blacktriangleleft$

$$\begin{aligned}\Sigma M: M_B &= Fd_B \\ &= 80 \text{ N} (.05 \text{ m}) \\ &= 4.0000 \text{ N} \cdot \text{m}\end{aligned}$$

or $\mathbf{M}_B = 4.00 \text{ N} \cdot \text{m} \curvearrowright \blacktriangleleft$

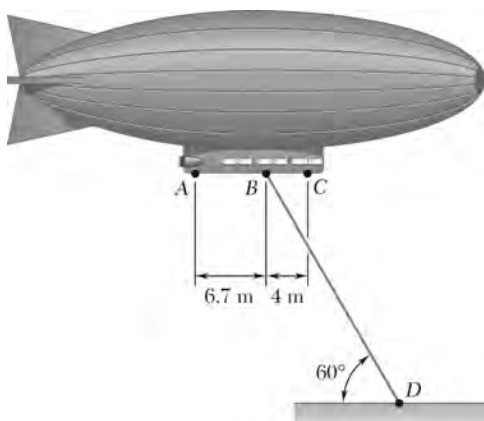
(b) If the two vertical forces are to be equivalent to $\mathbf{M}_B$, they must be a couple. Further, the sense of the moment of this couple must be counterclockwise.



Then, with F_C and F_D acting as shown,

$$\begin{aligned}\Sigma M: M_D &= F_C d \\ 4.0000 \text{ N} \cdot \text{m} &= F_C (.04 \text{ m}) \\ F_C &= 100.000 \text{ N} \quad \text{or } \mathbf{F}_C = 100.0 \text{ N} \downarrow \blacktriangleleft\end{aligned}$$

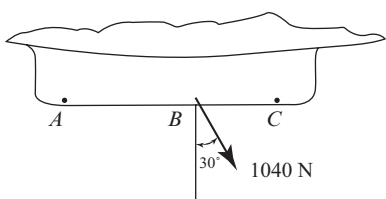
$$\begin{aligned}\Sigma F_y: 0 &= F_D - F_C \\ F_D &= 100.000 \text{ N} \quad \text{or } \mathbf{F}_D = 100.0 \text{ N} \uparrow \blacktriangleleft\end{aligned}$$



PROBLEM 3.83

A dirigible is tethered by a cable attached to its cabin at B . If the tension in the cable is 1040 N, replace the force exerted by the cable at B with an equivalent system formed by two parallel forces applied at A and C .

SOLUTION



Require the equivalent forces acting at A and C be parallel and at an angle of α with the vertical.

Then for equivalence,

$$\Sigma F_x: (1040 \text{ N}) \sin 30^\circ = F_A \sin \alpha + F_B \sin \alpha \quad (1)$$

$$\Sigma F_y: -(1040 \text{ N}) \cos 30^\circ = -F_A \cos \alpha - F_B \cos \alpha \quad (2)$$

Dividing Equation (1) by Equation (2),

$$\frac{(1040 \text{ N}) \sin 30^\circ}{-(1040 \text{ N}) \cos 30^\circ} = \frac{(F_A + F_B) \sin \alpha}{-(F_A + F_B) \cos \alpha}$$

Simplifying yields $\alpha = 30^\circ$

Based on

$$\Sigma M_C: [(1040 \text{ N}) \cos 30^\circ](4 \text{ m}) = (F_A \cos 30^\circ)(10.7 \text{ m})$$

$$F_A = 388.79 \text{ N}$$

or

$$\mathbf{F}_A = 389 \text{ N} \nearrow 60^\circ \blacktriangleleft$$

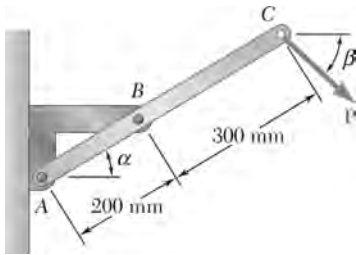
Based on

$$\Sigma M_A: -[(1040 \text{ N}) \cos 30^\circ](6.7 \text{ m}) = (F_C \cos 30^\circ)(10.7 \text{ m})$$

$$F_C = 651.21 \text{ N}$$

or

$$\mathbf{F}_C = 651 \text{ N} \nearrow 60^\circ \blacktriangleleft$$



PROBLEM 3.84

The force $\mathbf{P}$ has a magnitude of 250 N and is applied at the end C of a 500-mm rod AC attached to a bracket at A and B . Assuming $\alpha = 30^\circ$ and $\beta = 60^\circ$, replace $\mathbf{P}$ with (a) an equivalent force-couple system at B , (b) an equivalent system formed by two parallel forces applied at A and B .

SOLUTION

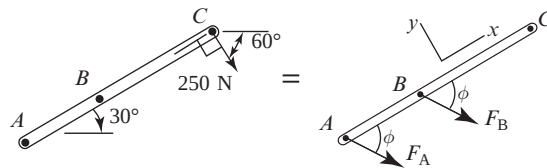
(a) Equivalence requires $\Sigma \mathbf{F}: \mathbf{F} = \mathbf{P}$ or $\mathbf{F} = 250 \text{ N} \nearrow 60^\circ$

$$\Sigma \mathbf{M}_B: M = -(0.3 \text{ m})(250 \text{ N}) = -75 \text{ N} \cdot \text{m}$$

The equivalent force-couple system at B is

$$\mathbf{F} = 250 \text{ N} \nearrow 60^\circ \qquad \mathbf{M} = 75.0 \text{ N} \cdot \text{m} \curvearrowright$$

(b) Require



Equivalence then requires

$$\Sigma F_x: 0 = F_A \cos \phi + F_B \cos \phi$$

$$F_A = -F_B \quad \text{or} \quad \cos \phi = 0$$

$$\Sigma F_y: -250 = -F_A \sin \phi - F_B \sin \phi$$

Now if

$$F_A = -F_B \Rightarrow -250 = 0 \quad \text{reject}$$

$$\cos \phi = 0$$

or

$$\phi = 90^\circ$$

and

$$F_A + F_B = 250$$

$$\text{Also} \quad \Sigma M_B: -(0.3 \text{ m})(250 \text{ N}) = (0.2 \text{ m})F_A$$

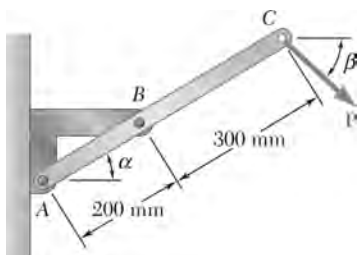
or

$$F_A = -375 \text{ N}$$

and

$$F_B = 625 \text{ N}$$

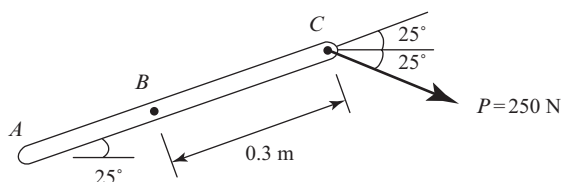
$$\mathbf{F}_A = 375 \text{ N} \nearrow 60^\circ \quad \mathbf{F}_B = 625 \text{ N} \nearrow 60^\circ \quad \blacktriangleleft$$



PROBLEM 3.85

Solve Problem 3.84, assuming $\alpha = \beta = 25^\circ$.

SOLUTION



(a) Equivalence requires

$$\Sigma \mathbf{F}: \mathbf{F}_B = \mathbf{P} \quad \text{or} \quad \mathbf{F}_B = 250 \text{ N} \searrow 25.0^\circ$$

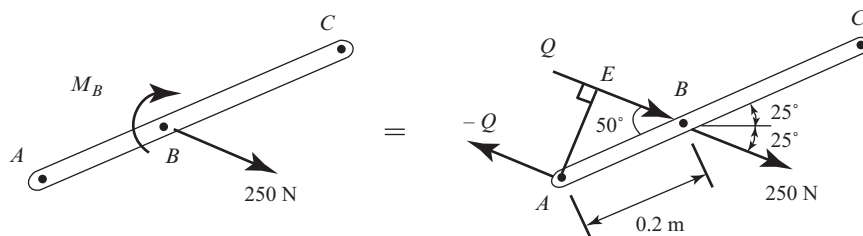
$$\Sigma \mathbf{M}_B: M_B = -(0.3 \text{ m})[(250 \text{ N})\sin 50^\circ] = -57.453 \text{ N} \cdot \text{m}$$

The equivalent force-couple system at B is

$$\mathbf{F}_B = 250 \text{ N} \searrow 25.0^\circ$$

$$\mathbf{M}_B = 57.5 \text{ N} \cdot \text{m} \curvearrowleft$$

(b) Require



Equivalence requires

$$M_B = d_{AE} Q \quad (0.3 \text{ m})[(250 \text{ N})\sin 50^\circ] \\ = [(0.2 \text{ m})\sin 50^\circ] Q$$

$$Q = 375 \text{ N}$$

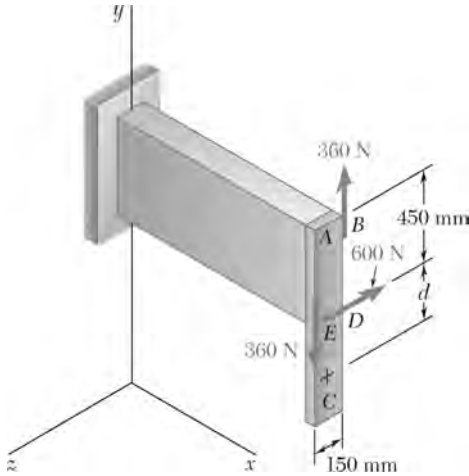
Adding the forces at B:

$$\mathbf{F}_A = 375 \text{ N} \searrow 25.0^\circ$$

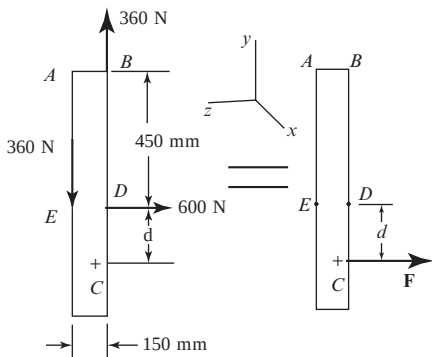
$$\mathbf{F}_B = 625 \text{ N} \searrow 25.0^\circ \curvearrowleft$$

PROBLEM 3.86

A force and a couple are applied as shown to the end of a cantilever beam. (a) Replace this system with a single force $\mathbf{F}$ applied at Point C, and determine the distance d from C to a line drawn through Points D and E. (b) Solve Part a if the directions of the two 360-N forces are reversed.



SOLUTION

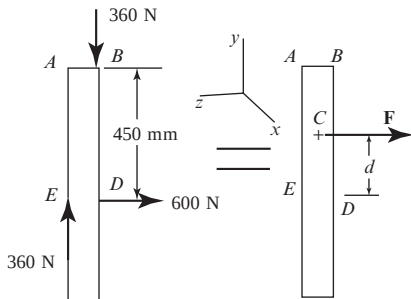


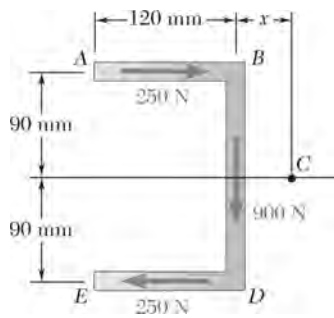
(a) We have $\Sigma \mathbf{F}$: $\mathbf{F} = (360 \text{ N})\mathbf{j} - (360 \text{ N})\mathbf{j} - (600 \text{ N})\mathbf{k}$
or $\mathbf{F} = -(600 \text{ N})\mathbf{k} \blacktriangleleft$

and ΣM_D : $(360 \text{ N})(0.15 \text{ m}) = (600 \text{ N})(d)$
 $d = 0.09 \text{ m}$
or $d = 90.0 \text{ mm below } ED \blacktriangleleft$

(b) We have from Part a $\mathbf{F} = -(600 \text{ N})\mathbf{k} \blacktriangleleft$

and ΣM_D : $-(360 \text{ N})(0.15 \text{ m}) = -(600 \text{ N})(d)$
 $d = 0.09 \text{ m}$
or $d = 90.0 \text{ mm above } ED \blacktriangleleft$



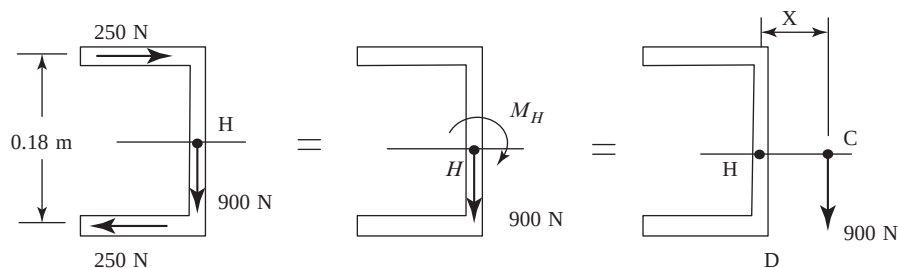


PROBLEM 3.87

The shearing forces exerted on the cross section of a steel channel can be represented by a 900-N vertical force and two 250-N horizontal forces as shown. Replace this force and couple with a single force **F** applied at Point C, and determine the distance x from C to line BD . (Point C is defined as the *shear center* of the section.)

SOLUTION

Replace the 250-N forces with a couple and move the 900-N force to Point C such that its moment about H is equal to the moment of the couple



$$M_H = (0.18)(250 \text{ N})$$

$$= 45 \text{ N} \cdot \text{m}$$

Then

$$M_H = x(900 \text{ N})$$

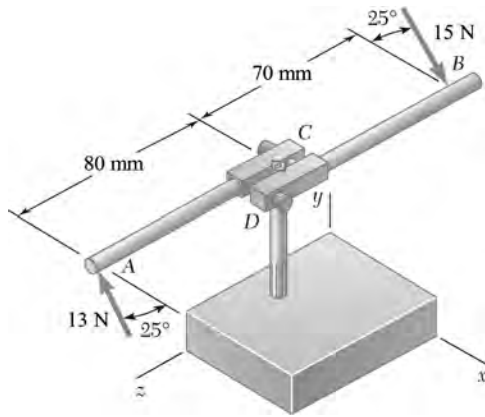
or

$$45 \text{ N} \cdot \text{m} = x(900 \text{ N})$$

$$x = 0.05 \text{ m}$$

$$\mathbf{F} = 900 \text{ N} \downarrow \quad x = 50.0 \text{ mm} \quad \blacktriangleleft$$

PROBLEM 3.88



While tapping a hole, a machinist applies the horizontal forces shown to the handle of the tap wrench. Show that these forces are equivalent to a single force, and specify, if possible, the point of application of the single force on the handle.

SOLUTION

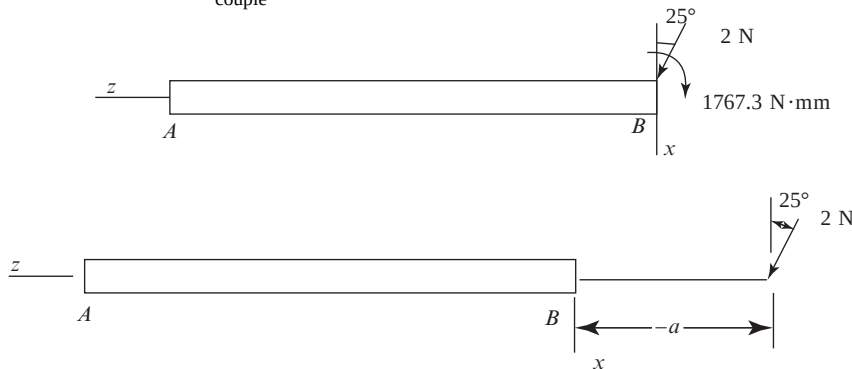
Since the forces at A and B are parallel, the force at B can be replaced with the sum of two forces with one of the forces equal in magnitude to the force at A except with an opposite sense, resulting in a force-couple.

Have $F_B = 15 \text{ N} - 13 \text{ N} = 2 \text{ N}$, where the 13 N force be part of the couple. Combining the two parallel forces,

$$M_{\text{couple}} = (13 \text{ N})[(80 \text{ mm} + 70 \text{ mm}) \cos 25^\circ] \\ = 1767.3 \text{ N} \cdot \text{mm}$$

and

$$\mathbf{M}_{\text{couple}} = 1767.3 \text{ N} \cdot \text{mm}$$



A single equivalent force will be located in the negative z-direction

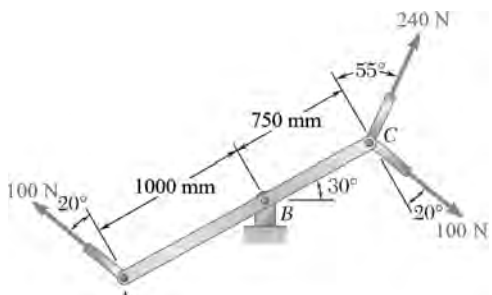
Based on $\Sigma M_B: -1767.3 \text{ N} \cdot \text{mm} = [(2 \text{ N}) \cos 25^\circ](a)$

$$a = 975 \text{ mm}$$

$$\mathbf{F}' = (2 \text{ N})(\cos 25^\circ \mathbf{i} + \sin 25^\circ \mathbf{k})$$

$\mathbf{F}' = (1.813 \text{ N})\mathbf{i} + (0.845 \text{ N})\mathbf{k}$ and is applied on an extension of handle BD at a distance of 975 mm to the right of B





PROBLEM 3.89

Three control rods attached to a lever ABC exert on it the forces shown. (a) Replace the three forces with an equivalent force-couple system at B . (b) Determine the single force that is equivalent to the force-couple system obtained in Part a, and specify its point of application on the lever.

SOLUTION

- (a) First note that the two 100-N forces form a couple. Then

$$\mathbf{F} = 240 \text{ N} \nearrow \theta$$

where

$$\theta = 180^\circ - (60^\circ + 55^\circ) = 65^\circ$$

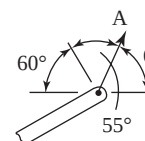
and

$$\begin{aligned} M &= \Sigma M_B \\ &= (750 \text{ mm})(240 \text{ N})\cos 55^\circ - (1750 \text{ mm})(100 \text{ N})\cos 20^\circ \\ &= -61202 \text{ N}\cdot\text{mm} \end{aligned}$$

The equivalent force-couple system at B is

$$\mathbf{F} = 240 \text{ N} \nearrow 65^\circ$$

$$\mathbf{M} = 61.2 \text{ N}\cdot\text{m} \curvearrowright$$



- (b) The single equivalent force $\mathbf{F}'$ is equal to $\mathbf{F}$. Further, since the sense of $\mathbf{M}$ is clockwise, $\mathbf{F}'$ must be applied between A and B . For equivalence,

$$\Sigma M_B: M = -aF'\cos 55^\circ$$

where a is the distance from B to the point of application of $\mathbf{F}'$. Then

$$-61202 \text{ N}\cdot\text{mm} = -a(240 \text{ N})\cos 55^\circ$$

or

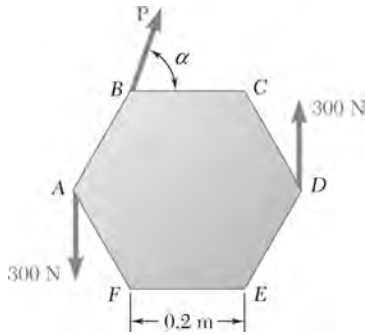
$$a = 444.6 \text{ mm}$$

$$\mathbf{F}' = 240 \text{ N} \nearrow 65.0^\circ$$

and is applied to the lever 445 mm

To the left of pin B

PROBLEM 3.90



A hexagonal plate is acted upon by the force $\mathbf{P}$ and the couple shown. Determine the magnitude and the direction of the smallest force $\mathbf{P}$ for which this system can be replaced with a single force at E .

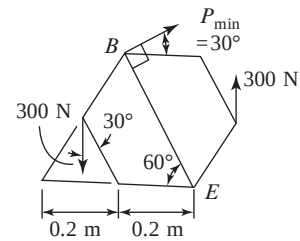
SOLUTION

From the statement of the problem, it follows that $\Sigma M_E = 0$ for the given force-couple system. Further, for $\mathbf{P}_{\min}$, must require that $\mathbf{P}$ be perpendicular to $\mathbf{r}_{B/E}$. Then

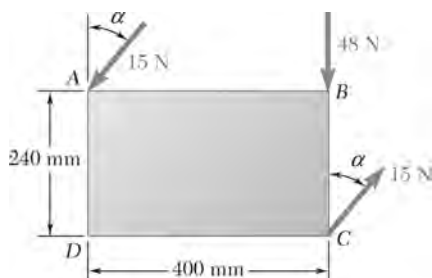
$$\begin{aligned}\Sigma M_E: & (0.2 \sin 30^\circ + 0.2) \text{ m} \times 300 \text{ N} \\ & + (0.2 \text{ m}) \sin 30^\circ \times 300 \text{ N} \\ & - (0.4 \text{ m}) P_{\min} = 0\end{aligned}$$

or

$$P_{\min} = 300 \text{ N}$$



$$\mathbf{P}_{\min} = 300 \text{ N} \quad \nearrow 30.0^\circ \quad \blacktriangleleft$$



PROBLEM 3.91

A rectangular plate is acted upon by the force and couple shown. This system is to be replaced with a single equivalent force. (a) For $\alpha = 40^\circ$, specify the magnitude and the line of action of the equivalent force. (b) Specify the value of α if the line of action of the equivalent force is to intersect line CD 300 mm to the right of D.

SOLUTION

- (a) The given force-couple system (F , M) at B is

$$\mathbf{F} = 48 \text{ N} \downarrow$$

and $M = \Sigma M_B = (0.4 \text{ m})(15 \text{ N}) \cos 40^\circ + (0.24 \text{ m})(15 \text{ N}) \sin 40^\circ$

or $\mathbf{M} = 6.9103 \text{ N} \cdot \text{m} \curvearrowright$

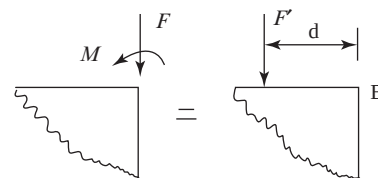
The single equivalent force F' is equal to $\mathbf{F}$. Further for equivalence

$$\Sigma M_B: M = dF'$$

or $6.9103 \text{ N} \cdot \text{m} = d \times 48 \text{ N}$

or $d = 0.14396 \text{ m}$

and the line of action of F' intersects line AB 144 mm to the right of A.



- (b) Following the solution to Part a but with $d = 0.1 \text{ m}$ and α unknown, have

$$\begin{aligned} \Sigma M_B: (0.4 \text{ m})(15 \text{ N}) \cos \alpha + (0.24 \text{ m})(15 \text{ N}) \sin \alpha \\ = (0.1 \text{ m})(48 \text{ N}) \end{aligned}$$

or $5 \cos \alpha + 3 \sin \alpha = 4$

Rearranging and squaring $25 \cos^2 \alpha = (4 - 3 \sin \alpha)^2$

Using $\cos^2 \alpha = 1 - \sin^2 \alpha$ and expanding

$$25(1 - \sin^2 \alpha) = 16 - 24 \sin \alpha + 9 \sin^2 \alpha$$

or $34 \sin^2 \alpha - 24 \sin \alpha - 9 = 0$

Then

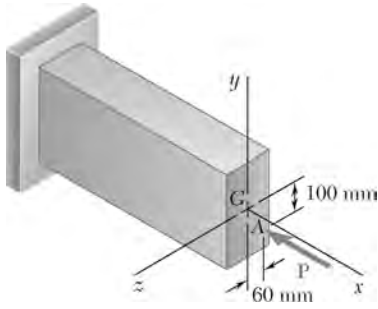
$$\sin \alpha = \frac{24 \pm \sqrt{(-24)^2 - 4(34)(-9)}}{2(34)}$$

$$\sin \alpha = 0.97686 \quad \text{or} \quad \sin \alpha = -0.27098$$

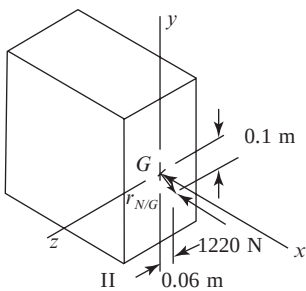
$$\alpha = 77.7^\circ \quad \text{or} \quad \alpha = -15.72^\circ$$

PROBLEM 3.92

An eccentric, compressive 1220-N force $\mathbf{P}$ is applied to the end of a cantilever beam. Replace $\mathbf{P}$ with an equivalent force-couple system at G .



SOLUTION



We have

$$\Sigma \mathbf{F}: -(1220 \text{ N})\mathbf{i} = \mathbf{F}$$

$$\mathbf{F} = -(1220 \text{ N})\mathbf{i} \quad \blacktriangleleft$$

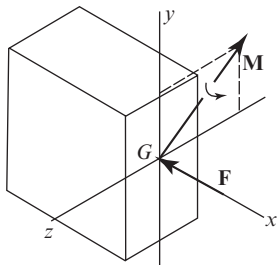
Also, we have

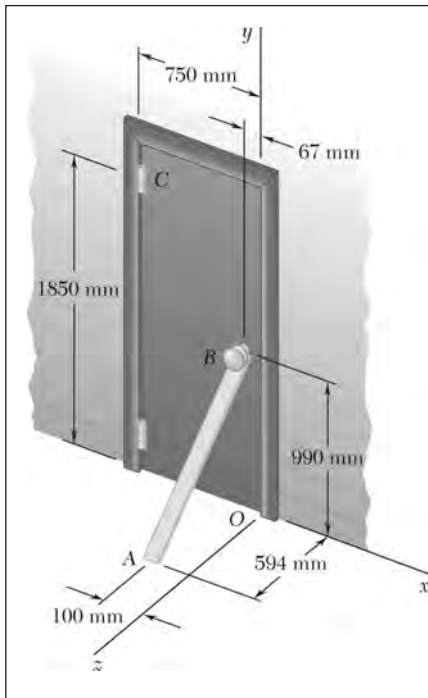
$$\Sigma \mathbf{M}_G: \mathbf{r}_{A/G} \times \mathbf{P} = \mathbf{M}$$

$$1220 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -1 & -0.06 \\ -1 & 0 & 0 \end{vmatrix} \text{ N} \cdot \text{m} = \mathbf{M}$$

$$\mathbf{M} = (1220 \text{ N} \cdot \text{m})[(-0.06)(-1)\mathbf{j} - (-0.1)(-1)\mathbf{k}]$$

$$\text{or } \mathbf{M} = (73.2 \text{ N} \cdot \text{m})\mathbf{j} - (122 \text{ N} \cdot \text{m})\mathbf{k} \quad \blacktriangleleft$$





PROBLEM 3.93

To keep a door closed, a wooden stick is wedged between the floor and the doorknob. The stick exerts at B a 175-N force directed along line AB . Replace that force with an equivalent force-couple system at C .

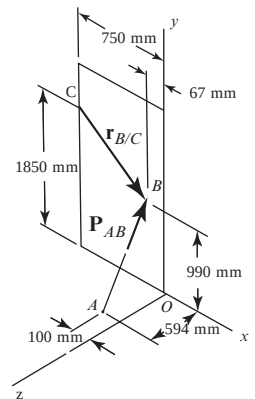
SOLUTION

We have

$$\Sigma \mathbf{F}: \mathbf{P}_{AB} = \mathbf{F}_C$$

where

$$\begin{aligned} \mathbf{P}_{AB} &= \lambda_{AB} P_{AB} \\ &= \frac{(33 \text{ mm})\mathbf{i} + (990 \text{ mm})\mathbf{j} - (594 \text{ mm})\mathbf{k}}{1155.00 \text{ mm}} (175 \text{ N}) \end{aligned}$$



$$\text{or } \mathbf{F}_C = (5.00 \text{ N})\mathbf{i} + (150 \text{ N})\mathbf{j} - (90.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

We have

$$\Sigma \mathbf{M}_C: \mathbf{r}_{B/C} \times \mathbf{P}_{AB} = \mathbf{M}_C$$

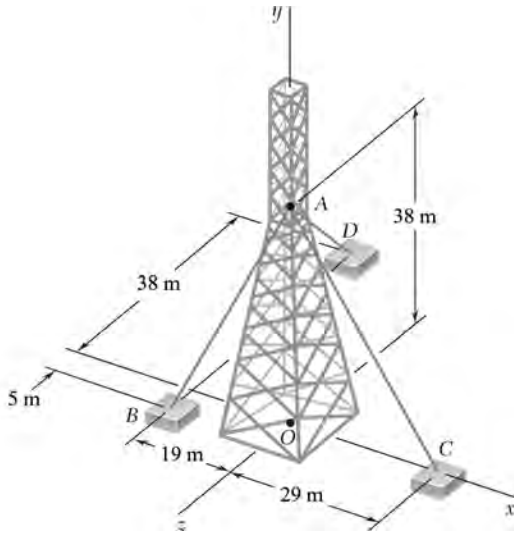
$$\mathbf{M}_C = 5 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.683 & -0.860 & 0 \\ 1 & 30 & -18 \end{vmatrix} \text{ N} \cdot \text{m}$$

$$\begin{aligned} &= (5) \{ (-0.860)(-18)\mathbf{i} - (0.683)(-18)\mathbf{j} \\ &\quad + [(0.683)(30) - (0.860)(1)]\mathbf{k} \} \end{aligned}$$

$$\text{or } \mathbf{M}_C = (77.4 \text{ N} \cdot \text{m})\mathbf{i} + (61.5 \text{ N} \cdot \text{m})\mathbf{j} + (106.8 \text{ N} \cdot \text{m})\mathbf{k} \quad \blacktriangleleft$$

PROBLEM 3.94

An antenna is guyed by three cables as shown. Knowing that the tension in cable AB is 1.44 kN, replace the force exerted at A by cable AB with an equivalent force-couple system at the center O of the base of the antenna.



SOLUTION

We have

$$d_{AB} = \sqrt{(-19)^2 + (-38)^2 + (5)^2} = 42.778 \text{ m}$$

Then

$$\begin{aligned} \mathbf{T}_{AB} &= \frac{1440 \text{ N}}{42.778} (-19\mathbf{i} - 38\mathbf{j} + 5\mathbf{k}) \\ &= -(639.58 \text{ N})\mathbf{i} - (1279.16 \text{ N})\mathbf{j} + (168.31 \text{ N})\mathbf{k} \end{aligned}$$

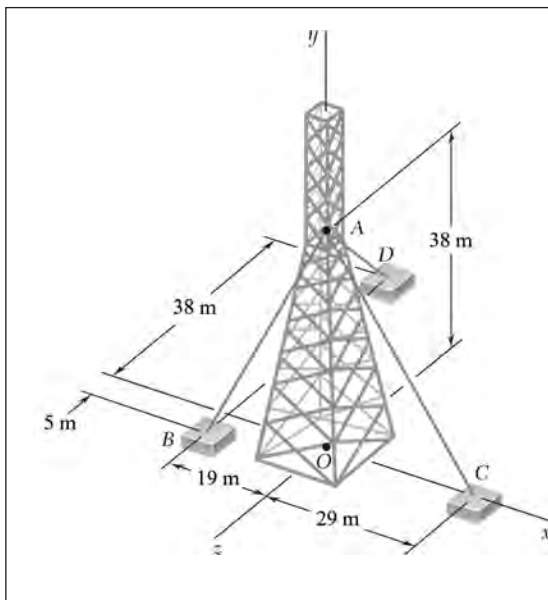
Now

$$\begin{aligned} \mathbf{M} &= \mathbf{M}_O = \mathbf{r}_{A/O} \times \mathbf{T}_{AB} \\ &= 38\mathbf{j} \times [-(639.58)\mathbf{i} - (1279.16)\mathbf{j} + (168.31)\mathbf{k}] \\ &= (6396 \text{ N}\cdot\text{m})\mathbf{i} + (24304 \text{ N}\cdot\text{m})\mathbf{k} \end{aligned}$$

The equivalent force-couple system at O is

$$\mathbf{F} = -(640 \text{ N})\mathbf{i} - (1279 \text{ N})\mathbf{j} + (168.3 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$\mathbf{M} = (6.40 \text{ kN}\cdot\text{m})\mathbf{i} + (24.3 \text{ kN}\cdot\text{m})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 3.95

An antenna is guyed by three cables as shown. Knowing that the tension in cable AD is 1.35 kN, replace the force exerted at A by cable AD with an equivalent force-couple system at the center O of the base of the antenna.

SOLUTION

We have
$$d_{AD} = \sqrt{(-19)^2 + (-38)^2 + (-38)^2}$$
$$= 57 \text{ m}$$

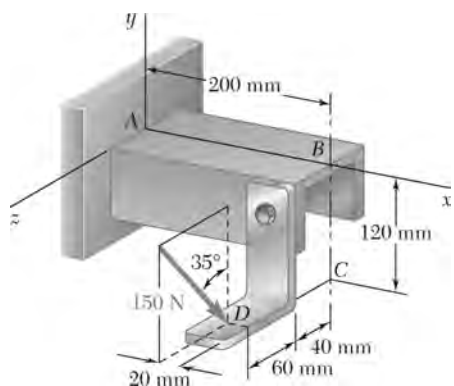
Then
$$\mathbf{T}_{AD} = \frac{1350 \text{ N}}{57} (-19\mathbf{i} - 38\mathbf{j} + 38\mathbf{k})$$
$$= 450 \text{ N}(-\mathbf{i} - 2\mathbf{j} + 2\mathbf{k})$$

Now
$$\mathbf{M} = \mathbf{M}_O = \mathbf{r}_{A/O} \times \mathbf{T}_{AD}$$
$$= 128\mathbf{j} \times 450(-\mathbf{i} - 2\mathbf{j} + 2\mathbf{k})$$
$$= -(115200 \text{ N}\cdot\text{m})\mathbf{i} + (57600 \text{ N}\cdot\text{m})\mathbf{k}$$

The equivalent force-couple system at O is

$$\mathbf{F} = -(450 \text{ N})\mathbf{i} - (900 \text{ N})\mathbf{j} - (900 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$\mathbf{M} = -(115.2 \text{ kN}\cdot\text{m})\mathbf{i} + (57.6 \text{ kN}\cdot\text{m})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 3.96

Replace the 150-N force with an equivalent force-couple system at A.

SOLUTION

Equivalence requires $\Sigma \mathbf{F}: \mathbf{F} = (150 \text{ N})(-\cos 35^\circ \mathbf{j} - \sin 35^\circ \mathbf{k})$
 $= -(122.873 \text{ N})\mathbf{j} - (86.036 \text{ N})\mathbf{k}$

$$\Sigma \mathbf{M}_A: \mathbf{M} = \mathbf{r}_{D/A} \times \mathbf{F}$$

where $\mathbf{r}_{D/A} = (0.18 \text{ m})\mathbf{i} - (0.12 \text{ m})\mathbf{j} + (0.1 \text{ m})\mathbf{k}$

Then
$$\mathbf{M} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.18 & -0.12 & 0.1 \\ 0 & -122.873 & -86.036 \end{vmatrix} \text{ N} \cdot \text{m}$$

$$= [(-0.12)(-86.036) - (0.1)(-122.873)]\mathbf{i}$$

$$+ [-(0.18)(-86.036)]\mathbf{j}$$

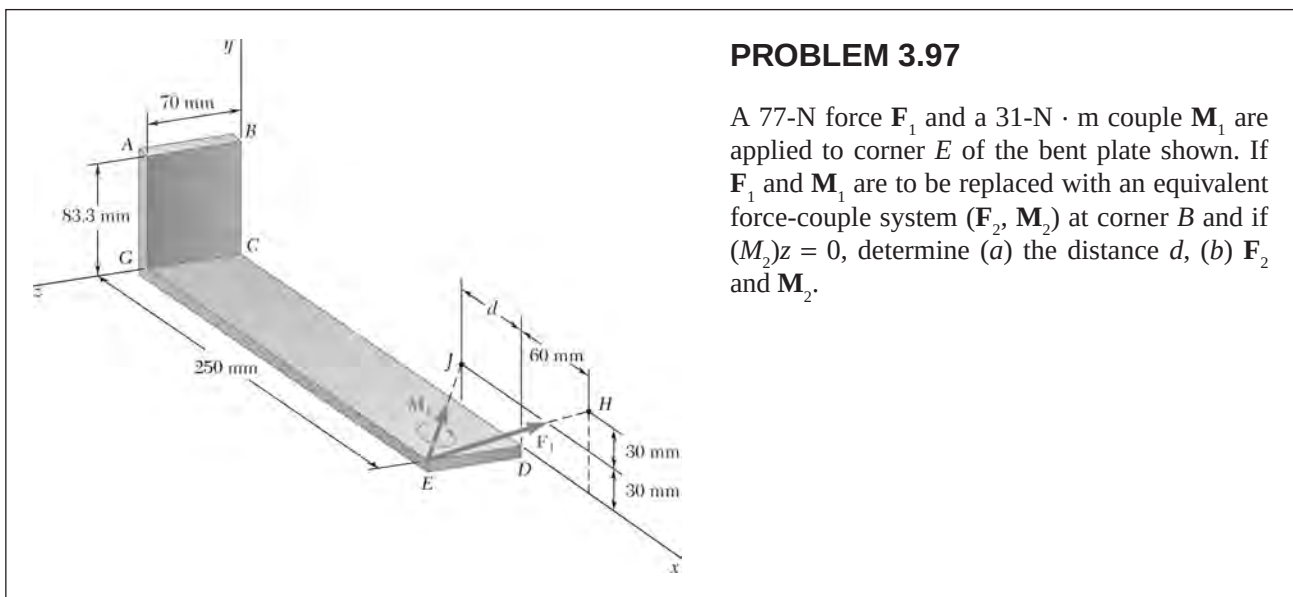
$$+ [(0.18)(-122.873)]\mathbf{k}$$

$$= (22.6 \text{ N} \cdot \text{m})\mathbf{i} + (15.49 \text{ N} \cdot \text{m})\mathbf{j} - (22.1 \text{ N} \cdot \text{m})\mathbf{k}$$

The equivalent force-couple system at A is

$$\mathbf{F} = -(122.9 \text{ N})\mathbf{j} - (86.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$\mathbf{M} = (22.6 \text{ N} \cdot \text{m})\mathbf{i} + (15.49 \text{ N} \cdot \text{m})\mathbf{j} - (22.1 \text{ N} \cdot \text{m})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 3.97

A 77-N force $\mathbf{F}_1$ and a 31-N · m couple $\mathbf{M}_1$ are applied to corner E of the bent plate shown. If $\mathbf{F}_1$ and $\mathbf{M}_1$ are to be replaced with an equivalent force-couple system ($\mathbf{F}_2$, $\mathbf{M}_2$) at corner B and if $(M_2)_z = 0$, determine (a) the distance d , (b) $\mathbf{F}_2$ and $\mathbf{M}_2$.

SOLUTION

(a) We have

$$\Sigma M_{Bz}: M_{2z} = 0$$

$$\mathbf{k} \cdot (\mathbf{r}_{H/B} \times \mathbf{F}_1) + M_{1z} = 0 \quad (1)$$

where

$$\mathbf{r}_{H/B} = (0.31 \text{ m})\mathbf{i} - (0.0233)\mathbf{j}$$

$$\begin{aligned} \mathbf{F}_1 &= \lambda_{EH} F_1 \\ &= \frac{(0.06 \text{ m})\mathbf{i} + (0.06 \text{ m})\mathbf{j} - (0.07 \text{ m})\mathbf{k}}{0.11 \text{ m}} (77 \text{ N}) \end{aligned}$$

$$= (42 \text{ N})\mathbf{i} + (42 \text{ N})\mathbf{j} - (49 \text{ N})\mathbf{k}$$

$$M_{1z} = \mathbf{k} \cdot \mathbf{M}_1$$

$$\begin{aligned} \mathbf{M}_1 &= \lambda_{EJ} M_1 \\ &= \frac{-d\mathbf{i} + (0.03 \text{ m})\mathbf{j} - (0.07 \text{ m})\mathbf{k}}{\sqrt{d^2 + 0.0058} \text{ m}} (31 \text{ N} \cdot \text{m}) \end{aligned}$$

Then from Equation (1),

$$\begin{vmatrix} 0 & 0 & 1 \\ 0.31 & -0.0233 & 0 \\ 42 & 42 & -49 \end{vmatrix} + \frac{(-0.07 \text{ m})(31 \text{ N} \cdot \text{m})}{\sqrt{d^2 + 0.0058}} = 0$$

Solving for d , Equation (1) reduces to

$$(13.0200 + 0.9786) - \frac{2.17 \text{ N} \cdot \text{m}}{\sqrt{d^2 + 0.0058}} = 0$$

From which

$$d = 0.1350 \text{ m}$$

$$\text{or } d = 135.0 \text{ mm} \quad \blacktriangleleft$$

PROBLEM 3.97 (Continued)

(b)

$$\mathbf{F}_2 = \mathbf{F}_1 = (42\mathbf{i} + 42\mathbf{j} - 49\mathbf{k})\text{N} \quad \text{or} \quad \mathbf{F}_2 = (42\text{ N})\mathbf{i} + (42\text{ N})\mathbf{j} - (49\text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$\mathbf{M}_2 = \mathbf{r}_{H/B} \times \mathbf{F}_1 + \mathbf{M}_1$$

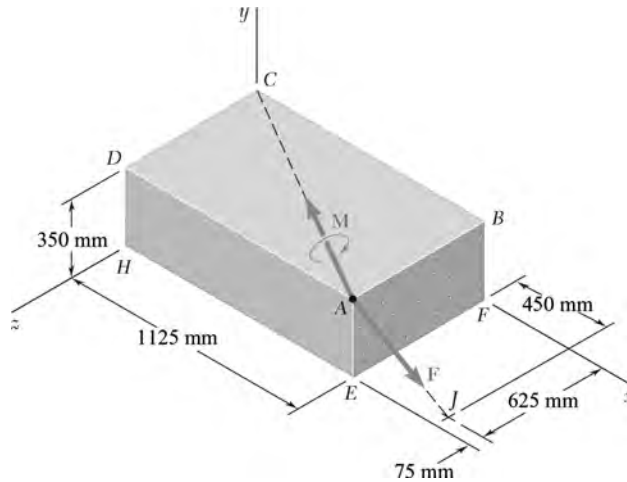
$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.31 & -0.0233 & 0 \\ 42 & 42 & -49 \end{vmatrix} + \frac{(0.1350)\mathbf{i} + 0.03\mathbf{j} - 0.07\mathbf{k}}{0.155000}(31\text{ N} \cdot \text{m})$$

$$= (1.14170\mathbf{i} + 15.1900\mathbf{j} + 13.9986\mathbf{k})\text{ N} \cdot \text{m}$$

$$+ (-27.000\mathbf{i} + 6.0000\mathbf{j} - 14.0000\mathbf{k})\text{ N} \cdot \text{m}$$

$$\mathbf{M}_2 = -(25.858\text{ N} \cdot \text{m})\mathbf{i} + (21.190\text{ N} \cdot \text{m})\mathbf{j}$$

$$\text{or } \mathbf{M}_2 = -(25.9\text{ N} \cdot \text{m})\mathbf{i} + (21.2\text{ N} \cdot \text{m})\mathbf{j} \quad \blacktriangleleft$$



PROBLEM 3.98

A 230 N force $\mathbf{F}$ and a 265 N·m couple $\mathbf{M}$ are applied to corner A of the block shown. Replace the given force-couple system with an equivalent force-couple system at corner H.

SOLUTION

We have

$$d_{AJ} = \sqrt{(450)^2 + (-350)^2 + (-75)^2} = 575 \text{ mm}$$

Then

$$\begin{aligned}\mathbf{F} &= \frac{230 \text{ N}}{575} (450\mathbf{i} - 350\mathbf{j} - 75\mathbf{k}) \\ &= (180 \text{ N})\mathbf{i} - (140 \text{ N})\mathbf{j} - (30 \text{ N})\mathbf{k}\end{aligned}$$

Also

$$d_{AC} = \sqrt{(-1125)^2 + (0)^2 + (-700)^2} = 1325 \text{ mm}$$

Then

$$\begin{aligned}\mathbf{M} &= \frac{265 \text{ N}\cdot\text{m}}{1325} (-1125\mathbf{i} - 700\mathbf{k}) \\ &= -(225 \text{ N}\cdot\text{m})\mathbf{i} - (140 \text{ N}\cdot\text{m})\mathbf{k}\end{aligned}$$

Now

$$\mathbf{M}' = \mathbf{M} + \mathbf{r}_{A/H} \times \mathbf{F}$$

where

$$\mathbf{r}_{A/H} = (1125 \text{ mm})\mathbf{i} + (350 \text{ mm})\mathbf{j}$$

Then

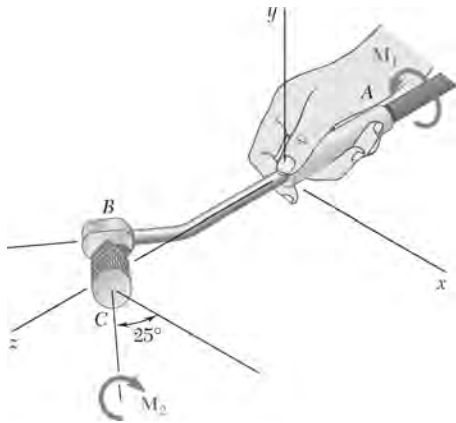
$$\mathbf{M}' = (-225\mathbf{i} - 140\mathbf{k}) + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1.125 & 0.35 & 0 \\ 180 & -140 & -30 \end{vmatrix} = (1.125 \text{ m})\mathbf{i} + (350 \text{ m})\mathbf{j}$$

$$\begin{aligned}\mathbf{M}' &= -225\mathbf{i} - 140\mathbf{k} - 10.5\mathbf{i} + 33.75\mathbf{j} - 220.5\mathbf{k} \\ &= -(235.5 \text{ N}\cdot\text{m})\mathbf{i} + (33.75 \text{ N}\cdot\text{m})\mathbf{j} - (360.5 \text{ N}\cdot\text{m})\mathbf{k}\end{aligned}$$

The equivalent force-couple system at H is

$$\mathbf{F}' = (180 \text{ N})\mathbf{i} - (140 \text{ N})\mathbf{j} - (30 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$\mathbf{M}' = -(235.5 \text{ N}\cdot\text{m})\mathbf{i} + (33.75 \text{ N}\cdot\text{m})\mathbf{j} - (360.5 \text{ N}\cdot\text{m})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 3.99

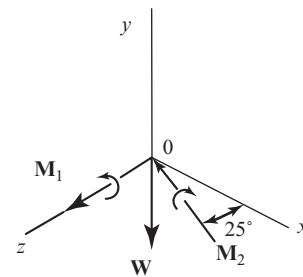
The handpiece for a miniature industrial grinder weighs 3 N, and its center of gravity is located on the y axis. The head of the handpiece is offset in the xz plane in such a way that line BC forms an angle of 25° with the x direction. Show that the weight of the handpiece and the two couples $\mathbf{M}_1$ and $\mathbf{M}_2$ can be replaced with a single equivalent force. Further, assuming that $M_1 = 85 \text{ N}\cdot\text{mm}$ and $M_2 = 81.25 \text{ N}\cdot\text{mm}$, determine (a) the magnitude and the direction of the equivalent force, (b) the point where its line of action intersects the xz plane.

SOLUTION

First assume that the given force $\mathbf{W}$ and couples $\mathbf{M}_1$ and $\mathbf{M}_2$ act at the origin.

Now $\mathbf{W} = -W\mathbf{j}$

and $\mathbf{M} = \mathbf{M}_1 + \mathbf{M}_2$
 $= -(M_2 \cos 25^\circ)\mathbf{i} + (M_1 - M_2 \sin 25^\circ)\mathbf{k}$



Note that since $\mathbf{W}$ and $\mathbf{M}$ are perpendicular, it follows that they can be replaced with a single equivalent force.

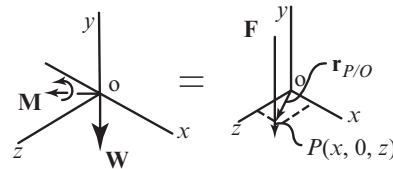
(a) We have $\mathbf{F} = \mathbf{W}$ or $\mathbf{F} = -W\mathbf{j} = -(3 \text{ N})\mathbf{j}$ or $\mathbf{F} = -(3\text{N})\mathbf{j} \blacktriangleleft$

(b) Assume that the line of action of $\mathbf{F}$ passes through Point $P(x, 0, z)$. Then for equivalence

$$\mathbf{M} = \mathbf{r}_{P/O} \times \mathbf{F}$$

where

$$\begin{aligned} \mathbf{r}_{P/O} &= x\mathbf{i} + z\mathbf{k} \\ &-(M_2 \cos 25^\circ)\mathbf{i} + (M_1 - M_2 \sin 25^\circ)\mathbf{k} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & 0 & z \\ 0 & -W & 0 \end{vmatrix} = (Wz)\mathbf{i} - (Wx)\mathbf{k} \end{aligned}$$



Equating the $\mathbf{i}$ and $\mathbf{k}$ coefficients, $z = \frac{-M_2 \cos 25^\circ}{W}$ and $x = -\left(\frac{M_1 - M_2 \sin 25^\circ}{W}\right)$

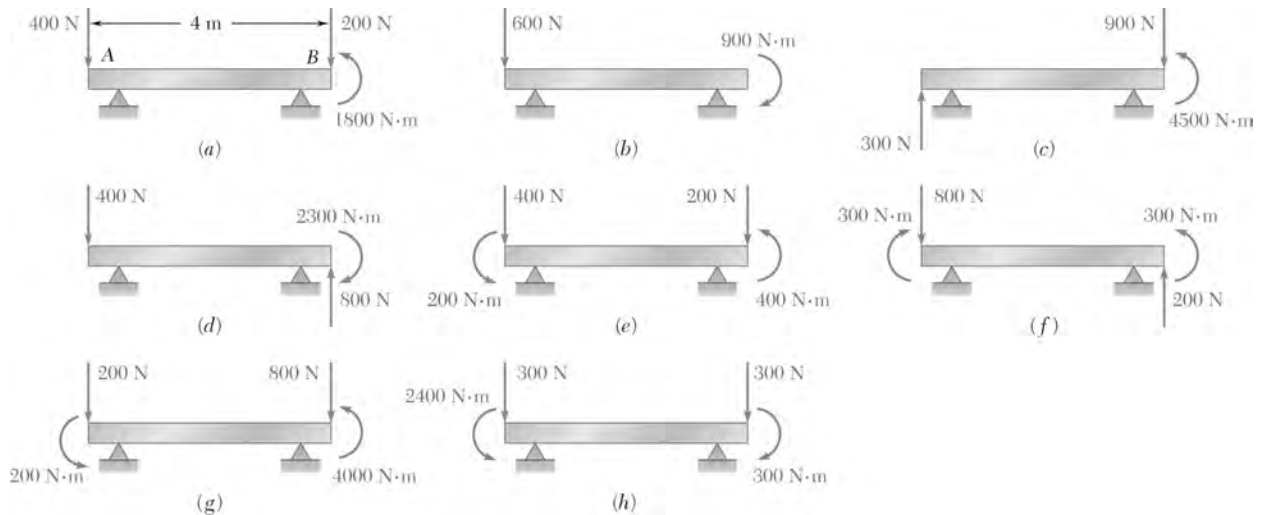
(b) For $W = 3 \text{ N}$ $M_1 = 85 \text{ N}\cdot\text{mm}$ $M_2 = 81.25 \text{ N}\cdot\text{mm}$

$$x = -\left(\frac{0.85 - 81.25 \sin 25^\circ}{-3}\right) = 16.887 \text{ mm} \quad \text{or} \quad x = 16.89 \text{ mm} \blacktriangleleft$$

$$z = \frac{-81.25 \cos 25^\circ}{3} = -24.546 \text{ mm} \quad \text{or} \quad z = -24.5 \text{ mm} \blacktriangleleft$$

PROBLEM 3.100

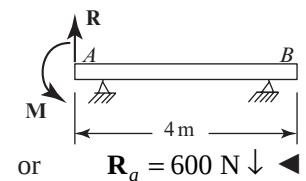
A 4-m-long beam is subjected to a variety of loadings. (a) Replace each loading with an equivalent force-couple system at end A of the beam. (b) Which of the loadings are equivalent?



SOLUTION

(a) (a) We have $\Sigma F_y: -400 \text{ N} - 200 \text{ N} = R_a$

and $\Sigma M_A: 1800 \text{ N} \cdot \text{m} - (200 \text{ N})(4 \text{ m}) = M_a$



or $M_a = 1000 \text{ N} \cdot \text{m} \curvearrowright$

(b) We have $\Sigma F_y: -600 \text{ N} = R_b$

and $\Sigma M_A: -900 \text{ N} \cdot \text{m} = M_b$

or $R_b = 600 \text{ N} \downarrow$

or $M_b = 900 \text{ N} \cdot \text{m} \curvearrowright$

(c) We have $\Sigma F_y: 300 \text{ N} - 900 \text{ N} = R_c$

and $\Sigma M_A: 4500 \text{ N} \cdot \text{m} - (900 \text{ N})(4 \text{ m}) = M_c$

or $R_c = 600 \text{ N} \downarrow$

or $M_c = 900 \text{ N} \cdot \text{m} \curvearrowright$

PROBLEM 3.100 (Continued)

(d) We have $\Sigma F_y: -400 \text{ N} + 800 \text{ N} = R_d$ or $R_d = 400 \text{ N} \uparrow \blacktriangleleft$

and $\Sigma M_A: (800 \text{ N})(4 \text{ m}) - 2300 \text{ N} \cdot \text{m} = M_d$ or $M_d = 900 \text{ N} \cdot \text{m} \curvearrowright \blacktriangleleft$

(e) We have $\Sigma F_y: -400 \text{ N} - 200 \text{ N} = R_e$ or $R_e = 600 \text{ N} \downarrow \blacktriangleleft$

and $\Sigma M_A: 200 \text{ N} \cdot \text{m} + 400 \text{ N} \cdot \text{m} - (200 \text{ N})(4 \text{ m}) = M_e$ or $M_e = 200 \text{ N} \cdot \text{m} \curvearrowleft \blacktriangleleft$

(f) We have $\Sigma F_y: -800 \text{ N} + 200 \text{ N} = R_f$ or $R_f = 600 \text{ N} \downarrow \blacktriangleleft$

and $\Sigma M_A: -300 \text{ N} \cdot \text{m} + 300 \text{ N} \cdot \text{m} + (200 \text{ N})(4 \text{ m}) = M_f$ or $M_f = 800 \text{ N} \cdot \text{m} \curvearrowright \blacktriangleleft$

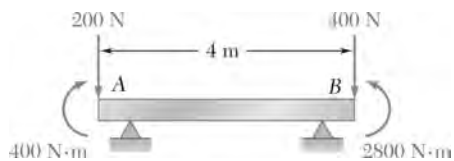
(g) We have $\Sigma F_y: -200 \text{ N} - 800 \text{ N} = R_g$ or $R_g = 1000 \text{ N} \downarrow \blacktriangleleft$

and $\Sigma M_A: 200 \text{ N} \cdot \text{m} + 4000 \text{ N} \cdot \text{m} - (800 \text{ N})(4 \text{ m}) = M_g$ or $M_g = 1000 \text{ N} \cdot \text{m} \curvearrowright \blacktriangleleft$

(h) We have $\Sigma F_y: -300 \text{ N} - 300 \text{ N} = R_h$ or $R_h = 600 \text{ N} \downarrow \blacktriangleleft$

and $\Sigma M_A: 2400 \text{ N} \cdot \text{m} - 300 \text{ N} \cdot \text{m} - (300 \text{ N})(4 \text{ m}) = M_h$ or $M_h = 900 \text{ N} \cdot \text{m} \curvearrowright \blacktriangleleft$

(b) Therefore, loadings (c) and (h) are equivalent. $\blacktriangleleft$



PROBLEM 3.101

A 4-m-long beam is loaded as shown. Determine the loading of Problem 3.100 which is equivalent to this loading.

SOLUTION

We have

$$\Sigma F_y: -200 \text{ N} - 400 \text{ N} = R$$

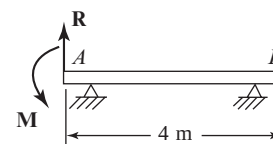
$$\text{or } R = 600 \text{ N} \downarrow \blacktriangleleft$$

and

$$\Sigma M_A: -400 \text{ N} \cdot \text{m} + 2800 \text{ N} \cdot \text{m} - (400 \text{ N})(4 \text{ m}) = M$$

or

$$M = 800 \text{ N} \cdot \text{m} \curvearrowright$$



Equivalent to case (f) of Problem 3.101 $\blacktriangleleft$

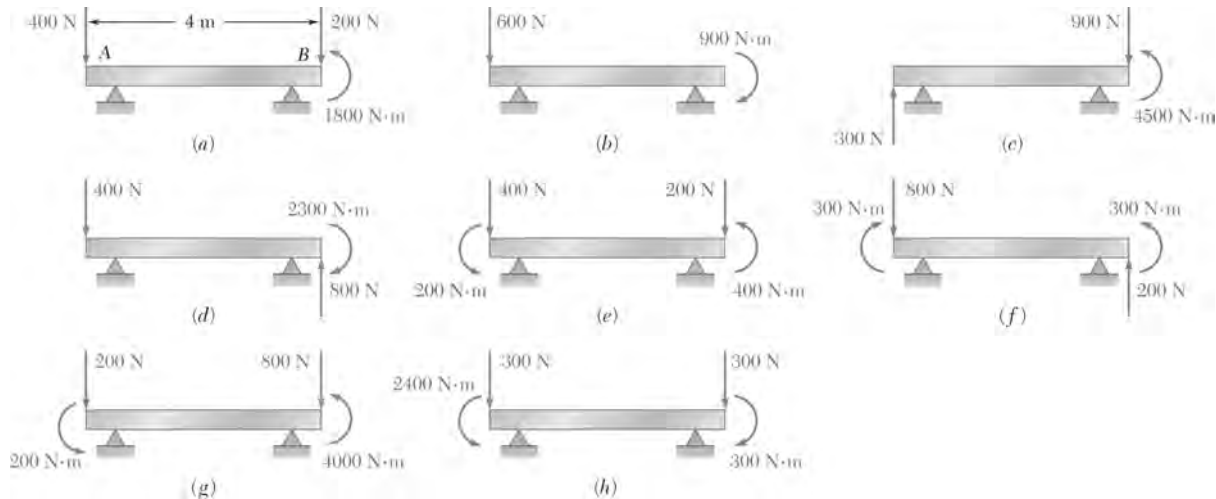
Problem 3.101 Equivalent force-couples at A

Case	R	M
(a)	600 N ↓	1000 N · m ↻
(b)	600 N ↓	900 N · m ↻
(c)	600 N ↓	900 N · m ↻
(d)	400 N ↑	900 N · m ↻
(e)	600 N ↓	200 N · m ↻
(f)	600 N ↓	800 N · m ↻
(g)	1000 N ↓	1000 N · m ↻
(h)	600 N ↓	900 N · m ↻

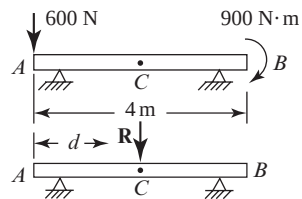
PROBLEM 3.102

Determine the single equivalent force and the distance from Point A to its line of action for the beam and loading of (a) Problem 3.101b, (b) Problem 3.101d, (c) Problem 3.101e.

PROBLEM 3.101 A 4-m-long beam is subjected to a variety of loadings. (a) Replace each loading with an equivalent force-couple system at end A of the beam. (b) Which of the loadings are equivalent?



SOLUTION



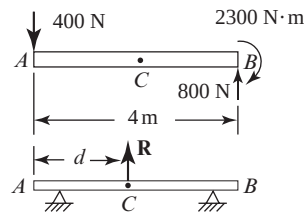
(a) For equivalent single force at distance d from A

We have $\Sigma F_y: -600 \text{ N} = R$

or $R = 600 \text{ N} \downarrow \blacktriangleleft$

and $\Sigma M_C: (600 \text{ N})(d) - 900 \text{ N} \cdot \text{m} = 0$

or $d = 1.500 \text{ m} \blacktriangleleft$

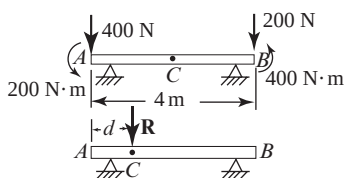


(b) We have $\Sigma F_y: -400 \text{ N} + 800 \text{ N} = R$

or $R = 400 \text{ N} \uparrow \blacktriangleleft$

and $\Sigma M_C: (400 \text{ N})(d) + (800 \text{ N})(4 - d) - 2300 \text{ N} \cdot \text{m} = 0$

or $d = 2.25 \text{ m} \blacktriangleleft$



(c) We have $\Sigma F_y: -400 \text{ N} - 200 \text{ N} = R$

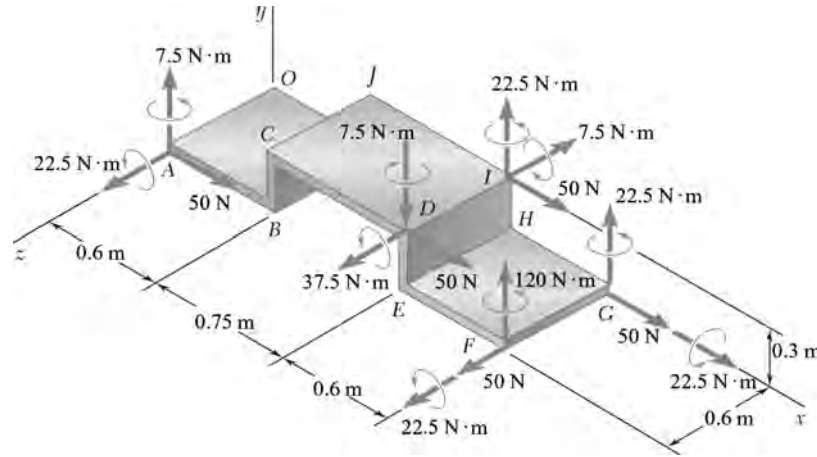
or $R = 600 \text{ N} \downarrow \blacktriangleleft$

and $\Sigma M_C: 200 \text{ N} \cdot \text{m} + (400 \text{ N})(d) - (200 \text{ N})(4 - d) + 400 \text{ N} \cdot \text{m} = 0$

or $d = 0.333 \text{ m} \blacktriangleleft$

PROBLEM 3.103

Five separate force-couple systems act at the corners of a piece of sheet metal, which has been bent into the shape shown. Determine which of these systems is equivalent to a force $\mathbf{F} = (50 \text{ N})\mathbf{i}$ and a couple of moment $\mathbf{M} = (22.5 \text{ N}\cdot\text{m})\mathbf{j} + (22.5 \text{ N}\cdot\text{m})\mathbf{k}$ located at the origin.



SOLUTION

First note that the force-couple system at F cannot be equivalent because of the direction of the force [The force of the other four systems is $(10 \text{ N})\mathbf{i}$]. Next move each of the systems to the origin O ; the forces remain unchanged.

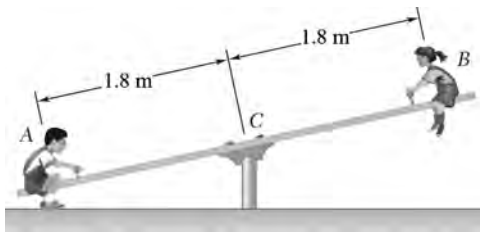
$$\begin{aligned} A: \quad \mathbf{M}_A = \Sigma \mathbf{M}_O &= (7.5 \text{ N}\cdot\text{m})\mathbf{j} + (22.5 \text{ N}\cdot\text{m})\mathbf{k} + (0.6 \text{ m})\mathbf{k} \times (50 \text{ N})\mathbf{i} \\ &= (37.5 \text{ N}\cdot\text{m})\mathbf{j} + (22.5 \text{ N}\cdot\text{m})\mathbf{k} \end{aligned}$$

$$\begin{aligned} D: \quad \mathbf{M}_D = \Sigma \mathbf{M}_O &= -(7.5 \text{ N}\cdot\text{m})\mathbf{j} + (37.5 \text{ N}\cdot\text{m})\mathbf{k} \\ &\quad + [(1.35 \text{ m})\mathbf{i} + (0.3 \text{ m})\mathbf{j} + (0.6 \text{ m})\mathbf{k}] \times (50 \text{ N})\mathbf{i} \\ &= (22.5 \text{ N}\cdot\text{m})\mathbf{j} + (22.5 \text{ N}\cdot\text{m})\mathbf{k} \end{aligned}$$

$$G: \quad \mathbf{M}_G = \Sigma \mathbf{M}_O = (22.5 \text{ N}\cdot\text{m})\mathbf{i} + (22.5 \text{ N}\cdot\text{m})\mathbf{j}$$

$$\begin{aligned} I: \quad \mathbf{M}_I = \Sigma \mathbf{M}_I &= (22.5 \text{ N}\cdot\text{m})\mathbf{j} - (7.5 \text{ N}\cdot\text{m})\mathbf{k} \\ &\quad + [(1.35 \text{ m})\mathbf{i} + (0.3 \text{ m})\mathbf{j}] \times (50 \text{ N})\mathbf{i} \\ &= (22.5 \text{ N}\cdot\text{m})\mathbf{j} - (22.5 \text{ N}\cdot\text{m})\mathbf{k} \end{aligned}$$

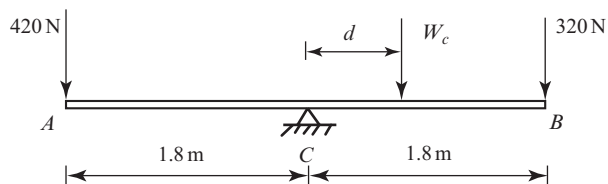
The equivalent force-couple system is the system at corner D .



PROBLEM 3.104

The weights of two children sitting at ends A and B of a seesaw are 420 N and 320 N , respectively. Where should a third child sit so that the resultant of the weights of the three children will pass through C if she weighs (a) 300 N , (b) 260 N .

SOLUTION



- (a) For the resultant weight to act at C , $\Sigma M_C = 0$ $W_C = 300\text{ N}$

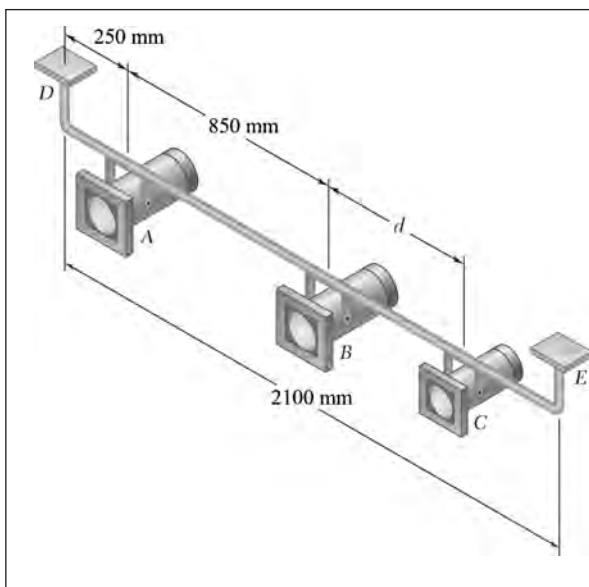
$$\text{Then } (420\text{ N})(1.8\text{ m}) - (300\text{ N})(d) - (320\text{ N})(1.8\text{ m}) = 0$$

$$d = 0.6\text{ m to the right of } C \quad \blacktriangleleft$$

- (b) For the resultant weight to act at C , $\Sigma M_C = 0$ $W_C = 260\text{ N}$

$$\text{Then } (420\text{ N})(1.8\text{ m}) - (260\text{ N})(d) - (320\text{ N})(1.8\text{ m}) = 0$$

$$d = 0.692\text{ m to the right of } C \quad \blacktriangleleft$$



PROBLEM 3.105

Three stage lights are mounted on a pipe as shown. The lights at A and B each weigh 20.5 N, while the one at C weighs 17.5 N. (a) If $d = 625$ mm, determine the distance from D to the line of action of the resultant of the weights of the three lights. (b) Determine the value of d so that the resultant of the weights passes through the midpoint of the pipe.

SOLUTION



For equivalence

$$\Sigma F_y: -20.5 - 20.5 - 17.5 = -R \quad \text{or} \quad R = 58.5 \text{ N} \downarrow$$

$$M_D: -(250 \text{ mm})(20.5 \text{ N}) - (1100 \text{ mm})(20.5 \text{ N}) - [(1100 + d) \text{ mm}](17.5 \text{ N}) = -(L \text{ mm})(58.5 \text{ N})$$

$$\text{or} \quad 46925 + 17.5d = 58.5L \quad (d, L \text{ in mm})$$

$$(a) \quad d = 625 \text{ mm}$$

$$\text{We have} \quad 46925 + (17.5)(625) = 58.5L \quad \text{or} \quad L = 989 \text{ mm}$$

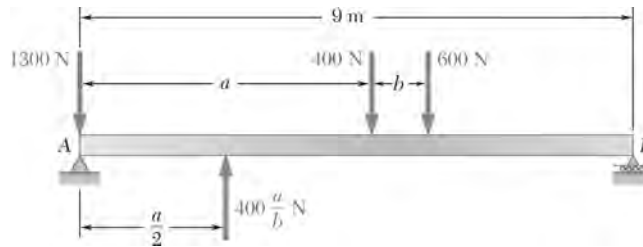
The resultant passes through a Point 989 mm to the right of D. ◀

$$(b) \quad L = 1050 \text{ mm}$$

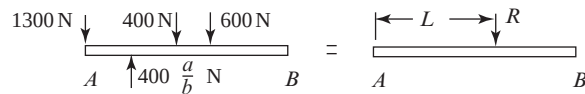
$$\text{We have} \quad 46925 + (17.5)d = (58.5)(1050) \quad \text{or} \quad d = 829 \text{ mm} \quad \blacktriangleleft$$

PROBLEM 3.106

A beam supports three loads of given magnitude and a fourth load whose magnitude is a function of position. If $b = 1.5$ m and the loads are to be replaced with a single equivalent force, determine (a) the value of a so that the distance from support A to the line of action of the equivalent force is maximum, (b) the magnitude of the equivalent force and its point of application on the beam.



SOLUTION



For equivalence

$$\Sigma F_y: -1300 + 400 \frac{a}{b} - 400 - 600 = -R$$

or

$$R = \left(2300 - 400 \frac{a}{b} \right) \text{ N} \quad (1)$$

$$\Sigma M_A: \frac{a}{2} \left(400 \frac{a}{b} \right) - a(400) - (a + b)(600) = -LR$$

or

$$L = \frac{1000a + 600b - 200 \frac{a^2}{b}}{2300 - 400 \frac{a}{b}}$$

Then with

$$b = 1.5 \text{ m} \quad L = \frac{10a + 9 - \frac{4}{3}a^2}{23 - \frac{8}{3}a} \quad (2)$$

where a, L are in m

(a) Find value of a to maximize L

$$\frac{dL}{da} = \frac{\left(10 - \frac{8}{3}a \right) \left(23 - \frac{8}{3}a \right) - \left(10a + 9 - \frac{4}{3}a^2 \right) \left(-\frac{8}{3} \right)}{\left(23 - \frac{8}{3}a \right)^2}$$

PROBLEM 3.106 (Continued)

$$\text{or} \quad 230 - \frac{184}{3}a - \frac{80}{3}a + \frac{64}{9}a^2 + \frac{80}{3}a + 24 - \frac{32}{9}a^2 = 0$$

$$\text{or} \quad 16a^2 - 276a + 1143 = 0$$

$$\text{Then} \quad a = \frac{276 \pm \sqrt{(-276)^2 - 4(16)(1143)}}{2(16)}$$

$$\text{or} \quad a = 10.3435 \text{ m} \quad \text{and} \quad a = 6.9065 \text{ m}$$

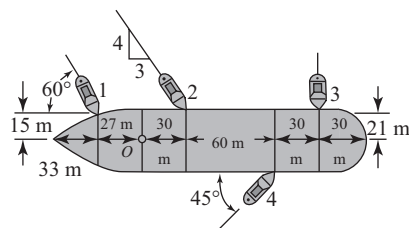
$$\text{Since} \quad AB = 9 \text{ m, } a \text{ must be less than 9 m} \quad a = 6.91 \text{ m} \quad \blacktriangleleft$$

$$(b) \quad \text{Using Eq. (1)} \quad R = 2300 - 400 \frac{6.9065}{1.5} \quad \text{or } R = 458 \text{ N} \quad \blacktriangleleft$$

$$\text{and using Eq. (2)} \quad L = \frac{10(6.9065) + 9 - \frac{4}{3}(6.9065)^2}{23 - \frac{8}{3}(6.9065)} = 3.16 \text{ m}$$

$$\mathbf{R} \text{ is applied } 3.16 \text{ m to the right of } A \quad \blacktriangleleft$$

PROBLEM 3.107



Four tugboats are used to bring an ocean liner to its pier. Each tugboat exerts a 25 kN force in the direction shown. Determine (a) the equivalent force-couple system at the foremast O , (b) the point on the hull where a single, more powerful tugboat should push to produce the same effect as the original four tugboats.

SOLUTION

- (a) **Force-Couple System at O .** Each of the given forces is resolved into components in the diagram shown (kip units are used). The force-couple system at O equivalent to the given system of forces consists of a force $\mathbf{R}$ and a couple $\mathbf{M}_O^R$ defined as follows:

$$\begin{aligned} R &= \Sigma F \\ &= (12.5\mathbf{i} - 21.65\mathbf{j}) + (15\mathbf{i} - 20\mathbf{j}) + (-25\mathbf{j}) + (17.68\mathbf{i} - 17.68\mathbf{j}) \\ &= (45.18\mathbf{i} - 48.97\mathbf{j}) \end{aligned}$$

$$\begin{aligned} \mathbf{M}_O^R &= \Sigma(\mathbf{r} \times \mathbf{F}) \\ &= (-27\mathbf{i} + 15\mathbf{j}) \times (12.5\mathbf{i} - 21.65\mathbf{j}) \\ &\quad + (30\mathbf{i} + 21\mathbf{j}) \times (15\mathbf{i} - 20\mathbf{j}) \\ &\quad + (120\mathbf{i} + 21\mathbf{j}) \times (-25\mathbf{j}) \\ &\quad + (90\mathbf{i} + 21\mathbf{j}) \times (17.68\mathbf{i} - 17.68\mathbf{j}) \\ &= (584.55 - 187.5 - 600 - 315 - 3000 + 1591.2 + 371.28)\mathbf{k} \\ &= -1555.47\mathbf{k} \end{aligned}$$

The equivalent force-couple system at O is thus,

$$\mathbf{R} = (45.2\text{ kN})\mathbf{i} - (49.0\text{ kN})\mathbf{j} \quad \mathbf{M}_O^R = (1555\text{ kN}\cdot\text{m})\mathbf{k}$$

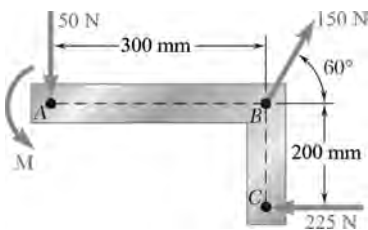
or $\mathbf{R} = 666\text{ kN} \searrow 47.3^\circ \quad \mathbf{M}_O^R = 1400.9\text{ kN}\cdot\text{m} \curvearrowleft$

- (b) **Single Tugboat.** The force exerted by a single tugboat must be equal to $\mathbf{R}$, and its point of application A must be such that the moment of $\mathbf{R}$ about O is equal to $\mathbf{M}_O^R$. Observing that the position vector of A is

$$\mathbf{r} = x\mathbf{i} + 21\mathbf{j}$$

we write $\mathbf{r} \times \mathbf{R} = \mathbf{M}_O^R$

$$\begin{aligned} (x\mathbf{i} + 21\mathbf{j}) \times (45.18\mathbf{i} - 48.97\mathbf{j}) &= -1555.47\mathbf{k} \\ -x(48.97)\mathbf{k} - 948.78\mathbf{k} &= -1555.47\mathbf{k} \quad x = 12.39\text{ m} \end{aligned}$$



PROBLEM 3.108

A couple of magnitude $M = 7 \text{ N}\cdot\text{m}$ and the three forces shown are applied to an angle bracket. (a) Find the resultant of this system of forces. (b) Locate the points where the line of action of the resultant intersects line AB and line BC.

SOLUTION

(a) We have $\Sigma \mathbf{F}$: $\mathbf{R} = (-50\mathbf{j}) + (150 \cos 60^\circ)\mathbf{i}$
 $+ 150 \sin 60^\circ\mathbf{j} + (-225\mathbf{i})$
 $= -(150 \text{ N})\mathbf{i} + (79.9 \text{ N})\mathbf{j}$

or $\mathbf{R} = 170 \text{ N} \nearrow 28.0^\circ \blacktriangleleft$

(b) First reduce the given forces and couple to an equivalent force-couple system $(\mathbf{R}, \mathbf{M}_B)$ at B.

We have ΣM_B : $M_B = (7000 \text{ N}\cdot\text{mm}) + (300 \text{ mm})(50 \text{ N}) - (200 \text{ mm})(225 \text{ N})$
 $= -23000 \text{ N}\cdot\text{mm}$

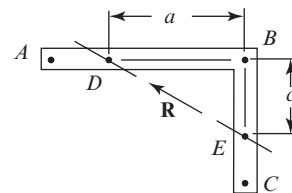
Then with $\mathbf{R}$ at D ΣM_B : $-23000 \text{ N}\cdot\text{m} = a(79.9 \text{ N})$

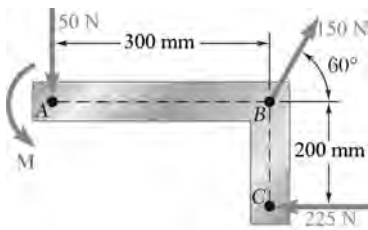
or $a = 288 \text{ mm}$

and with $\mathbf{R}$ at E ΣM_B : $-23000 \text{ N}\cdot\text{m} = c(150 \text{ N})$

or $c = 153.3 \text{ mm}$

The line of action of $\mathbf{R}$ intersects line AB 288 mm to the left of B and intersects line BC 153.3 mm below B. $\blacktriangleleft$





PROBLEM 3.109

A couple $\mathbf{M}$ and the three forces shown are applied to an angle bracket. Find the moment of the couple if the line of action of the resultant of the force system is to pass through (a) Point A, (b) Point B, (c) Point C.

SOLUTION

In each case, must have $\mathbf{M}_i^R = 0$

$$(a) \quad +\circlearrowleft M_A^B = \Sigma M_A = M + (300 \text{ mm})[(150 \text{ N})\sin 60^\circ] - (200 \text{ mm})(225 \text{ N}) = 0$$

$$M = +6028.9 \text{ N} \cdot \text{mm}$$

$$\mathbf{M} = 6.03 \text{ N} \cdot \text{m} \quad \circlearrowleft \blacktriangleleft$$

$$(b) \quad +\circlearrowleft M_B^R = \Sigma M_B = M + (300 \text{ mm})(50 \text{ N}) - (200 \text{ mm})(225 \text{ N}) = 0$$

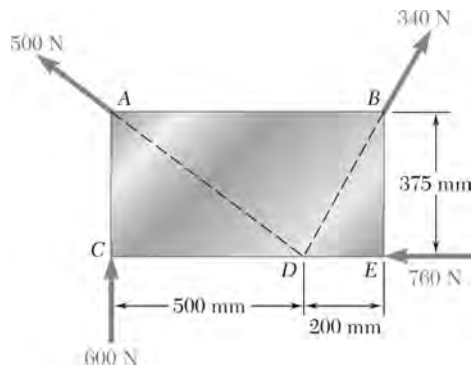
$$M = +30000 \text{ N} \cdot \text{mm}$$

$$\mathbf{M} = 30 \text{ N} \cdot \text{m} \quad \circlearrowleft \blacktriangleleft$$

$$(c) \quad +\circlearrowleft M_C^R = \Sigma M_C = M + (300 \text{ mm})(50 \text{ N}) - (200 \text{ mm})[(150 \text{ N})\cos 60^\circ] = 0$$

$$M = 0$$

$$\mathbf{M} = 0 \quad \blacktriangleleft$$



PROBLEM 3.110

Four forces act on a 700×375 -mm plate as shown. (a) Find the resultant of these forces. (b) Locate the two points where the line of action of the resultant intersects the edge of the plate.

SOLUTION

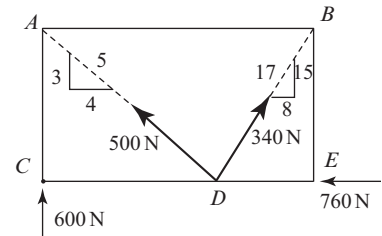
(a)

$$\begin{aligned} \mathbf{R} &= \Sigma \mathbf{F} \\ &= (-400 \text{ N} + 160 \text{ N} - 760 \text{ N})\mathbf{i} \\ &\quad + (600 \text{ N} + 300 \text{ N} + 300 \text{ N})\mathbf{j} \\ &= -(1000 \text{ N})\mathbf{i} + (1200 \text{ N})\mathbf{j} \end{aligned}$$

$$\begin{aligned} R &= \sqrt{(1000 \text{ N})^2 + (1200 \text{ N})^2} \\ &= 1562.09 \text{ N} \end{aligned}$$

$$\begin{aligned} \tan \theta &= \left(-\frac{1200 \text{ N}}{1000 \text{ N}} \right) \\ &= -1.20000 \\ \theta &= -50.194^\circ \end{aligned}$$

$$\mathbf{R} = 1562 \text{ N} \nearrow 50.2^\circ \blacktriangleleft$$

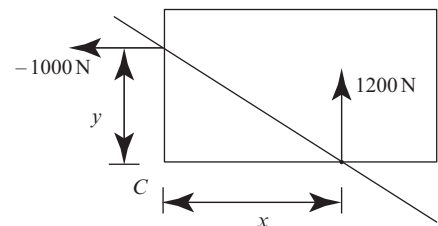


(b)

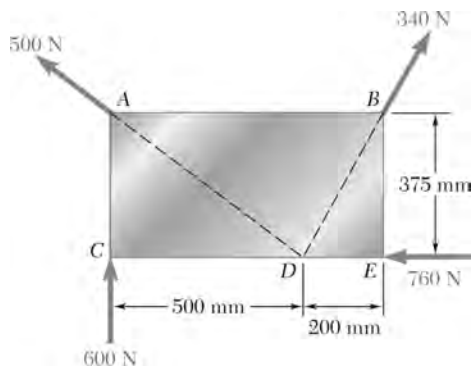
$$\begin{aligned} \mathbf{M}_C^R &= \Sigma \mathbf{r} \times \mathbf{F} \\ &= (0.5 \text{ m})\mathbf{i} \times (300 \text{ N} + 300 \text{ N})\mathbf{j} \\ &= (300 \text{ N} \cdot \text{m})\mathbf{k} \end{aligned}$$

$$\begin{aligned} (300 \text{ N} \cdot \text{m})\mathbf{k} &= x\mathbf{i} \times (1200 \text{ N})\mathbf{j} \\ x &= 0.25000 \text{ m} \\ x &= 250 \text{ mm} \end{aligned}$$

$$\begin{aligned} (300 \text{ N} \cdot \text{m}) &= y\mathbf{j} \times (-1000 \text{ N})\mathbf{i} \\ y &= 0.30000 \text{ m} \\ y &= 300 \text{ mm} \end{aligned}$$



Intersection 250 mm to right of C and 300 mm above C ◀



PROBLEM 3.111

Solve Problem 3.110, assuming that the 760-N force is directed to the right.

PROBLEM 3.110 Four forces act on a 700×375 -mm plate as shown. (a) Find the resultant of these forces. (b) Locate the two points where the line of action of the resultant intersects the edge of the plate.

SOLUTION

(a)

$$\mathbf{R} = \Sigma \mathbf{F}$$

$$\begin{aligned} &= (-400 \text{ N} + 160 \text{ N} + 760 \text{ N})\mathbf{i} \\ &\quad + (600 \text{ N} + 300 \text{ N} + 300 \text{ N})\mathbf{j} \\ &= (520 \text{ N})\mathbf{i} + (1200 \text{ N})\mathbf{j} \end{aligned}$$

$$R = \sqrt{(520 \text{ N})^2 + (1200 \text{ N})^2} = 1307.82 \text{ N}$$

$$\tan \theta = \left(\frac{1200 \text{ N}}{520 \text{ N}} \right) = 2.3077$$

$$\theta = 66.5714^\circ$$

(b)

$$\mathbf{M}_C^R = \Sigma \mathbf{r} \times \mathbf{F}$$

$$\begin{aligned} &= (0.5 \text{ m})\mathbf{i} \times (300 \text{ N} + 300 \text{ N})\mathbf{j} \\ &= (300 \text{ N} \cdot \text{m})\mathbf{k} \end{aligned}$$

$$(300 \text{ N} \cdot \text{m})\mathbf{k} = x\mathbf{i} \times (1200 \text{ N})\mathbf{j}$$

$$x = 0.25000 \text{ m}$$

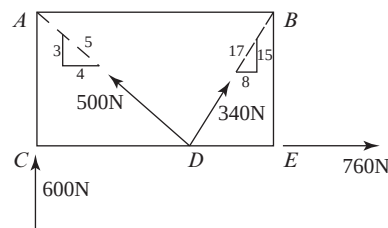
or $x = 0.250 \text{ mm}$

$$(300 \text{ N} \cdot \text{m})\mathbf{k} = [x'\mathbf{i} + (0.375 \text{ m})\mathbf{j}] \times [(520 \text{ N})\mathbf{i} + (1200 \text{ N})\mathbf{j}]$$

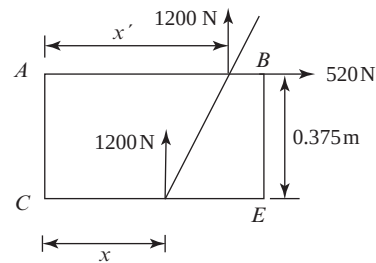
$$= (1200x' - 195)\mathbf{k}$$

$$x' = 0.41250 \text{ m}$$

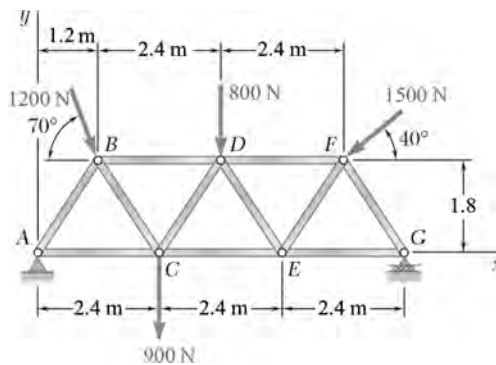
or $x' = 412.5 \text{ mm}$



$$\mathbf{R} = 1308 \text{ N} \nearrow 66.6^\circ \blacktriangleleft$$



Intersection 412 mm to the right of A and 250 mm to the right of C ◀



PROBLEM 3.112

A truss supports the loading shown. Determine the equivalent force acting on the truss and the point of intersection of its line of action with a line drawn through Points A and G.

SOLUTION

We have

$$\mathbf{R} = \Sigma \mathbf{F}$$

$$\begin{aligned} \mathbf{R} &= (1200 \text{ N})(\cos 70^\circ \mathbf{i} - \sin 70^\circ \mathbf{j}) - (800 \text{ N})\mathbf{j} \\ &\quad + (1500 \text{ N})(-\cos 40^\circ \mathbf{i} - \sin 40^\circ \mathbf{j}) - (900 \text{ N})\mathbf{j} \end{aligned}$$

$$\mathbf{R} = -(738.64 \text{ N})\mathbf{i} - (3791.8 \text{ N})\mathbf{j}$$

$$R = \sqrt{R_x^2 + R_y^2}$$

$$= \sqrt{(-738.6)^2 + (3791.8)^2}$$

$$= 3863 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{R_y}{R_x} \right)$$

$$= \tan^{-1} \left(\frac{-3791.8}{-738.6} \right)$$

$$= 78.977^\circ$$

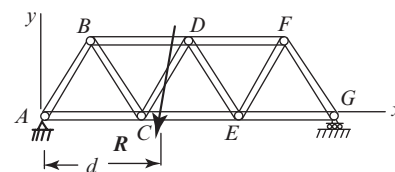
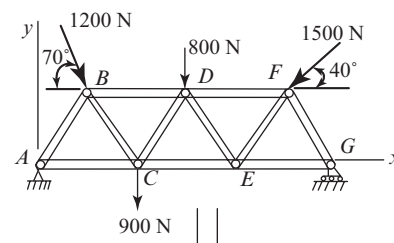
$$\text{or } \mathbf{R} = 3860 \text{ N} \nearrow 79.0^\circ \blacktriangleleft$$

We have $\Sigma M_A = dR_y$

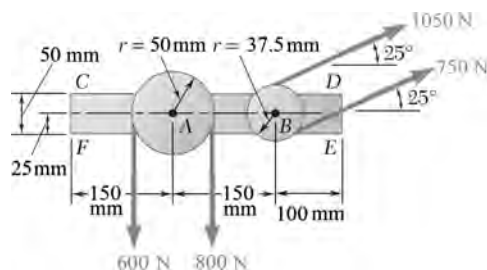
$$\begin{aligned} \text{where } \Sigma M_A &= -[1200 \text{ N} \cos 70^\circ](1.8 \text{ m}) - [1200 \text{ N} \sin 70^\circ](1.2 \text{ m}) \\ &\quad - (800 \text{ N})(3.6 \text{ m}) + [1500 \text{ N} \cos 40^\circ](1.8 \text{ m}) \\ &\quad - [1500 \text{ N} \sin 40^\circ](6.0 \text{ m}) - (900 \text{ N})(2.4 \text{ m}) \\ &= -10848.75 \text{ N} \cdot \text{m} \end{aligned}$$

$$d = \frac{-10848.75 \text{ N} \cdot \text{m}}{-3791.8 \text{ N}}$$

$$= 2.861 \text{ m}$$



$$\text{or } d = 2.86 \text{ m to the right of A} \blacktriangleleft$$



PROBLEM 3.113

Pulleys A and B are mounted on bracket $CDEF$. The tension on each side of the two belts is as shown. Replace the four forces with a single equivalent force, and determine where its line of action intersects the bottom edge of the bracket.

SOLUTION

Equivalent force-couple at A due to belts on pulley A

$$\text{We have } \Sigma \mathbf{F}: -600 \text{ N} - 800 \text{ N} = R_A$$

$$\mathbf{R}_A = 1400 \text{ N} \downarrow$$

$$\text{We have } \Sigma \mathbf{M}_A: -200 \text{ N}(50 \text{ mm}) = M_A$$

$$\mathbf{M}_A = 10000 \text{ N} \cdot \text{mm} \curvearrowright$$

Equivalent force-couple at B due to belts on pulley B

$$\text{We have } \Sigma \mathbf{F}: (1050 \text{ N} + 750 \text{ N}) \nearrow 25^\circ = \mathbf{R}_B$$

$$\mathbf{R}_B = 1800 \text{ N} \nearrow 25^\circ$$

$$\text{We have } \Sigma \mathbf{M}_B: -300 \text{ N}(37.5 \text{ mm}) = M_B$$

$$\mathbf{M}_B = 11250 \text{ N} \cdot \text{mm}$$

Equivalent force-couple at F

$$\text{We have } \Sigma \mathbf{F}: \mathbf{R}_F = (-1400 \text{ N})\mathbf{j} + (1800 \text{ N})(\cos 25^\circ \mathbf{i} + \sin 25^\circ \mathbf{j})$$

$$= (1631.35 \text{ N})\mathbf{i} - (639.285 \text{ N})\mathbf{j}$$

$$R = R_F$$

$$= \sqrt{R_{Fx}^2 + R_{Fy}^2}$$

$$= \sqrt{(1631.35)^2 + (-639.285)^2}$$

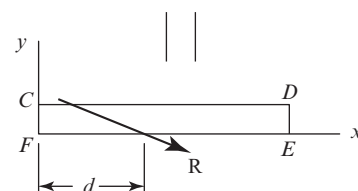
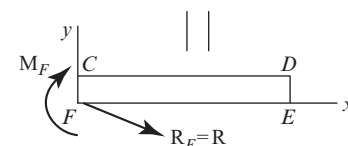
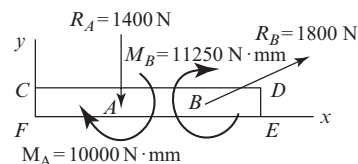
$$= 1752.13 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{R_{Fy}}{R_{Fx}} \right)$$

$$= \tan^{-1} \left(\frac{-639.285}{1631.35} \right)$$

$$= -21.399^\circ$$

$$\text{or } \mathbf{R}_F = \mathbf{R} = 1752 \text{ N} \nearrow 21.4^\circ \blacktriangleleft$$



PROBLEM 3.113 (Continued)

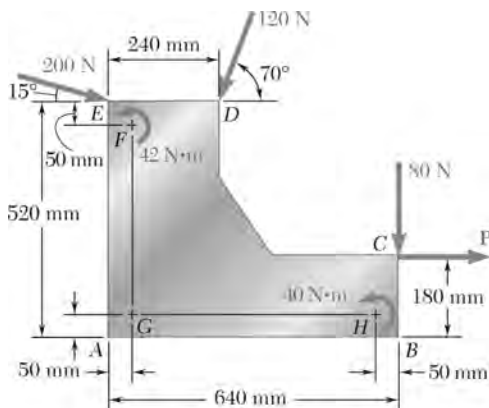
We have $\Sigma \mathbf{M}_F$:

$$\begin{aligned} M_F &= -(1400 \text{ N})(150 \text{ mm}) - 10,000 \text{ N} \cdot \text{mm} \\ &\quad - [(1800 \text{ N}) \cos 25^\circ](25 \text{ mm}) \\ &\quad + [(1800 \text{ N}) \sin 25^\circ](300 \text{ mm}) - 11250 \text{ N} \cdot \text{mm} \\ \mathbf{M}_F &= -(43820 \text{ N} \cdot \text{mm})\mathbf{k} \end{aligned}$$

To determine where a single resultant force will intersect line FE ,

$$\begin{aligned} M_F &= dR_y \\ d &= \frac{M_F}{R_y} \\ &= \frac{-43820}{-639.285} \\ &= 68.545 \text{ mm} \end{aligned}$$

$$\text{or } d = 68.6 \text{ mm} \quad \blacktriangleleft$$

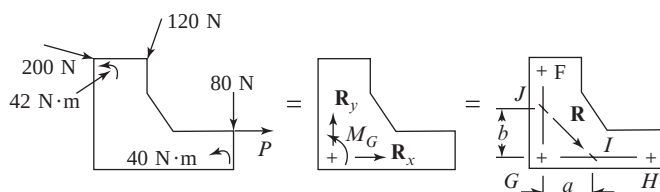


PROBLEM 3.114

A machine component is subjected to the forces and couples shown. The component is to be held in place by a single rivet that can resist a force but not a couple. For $P = 0$, determine the location of the rivet hole if it is to be located (a) on line FG , (b) on line GH .

SOLUTION

We have



First replace the applied forces and couples with an equivalent force-couple system at G .

$$\text{Thus } \Sigma F_x: 200 \cos 15^\circ - 120 \cos 70^\circ + P = R_x$$

$$\text{or } R_x = (152.142 + P) \text{ N}$$

$$\Sigma F_y: -200 \sin 15^\circ - 120 \sin 70^\circ - 80 = R_y$$

$$\text{or } R_y = -244.53 \text{ N}$$

$$\begin{aligned} \Sigma M_G: & -(0.47 \text{ m})(200 \text{ N}) \cos 15^\circ + (0.05 \text{ m})(200 \text{ N}) \sin 15^\circ \\ & + (0.47 \text{ m})(120 \text{ N}) \cos 70^\circ - (0.19 \text{ m})(120 \text{ N}) \sin 70^\circ \\ & - (0.13 \text{ m})(P \text{ N}) - (0.59 \text{ m})(80 \text{ N}) + 42 \text{ N} \cdot \text{m} \\ & + 40 \text{ N} \cdot \text{m} = M_G \end{aligned}$$

$$\text{or } M_G = -(55.544 + 0.13P) \text{ N} \cdot \text{m} \quad (1)$$

Setting $P = 0$ in Eq. (1):

$$\text{Now with } \mathbf{R} \text{ at } I \quad \Sigma M_G: -55.544 \text{ N} \cdot \text{m} = -a(244.53 \text{ N})$$

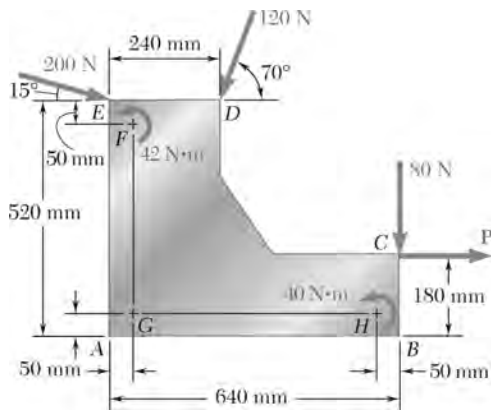
$$\text{or } a = 0.227 \text{ m}$$

$$\text{and with } \mathbf{R} \text{ at } J \quad \Sigma M_G: -55.544 \text{ N} \cdot \text{m} = -b(152.142 \text{ N})$$

$$\text{or } b = 0.365 \text{ m}$$

(a) The rivet hole is 0.365 m above G . ◀

(b) The rivet hole is 0.227 m to the right of G . ◀



PROBLEM 3.115

Solve Problem 3.114, assuming that $P = 60$ N.

PROBLEM 3.115 A machine component is subjected to the forces and couples shown. The component is to be held in place by a single rivet that can resist a force but not a couple. For $P = 0$, determine the location of the rivet hole if it is to be located (a) on line FG , (b) on line GH .

SOLUTION

See the solution to Problem 3.115 leading to the development of Equation (1)

$$M_G = -(55.544 + 0.13P) \text{ N} \cdot \text{m}$$

and $R_x = (152.142 + P) \text{ N}$

For $P = 60 \text{ N}$

We have $R_x = (152.142 + 60)$
 $= 212.14 \text{ N}$

$$M_G = -[55.544 + 0.13(60)]$$

$$= -63.344 \text{ N} \cdot \text{m}$$

Then with $\mathbf{R}$ at I $\Sigma M_G: -63.344 \text{ N} \cdot \text{m} = -a(244.53 \text{ N})$

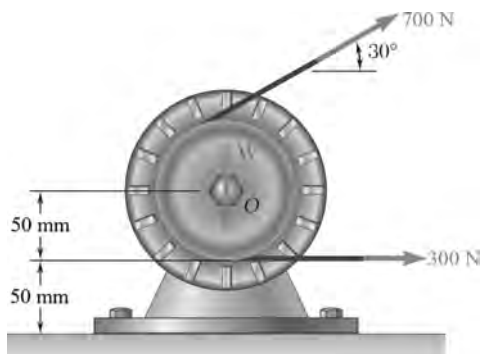
or $a = 0.259 \text{ m}$

and with $\mathbf{R}$ at J $\Sigma M_G: -63.344 \text{ N} \cdot \text{m} = -b(212.14 \text{ N})$

or $b = 0.299 \text{ m}$

(a) The rivet hole is 0.299 m above G . ◀

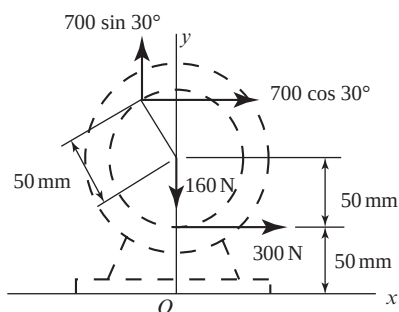
(b) The rivet hole is 0.259 m to the right of G . ◀



PROBLEM 3.116

A 160 N motor is mounted on the floor. Find the resultant of the weight and the forces exerted on the belt, and determine where the line of action of the resultant intersects the floor.

SOLUTION



We have

$$\Sigma \mathbf{F}: (300 \text{ N})\mathbf{i} - (160 \text{ N})\mathbf{j} + (700 \text{ N})(\cos 30^\circ \mathbf{i} + \sin 30^\circ \mathbf{j}) = \mathbf{R}$$

$$\mathbf{R} = (906.22 \text{ N})\mathbf{i} + (190 \text{ N})\mathbf{j}$$

$$\text{or } \mathbf{R} = 926 \text{ N } \nearrow 11.84^\circ \blacktriangleleft$$

We have

$$\Sigma M_O: \Sigma M_O = xR_y$$

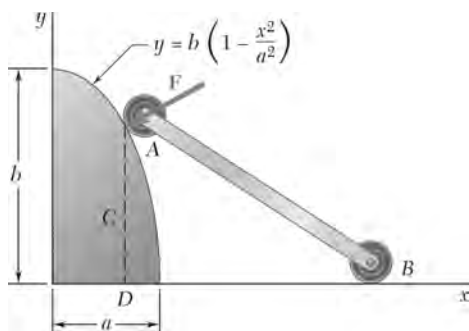
$$-[(700 \text{ N})\cos 30^\circ][(100 + 50\cos 30^\circ)\text{mm}] - [(700 \text{ N})\sin 30^\circ][(50 \text{ mm})\sin 30^\circ] - (300 \text{ N})(50 \text{ mm}) = x(190 \text{ N})$$

$$x = \frac{1}{190}(-86871 - 8750 - 15000) \text{ N} \cdot \text{mm}$$

and

$$x = -582.21 \text{ mm}$$

Or, resultant intersects the base (x axis) 582 mm to the left of the vertical centerline (y axis) of the motor. $\blacktriangleleft$



PROBLEM 3.117

As follower AB rolls along the surface of member C, it exerts a constant force $\mathbf{F}$ perpendicular to the surface. (a) Replace $\mathbf{F}$ with an equivalent force-couple system at the Point D obtained by drawing the perpendicular from the point of contact to the x axis. (b) For $a = 1$ m and $b = 2$ m, determine the value of x for which the moment of the equivalent force-couple system at D is maximum.

SOLUTION

(a) The slope of any tangent to the surface of member C is

$$\frac{dy}{dx} = \frac{d}{dx} \left[b \left(1 - \frac{x^2}{a^2} \right) \right] = -\frac{2b}{a^2} x$$

Since the force $\mathbf{F}$ is perpendicular to the surface,

$$\tan \alpha = -\left(\frac{dy}{dx} \right)^{-1} = \frac{a^2}{2b} \left(\frac{1}{x} \right)$$

For equivalence

$$\Sigma F: \mathbf{F} = \mathbf{R}$$

$$\Sigma M_D: (F \cos \alpha)(y_A) = M_D$$

where

$$\cos \alpha = \frac{2bx}{\sqrt{(a^2)^2 + (2bx)^2}}$$

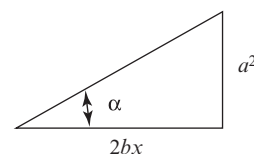
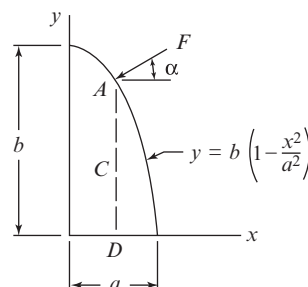
$$y_A = b \left(1 - \frac{x^2}{a^2} \right)$$

$$M_D = \frac{2Fb^2 \left(x - \frac{x^3}{a^2} \right)}{\sqrt{a^4 + 4b^2 x^2}}$$

Therefore, the equivalent force-couple system at D is

$$\mathbf{R} = F \nearrow \tan^{-1} \left(\frac{a^2}{2bx} \right) \leftarrow$$

$$\mathbf{M} = \frac{2Fb^2 \left(x - \frac{x^3}{a^2} \right)}{\sqrt{a^4 + 4b^2 x^2}} \leftarrow$$



PROBLEM 3.117 (Continued)

(b) To maximize M , the value of x must satisfy $\frac{dM}{dx} = 0$

where, for $a = 1 \text{ m}$, $b = 2 \text{ m}$

$$M = \frac{8F(x - x^3)}{\sqrt{1 + 16x^2}}$$

$$\frac{dM}{dx} = 8F \frac{\sqrt{1 + 16x^2}(1 - 3x^2) - (x - x^3) \left[\frac{1}{2}(32x)(1 + 16x^2)^{-1/2} \right]}{(1 + 16x^2)} = 0$$

$$(1 + 16x^2)(1 - 3x^2) - 16x(x - x^3) = 0$$

or

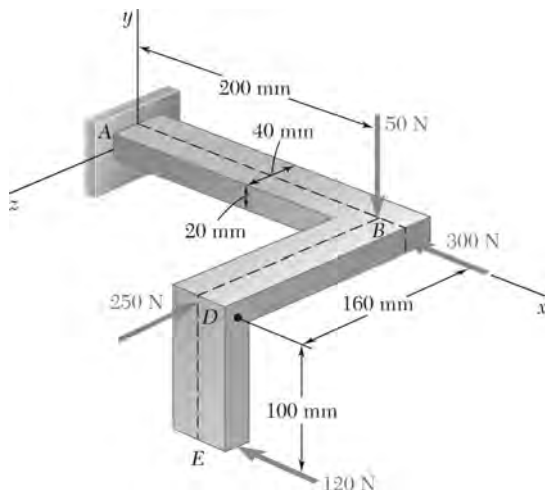
$$32x^4 + 3x^2 - 1 = 0$$

$$x^2 = \frac{-3 \pm \sqrt{9 - 4(32)(-1)}}{2(32)} = 0.136011 \text{ m}^2 \quad \text{and} \quad -0.22976 \text{ m}^2$$

Using the positive value of x^2

$$x = 0.36880 \text{ m}$$

$$\text{or } x = 369 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 3.118

Four forces are applied to the machine component ABDE as shown. Replace these forces by an equivalent force-couple system at A.

SOLUTION

$$\mathbf{R} = -(50 \text{ N})\mathbf{j} - (300 \text{ N})\mathbf{i} - (120 \text{ N})\mathbf{i} - (250 \text{ N})\mathbf{k}$$

$$\mathbf{R} = -(420 \text{ N})\mathbf{i} - (50 \text{ N})\mathbf{j} - (250 \text{ N})\mathbf{k}$$

$$\mathbf{r}_B = (0.2 \text{ m})\mathbf{i}$$

$$\mathbf{r}_D = (0.2 \text{ m})\mathbf{i} + (0.16 \text{ m})\mathbf{k}$$

$$\mathbf{r}_E = (0.2 \text{ m})\mathbf{i} - (0.1 \text{ m})\mathbf{j} + (0.16 \text{ m})\mathbf{k}$$

$$\begin{aligned} \mathbf{M}_A^R &= \mathbf{r}_B \times [-(300 \text{ N})\mathbf{i} - (50 \text{ N})\mathbf{j}] \\ &\quad + \mathbf{r}_D \times (-250 \text{ N})\mathbf{k} + \mathbf{r}_E \times (-120 \text{ N})\mathbf{i} \end{aligned}$$

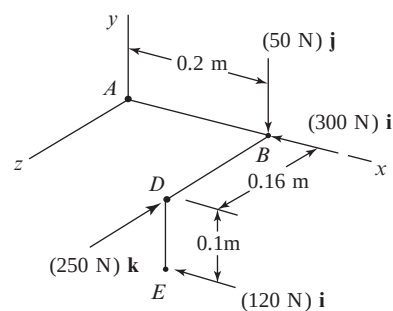
$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.2 \text{ m} & 0 & 0 \\ -300 \text{ N} & -50 \text{ N} & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.2 \text{ m} & 0 & 0.16 \text{ m} \\ 0 & 0 & -250 \text{ N} \end{vmatrix}$$

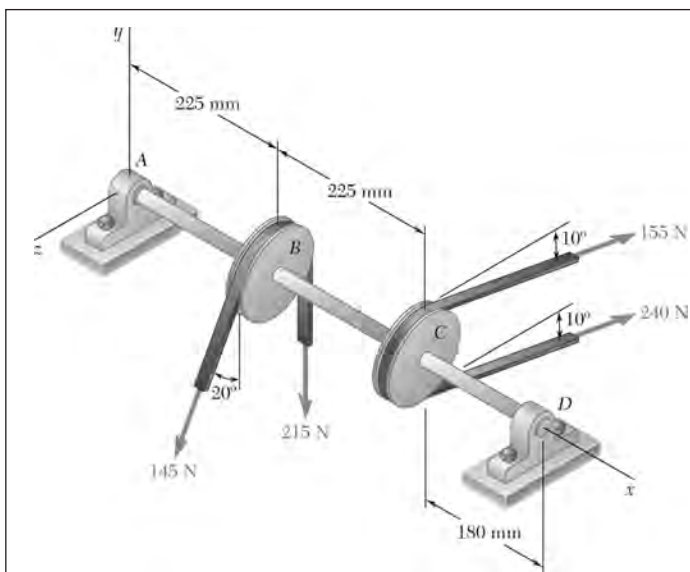
$$+ \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.2 \text{ m} & -0.1 \text{ m} & 0.16 \text{ m} \\ -120 \text{ N} & 0 & 0 \end{vmatrix}$$

$$= -(10 \text{ N} \cdot \text{m})\mathbf{k} + (50 \text{ N} \cdot \text{m})\mathbf{j} - (19.2 \text{ N} \cdot \text{m})\mathbf{j} - (12 \text{ N} \cdot \text{m})\mathbf{k}$$

Force-couple system at A is

$$\mathbf{R} = -(420 \text{ N})\mathbf{i} - (50 \text{ N})\mathbf{j} - (250 \text{ N})\mathbf{k} \quad \mathbf{M}_A^R = (30.8 \text{ N} \cdot \text{m})\mathbf{j} - (220 \text{ N} \cdot \text{m})\mathbf{k} \quad \blacktriangleleft$$





PROBLEM 3.119

Two 150-mm-diameter pulleys are mounted on line shaft AD . The belts at B and C lie in vertical planes parallel to the yz plane. Replace the belt forces shown with an equivalent force-couple system at A .

SOLUTION

Equivalent force-couple at each pulley

Pulley B

$$\mathbf{R}_B = (145 \text{ N})(-\cos 20^\circ \mathbf{j} + \sin 20^\circ \mathbf{k}) - 215 \text{ N} \mathbf{j}$$

$$= -(351.26 \text{ N}) \mathbf{j} + (49.593 \text{ N}) \mathbf{k}$$

$$\mathbf{M}_B = -(215 \text{ N} - 145 \text{ N})(0.075 \text{ m}) \mathbf{i}$$

$$= -(5.25 \text{ N} \cdot \text{m}) \mathbf{i}$$

Pulley C

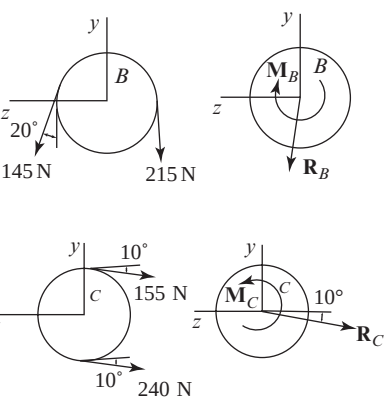
$$\mathbf{R}_C = (155 \text{ N} + 240 \text{ N})(-\sin 10^\circ \mathbf{j} - \cos 10^\circ \mathbf{k})$$

$$= -(68.591 \text{ N}) \mathbf{j} - (389.00 \text{ N}) \mathbf{k}$$

$$\mathbf{M}_C = (240 \text{ N} - 155 \text{ N})(0.075 \text{ m}) \mathbf{i}$$

$$= (6.3750 \text{ N} \cdot \text{m}) \mathbf{i}$$

Then $\mathbf{R} = \mathbf{R}_B + \mathbf{R}_C = -(419.85 \text{ N}) \mathbf{j} - (339.41) \mathbf{k}$



or $\mathbf{R} = (420 \text{ N}) \mathbf{j} - (339 \text{ N}) \mathbf{k} \blacktriangleleft$

$$\mathbf{M}_A = \mathbf{M}_B + \mathbf{M}_C + \mathbf{r}_{B/A} \times \mathbf{R}_B + \mathbf{r}_{C/A} \times \mathbf{R}_C$$

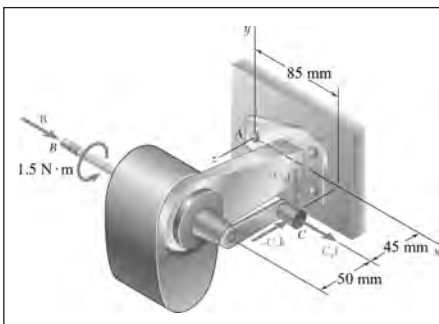
$$= -(5.25 \text{ N} \cdot \text{m}) \mathbf{i} + (6.3750 \text{ N} \cdot \text{m}) \mathbf{i} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.225 & 0 & 0 \\ 0 & -351.26 & 49.593 \end{vmatrix} \text{ N} \cdot \text{m}$$

$$+ \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.45 & 0 & 0 \\ 0 & -68.591 & -389.00 \end{vmatrix} \text{ N} \cdot \text{m}$$

$$= (1.12500 \text{ N} \cdot \text{m}) \mathbf{i} + (163.892 \text{ N} \cdot \text{m}) \mathbf{j} - (109.899 \text{ N} \cdot \text{m}) \mathbf{k}$$

or $\mathbf{M}_A = (1.125 \text{ N} \cdot \text{m}) \mathbf{i} + (163.9 \text{ N} \cdot \text{m}) \mathbf{j} - (109.9 \text{ N} \cdot \text{m}) \mathbf{k} \blacktriangleleft$

PROPRIETARY MATERIAL. © 2010 The McGraw-Hill Companies, Inc. All rights reserved. No part of this Manual may be displayed, reproduced or distributed in any form or by any means, without the prior written permission of the publisher, or used beyond the limited distribution to teachers and educators permitted by McGraw-Hill for their individual course preparation. If you are a student using this Manual, you are using it without permission.



PROBLEM 3.120

While using a pencil sharpener, a student applies the forces and couple shown. (a) Determine the forces exerted at B and C knowing that these forces and the couple are equivalent to a force-couple system at A consisting of the force $\mathbf{R} = (13\text{N})\mathbf{i} + R_y\mathbf{j} - (3.5\text{N})\mathbf{k}$ and the couple $\mathbf{M}_A^R = M_x\mathbf{i} + (1.5\text{N}\cdot\text{m})\mathbf{j} - (1.08\text{N}\cdot\text{m})\mathbf{k}$. (b) Find the corresponding values of R_y and M_x .

SOLUTION

(a) From the statement of the problem, equivalence requires

$$\Sigma \mathbf{F}: \mathbf{B} + \mathbf{C} = \mathbf{R}$$

$$\text{or} \quad \Sigma F_x: B_x + C_x = 13 \text{ N} \quad (1)$$

$$\Sigma F_y: -C_y = R_y \quad (2)$$

$$\Sigma F_z: -C_z = -3.5 \text{ N} \quad \text{or} \quad C_z = 3.5 \text{ N}$$

$$\text{and} \quad \Sigma \mathbf{M}_A: (\mathbf{r}_{B/A} \times \mathbf{B} + \mathbf{M}_B) + \mathbf{r}_{C/A} \times \mathbf{C} = \mathbf{M}_A^R$$

$$\text{or} \quad \Sigma M_x: (1500 \text{ N}\cdot\text{mm}) + (45 \text{ mm})(C_y) = M_x \quad (3)$$

$$\Sigma M_y: (95 \text{ mm})(B_x) + (45 \text{ mm})(C_x) + (85 \text{ mm})(3.5 \text{ N}) = 1500 \text{ N}\cdot\text{mm}$$

$$\text{or} \quad 95B_x + 45C_x = 1202.5$$

$$\text{Using Eq. (1)} \quad 95B_x + 45(13 - B_x) = 1202.5$$

$$\text{or} \quad B_x = 12.35 \text{ N}$$

$$\text{and} \quad C_x = 0.65 \text{ N}$$

$$\Sigma M_z: -(85 \text{ mm})(C_y) = -1080 \text{ N}\cdot\text{mm}$$

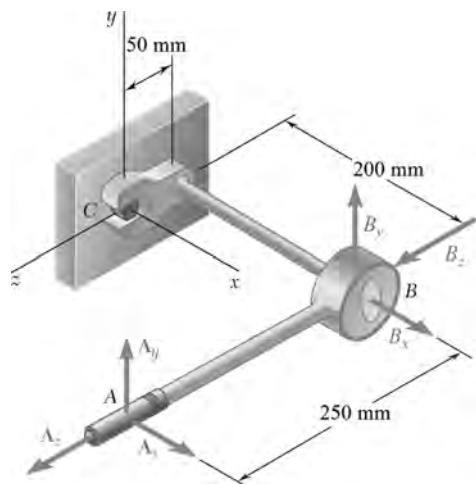
$$\text{or} \quad C_y = 12.706 \text{ N}$$

$$\mathbf{B} = (12.35 \text{ N})\mathbf{i} \quad \mathbf{C} = (0.65 \text{ N})\mathbf{i} - (12.71 \text{ N})\mathbf{j} - (3.5 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$(b) \quad \text{Eq. (2)} \Rightarrow \quad R_y = -12.71 \text{ N} \quad \blacktriangleleft$$

$$\begin{aligned} \text{Using Eq. (3)} \quad 1500 + (45)(12.706) &= M_x \\ &= 2071.77 \text{ N}\cdot\text{mm} \end{aligned}$$

$$\text{or} \quad M_x = 2.07 \text{ N}\cdot\text{m} \quad \blacktriangleleft$$

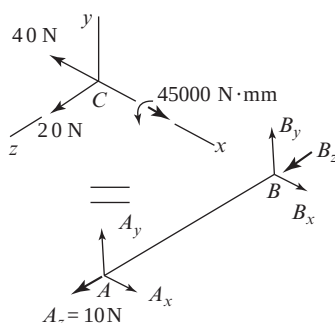


PROBLEM 3.121

A mechanic uses a crowfoot wrench to loosen a bolt at C. The mechanic holds the socket wrench handle at Points A and B and applies forces at these points. Knowing that these forces are equivalent to a force-couple system at C consisting of the force $\mathbf{C} = (40 \text{ N})\mathbf{i} + (20 \text{ N})\mathbf{k}$ and the couple $\mathbf{M}_C = (45 \text{ N} \cdot \text{m})\mathbf{i}$, determine the forces applied at A and at B when $A_z = 10 \text{ N}$.

SOLUTION

We have



or

$$\Sigma \mathbf{F}: \quad \mathbf{A} + \mathbf{B} = \mathbf{C}$$

$$F_x: \quad A_x + B_x = 40 \text{ N}$$

$$B_x = -(A_x + 40 \text{ N}) \quad (1)$$

$$\Sigma F_y: \quad A_y + B_y = 0$$

or

$$A_y = -B_y \quad (2)$$

$$\Sigma F_z: \quad 10 \text{ N} + B_z = 20 \text{ N}$$

or

$$B_z = 10 \text{ N} \quad (3)$$

We have

$$\Sigma \mathbf{M}_C: \quad \mathbf{r}_{B/C} \times \mathbf{B} + \mathbf{r}_{A/C} \times \mathbf{A} = \mathbf{M}_C$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 200 & 0 & -50 \\ B_x & B_y & 10 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 200 & 0 & 200 \\ A_x & A_y & 10 \end{vmatrix} \text{ N} \cdot \text{mm} = (45000 \text{ N} \cdot \text{mm})\mathbf{i}$$

or

$$(50B_y - 200A_y)\mathbf{i} + (50B_x - 2000 + 200A_x - 2000)\mathbf{j}$$

$$+ (200B_y + 200A_y)\mathbf{k} = (45000 \text{ N} \cdot \text{mm})\mathbf{i}$$

From

$$\mathbf{i}\text{-coefficient} \quad 50B_y - 200A_y = 45000 \quad (4)$$

$$\mathbf{j}\text{-coefficient} \quad 50B_x + 200A_x = 4000 \quad (5)$$

$$\mathbf{k}\text{-coefficient} \quad 200B_y + 200A_y = 0 \quad (6)$$

PROBLEM 3.121 (Continued)

From Equations (2) and (4): $50B_y - 200(-B_y) = 45000$

$$B_y = 180 \text{ N} \quad A_y = -180 \text{ N}$$

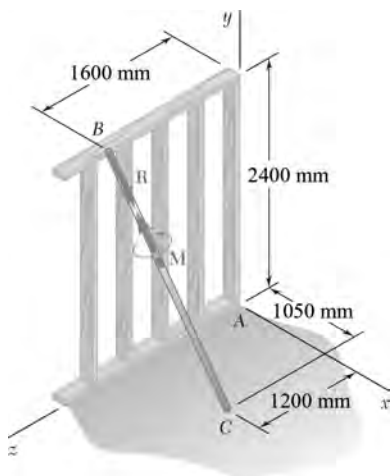
From Equations (1) and (5): $50(-A_x - 40) + 200A_x = 4000$

$$A_x = 40 \text{ N}$$

From Equation (1): $B_x = -(40 + 40) = -80 \text{ N}$

$$\mathbf{A} = (40 \text{ N})\mathbf{i} - (180 \text{ N})\mathbf{j} + (10 \text{ N})\mathbf{k}$$

$$\mathbf{B} = -(80 \text{ N})\mathbf{i} + (180 \text{ N})\mathbf{j} + (10 \text{ N})\mathbf{k}$$



PROBLEM 3.122

As an adjustable brace BC is used to bring a wall into plumb, the force-couple system shown is exerted on the wall. Replace this force-couple system with an equivalent force-couple system at A if $R = 106 \text{ N}$ and $M = 20 \text{ N} \cdot \text{m}$.

SOLUTION

We have $\Sigma \mathbf{F}: \mathbf{R} = \mathbf{R}_A = R\lambda_{BC}$

where $\lambda_{BC} = \frac{(1050 \text{ mm})\mathbf{i} - (2400 \text{ mm})\mathbf{j} - (400 \text{ mm})\mathbf{k}}{2650 \text{ mm}}$

$$\mathbf{R}_A = \frac{106 \text{ N}}{2650} (1050\mathbf{i} - 2400\mathbf{j} - 400\mathbf{k})$$

or $\mathbf{R}_A = (42 \text{ N})\mathbf{i} - (96 \text{ N})\mathbf{j} - (16 \text{ N})\mathbf{k}$

We have $\Sigma \mathbf{M}_A: \mathbf{r}_{C/A} \times \mathbf{R} + \mathbf{M} = \mathbf{M}_A$

where $\mathbf{r}_{C/A} = (1050 \text{ mm})\mathbf{i} + (1200 \text{ mm})\mathbf{k}$
 $= 1.05 \text{ m}\mathbf{j} + 1.2 \text{ m}\mathbf{k}$

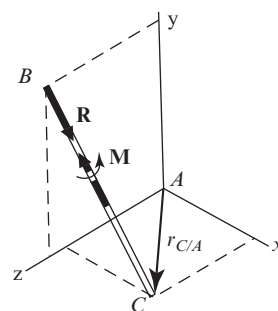
$$\mathbf{R} = (42 \text{ N})\mathbf{i} - (96 \text{ N})\mathbf{j} - (16 \text{ N})\mathbf{k}$$

$$\begin{aligned} \mathbf{M} &= -\lambda_{BC} M \\ &= \frac{-1050\mathbf{i} + 2400\mathbf{j} + 400\mathbf{k}}{2650} (20 \text{ N} \cdot \text{m}) \\ &= -(7.92 \text{ N} \cdot \text{m})\mathbf{i} + (18.113 \text{ N} \cdot \text{m})\mathbf{j} + (3.019 \text{ N} \cdot \text{m})\mathbf{k} \end{aligned}$$

Then $\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1.05 & 0 & 1.2 \\ 42 & -96 & -16 \end{vmatrix} \text{ N} \cdot \text{m} + (-7.924\mathbf{i} + 18.113\mathbf{j} + 3.019\mathbf{k}) \text{ N} \cdot \text{m}$

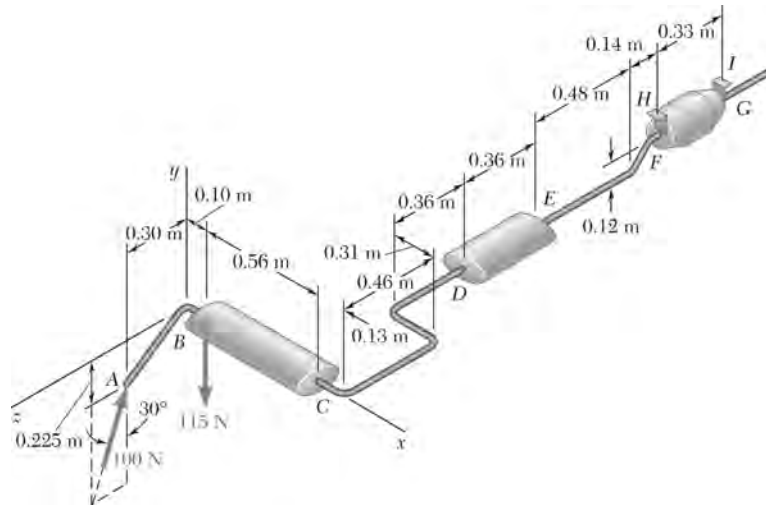
$$\mathbf{M}_A = (107.28 \text{ N} \cdot \text{m})\mathbf{i} + (85.313 \text{ N} \cdot \text{m})\mathbf{j} - (97.781 \text{ N} \cdot \text{m})\mathbf{k}$$

$$\text{or } \mathbf{M}_A = (107.3 \text{ N} \cdot \text{m})\mathbf{i} + (85.3 \text{ N} \cdot \text{m})\mathbf{j} - (97.8 \text{ N} \cdot \text{m})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 3.123

A mechanic replaces a car's exhaust system by firmly clamping the catalytic converter FG to its mounting brackets H and I and then loosely assembling the mufflers and the exhaust pipes. To position the tailpipe AB , he pushes in and up at A while pulling down at B . (a) Replace the given force system with an equivalent force-couple system at D . (b) Determine whether pipe CD tends to rotate clockwise or counterclockwise relative to muffler DE , as viewed by the mechanic.



SOLUTION

(a) Equivalence requires

$$\begin{aligned}\Sigma \mathbf{F}: \quad \mathbf{R} &= \mathbf{A} + \mathbf{B} \\ &= (100 \text{ N})(\cos 30^\circ \mathbf{j} - \sin 30^\circ \mathbf{k}) - (115 \text{ N})\mathbf{j} \\ &= -(28.4 \text{ N})\mathbf{j} - (50 \text{ N})\mathbf{k}\end{aligned}$$

and

$$\Sigma \mathbf{M}_D: \quad \mathbf{M}_D = \mathbf{r}_{A/D} \times \mathbf{F}_A + \mathbf{r}_{B/D} \times \mathbf{F}_B$$

where

$$\mathbf{r}_{A/D} = -(0.48 \text{ m})\mathbf{i} - (0.225 \text{ m})\mathbf{j} + (1.12 \text{ m})\mathbf{k}$$

$$\mathbf{r}_{B/D} = -(0.38 \text{ m})\mathbf{i} + (0.82 \text{ m})\mathbf{k}$$

Then

$$\begin{aligned}\mathbf{M}_D &= 100 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -0.48 & -0.225 & 1.12 \\ 0 & \cos 30^\circ & -\sin 30^\circ \end{vmatrix} + 115 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -0.38 & 0 & 0.82 \\ 0 & -1 & 0 \end{vmatrix} \\ &= 100[(0.225 \sin 30^\circ - 1.12 \cos 30^\circ)\mathbf{i} + (-0.48 \sin 30^\circ)\mathbf{j} \\ &\quad + (-0.48 \cos 30^\circ)\mathbf{k}] + 115[(0.82)\mathbf{i} + (0.38)\mathbf{k}] \\ &= 8.56\mathbf{i} - 24.0\mathbf{j} + 2.13\mathbf{k}\end{aligned}$$

PROBLEM 3.123 (Continued)

The equivalent force-couple system at D is

$$\mathbf{R} = -(28.4 \text{ N})\mathbf{j} - (50.0 \text{ N})\mathbf{k}$$



$$M_D = (8.56 \text{ N} \cdot \text{m})\mathbf{i} - (24.0 \text{ N} \cdot \text{m})\mathbf{j} + (2.13 \text{ N} \cdot \text{m})\mathbf{k}$$

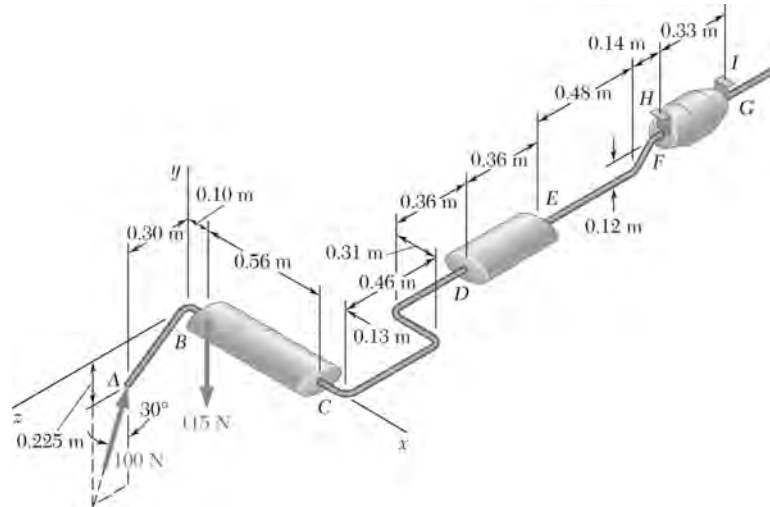


(b) Since $(M_D)_z$ is positive, pipe CD will tend to rotate counterclockwise relative to muffler DE .



PROBLEM 3.124

For the exhaust system of Problem 3.123, (a) replace the given force system with an equivalent force-couple system at F , where the exhaust pipe is connected to the catalytic converter, (b) determine whether pipe EF tends to rotate clockwise or counterclockwise, as viewed by the mechanic.



SOLUTION

(a) Equivalence requires

$$\begin{aligned}\Sigma \mathbf{F}: \quad \mathbf{R} &= \mathbf{A} + \mathbf{B} \\ &= (100 \text{ N})(\cos 30^\circ \mathbf{j} - \sin 30^\circ \mathbf{k}) - (115 \text{ N})\mathbf{j} \\ &= -(28.4 \text{ N})\mathbf{j} - (50 \text{ N})\mathbf{k}\end{aligned}$$

and

$$\mathbf{M}_F: \quad \mathbf{M}_F = \mathbf{r}_{A/F} \times \mathbf{A} + \mathbf{r}_{B/F} \times \mathbf{B}$$

where

$$\mathbf{r}_{A/F} = -(0.48 \text{ m})\mathbf{i} - (0.345 \text{ m})\mathbf{j} + (2.10 \text{ m})\mathbf{k}$$

$$\mathbf{r}_{B/F} = -(0.38 \text{ m})\mathbf{i} - (0.12 \text{ m})\mathbf{j} + (1.80 \text{ m})\mathbf{k}$$

Then

$$\mathbf{M}_F = 100 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -0.48 & -0.345 & 2.10 \\ 0 & \cos 30^\circ & -\sin 30^\circ \end{vmatrix} + 115 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -0.38 & 0.12 & 1.80 \\ 0 & -1 & 0 \end{vmatrix}$$

$$\begin{aligned}\mathbf{M}_F &= 100[(0.345 \sin 30^\circ - 2.10 \cos 30^\circ)\mathbf{i} + (-0.48 \sin 30^\circ)\mathbf{j} \\ &\quad + (-0.48 \cos 30^\circ)\mathbf{k}] + 115[(1.80)\mathbf{i} + (0.38)\mathbf{k}] \\ &= 42.4\mathbf{i} - 24.0\mathbf{j} + 2.13\mathbf{k}\end{aligned}$$

PROBLEM 3.124 (Continued)

The equivalent force-couple system at F is

$$\mathbf{R} = -(28.4 \text{ N})\mathbf{j} - (50 \text{ N})\mathbf{k}$$

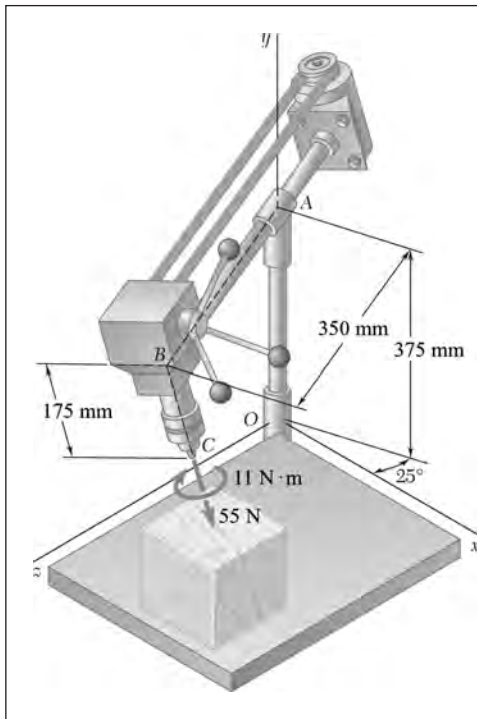


$$\mathbf{M}_F = (42.4 \text{ N} \cdot \text{m})\mathbf{i} - (24.0 \text{ N} \cdot \text{m})\mathbf{j} + (2.13 \text{ N} \cdot \text{m})\mathbf{k}$$



(b) Since $(M_F)_z$ is positive, pipe EF will tend to rotate counterclockwise relative to the mechanic.





PROBLEM 3.125

The head-and-motor assembly of a radial drill press was originally positioned with arm AB parallel to the z axis and the axis of the chuck and bit parallel to the y axis. The assembly was then rotated 25° about the y axis and 20° about the centerline of the horizontal arm AB , bringing it into the position shown. The drilling process was started by switching on the motor and rotating the handle to bring the bit into contact with the workpiece. Replace the force and couple exerted by the drill press with an equivalent force-couple system at the center O of the base of the vertical column.

SOLUTION

We have $\mathbf{R} = \mathbf{F} = (55 \text{ N})[(\sin 20^\circ \cos 25^\circ)\mathbf{i} - (\cos 20^\circ)\mathbf{j} - (\sin 20^\circ \sin 25^\circ)\mathbf{k}]$
 $= (17.0485 \text{ N})\mathbf{i} - (51.683 \text{ N})\mathbf{j} - (7.9499 \text{ N})\mathbf{k}$

or $\mathbf{R} = (17.05 \text{ N})\mathbf{i} - (51.7 \text{ N})\mathbf{j} - (7.95 \text{ N})\mathbf{k}$

We have $\mathbf{M}_O = \mathbf{r}_{B/O} \times \mathbf{F} + \mathbf{M}_C$

where

$$\mathbf{r}_{B/O} = [(350 \text{ mm})\sin 25^\circ]\mathbf{i} + (375 \text{ mm})\mathbf{j} + [(350 \text{ mm})\cos 25^\circ]\mathbf{k}$$

$$= (147.9175 \text{ mm})\mathbf{i} + (375 \text{ mm})\mathbf{j} + (317.208 \text{ mm})\mathbf{k}$$

$$\mathbf{r}_{B/O} = [(0.1479 \text{ m})\mathbf{i} + (0.375 \text{ m})\mathbf{j} + (0.3172 \text{ m})\mathbf{k}]$$

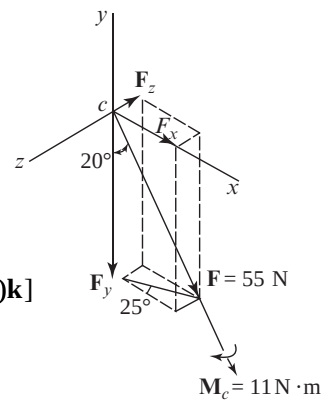
$$\mathbf{M}_C = (11 \text{ N} \cdot \text{m})[(\sin 20^\circ \cos 25^\circ)\mathbf{i} - (\cos 20^\circ)\mathbf{j} - (\sin 20^\circ \sin 25^\circ)\mathbf{k}]$$

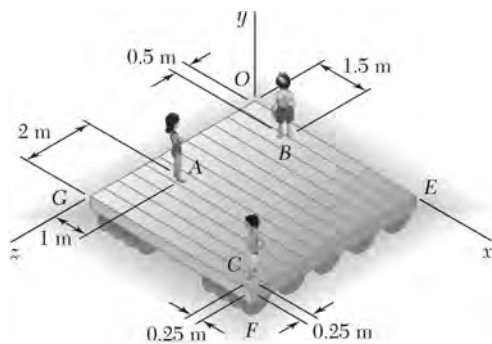
$$= (3.4097 \text{ N} \cdot \text{m})\mathbf{i} - (10.337 \text{ N} \cdot \text{m})\mathbf{j} - (1.59 \text{ N} \cdot \text{m})\mathbf{k}$$

$$\mathbf{M}_O = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.1479 & 0.375 & 0.3172 \\ 17.048 & -51.683 & -7.9499 \end{vmatrix} \text{ N} \cdot \text{m} + (3.4097\mathbf{i} - 10.337\mathbf{j} - 1.59)\text{ N} \cdot \text{m}$$

$$= (16.462 \text{ N} \cdot \text{m})\mathbf{i} - (3.6165 \text{ N} \cdot \text{m})\mathbf{j} - (15.627 \text{ N} \cdot \text{m})\mathbf{k}$$

or $\mathbf{M}_O = (16.46 \text{ N} \cdot \text{m})\mathbf{i} - (3.62 \text{ N} \cdot \text{m})\mathbf{j} - (15.63 \text{ N} \cdot \text{m})\mathbf{k}$

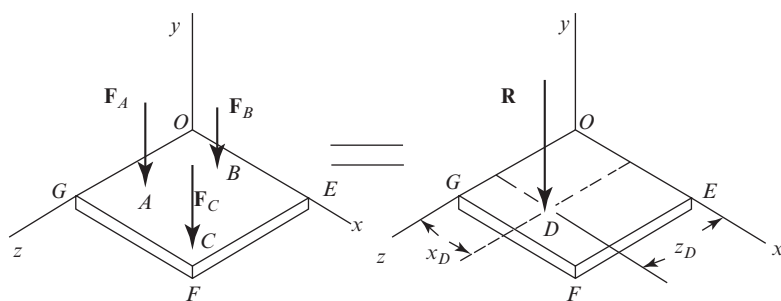




PROBLEM 3.126

Three children are standing on a 5×5-m raft. If the weights of the children at Points A, B, and C are 375 N, 260 N, and 400 N, respectively, determine the magnitude and the point of application of the resultant of the three weights.

SOLUTION



We have

$$\begin{aligned}\Sigma \mathbf{F}: \quad \mathbf{F}_A + \mathbf{F}_B + \mathbf{F}_C &= \mathbf{R} \\ -(375 \text{ N})\mathbf{j} - (260 \text{ N})\mathbf{j} - (400 \text{ N})\mathbf{j} &= \mathbf{R} \\ -(1035 \text{ N})\mathbf{j} &= \mathbf{R}\end{aligned}$$

$$\text{or } R = 1035 \text{ N} \quad \blacktriangleleft$$

We have

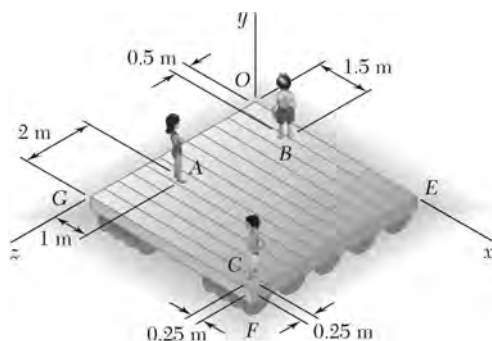
$$\begin{aligned}\Sigma M_x: \quad F_A(z_A) + F_B(z_B) + F_C(z_C) &= R(z_D) \\ (375 \text{ N})(3 \text{ m}) + (260 \text{ N})(0.5 \text{ m}) + (400 \text{ N})(4.75 \text{ m}) &= (1035 \text{ N})(z_D) \\ z_D &= 3.0483 \text{ m}\end{aligned}$$

$$\text{or } z_D = 3.05 \text{ m} \quad \blacktriangleleft$$

We have

$$\begin{aligned}\Sigma M_z: \quad F_A(x_A) + F_B(x_B) + F_C(x_C) &= R(x_D) \\ 375 \text{ N}(1 \text{ m}) + (260 \text{ N})(1.5 \text{ m}) + (400 \text{ N})(4.75 \text{ m}) &= (1035 \text{ N})(x_D) \\ x_D &= 2.5749 \text{ m}\end{aligned}$$

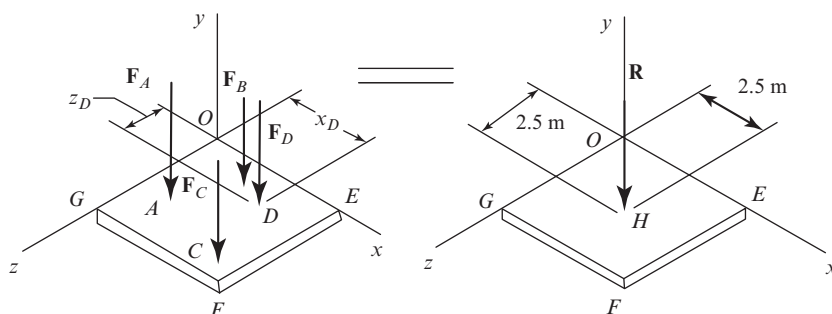
$$\text{or } x_D = 2.57 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.127

Three children are standing on a 5×5-m raft. The weights of the children at Points A, B, and C are 375 N, 260 N, and 400 N, respectively. If a fourth child of weight 425 N climbs onto the raft, determine where she should stand if the other children remain in the positions shown and the line of action of the resultant of the four weights is to pass through the center of the raft.

SOLUTION



We have

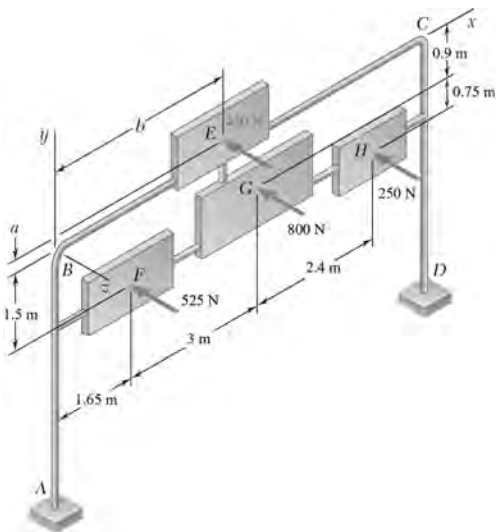
$$\begin{aligned}\Sigma \mathbf{F}: \quad \mathbf{F}_A + \mathbf{F}_B + \mathbf{F}_C &= \mathbf{R} \\ -(375 \text{ N})\mathbf{j} - (260 \text{ N})\mathbf{j} - (400 \text{ N})\mathbf{j} - (425 \text{ N})\mathbf{j} &= \mathbf{R} \\ \mathbf{R} &= -(1460 \text{ N})\mathbf{j}\end{aligned}$$

We have

$$\begin{aligned}\Sigma M_x: \quad F_A(z_A) + F_B(z_B) + F_C(z_C) + F_D(z_D) &= R(z_H) \\ (375 \text{ N})(3 \text{ m}) + (260 \text{ N})(0.5 \text{ m}) + (400 \text{ N})(4.75 \text{ m}) \\ + (425 \text{ N})(z_D) &= (1460 \text{ N})(2.5 \text{ m}) \\ z_D &= 1.16471 \text{ m} \qquad \text{or } z_D = 1.165 \text{ m} \quad \blacktriangleleft\end{aligned}$$

We have

$$\begin{aligned}\Sigma M_z: \quad F_A(x_A) + F_B(x_B) + F_C(x_C) + F_D(x_D) &= R(x_H) \\ (375 \text{ N})(1 \text{ m}) + (260 \text{ N})(1.5 \text{ m}) + (400 \text{ N})(4.75 \text{ m}) \\ + (425 \text{ N})(x_D) &= (1460 \text{ N})(2.5 \text{ m}) \\ x_D &= 2.3235 \text{ m} \qquad \text{or } x_D = 2.32 \text{ m} \quad \blacktriangleleft\end{aligned}$$

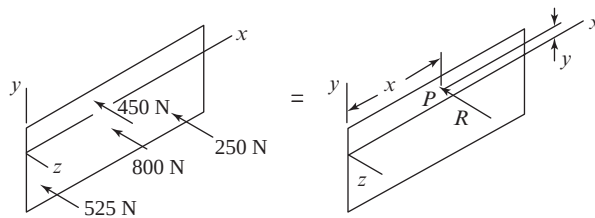


PROBLEM 3.128

Four signs are mounted on a frame spanning a highway, and the magnitudes of the horizontal wind forces acting on the signs are as shown. Determine the magnitude and the point of application of the resultant of the four wind forces when $a = 0.3 \text{ m}$ and $b = 4 \text{ m}$.

SOLUTION

We have



Assume that the resultant $\mathbf{R}$ is applied at Point P whose coordinates are $(x, y, 0)$. Equivalence then requires

$$\Sigma F_z: -525 - 450 - 800 - 250 = -R$$

$$\text{or } R = 2025 \text{ N} \quad \blacktriangleleft$$

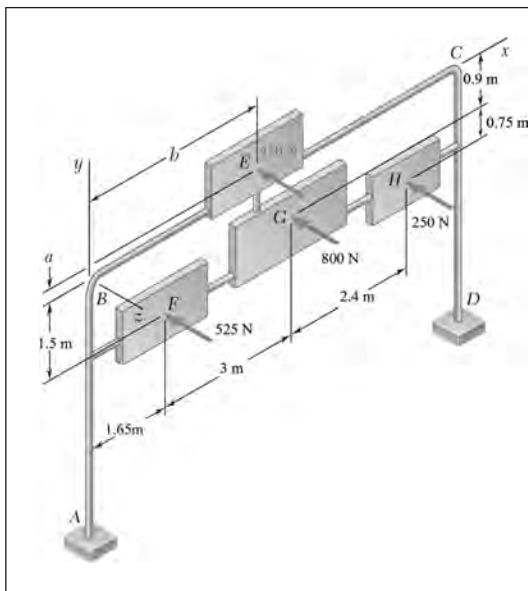
$$\Sigma M_x: (1.5 \text{ m})(525 \text{ N}) - (0.3 \text{ m})(450 \text{ N}) + (0.9 \text{ m})(800 \text{ N}) + (1.65 \text{ m})(250 \text{ N}) = -y(2025 \text{ N})$$

$$\text{or } y = -0.8815 \text{ m}$$

$$\Sigma M_y: (1.65 \text{ m})(525 \text{ N}) + (4 \text{ m})(450 \text{ N}) + (4.65 \text{ m})(800 \text{ N}) + (7.05 \text{ m})(250 \text{ N}) = -x(2025 \text{ N})$$

$$\text{or } x = 4.02 \text{ N} \cdot \text{m}$$

$\mathbf{R}$ acts $4.02 \text{ N} \cdot \text{m}$ to the right of member AB and 0.881 m below member BC . $\blacktriangleleft$



PROBLEM 3.129

Four signs are mounted on a frame spanning a highway, and the magnitudes of the horizontal wind forces acting on the signs are as shown. Determine a and b so that the point of application of the resultant of the four forces is at G .

SOLUTION

Since $\mathbf{R}$ acts at G , equivalence then requires that $\Sigma \mathbf{M}_G$ of the applied system of forces also be zero. Then

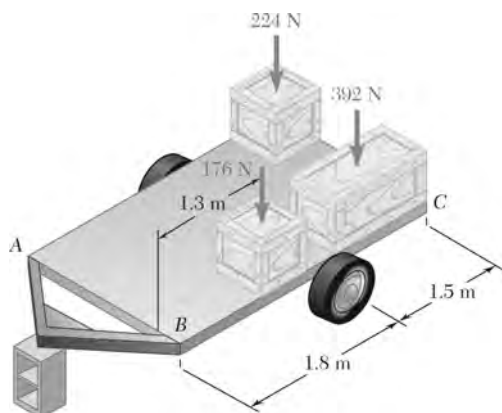
at

$$G: \Sigma M_x: -(a + 0.9) \text{ m} \times (450 \text{ N}) + (0.6 \text{ m})(525 \text{ N}) + (0.75 \text{ m})(250 \text{ N}) = 0$$

$$\text{or } a = 0.217 \text{ m} \quad \blacktriangleleft$$

$$\Sigma M_y: -(3 \text{ m})(525 \text{ N}) - (4.65 - b) \text{ m} \times (450 \text{ N}) + (2.4 \text{ m})(250 \text{ N}) = 0$$

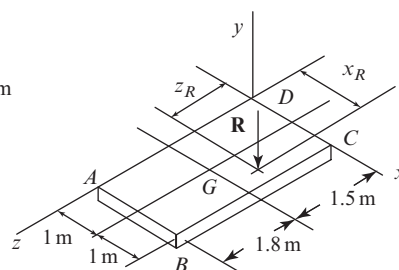
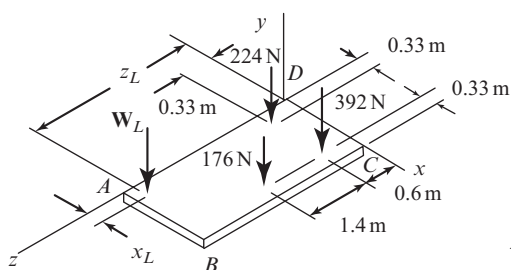
$$\text{or } b = 6.82 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.130*

A group of students loads a 2×3.3 -m flatbed trailer with two $0.66 \times 0.66 \times 0.66$ -m boxes and one $0.66 \times 0.66 \times 1.2$ -m box. Each of the boxes at the rear of the trailer is positioned so that it is aligned with both the back and a side of the trailer. Determine the smallest load the students should place in a second $0.66 \times 0.66 \times 1.2$ -m box and where on the trailer they should secure it, without any part of the box overhanging the sides of the trailer, if each box is uniformly loaded and the line of action of the resultant of the weights of the four boxes is to pass through the point of intersection of the centerlines of the trailer and the axle. (Hint: Keep in mind that the box may be placed either on its side or on its end.)

SOLUTION



For the smallest weight on the trailer so that the resultant force of the four weights acts over the axle at the intersection with the centerline of the trailer, the added $0.66 \times 0.66 \times 1.2$ -m box should be placed adjacent to one of the edges of the trailer with the 0.66×0.66 -m side on the bottom. The edges to be considered are based on the location of the resultant for the three given weights.

We have $\Sigma \mathbf{F}: -(224 \text{ N})\mathbf{j} - (392 \text{ N})\mathbf{j} - (176 \text{ N})\mathbf{j} = \mathbf{R}$

$$\mathbf{R} = -(792 \text{ N})\mathbf{j}$$

We have $\Sigma M_z: -(224 \text{ N})(0.33 \text{ m}) - (392 \text{ N})(1.67 \text{ m}) - (176 \text{ N})(1.67 \text{ m}) = (-792 \text{ N})(x)$

$$x_R = 1.29101 \text{ m}$$

We have $\Sigma M_x: (224 \text{ N})(0.33 \text{ m}) + (392 \text{ N})(0.6 \text{ m}) + (176 \text{ N})(2.0 \text{ m}) = (792 \text{ N})(z)$

$$z_R = 0.83475 \text{ m}$$

From the statement of the problem, it is known that the resultant of $\mathbf{R}$ from the original loading and the lightest load $\mathbf{W}$ passes through G , the point of intersection of the two centerlines. Thus, $\Sigma \mathbf{M}_G = 0$.

Further, since the lightest load $\mathbf{W}$ is to be as small as possible, the fourth box should be placed as far from G as possible without the box overhanging the trailer. These two requirements imply

$$(0.33 \text{ m} \leq x \leq 1 \text{ m})(1.5 \text{ m} \leq z \leq 2.97 \text{ m})$$

PROBLEM 3.130* (Continued)

With $x_L = 0.33 \text{ m}$

at $G: \Sigma M_Z: (1 - 0.33) \text{ m} \times W_L - (1.29101 - 1) \text{ m} \times (792 \text{ N}) = 0$

or $W_L = 344.00 \text{ N}$

Now must check if this is physically possible,

at $G: \Sigma M_x: (Z_L - 1.5) \text{ m} \times (344 \text{ N}) - (1.5 - 0.83475) \text{ m} \times (792 \text{ N}) = 0$

or $Z_L = 3.032 \text{ m}$

which is **not** acceptable.

With $Z_L = 2.97 \text{ m}$:

at $G: \Sigma M_x: (2.97 - 1.5) \text{ m} \times W_L - (1.5 - 0.83475) \text{ m} \times (792 \text{ N}) = 0$

or $W_L = 358.42 \text{ N}$

Now check if this is physically possible

at $G: \Sigma M_z: (1 - X_L) \text{ m} \times (358.42 \text{ N}) - (1.29101 - 1) \text{ m} \times (792 \text{ N}) = 0$

or $X_L = 0.357 \text{ m}$ ok!

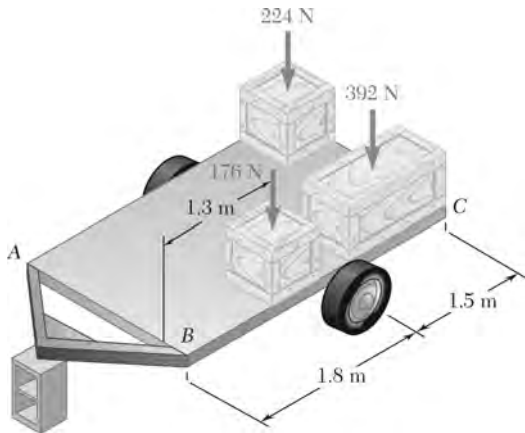
The minimum weight of the fourth box is $W_L = 358 \text{ N}$ ◀

And it is placed on end (A $0.66 \times 0.66\text{-m}$ side down) along side AB with the center of the box 0.357 m from side AD . ▶

PROBLEM 3.131*

Solve Problem 3.131 if the students want to place as much weight as possible in the fourth box and at least one side of the box must coincide with a side of the trailer.

PROBLEM 3.130* A group of students loads a 2×3.3 -m flatbed trailer with two $0.66 \times 0.66 \times 0.66$ -m boxes and one $0.66 \times 0.66 \times 1.2$ -m box. Each of the boxes at the rear of the trailer is positioned so that it is aligned with both the back and a side of the trailer. Determine the smallest load the students should place in a second $0.66 \times 0.66 \times 1.2$ -m box and where on the trailer they should secure it, without any part of the box overhanging the sides of the trailer, if each box is uniformly loaded and the line of action of the resultant of the weights of the four boxes is to pass through the point of intersection of the centerlines of the trailer and the axle. (*Hint: Keep in mind that the box may be placed either on its side or on its end.*)



SOLUTION

First replace the three known loads with a single equivalent force $\mathbf{R}$ applied at coordinate $(X_R, 0, Z_R)$

Equivalence requires

$$\Sigma F_y: -224 - 392 - 176 = -R$$

or

$$\mathbf{R} = 792 \text{ N} \downarrow$$

$$\Sigma M_x: (0.33 \text{ m})(224 \text{ N}) + (0.6 \text{ m})(392 \text{ N}) + (2 \text{ m})(176 \text{ N}) = z_R(792 \text{ N})$$

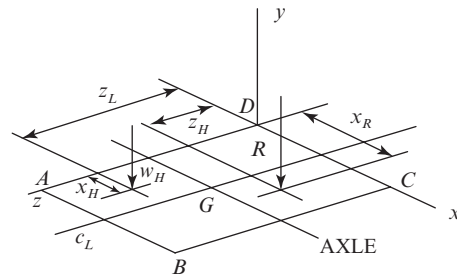
or

$$z_R = 0.83475 \text{ m}$$

$$\Sigma M_z: -(0.33 \text{ m})(224 \text{ N}) - (1.67 \text{ m})(392 \text{ N}) - (1.67 \text{ m})(176 \text{ N}) = x_R(792 \text{ N})$$

or

$$x_R = 1.29101 \text{ m}$$



From the statement of the problem, it is known that the resultant of $\mathbf{R}$ and the heaviest loads $\mathbf{W}_H$ passes through G , the point of intersection of the two centerlines. Thus,

$$\Sigma \mathbf{M}_G = 0$$

Further, since $\mathbf{W}_H$ is to be as large as possible, the fourth box should be placed as close to G as possible while keeping one of the sides of the box coincident with a side of the trailer. Thus, the two limiting cases are

$$x_H = 0.6 \text{ m} \quad \text{or} \quad z_H = 2.7 \text{ m}$$

PROBLEM 3.131* (Continued)

Now consider these two possibilities

With $x_H = 0.6$ m:

$$\text{at } G: \Sigma M_Z: (1 - 0.6) \text{ m} \times W_H - (1.29101 - 1) \text{ m} \times (792 \text{ N}) = 0$$

$$\text{or } W_H = 576.20 \text{ N}$$

Checking if this is physically possible

$$\text{at } G: \Sigma M_x: (z_H - 1.5) \text{ m} \times (576.20 \text{ N}) - (1.5 - 0.83475) \text{ m} \times (792 \text{ N}) = 0$$

$$\text{or } z_H = 2.414 \text{ m}$$

which is acceptable.

With $z_H = 2.7$ m

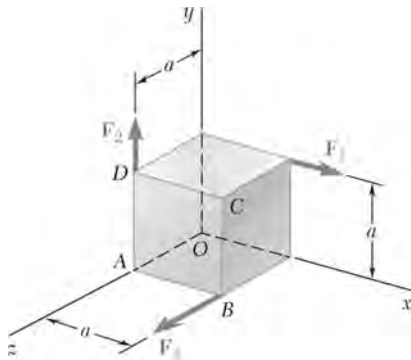
$$\text{at } G: \Sigma M_x: (2.7 - 1.5) W_H - (1.5 - 0.83475) \text{ m} \times (792 \text{ N}) = 0$$

$$\text{or } W_H = 439 \text{ N}$$

Since this is less than the first case, the maximum weight of the fourth box is

$$W_H = 576 \text{ N} \quad \blacktriangleleft$$

and it is placed with a 0.66×1.2 -m side down, a 0.66-m edge alongside AD , and the center 2.41 m from side DC . $\blacktriangleleft$



PROBLEM 3.132

Three forces of the same magnitude P act on a cube of side a as shown. Replace the three forces by an equivalent wrench and determine (a) the magnitude and direction of the resultant force $\mathbf{R}$, (b) the pitch of the wrench, (c) the axis of the wrench.

SOLUTION

Force-couple system at O :

$$\begin{aligned}\mathbf{R} &= P\mathbf{i} + P\mathbf{j} + P\mathbf{k} = P(\mathbf{i} + \mathbf{j} + \mathbf{k}) \\ \mathbf{M}_O^R &= a\mathbf{j} \times P\mathbf{i} + a\mathbf{k} \times P\mathbf{j} + a\mathbf{i} \times P\mathbf{k} \\ &= -Pa\mathbf{k} - Pa\mathbf{i} - Pa\mathbf{j} \\ \mathbf{M}_O^R &= -Pa(\mathbf{i} + \mathbf{j} + \mathbf{k})\end{aligned}$$

Since $\mathbf{R}$ and $\mathbf{M}_O^R$ have the same direction, they form a wrench with $\mathbf{M}_1 = \mathbf{M}_O^R$. Thus, the axis of the wrench is the diagonal OA . We note that

$$\cos \theta_x = \cos \theta_y = \cos \theta_z = \frac{a}{a\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$R = P\sqrt{3} \quad \theta_x = \theta_y = \theta_z = 54.7^\circ$$

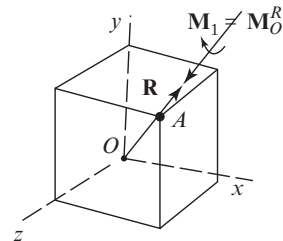
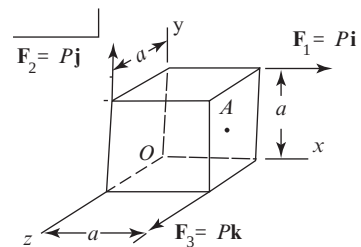
$$M_1 = M_O^R = -Pa\sqrt{3}$$

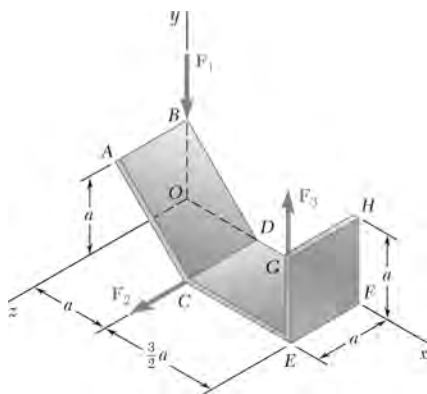
$$\text{Pitch} = p = \frac{M_1}{R} = \frac{-Pa\sqrt{3}}{P\sqrt{3}} = -a$$

$$(a) \quad R = P\sqrt{3} \quad \theta_x = \theta_y = \theta_z = 54.7^\circ$$

$$(b) \quad -a$$

$$(c) \quad \text{Axis of the wrench is diagonal } OA$$





PROBLEM 3.133*

A piece of sheet metal is bent into the shape shown and is acted upon by three forces. If the forces have the same magnitude P , replace them with an equivalent wrench and determine (a) the magnitude and the direction of the resultant force $\mathbf{R}$, (b) the pitch of the wrench, (c) the axis of the wrench.

SOLUTION

First reduce the given forces to an equivalent force-couple system $(\mathbf{R}, \mathbf{M}_O^R)$ at the origin.

We have

$$\Sigma \mathbf{F}: -P\mathbf{j} + P\mathbf{j} + P\mathbf{k} = \mathbf{R}$$

or

$$\mathbf{R} = P\mathbf{k}$$

$$\Sigma \mathbf{M}_O: -(aP)\mathbf{j} + \left[-(aP)\mathbf{i} + \left(\frac{5}{2}aP \right)\mathbf{k} \right] = \mathbf{M}_O^R$$

or

$$\mathbf{M}_O^R = aP \left(-\mathbf{i} - \mathbf{j} + \frac{5}{2}\mathbf{k} \right)$$

(a) Then for the wrench

$$R = P \quad \blacktriangleleft$$

and

$$\lambda_{\text{axis}} = \frac{\mathbf{R}}{R} = \mathbf{k}$$

$$\cos \theta_x = 0 \quad \cos \theta_y = 0 \quad \cos \theta_z = 1$$

or

$$\theta_x = 90^\circ \quad \theta_y = 90^\circ \quad \theta_z = 0^\circ \quad \blacktriangleleft$$

(b) Now

$$\begin{aligned} M_1 &= \lambda_{\text{axis}} \cdot \mathbf{M}_O^R \\ &= \mathbf{k} \cdot aP \left(-\mathbf{i} - \mathbf{j} + \frac{5}{2}\mathbf{k} \right) \\ &= \frac{5}{2}aP \end{aligned}$$

Then

$$P = \frac{M_1}{R} = \frac{\frac{5}{2}aP}{P} \quad \text{or} \quad P = \frac{5}{2}a \quad \blacktriangleleft$$

PROBLEM 3.133* (Continued)

- (c) The components of the wrench are $(\mathbf{R}, \mathbf{M}_1)$, where $\mathbf{M}_1 = M_1 \lambda_{\text{axis}}$, and the axis of the wrench is assumed to intersect the xy plane at Point Q whose coordinates are $(x, y, 0)$. Thus require

$$\mathbf{M}_z = \mathbf{r}_Q \times \mathbf{R}_R$$

Where

$$\mathbf{M}_z = \mathbf{M}_O \times \mathbf{M}_1$$

Then

$$aP \left(-\mathbf{i} - \mathbf{j} + \frac{5}{2}\mathbf{k} \right) - \frac{5}{2}aP\mathbf{k} = (x\mathbf{i} + y\mathbf{j}) + P\mathbf{k}$$

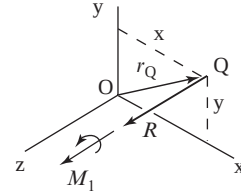
Equating coefficients

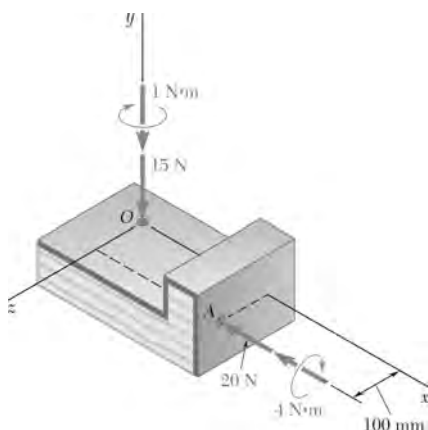
$$\mathbf{i}: -aP = yP \quad \text{or} \quad y = -a$$

$$\mathbf{j}: -aP = -xP \quad \text{or} \quad x = a$$

The axis of the wrench is parallel to the z axis and intersects the xy plane at

$$x = a, y = -a. \quad \blacktriangleleft$$





PROBLEM 3.134*

The forces and couples shown are applied to two screws as a piece of sheet metal is fastened to a block of wood. Reduce the forces and the couples to an equivalent wrench and determine (a) the resultant force $\mathbf{R}$, (b) the pitch of the wrench, (c) the point where the axis of the wrench intersects the xz plane.

SOLUTION

First, reduce the given force system to a force-couple system.

We have $\Sigma \mathbf{F}: -(20 \text{ N})\mathbf{i} - (15 \text{ N})\mathbf{j} = \mathbf{R} \quad R = 25 \text{ N}$

We have $\Sigma \mathbf{M}_O: \Sigma(\mathbf{r}_O \times \mathbf{F}) + \Sigma \mathbf{M}_C = \mathbf{M}_O^R$

$$\mathbf{M}_O^R = -20 \text{ N}(0.1 \text{ m})\mathbf{j} - (4 \text{ N} \cdot \text{m})\mathbf{i} - (1 \text{ N} \cdot \text{m})\mathbf{j}$$

$$= -(4 \text{ N} \cdot \text{m})\mathbf{i} - (3 \text{ N} \cdot \text{m})\mathbf{j}$$

(a)

$$\mathbf{R} = -(20.0 \text{ N})\mathbf{i} - (15.0 \text{ N})\mathbf{j} \quad \blacktriangleleft$$

(b) We have

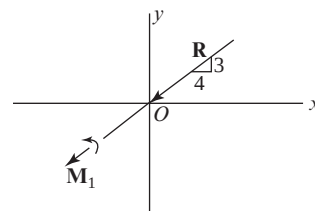
$$M_1 = \lambda_R \cdot \mathbf{M}_O^R \quad \lambda = \frac{R}{M_1}$$

$$= (-0.8\mathbf{i} - 0.6\mathbf{j}) \cdot [-(4 \text{ N} \cdot \text{m})\mathbf{i} - (3 \text{ N} \cdot \text{m})\mathbf{j}]$$

$$= 5 \text{ N} \cdot \text{m}$$

Pitch

$$p = \frac{M_1}{R} = \frac{5 \text{ N} \cdot \text{m}}{25 \text{ N}} = 0.200 \text{ m}$$



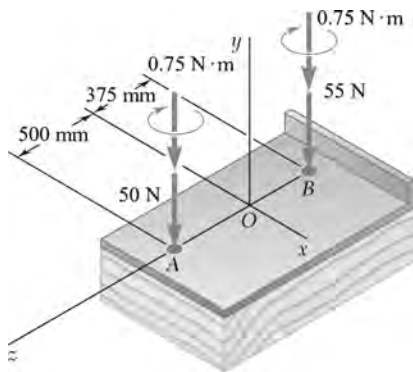
$$\text{or } p = 0.200 \text{ m} \quad \blacktriangleleft$$

(c) From above note that

$$\mathbf{M}_1 = \mathbf{M}_O^R$$

Therefore, the axis of the wrench goes through the origin. The line of action of the wrench lies in the xy plane with a slope of

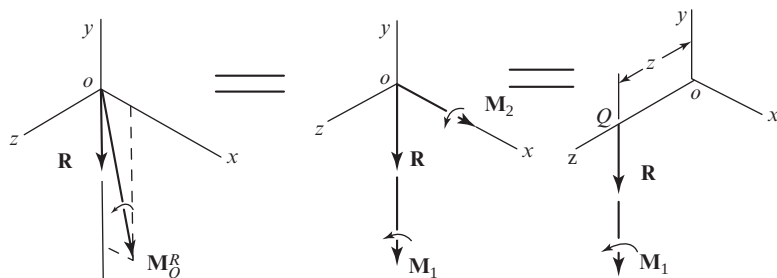
$$y = \frac{3}{4}x \quad \blacktriangleleft$$



PROBLEM 3.135*

The forces and couples shown are applied to two screws as a piece of sheet metal is fastened to a block of wood. Reduce the forces and the couples to an equivalent wrench and determine (a) the resultant force $\mathbf{R}$, (b) the pitch of the wrench, (c) the point where the axis of the wrench intersects the xz plane.

SOLUTION



First, reduce the given force system to a force-couple at the origin.

We have $\Sigma \mathbf{F}: -(50 \text{ N})\mathbf{j} - (55 \text{ N})\mathbf{j} = \mathbf{R}$

$$\mathbf{R} = -(105 \text{ N})\mathbf{j}$$

We have $\Sigma \mathbf{M}_O: \Sigma (\mathbf{r}_O \times \mathbf{F}) + \Sigma \mathbf{M}_C = \mathbf{M}_O^R$

$$\begin{aligned} \mathbf{M}_O^R &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 500 \\ 0 & -50 & 0 \end{vmatrix} \text{ N} \cdot \text{mm} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & -375 \\ 0 & -55 & 0 \end{vmatrix} \text{ N} \cdot \text{mm} - (1.5 \times 1000 \text{ N} \cdot \text{mm})\mathbf{j} \\ &= (4375 \text{ N} \cdot \text{mm})\mathbf{i} - (1500 \text{ N} \cdot \text{mm})\mathbf{j} \end{aligned}$$

$$(a) \quad \mathbf{R} = -(105 \text{ N})\mathbf{j} \quad \text{or } \mathbf{R} = -(105 \text{ N})\mathbf{j} \quad \blacktriangleleft$$

$$\begin{aligned} (b) \quad \text{We have} \quad M_1 &= \boldsymbol{\lambda}_R \cdot \mathbf{M}_O^R \quad \boldsymbol{\lambda}_R = \frac{\mathbf{R}}{R} \\ &= (-\mathbf{j}) \cdot [(4375 \text{ N} \cdot \text{mm})\mathbf{i} - (1500 \text{ N} \cdot \text{mm})\mathbf{j}] \\ &= 1500 \text{ N} \cdot \text{mm} \quad \text{and} \quad \mathbf{M}_1 = -(1500 \text{ N} \cdot \text{mm})\mathbf{j} \end{aligned}$$

$$\text{and pitch} \quad p = \frac{M_1}{R} = \frac{1500 \text{ N} \cdot \text{mm}}{105 \text{ N}} = 14.2857 \text{ mm} \quad \text{or } p = 14.29 \text{ mm} \quad \blacktriangleleft$$

PROBLEM 3.135* (Continued)

(c) We have

$$\mathbf{M}_O^R = \mathbf{M}_1 + \mathbf{M}_2$$

$$\mathbf{M}_2 = \mathbf{M}_O^R - \mathbf{M}_1 = (4375 \text{ N} \cdot \text{mm})\mathbf{i}$$

Require

$$\mathbf{M}_2 = \mathbf{r}_{Q/O} \times \mathbf{R}$$

$$(4375 \text{ N} \cdot \text{mm})\mathbf{i} = (x\mathbf{i} + z\mathbf{k}) \times [-(105 \text{ N})\mathbf{j}]$$

$$4375\mathbf{i} = -(105x)\mathbf{k} + (105z)\mathbf{i}$$

From $\mathbf{i}$:

$$4375 = 105z$$

$$z = 41.667 \text{ mm.}$$

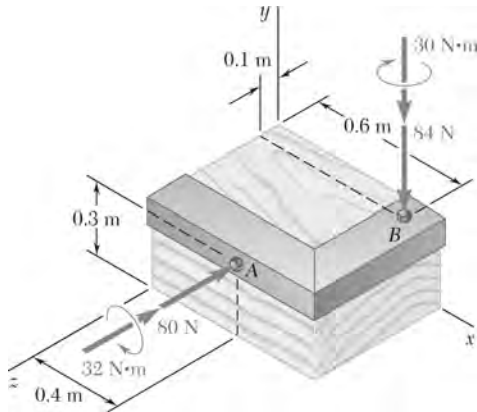
From $\mathbf{k}$:

$$0 = -105x$$

$$x = 0$$

The axis of the wrench is parallel to the y axis and intersects the xz plane at $x = 0$, $z = 41.7 \text{ mm}$

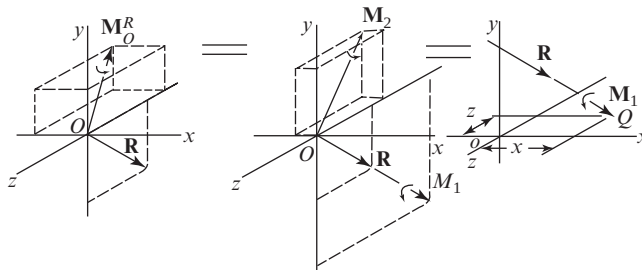




PROBLEM 3.136*

Two bolts at A and B are tightened by applying the forces and couples shown. Replace the two wrenches with a single equivalent wrench and determine (a) the resultant $\mathbf{R}$, (b) the pitch of the single equivalent wrench, (c) the point where the axis of the wrench intersects the xz plane.

SOLUTION



First, reduce the given force system to a force-couple at the origin.

We have $\Sigma \mathbf{F}: -(84 \text{ N})\mathbf{j} - (80 \text{ N})\mathbf{k} = \mathbf{R} \quad R = 116 \text{ N}$

and $\Sigma \mathbf{M}_O: \Sigma (\mathbf{r}_O \times \mathbf{F}) + \Sigma \mathbf{M}_C = \mathbf{M}_O^R$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.6 & 0 & 0.1 \\ 0 & 84 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.4 & 0.3 & 0 \\ 0 & 0 & 80 \end{vmatrix} + (-30\mathbf{j} - 32\mathbf{k}) \text{ N} \cdot \text{m} = \mathbf{M}_O^R$$

$$\mathbf{M}_O^R = -(15.6 \text{ N} \cdot \text{m})\mathbf{i} + (2 \text{ N} \cdot \text{m})\mathbf{j} - (82.4 \text{ N} \cdot \text{m})\mathbf{k}$$

(a) $\mathbf{R} = -(84.0 \text{ N})\mathbf{j} - (80.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$

(b) We have $M_1 = \lambda_R \cdot \mathbf{M}_O^R \quad \lambda_R = \frac{\mathbf{R}}{R}$

$$= -\frac{-84\mathbf{j} - 80\mathbf{k}}{116} \cdot [-(15.6 \text{ N} \cdot \text{m})\mathbf{i} + (2 \text{ N} \cdot \text{m})\mathbf{j} - (82.4 \text{ N} \cdot \text{m})\mathbf{k}]$$

$$= 55.379 \text{ N} \cdot \text{m}$$

and $\mathbf{M}_1 = M_1 \lambda_R = -(40.102 \text{ N} \cdot \text{m})\mathbf{j} - (38.192 \text{ N} \cdot \text{m})\mathbf{k}$

Then pitch $p = \frac{M_1}{R} = \frac{55.379 \text{ N} \cdot \text{m}}{116 \text{ N}} = 0.47741 \text{ m} \quad \text{or } p = 0.477 \text{ m} \quad \blacktriangleleft$

PROBLEM 3.136* (Continued)

(c) We have $\mathbf{M}_O^R = \mathbf{M}_1 + \mathbf{M}_2$
 $\mathbf{M}_2 = \mathbf{M}_O^R - \mathbf{M}_1 = [(-15.6\mathbf{i} + 2\mathbf{j} - 82.4\mathbf{k}) - (40.102\mathbf{j} - 38.192\mathbf{k})] \text{ N} \cdot \text{m}$
 $= -(15.6 \text{ N} \cdot \text{m})\mathbf{i} + (42.102 \text{ N} \cdot \text{m})\mathbf{j} - (44.208 \text{ N} \cdot \text{m})\mathbf{k}$

Require $\mathbf{M}_2 = \mathbf{r}_{Q/O} \times \mathbf{R}$
 $(-15.6\mathbf{i} + 42.102\mathbf{j} - 44.208\mathbf{k}) = (x\mathbf{i} + z\mathbf{k}) \times (84\mathbf{j} - 80\mathbf{k})$
 $= (84z)\mathbf{i} + (80x)\mathbf{j} - (84x)\mathbf{k}$

From $\mathbf{i}$: $-15.6 = 84z$
 $z = -0.185714 \text{ m}$

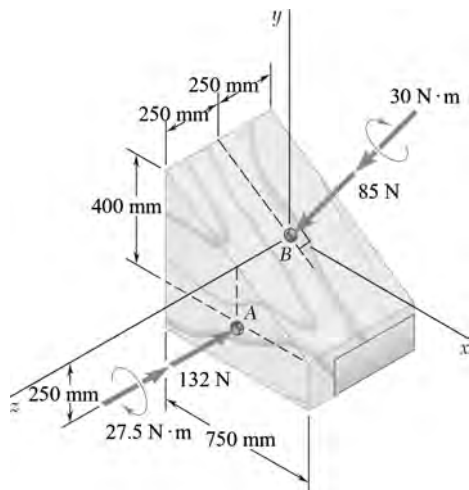
or $z = -0.1857 \text{ m}$

From $\mathbf{k}$: $-44.208 = -84x$
 $x = 0.52629 \text{ m}$

or $x = 0.526 \text{ m}$

The axis of the wrench intersects the xz plane at

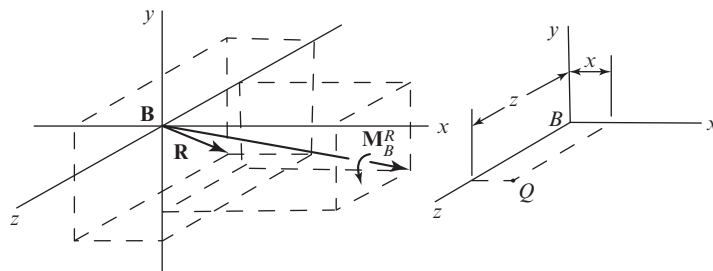
$$x = 0.526 \text{ m} \quad y = 0 \quad z = -0.1857 \text{ m} \quad \blacktriangleleft$$



PROBLEM 3.137*

Two bolts at A and B are tightened by applying the forces and couples shown. Replace the two wrenches with a single equivalent wrench and determine (a) the resultant $\mathbf{R}$, (b) the pitch of the single equivalent wrench, (c) the point where the axis of the wrench intersects the xz plane.

SOLUTION



First, reduce the given force system to a force-couple at the origin at B.

(a) We have
$$\Sigma \mathbf{F}: -(132 \text{ N})\mathbf{k} - (85 \text{ N})\left(\frac{200}{425}\mathbf{i} + \frac{375}{425}\mathbf{j}\right) = \mathbf{R}$$

$$\mathbf{R} = -(40 \text{ N})\mathbf{i} - (75 \text{ N})\mathbf{j} - (132 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

and $R = 157 \text{ N}$

We have $\Sigma \mathbf{M}_B: \mathbf{r}_{A/B} \times \mathbf{F}_A + \mathbf{M}_A + \mathbf{M}_B = \mathbf{M}_B^R$

$$\mathbf{M}_B^R = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -250 & 0 \\ 0 & 0 & -132 \end{vmatrix} - 27500\mathbf{k} - 30000\left(\frac{200}{425}\mathbf{i} + \frac{375}{425}\mathbf{j}\right)$$

$$= 33000\mathbf{i} - 27500\mathbf{k} - 14117.6\mathbf{i} - 26470.6\mathbf{j}$$

$$\mathbf{M}_B^R = (18882.4 \text{ N} \cdot \text{mm})\mathbf{i} - (26470.6 \text{ N} \cdot \text{mm})\mathbf{j} - (27500 \text{ N} \cdot \text{mm})\mathbf{k}$$

$$= \mathbf{M}_B^R = (18.88 \text{ N} \cdot \text{m})\mathbf{i} - (26.5 \text{ N} \cdot \text{m})\mathbf{j} - (27.5 \text{ N} \cdot \text{m})\mathbf{k}$$

PROBLEM 3.137* (Continued)

(b) We have $M_1 = \lambda_R \cdot \mathbf{M}_O^R \quad \lambda_R = \frac{\mathbf{R}}{R}$

$$= \frac{-40\mathbf{i} - 75\mathbf{j} - 132\mathbf{k}}{157} \cdot [(18.88 \text{ N} \cdot \text{m})\mathbf{i} - (26.5 \text{ N} \cdot \text{m})\mathbf{j} - (27.5 \text{ N} \cdot \text{m})\mathbf{k}]$$

$$= -4.81 + 12.66 + 23.1 = 30.95 \text{ N} \cdot \text{m}$$

and $\mathbf{M}_1 = M_1 \lambda_R = -(7.885 \text{ N} \cdot \text{m})\mathbf{i} - (14.785 \text{ N} \cdot \text{m})\mathbf{j} - (26.021 \text{ N} \cdot \text{m})\mathbf{k}$

Then pitch $p = \frac{M_1}{R} = \frac{30.95 \text{ N} \cdot \text{m}}{157 \text{ N}} \quad \text{or } p = 197.1 \text{ mm} \quad \blacktriangleleft$

(c) We have $\mathbf{M}_B^R = \mathbf{M}_1 + \mathbf{M}_2$

$$\mathbf{M}_2 = \mathbf{M}_B^R - \mathbf{M}_1 = (18.88\mathbf{i} - 26.5\mathbf{j} - 27.5\mathbf{k}) - (-7.885\mathbf{i} - 14.785\mathbf{j} - 26.021\mathbf{k})$$

$$= (26.765 \text{ N} \cdot \text{m})\mathbf{i} - (11.715 \text{ N} \cdot \text{m})\mathbf{j} - (1.479 \text{ N} \cdot \text{m})\mathbf{k}$$

Require $\mathbf{M}_2 = \mathbf{r}_{Q/B} \times \mathbf{R}$

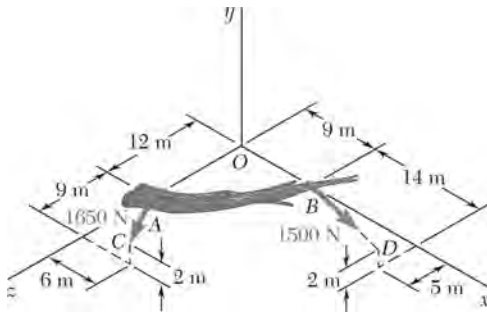
$$26.765\mathbf{i} - 11.715\mathbf{j} - 1.479\mathbf{k} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & 0 & z \\ -40 & -75 & -132 \end{vmatrix}$$

$$= (75z)\mathbf{i} - (40z)\mathbf{j} + (132x)\mathbf{j} - (75x)\mathbf{k}$$

From $\mathbf{i}$: $26.765 = 75z \quad z = 0.357 \text{ m} = 357 \text{ mm}$

From $\mathbf{k}$: $-1.479 = -75x \quad x = 0.01972 \text{ m} = 19.72 \text{ mm}$

The axis of the wrench intersects the xz plane at $x = 19.72 \text{ mm} \quad y = 0 \quad z = 357 \text{ mm} \quad \blacktriangleleft$



PROBLEM 3.138*

Two ropes attached at A and B are used to move the trunk of a fallen tree. Replace the forces exerted by the ropes with an equivalent wrench and determine (a) the resultant force $\mathbf{R}$, (b) the pitch of the wrench, (c) the point where the axis of the wrench intersects the yz plane.

SOLUTION

(a) First replace the given forces with an equivalent force-couple system $(\mathbf{R}, \mathbf{M}_O^R)$ at the origin.

We have

$$d_{AC} = \sqrt{(6)^2 + (2)^2 + (9)^2} = 11 \text{ m}$$

$$d_{BD} = \sqrt{(14)^2 + (2)^2 + (5)^2} = 15 \text{ m}$$

Then

$$\begin{aligned} T_{AC} &= \frac{1650 \text{ N}}{11} = (6\mathbf{i} + 2\mathbf{j} + 9\mathbf{k}) \\ &= (900 \text{ N})\mathbf{i} + (300 \text{ N})\mathbf{j} + (1350 \text{ N})\mathbf{k} \end{aligned}$$

and

$$\begin{aligned} T_{BD} &= \frac{1500 \text{ N}}{15} = (14\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}) \\ &= (1400 \text{ N})\mathbf{i} + (200 \text{ N})\mathbf{j} + (500 \text{ N})\mathbf{k} \end{aligned}$$

Equivalence then requires

$$\begin{aligned} \Sigma \mathbf{F}: \quad \mathbf{R} &= \mathbf{T}_{AC} + \mathbf{T}_{BD} \\ &= (900\mathbf{i} + 300\mathbf{j} + 1350\mathbf{k}) \\ &\quad + (1400\mathbf{i} + 200\mathbf{j} + 500\mathbf{k}) \\ &= (2300 \text{ N})\mathbf{i} + (500 \text{ N})\mathbf{j} + (1850 \text{ N})\mathbf{k} \end{aligned}$$

$$\begin{aligned} \Sigma \mathbf{M}_O: \quad \mathbf{M}_O^R &= \mathbf{r}_A \times \mathbf{T}_{AC} + \mathbf{r}_B \times \mathbf{T}_{BD} \\ &= (12 \text{ m})\mathbf{k} \times [(900 \text{ N})\mathbf{i} + (300 \text{ N})\mathbf{j} + (1350 \text{ N})\mathbf{k}] \\ &\quad + (9 \text{ m})\mathbf{i} \times [(1400 \text{ N})\mathbf{i} + (200 \text{ N})\mathbf{j} + (500 \text{ N})\mathbf{k}] \\ &= -(3600)\mathbf{i} + (10800 - 4500)\mathbf{j} + (1800)\mathbf{k} \\ &= -(3600 \text{ N} \cdot \text{m})\mathbf{i} + (6300 \text{ N} \cdot \text{m})\mathbf{j} + (1800 \text{ N} \cdot \text{m})\mathbf{k} \end{aligned}$$

The components of the wrench are $(\mathbf{R}, \mathbf{M}_1)$, where

$$\mathbf{R} = (2300 \text{ N})\mathbf{i} + (500 \text{ N})\mathbf{j} + (1850 \text{ N})\mathbf{k}$$



PROBLEM 3.138* (Continued)

(b) We have

$$R = 100 \sqrt{(23)^2 + (5)^2 + (18.5)^2} = 2993.7 \text{ N}$$

Let

$$\lambda_{\text{axis}} = \frac{\mathbf{R}}{R} = \frac{1}{29.937} (23\mathbf{i} + 5\mathbf{j} + 18.5\mathbf{k})$$

Then

$$\begin{aligned} M_1 &= \lambda_{\text{axis}} \cdot \mathbf{M}_O^R \\ &= \frac{1}{29.937} (23\mathbf{i} + 5\mathbf{j} + 18.5\mathbf{k}) \cdot (-3600\mathbf{i} + 6300\mathbf{j} + 1800\mathbf{k}) \\ &= \frac{1}{0.29937} [(23)(-36) + (5)(63) + (18.5)(18)] \\ &= -601.26 \text{ N} \cdot \text{m} \end{aligned}$$

Finally

$$P = \frac{M_1}{R} = \frac{-601.26 \text{ N} \cdot \text{m}}{2993.7 \text{ N}}$$

$$\text{or } P = -0.201 \text{ m} \quad \blacktriangleleft$$

(c) We have

$$\begin{aligned} M_1 &= M_1 \lambda_{\text{axis}} \\ &= (-601.26 \text{ N} \cdot \text{m}) \times \frac{1}{29.937} (23\mathbf{i} + 5\mathbf{j} + 18.5\mathbf{k}) \end{aligned}$$

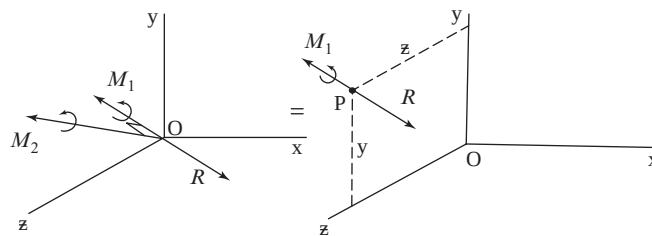
or

$$\mathbf{M}_1 = -(461.93 \text{ N} \cdot \text{m})\mathbf{i} - (100.421 \text{ N} \cdot \text{m})\mathbf{j} - (371.56 \text{ N} \cdot \text{m})\mathbf{k}$$

Now

$$\begin{aligned} \mathbf{M}_2 &= \mathbf{M}_O^R - \mathbf{M}_1 \\ &= (-3600\mathbf{i} + 6300\mathbf{j} + 1800\mathbf{k}) \\ &\quad - (-461.93\mathbf{i} - 100.421\mathbf{j} - 371.56\mathbf{k}) \\ &= -(3138.1 \text{ N} \cdot \text{m})\mathbf{i} + (6400.4 \text{ N} \cdot \text{m})\mathbf{j} + (2171.6 \text{ N} \cdot \text{m})\mathbf{k} \end{aligned}$$

For equivalence



PROBLEM 3.138* (Continued)

Thus require

$$\mathbf{M}_2 = \mathbf{r}_P \times \mathbf{R} \quad \mathbf{r} = (y\mathbf{j} + z\mathbf{k})$$

Substituting

$$-3138.1\mathbf{i} + 6400.4\mathbf{j} + 2171.6\mathbf{k} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & y & z \\ 2300 & 500 & 1850 \end{vmatrix}$$

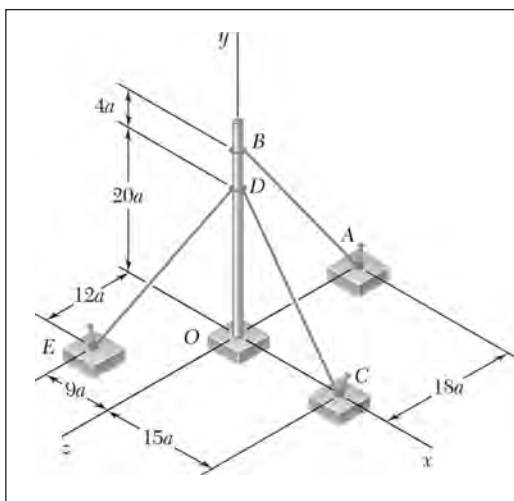
Equating coefficients

$$\mathbf{j}: 6400.4 = 2300z \quad \text{or} \quad z = 2.78 \text{ m}$$

$$\mathbf{k}: 2171.6 = -2300y \quad \text{or} \quad y = -0.944 \text{ m}$$

The axis of the wrench intersects the yz plane at $y = -0.944 \text{ m}$ $z = 2.78 \text{ m}$

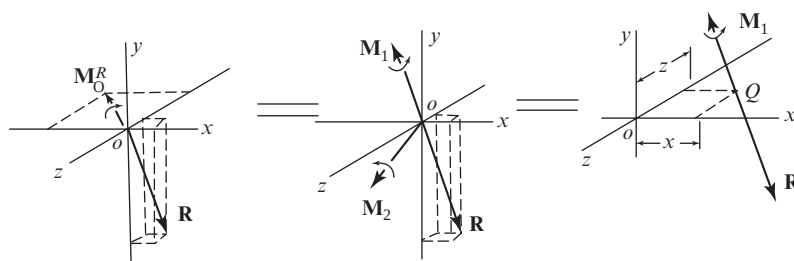




PROBLEM 3.139*

A flagpole is guyed by three cables. If the tensions in the cables have the same magnitude P , replace the forces exerted on the pole with an equivalent wrench and determine (a) the resultant force $\mathbf{R}$, (b) the pitch of the wrench, (c) the point where the axis of the wrench intersects the xz plane.

SOLUTION



- (a) First reduce the given force system to a force-couple at the origin.

We have

$$\Sigma \mathbf{F}: P\lambda_{BA} + P\lambda_{DC} + P\lambda_{DE} = \mathbf{R}$$

$$\mathbf{R} = P \left[\left(\frac{4}{5} \mathbf{j} - \frac{3}{5} \mathbf{k} \right) + \left(\frac{3}{5} \mathbf{i} - \frac{4}{5} \mathbf{j} \right) + \left(\frac{-9}{25} \mathbf{i} - \frac{4}{5} \mathbf{j} + \frac{12}{25} \mathbf{k} \right) \right]$$

$$\mathbf{R} = \frac{3P}{25} (2\mathbf{i} - 20\mathbf{j} - \mathbf{k}) \quad \blacktriangleleft$$

$$R = \frac{3P}{25} \sqrt{(2)^2 + (20)^2 + (1)^2} = \frac{27\sqrt{5}}{25} P$$

We have

$$\Sigma \mathbf{M}: \Sigma (\mathbf{r}_O \times P) = \mathbf{M}_O^R$$

$$(24a)\mathbf{j} \times \left(\frac{-4P}{5} \mathbf{j} - \frac{3P}{5} \mathbf{k} \right) + (20a)\mathbf{j} \times \left(\frac{3P}{5} \mathbf{i} - \frac{4P}{5} \mathbf{j} \right) + (20a)\mathbf{j} \times \left(\frac{-9P}{25} \mathbf{i} - \frac{4P}{5} \mathbf{j} + \frac{12P}{25} \mathbf{k} \right) = \mathbf{M}_O^R$$

$$\mathbf{M}_O^R = \frac{24Pa}{5} (-\mathbf{i} - \mathbf{k})$$

PROBLEM 3.139* (Continued)

(b) We have $M_1 = \lambda_R \cdot \mathbf{M}_O^R$

where $\lambda_R = \frac{\mathbf{R}}{R} = \frac{3P}{25} (2\mathbf{i} - 20\mathbf{j} - \mathbf{k}) \frac{25}{27\sqrt{5}P} = \frac{1}{9\sqrt{5}} (2\mathbf{i} - 20\mathbf{j} - \mathbf{k})$

Then $M_1 = \frac{1}{9\sqrt{5}} (2\mathbf{i} - 20\mathbf{j} - \mathbf{k}) \cdot \frac{24Pa}{5} (-\mathbf{i} - \mathbf{k}) = \frac{-8Pa}{15\sqrt{5}}$

and pitch $p = \frac{M_1}{R} = \frac{-8Pa}{15\sqrt{5}} \left(\frac{25}{27\sqrt{5}P} \right) = \frac{-8a}{81}$ or $p = -0.0988a$ ◀

(c) $\mathbf{M}_1 = M_1 \lambda_R = \frac{-8Pa}{15\sqrt{5}} \left(\frac{1}{9\sqrt{5}} \right) (2\mathbf{i} - 20\mathbf{j} - \mathbf{k}) = \frac{8Pa}{675} (-2\mathbf{i} + 20\mathbf{j} + \mathbf{k})$

Then $\mathbf{M}_2 = \mathbf{M}_O^R - \mathbf{M}_1 = \frac{24Pa}{5} (-\mathbf{i} - \mathbf{k}) - \frac{8Pa}{675} (-2\mathbf{i} + 20\mathbf{j} + \mathbf{k}) = \frac{8Pa}{675} (-430\mathbf{i} - 20\mathbf{j} - 406\mathbf{k})$

Require $\mathbf{M}_2 = \mathbf{r}_{Q/O} \times \mathbf{R}$

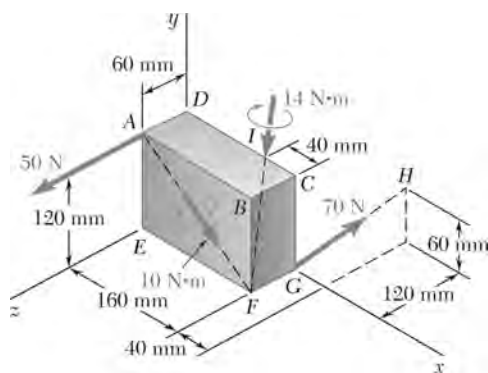
$$\begin{aligned} \left(\frac{8Pa}{675} \right) (-403\mathbf{i} - 20\mathbf{j} - 406\mathbf{k}) &= (x\mathbf{i} + z\mathbf{k}) \times \left(\frac{3P}{25} \right) (2\mathbf{i} - 20\mathbf{j} - \mathbf{k}) \\ &= \left(\frac{3P}{25} \right) [20z\mathbf{i} + (x + 2z)\mathbf{j} - 20x\mathbf{k}] \end{aligned}$$

From $\mathbf{i}$: $8(-403) \frac{Pa}{675} = 20z \left(\frac{3P}{25} \right) \quad z = -1.99012a$

From $\mathbf{k}$: $8(-406) \frac{Pa}{675} = -20x \left(\frac{3P}{25} \right) \quad x = 2.0049a$

The axis of the wrench intersects the xz plane at

$x = 2.00a, z = -1.990a$ ◀



PROBLEM 3.140*

Determine whether the force-and-couple system shown can be reduced to a single equivalent force $\mathbf{R}$. If it can, determine $\mathbf{R}$ and the point where the line of action of $\mathbf{R}$ intersects the yz plane. If it cannot be so reduced, replace the given system with an equivalent wrench and determine its resultant, its pitch, and the point where its axis intersects the yz plane.

SOLUTION

First, reduce the given force system to a force-couple at the origin.

We have $\Sigma \mathbf{F}: \mathbf{F}_A + \mathbf{F}_G = \mathbf{R}$

$$\begin{aligned}\mathbf{R} &= (50 \text{ N})\mathbf{k} + 70 \text{ N} \left[\frac{(40 \text{ mm})\mathbf{i} + (60 \text{ mm})\mathbf{j} - (120 \text{ mm})\mathbf{k}}{140 \text{ mm}} \right] \\ &= (20 \text{ N})\mathbf{i} + (30 \text{ N})\mathbf{j} - (10 \text{ N})\mathbf{k}\end{aligned}$$

and $R = 37.417 \text{ N}$

We have $\Sigma \mathbf{M}_O: \Sigma (\mathbf{r}_O \times \mathbf{F}) + \Sigma \mathbf{M}_C = \mathbf{M}_O^R$

$$\begin{aligned}\mathbf{M}_O^R &= [(0.12 \text{ m})\mathbf{j} \times (50 \text{ N})\mathbf{k}] + \{ (0.16 \text{ m})\mathbf{i} \times [(20 \text{ N})\mathbf{i} + (30 \text{ N})\mathbf{j} - (10 \text{ N})\mathbf{k}] \} \\ &\quad + (10 \text{ N} \cdot \text{m}) \left[\frac{(160 \text{ mm})\mathbf{i} - (120 \text{ mm})\mathbf{j}}{200 \text{ mm}} \right] \\ &\quad + (14 \text{ N} \cdot \text{m}) \left[\frac{(40 \text{ mm})\mathbf{i} - (120 \text{ mm})\mathbf{j} + (60 \text{ mm})\mathbf{k}}{140 \text{ mm}} \right] \\ \mathbf{M}_O^R &= (18 \text{ N} \cdot \text{m})\mathbf{i} - (8.4 \text{ N} \cdot \text{m})\mathbf{j} + (10.8 \text{ N} \cdot \text{m})\mathbf{k}\end{aligned}$$

To be able to reduce the original forces and couples to a single equivalent force, $\mathbf{R}$ and $\mathbf{M}$ must be perpendicular. Thus, $\mathbf{R} \cdot \mathbf{M} = 0$.

Substituting

$$(20\mathbf{i} + 30\mathbf{j} - 10\mathbf{k}) \cdot (18\mathbf{i} - 8.4\mathbf{j} + 10.8\mathbf{k}) \stackrel{?}{=} 0$$

or

$$(20)(18) + (30)(-8.4) + (-10)(10.8) \stackrel{?}{=} 0$$

or

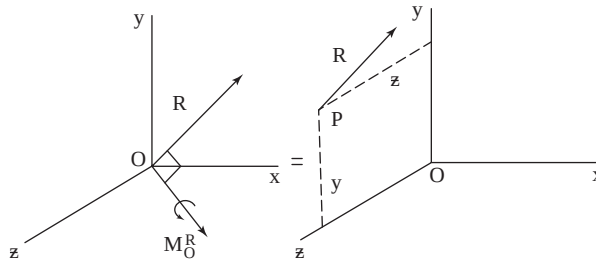
$$0 \stackrel{?}{=} 0$$

$\mathbf{R}$ and $\mathbf{M}$ are perpendicular so that the given system can be reduced to the single equivalent force

$$\mathbf{R} = (20.0 \text{ N})\mathbf{i} + (30.0 \text{ N})\mathbf{j} - (10.00 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

PROBLEM 3.140* (Continued)

Then for equivalence



Thus require

$$\mathbf{M}_O^R = \mathbf{r}_p \times \mathbf{R} \quad \mathbf{r}_p = y\mathbf{j} + z\mathbf{k}$$

Substituting

$$18\mathbf{i} - 8.4\mathbf{j} + 10.8\mathbf{k} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & y & z \\ 20 & 30 & -10 \end{vmatrix}$$

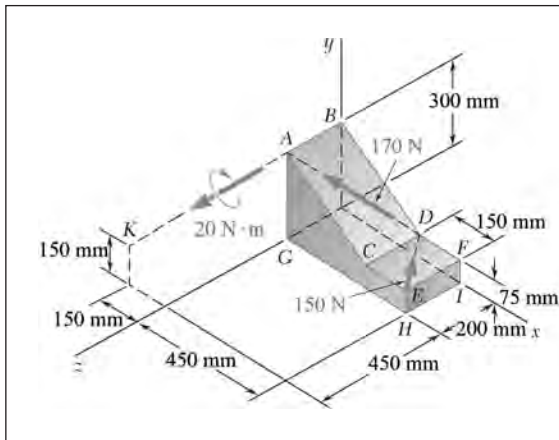
Equating coefficients

$$\mathbf{j}: -8.4 = 20z \quad \text{or} \quad z = -0.42 \text{ m}$$

$$\mathbf{k}: 10.8 = -20y \quad \text{or} \quad y = -0.54 \text{ m}$$

The line of action of $\mathbf{R}$ intersects the yz plane at $x = 0 \quad y = -0.540 \text{ m} \quad z = -0.420 \text{ m}$





PROBLEM 3.141*

Determine whether the force-and-couple system shown can be reduced to a single equivalent force $\mathbf{R}$. If it can, determine $\mathbf{R}$ and the point where the line of action of $\mathbf{R}$ intersects the yz plane. If it cannot be so reduced, replace the given system with an equivalent wrench and determine its resultant, its pitch, and the point where its axis intersects the yz plane.

SOLUTION

First determine the resultant of the forces at D . We have

$$d_{DA} = \sqrt{-(300)^2 + (225)^2 + (200)^2} = 425 \text{ mm}$$

$$d_{ED} = \sqrt{(-150)^2 + (0)^2 + (-200)^2} = 250 \text{ mm}$$

Then

$$\begin{aligned} \mathbf{F}_{DA} &= \frac{170 \text{ N}}{425} = [(-300)\mathbf{i} + (225)\mathbf{j} + (200)\mathbf{k}] \\ &= (-120 \text{ N})\mathbf{i} + (90 \text{ N})\mathbf{j} + (80 \text{ N})\mathbf{k} \end{aligned}$$

and

$$\begin{aligned} \mathbf{F}_{ED} &= \frac{150 \text{ N}}{250} = (-150\mathbf{i} - 200\mathbf{k}) \\ &= (-90 \text{ N})\mathbf{i} - (120 \text{ N})\mathbf{k} \end{aligned}$$

Then

$$\Sigma \mathbf{F}: \mathbf{R} = \mathbf{F}_{DA} + \mathbf{F}_{ED} = -(210 \text{ N})\mathbf{i} + (90 \text{ N})\mathbf{j} - (40 \text{ N})\mathbf{k}$$

For the applied couple

$$d_{AK} = \sqrt{(-150)^2 + (-150)^2 + (450)^2} = 150\sqrt{11} \text{ mm}$$

Then

$$\begin{aligned} \mathbf{M} &= \frac{20000 \text{ N} \cdot \text{mm}}{150\sqrt{11}} (-150\mathbf{i} - 150\mathbf{j} + 450\mathbf{k}) \\ &= \frac{20000}{\sqrt{11}} (-\mathbf{i} - \mathbf{j} + 3\mathbf{k}) \text{ N} \cdot \text{mm} \end{aligned}$$

To be able to reduce the original forces and couple to a single equivalent force, $\mathbf{R}$ and $\mathbf{M}$ must be perpendicular. Thus

$$\mathbf{R} \cdot \mathbf{M} \stackrel{?}{=} 0$$

PROBLEM 3.141* (Continued)

Substituting

$$(-210\mathbf{i} + 90\mathbf{j} - 40\mathbf{k}) \cdot \frac{20000}{\sqrt{11}}(-\mathbf{i} - \mathbf{j} + 3\mathbf{k}) \stackrel{?}{=} 0$$

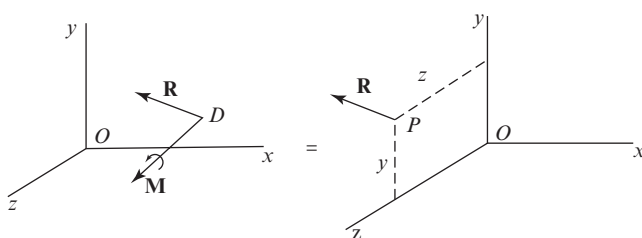
$$\text{or} \quad \frac{20000}{\sqrt{11}}[(-210)(-1) + (90)(-1) + (-40)(3)] \stackrel{?}{=} 0$$

$$\text{or} \quad 0 \stackrel{?}{=} 0$$

R and **M** are perpendicular so that the given system can be reduced to the single equivalent force

$$\mathbf{R} = -(210 \text{ N})\mathbf{i} + (90 \text{ N})\mathbf{j} - (40 \text{ N})\mathbf{k}$$

Then for equivalence



Thus require

$$\mathbf{M} = \mathbf{r}_{P/D} \times \mathbf{R}$$

where

$$\mathbf{r}_{P/D} = -(300 \text{ mm})\mathbf{i} + [(y - 75)\text{mm}]\mathbf{j} + (z \text{ mm})\mathbf{k}$$

Substituting

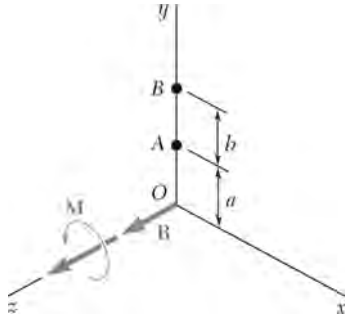
$$\begin{aligned} \frac{20000}{\sqrt{11}}(-\mathbf{i} - \mathbf{j} + 3\mathbf{k}) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -300 & (y - 75) & z \\ -210 & 90 & -40 \end{vmatrix} \\ &= [(y - 75)(-40) - z(90)]\mathbf{i} \\ &\quad + [(z)(-210) - (-300)(-40)]\mathbf{j} \\ &\quad + [(-300)(90) - (y - 75)(-210)]\mathbf{k} \end{aligned}$$

Equating coefficients

$$\mathbf{j}: -\frac{20000}{\sqrt{11}} = -210z - 12000 \quad \text{or} \quad z = -28.4 \text{ mm}$$

$$\mathbf{k}: \frac{60000}{\sqrt{11}} = -27000 + 210(y - 75) \quad \text{or} \quad y = 290 \text{ mm}$$

The line of action of **R** intersects the yz plane at $x = 0$ $y = 290 \text{ mm}$ $z = -28.4 \text{ mm}$



PROBLEM 3.142*

Replace the wrench shown with an equivalent system consisting of two forces perpendicular to the y axis and applied respectively at A and B .

SOLUTION

Express the forces at A and B as

$$\mathbf{A} = A_x \mathbf{i} + A_z \mathbf{k}$$

$$\mathbf{B} = B_x \mathbf{i} + B_z \mathbf{k}$$

Then, for equivalence to the given force system

$$\Sigma F_x: A_x + B_x = 0 \quad (1)$$

$$\Sigma F_z: A_z + B_z = R \quad (2)$$

$$\Sigma M_x: A_z(a) + B_z(a+b) = 0 \quad (3)$$

$$\Sigma M_z: -A_x(a) - B_x(a+b) = M \quad (4)$$

From Equation (1),

$$B_x = -A_x$$

Substitute into Equation (4)

$$-A_x(a) + A_x(a+b) = M$$

$$A_x = \frac{M}{b} \quad \text{and} \quad B_x = -\frac{M}{b}$$

From Equation (2),

$$B_z = R - A_z$$

and Equation (3),

$$A_z a + (R - A_z)(a+b) = 0$$

$$A_z = R \left(1 + \frac{a}{b} \right)$$

and

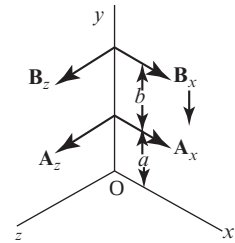
$$B_z = R - R \left(1 + \frac{a}{b} \right)$$

$$B_z = -\frac{a}{b} R$$

Then

$$\mathbf{A} = \left(\frac{M}{b} \right) \mathbf{i} + R \left(1 + \frac{a}{b} \right) \mathbf{k} \quad \blacktriangleleft$$

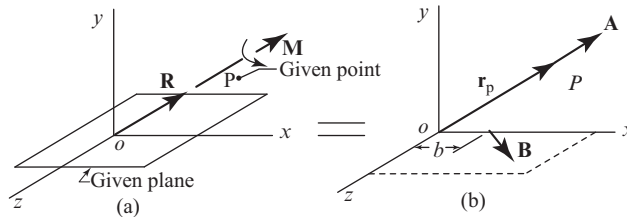
$$\mathbf{B} = -\left(\frac{M}{b} \right) \mathbf{i} - \left(\frac{a}{b} R \right) \mathbf{k} \quad \blacktriangleleft$$



PROBLEM 3.143*

Show that, in general, a wrench can be replaced with two forces chosen in such a way that one force passes through a given point while the other force lies in a given plane.

SOLUTION



First, choose a coordinate system so that the xy plane coincides with the given plane. Also, position the coordinate system so that the line of action of the wrench passes through the origin as shown in Figure *a*. Since the orientation of the plane and the components $(\mathbf{R}, \mathbf{M})$ of the wrench are known, it follows that the scalar components of $\mathbf{R}$ and $\mathbf{M}$ are known relative to the shown coordinate system.

A force system to be shown as equivalent is illustrated in Figure *b*. Let $\mathbf{A}$ be the force passing through the given Point P and $\mathbf{B}$ be the force that lies in the given plane. Let b be the x -axis intercept of $\mathbf{B}$.

The known components of the wrench can be expressed as

$$\mathbf{R} = R_x \mathbf{i} + R_y \mathbf{j} + R_z \mathbf{k} \quad \text{and} \quad \mathbf{M} = M_x \mathbf{i} + M_y \mathbf{j} + M_z \mathbf{k}$$

while the unknown forces $\mathbf{A}$ and $\mathbf{B}$ can be expressed as

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k} \quad \text{and} \quad \mathbf{B} = B_x \mathbf{i} + B_z \mathbf{k}$$

Since the position vector of Point P is given, it follows that the scalar components (x, y, z) of the position vector $\mathbf{r}_p$ are also known.

Then, for equivalence of the two systems

$$\Sigma F_x: R_x = A_x + B_x \quad (1)$$

$$\Sigma F_y: R_y = A_y \quad (2)$$

$$\Sigma F_z: R_z = A_z + B_z \quad (3)$$

$$\Sigma M_x: M_x = yA_z - zA_y \quad (4)$$

$$\Sigma M_y: M_y = zA_x - xA_z - bB_z \quad (5)$$

$$\Sigma M_z: M_z = xA_y - yA_x \quad (6)$$

Based on the above six independent equations for the six unknowns $(A_x, A_y, A_z, B_x, B_z, b)$, there exists a unique solution for $\mathbf{A}$ and $\mathbf{B}$.

From Equation (2)

$$A_y = R_y \quad \blacktriangleleft$$

PROBLEM 3.143* (Continued)

Equation (6)

$$A_x = \left(\frac{1}{y}\right)(xR_y - M_z) \quad \blacktriangleleft$$

Equation (1)

$$B_x = R_x - \left(\frac{1}{y}\right)(xR_y - M_z) \quad \blacktriangleleft$$

Equation (4)

$$A_z = \left(\frac{1}{y}\right)(M_x + zR_y) \quad \blacktriangleleft$$

Equation (3)

$$B_z = R_z - \left(\frac{1}{y}\right)(M_x + zR_y) \quad \blacktriangleleft$$

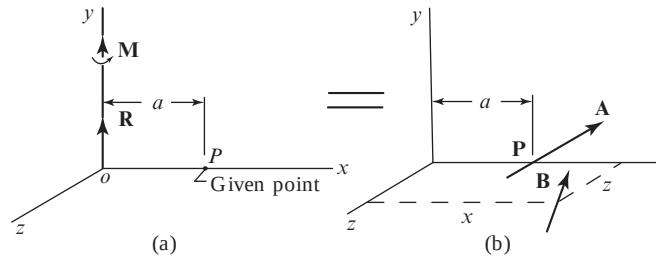
Equation (5)

$$b = \frac{(xM_x + yM_y + zM_z)}{(M_x - yR_z + zR_y)} \quad \blacktriangleleft$$

PROBLEM 3.144*

Show that a wrench can be replaced with two perpendicular forces, one of which is applied at a given point.

SOLUTION



First, observe that it is always possible to construct a line perpendicular to a given line so that the constructed line also passes through a given point. Thus, it is possible to align one of the coordinate axes of a rectangular coordinate system with the axis of the wrench while one of the other axes passes through the given point.

See Figures *a* and *b*.

We have $\mathbf{R} = R\mathbf{j}$ and $\mathbf{M} = M\mathbf{j}$ and are known.

The unknown forces $\mathbf{A}$ and $\mathbf{B}$ can be expressed as

$$\mathbf{A} = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k} \quad \text{and} \quad \mathbf{B} = B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k}$$

The distance a is known. It is assumed that force $\mathbf{B}$ intersects the xz plane at $(x, 0, z)$. Then for equivalence

$$\Sigma F_x: 0 = A_x + B_x \quad (1)$$

$$\Sigma F_y: R = A_y + B_y \quad (2)$$

$$\Sigma F_z: 0 = A_z + B_z \quad (3)$$

$$\Sigma M_x: 0 = -zB_y \quad (4)$$

$$\Sigma M_y: M = -aA_z - xB_z + zB_x \quad (5)$$

$$\Sigma M_z: 0 = aA_y + zB_y \quad (6)$$

Since $\mathbf{A}$ and $\mathbf{B}$ are made perpendicular,

$$\mathbf{A} \cdot \mathbf{B} = 0 \quad \text{or} \quad A_xB_x + A_yB_y + A_zB_z = 0 \quad (7)$$

There are eight unknowns: $A_x, A_y, A_z, B_x, B_y, B_z, x, z$

But only seven independent equations. Therefore, *there exists an infinite number of solutions.*

Next consider Equation (4): $0 = -zB_y$

If $B_y = 0$, Equation (7) becomes $A_xB_x + A_zB_z = 0$

Using Equations (1) and (3) this equation becomes $A_x^2 + A_z^2 = 0$

PROBLEM 3.144* (Continued)

Since the components of $\mathbf{A}$ must be real, a nontrivial solution is not possible. Thus, it is required that $B_y \neq 0$, so that from Equation (4), $z = 0$.

To obtain one possible solution, arbitrarily let $A_x = 0$.

(Note: Setting A_y , A_z , or B_z equal to zero results in unacceptable solutions.)

The defining equations then become

$$0 = B_x \quad (1)'$$

$$R = A_y + B_y \quad (2)$$

$$0 = A_z + B_z \quad (3)$$

$$M = -aA_z - xB_z \quad (5)'$$

$$0 = aA_y + xB_y \quad (6)$$

$$A_y B_y + A_z B_z = 0 \quad (7)'$$

Then Equation (2) can be written $A_y = R - B_y$

Equation (3) can be written $B_z = -A_z$

Equation (6) can be written $x = -\frac{aA_y}{B_y}$

Substituting into Equation (5)',

$$M = -aA_z - \left(-a\frac{R - B_y}{B_y}\right)(-A_z)$$

or

$$A_z = -\frac{M}{aR} B_y \quad (8)$$

Substituting into Equation (7)',

$$(R - B_y)B_y + \left(-\frac{M}{aR} B_y\right)\left(\frac{M}{aR} B_y\right) = 0$$

or

$$B_y = \frac{a^2 R^3}{a^2 R^2 + M^2}$$

Then from Equations (2), (8), and (3)

$$A_y = R - \frac{a^2 R^2}{a^2 R^2 + M^2} = \frac{RM^2}{a^2 R^2 + M^2}$$

$$A_z = -\frac{M}{aR} \left(\frac{a^2 R^3}{a^2 R^2 + M^2} \right) = -\frac{aR^2 M}{a^2 R^2 + M^2}$$

$$B_z = \frac{aR^2 M}{a^2 R^2 + M^2}$$

PROBLEM 3.144* (Continued)

In summary

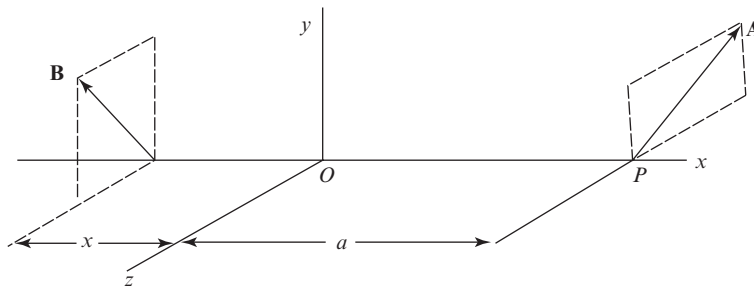
$$\mathbf{A} = \frac{RM}{a^2 R^2 + M^2} (M\mathbf{j} - aR\mathbf{k}) \quad \blacktriangleleft$$

$$\mathbf{B} = \frac{aR^2}{a^2 R^2 + M^2} (aR\mathbf{j} + M\mathbf{k}) \quad \blacktriangleleft$$

Which shows that it is possible to replace a wrench with two perpendicular forces, one of which is applied at a given point.

Lastly, if $R > 0$ and $M > 0$, it follows from the equations found for $\mathbf{A}$ and $\mathbf{B}$ that $A_y > 0$ and $B_y > 0$.

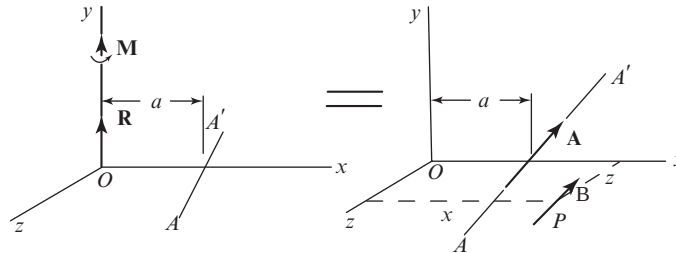
From Equation (6), $x < 0$ (assuming $a > 0$). Then, as a consequence of letting $A_x = 0$, force $\mathbf{A}$ lies in a plane parallel to the yz plane and to the right of the origin, while force $\mathbf{B}$ lies in a plane parallel to the yz plane but to the left of the origin, as shown in the figure below.



PROBLEM 3.145*

Show that a wrench can be replaced with two forces, one of which has a prescribed line of action.

SOLUTION



First, choose a rectangular coordinate system where one axis coincides with the axis of the wrench and another axis intersects the prescribed line of action (AA'). Note that it has been assumed that the line of action of force **B** intersects the xz plane at Point $P(x, 0, z)$. Denoting the known direction of line AA' by

$$\lambda_A = \lambda_x \mathbf{i} + \lambda_y \mathbf{j} + \lambda_z \mathbf{k}$$

it follows that force **A** can be expressed as

$$\mathbf{A} = A\lambda_A = A(\lambda_x \mathbf{i} + \lambda_y \mathbf{j} + \lambda_z \mathbf{k})$$

Force **B** can be expressed as

$$\mathbf{B} = B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}$$

Next, observe that since the axis of the wrench and the prescribed line of action AA' are known, it follows that the distance a can be determined. In the following solution, it is assumed that a is known.

Then, for equivalence

$$\Sigma F_x: 0 = A\lambda_x + B_x \quad (1)$$

$$\Sigma F_y: R = A\lambda_y + B_y \quad (2)$$

$$\Sigma F_z: 0 = A\lambda_z + B_z \quad (3)$$

$$\Sigma M_x: 0 = -zB_y \quad (4)$$

$$\Sigma M_y: M = -aA\lambda_z + zB_x - xB_z \quad (5)$$

$$\Sigma M_x: 0 = -aA\lambda_y + xB_y \quad (6)$$

Since there are six unknowns (A, B_x, B_y, B_z, x, z) and six independent equations, it will be possible to obtain a solution.

PROBLEM 3.145* (Continued)

Case 1: Let $z = 0$ to satisfy Equation (4)

Now Equation (2) $A\lambda_y = R - B_y$

Equation (3) $B_z = -A\lambda_z$

Equation (6) $x = -\frac{aA\lambda_y}{B_y} = -\left(\frac{a}{B_y}\right)(R - B_y)$

Substitution into Equation (5)

$$M = -aA\lambda_z - \left[-\left(\frac{a}{B_y}\right)(R - B_y)(-A\lambda_z) \right]$$

$$A = -\frac{1}{\lambda_z} \left(\frac{M}{aR} \right) B_y$$

Substitution into Equation (2)

$$R = -\frac{1}{\lambda_z} \left(\frac{M}{aR} \right) B_y \lambda_y + B_y$$

$$B_y = \frac{\lambda_z a R^2}{\lambda_z a R - \lambda_y M}$$

Then

$$A = -\frac{MR}{\lambda_z a R - \lambda_y M} = \frac{R}{\lambda_y - \frac{aR}{M} \lambda_z}$$

$$B_x = -A\lambda_x = \frac{\lambda_x MR}{\lambda_z a R - \lambda_y M}$$

$$B_z = -A\lambda_z = \frac{\lambda_z MR}{\lambda_z a R - \lambda_y M}$$

In summary

$$\mathbf{A} = \frac{P}{\lambda_y - \frac{aR}{M} \lambda_z} \lambda_A \mathbf{e}_A \quad \blacktriangleleft$$

$$\mathbf{B} = \frac{R}{\lambda_z a R - \lambda_y M} (\lambda_x M \mathbf{i} + \lambda_z a R \mathbf{j} + \lambda_z M \mathbf{k}) \quad \blacktriangleleft$$

and

$$x = a \left(1 - \frac{R}{B_y} \right)$$

$$= a \left[1 - R \left(\frac{\lambda_z a R - \lambda_y M}{\lambda_z a R^2} \right) \right]$$

$$\text{or } x = \frac{\lambda_y M}{\lambda_z R} \quad \blacktriangleleft$$

Note that for this case, the lines of action of both $\mathbf{A}$ and $\mathbf{B}$ intersect the x axis.

PROBLEM 3.145* (Continued)

Case 2: Let $B_y = 0$ to satisfy Equation (4)

Now Equation (2) $A = \frac{R}{\lambda_y}$

Equation (1) $B_x = -R \left(\frac{\lambda_x}{\lambda_y} \right)$

Equation (3) $B_z = -R \left(\frac{\lambda_z}{\lambda_y} \right)$

Equation (6) $aA\lambda_y = 0$ which requires $a = 0$

Substitution into Equation (5)

$$M = z \left[-R \left(\frac{\lambda_x}{\lambda_y} \right) \right] - x \left[-R \left(\frac{\lambda_z}{\lambda_y} \right) \right] \quad \text{or} \quad \lambda_z x - \lambda_x z = \left(\frac{M}{R} \right) \lambda_y$$

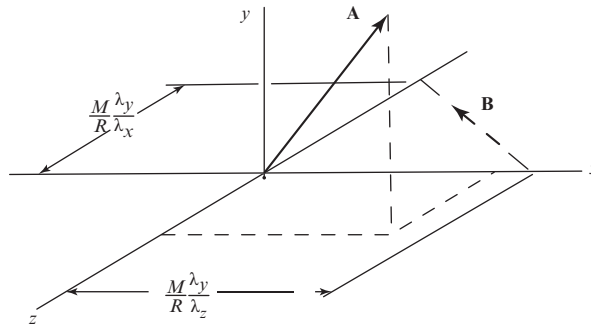
This last expression is the equation for the line of action of force **B**.

In summary

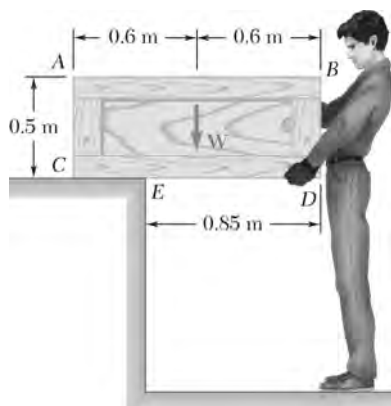
$$\mathbf{A} = \left(\frac{R}{\lambda_y} \right) \lambda_A$$

$$\mathbf{B} = \left(\frac{R}{\lambda_y} \right) (-\lambda_x \mathbf{i} - \lambda_z \mathbf{k})$$

Assuming that $\lambda_x, \lambda_y, \lambda_z > 0$, the equivalent force system is as shown below.



Note that the component of **A** in the xz plane is parallel to **B**.



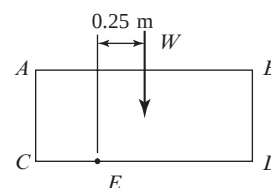
PROBLEM 3.146

A crate of mass 80 kg is held in the position shown. Determine (a) the moment produced by the weight $\mathbf{W}$ of the crate about E , (b) the smallest force applied at B that creates a moment of equal magnitude and opposite sense about E .

SOLUTION

- (a) By definition $W = mg = 80 \text{ kg}(9.81 \text{ m/s}^2) = 784.8 \text{ N}$

We have $\Sigma M_E: M_E = (784.8 \text{ N})(0.25 \text{ m})$



$$\mathbf{M}_E = 196.2 \text{ N} \cdot \text{m} \curvearrowleft$$

- (b) For the force at B to be the smallest, resulting in a moment ($\mathbf{M}_E$) about E , the line of action of force $\mathbf{F}_B$ must be perpendicular to the line connecting E to B . The sense of $\mathbf{F}_B$ must be such that the force produces a counterclockwise moment about E .

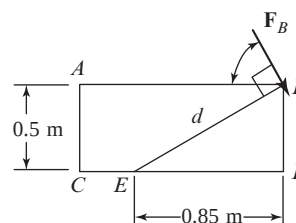
Note: $d = \sqrt{(0.85 \text{ m})^2 + (0.5 \text{ m})^2} = 0.98615 \text{ m}$

We have $\Sigma M_E: 196.2 \text{ N} \cdot \text{m} = F_B(0.98615 \text{ m})$

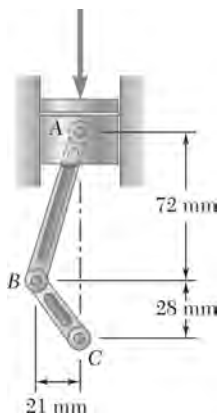
$$F_B = 198.954 \text{ N}$$

and $\theta = \tan^{-1}\left(\frac{0.85 \text{ m}}{0.5 \text{ m}}\right) = 59.534^\circ$

or



$$\mathbf{F}_B = 199.0 \text{ N} \searrow 59.5^\circ \curvearrowleft$$

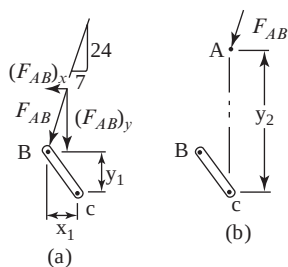


PROBLEM 3.147

It is known that the connecting rod AB exerts on the crank BC a 1.5-kN force directed down and to the left along the centerline of AB . Determine the moment of the force about C .

SOLUTION

Using (a)



$$\begin{aligned}
 M_C &= y_1(F_{AB})_x + x_1(F_{AB})_y \\
 &= (0.028 \text{ m})\left(\frac{7}{25} \times 1500 \text{ N}\right) + (0.021 \text{ m})\left(\frac{24}{25} \times 1500 \text{ N}\right) \\
 &= 42 \text{ N} \cdot \text{m}
 \end{aligned}$$

or $\mathbf{M}_C = 42.0 \text{ N} \cdot \text{m} \curvearrowleft$

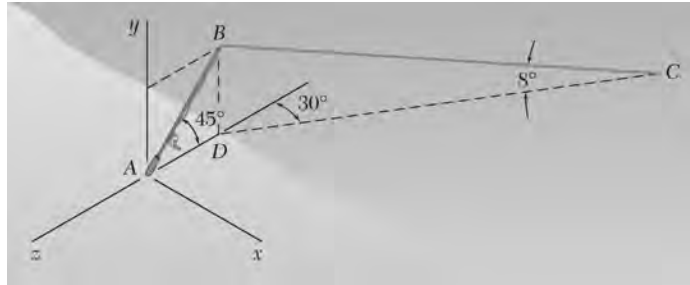
Using (b)

$$\begin{aligned}
 M_C &= y_2(F_{AB})_x \\
 &= (0.1 \text{ m})\left(\frac{7}{25} \times 1500 \text{ N}\right) = 42 \text{ N} \cdot \text{m}
 \end{aligned}$$

or $\mathbf{M}_C = 42.0 \text{ N} \cdot \text{m} \curvearrowleft$

PROBLEM 3.148

A 1.8 m-long fishing rod AB is securely anchored in the sand of a beach. After a fish takes the bait, the resulting force in the line is 30 N. Determine the moment about A of the force exerted by the line at B .



SOLUTION

We have $T_{xz} = (30 \text{ N}) \cos 8^\circ = 29.708 \text{ N}$

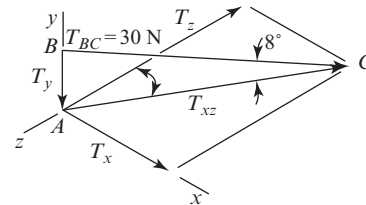
Then $T_x = T_{xz} \sin 30^\circ = 14.854 \text{ N}$
 $T_y = -T_{BC} \sin 8^\circ = -4.1752 \text{ N}$
 $T_z = -T_{xz} \cos 30^\circ = -25.728 \text{ N}$

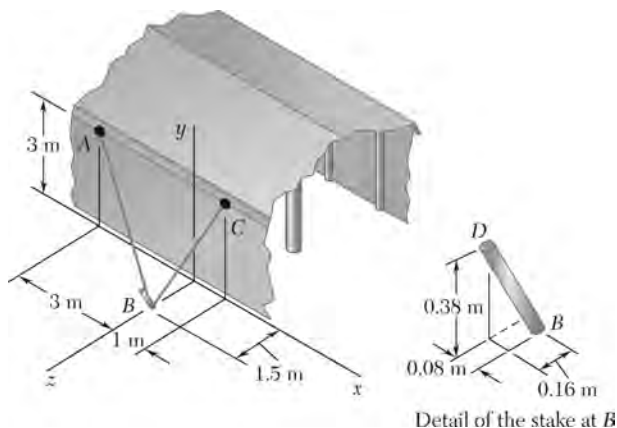
Now $\mathbf{M}_A = \mathbf{r}_{B/A} \times \mathbf{T}_{BC}$

where $\mathbf{r}_{B/A} = (1.8 \sin 45^\circ) \mathbf{j} - (1.8 \cos 45^\circ) \mathbf{k}$
 $= \frac{1.8 \text{ m}}{\sqrt{2}} (\mathbf{j} - \mathbf{k})$

Then $\mathbf{M}_A = \frac{1.8}{\sqrt{2}} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & -1 \\ 14.854 & -4.1752 & -25.728 \end{vmatrix}$
 $= \frac{1.8}{\sqrt{2}} (-25.728 - 4.1752) \mathbf{i} - \frac{1.8}{\sqrt{2}} (14.854) \mathbf{j} - \frac{1.8}{\sqrt{2}} (14.854) \mathbf{k}$

or $\mathbf{M}_A = -(38.1 \text{ N} \cdot \text{m}) \mathbf{i} - (18.91 \text{ N} \cdot \text{m}) \mathbf{j} - (18.91 \text{ N} \cdot \text{m}) \mathbf{k}$





PROBLEM 3.149

Ropes AB and BC are two of the ropes used to support a tent. The two ropes are attached to a stake at B . If the tension in rope AB is 540 N, determine (a) the angle between rope AB and the stake, (b) the projection on the stake of the force exerted by rope AB at Point B .

SOLUTION

First note $BA = \sqrt{(-3)^2 + (3)^2 + (-1.5)^2} = 4.5 \text{ m}$
 $BD = \sqrt{(-0.08)^2 + (0.38)^2 + (0.16)^2} = 0.42 \text{ m}$

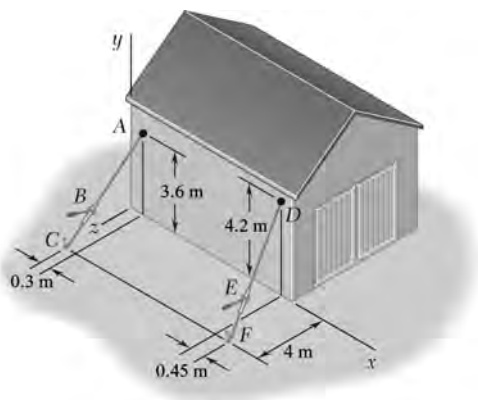
Then $\mathbf{T}_{BA} = \frac{T_{BA}}{4.5}(-3\mathbf{i} + 3\mathbf{j} - 1.5\mathbf{k})$
 $= \frac{T_{BA}}{3}(-2\mathbf{i} + 2\mathbf{j} - \mathbf{k})$
 $\lambda_{BD} = \frac{BD}{BD} = \frac{1}{0.42}(-0.08\mathbf{i} + 0.38\mathbf{j} + 0.16\mathbf{k})$
 $= \frac{1}{21}(-4\mathbf{i} + 19\mathbf{j} + 8\mathbf{k})$

(a) We have $\mathbf{T}_{BA} \cdot \lambda_{BD} = T_{BA} \cos \theta$
 or $\frac{T_{BA}}{3}(-2\mathbf{i} + 2\mathbf{j} - \mathbf{k}) \cdot \frac{1}{21}(-4\mathbf{i} + 19\mathbf{j} + 8\mathbf{k}) = T_{BA} \cos \theta$
 or $\cos \theta = \frac{1}{63}[(-2)(-4) + (2)(19) + (-1)(8)]$
 $= 0.60317$

or $\theta = 52.9^\circ \blacktriangleleft$

(b) We have $(T_{BA})_{BD} = \mathbf{T}_{BA} \cdot \lambda_{BD}$
 $= T_{BA} \cos \theta$
 $= (540 \text{ N})(0.60317)$

or $(T_{BA})_{BD} = 326 \text{ N} \blacktriangleleft$



PROBLEM 3.150

A farmer uses cables and winch pullers B and E to plumb one side of a small barn. If it is known that the sum of the moments about the x axis of the forces exerted by the cables on the barn at Points A and D is equal to $7092 \text{ N}\cdot\text{m}$, determine the magnitude of T_{DE} when $T_{AB} = 1.275 \text{ kN}$.

SOLUTION

The moment about the x axis due to the two cable forces can be found using the z components of each force acting at their intersection with the xy -plane (A and D). The x components of the forces are parallel to the x axis, and the y components of the forces intersect the x axis. Therefore, neither the x nor y components produce a moment about the x axis.

We have $\Sigma M_x: (T_{AB})_z(y_A) + (T_{DE})_z(y_D) = M_x$

where

$$\begin{aligned} (T_{AB})_z &= \mathbf{k} \cdot \mathbf{T}_{AB} \\ &= \mathbf{k} \cdot (T_{AB} \lambda_{AB}) - 0.3\mathbf{i} \\ &= \mathbf{k} \cdot \left[1275 \text{ N} \left(\frac{-0.3\mathbf{i} - 3.6\mathbf{j} + 4\mathbf{k}}{5.390} \right) \right] \\ &= 946.2 \text{ N} \end{aligned}$$

$$\begin{aligned} (T_{DE})_z &= \mathbf{k} \cdot \mathbf{T}_{DE} \\ &= \mathbf{k} \cdot (T_{DE} \lambda_{DE}) \\ &= \mathbf{k} \cdot \left[T_{DE} \left(\frac{0.45\mathbf{i} - 4.2\mathbf{j} + 4\mathbf{k}}{5.817} \right) \right] \\ &= 0.6876 T_{DE} \end{aligned}$$

$$y_A = 3.6 \text{ m}$$

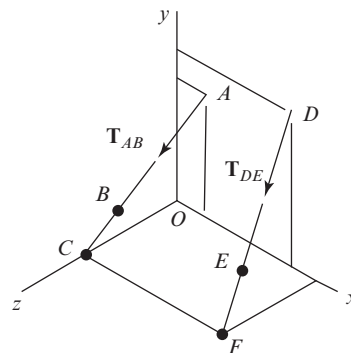
$$y_D = 4.2 \text{ m}$$

$$M_x = 7092 \text{ N} \cdot \text{m}$$

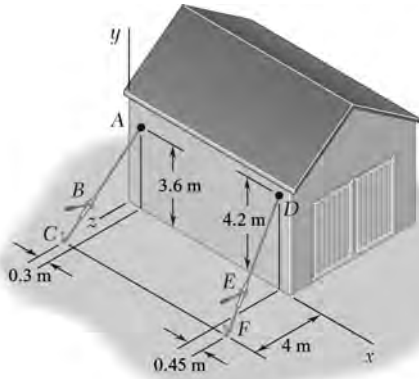
$$(946.2 \text{ N})(3.6 \text{ m}) + (0.6876 T_{DE})(4.2 \text{ m}) = 7092 \text{ N} \cdot \text{m}$$

and

$$T_{DE} = 1276.24 \text{ N}$$



$$\text{or } T_{DE} = 1.276 \text{ kN} \blacktriangleleft$$



PROBLEM 3.151

Solve Problem 3.150 when the tension in cable AB is 1.53 kN.

PROBLEM 3.150 A farmer uses cables and winch pullers B and E to plumb one side of a small barn. If it is known that the sum of the moments about the x axis of the forces exerted by the cables on the barn at Points A and D is equal to 7092 N·m, determine the magnitude of T_{DE} when $T_{AB} = 1.275$ kN

SOLUTION

The moment about the x axis due to the two cable forces can be found using the z components of each force acting at their intersection with the xy plane (A and D). The x components of the forces are parallel to the x axis, and the y components of the forces intersect the x axis. Therefore, neither the x or y components produce a moment about the x axis.

$$\text{We have} \quad \Sigma M_x: (T_{AB})_z(y_A) + (T_{DE})_z(y_D) = M_x$$

$$\begin{aligned} \text{where} \quad (T_{AB})_z &= \mathbf{k} \cdot \mathbf{T}_{AB} \\ &= \mathbf{k} \cdot (T_{AB} \lambda_{AB}) \\ &= \mathbf{k} \cdot \left[1530 \text{ N} \left(\frac{-0.3\mathbf{i} - 3.6\mathbf{j} + 4\mathbf{k}}{5.390} \right) \right] \\ &= 1135.44 \text{ N} \end{aligned}$$

$$\begin{aligned} (T_{DE})_z &= \mathbf{k} \cdot \mathbf{T}_{DE} \\ &= \mathbf{k} \cdot (T_{DE} \lambda_{DE}) \\ &= \mathbf{k} \cdot \left[T_{DE} \left(\frac{0.45\mathbf{i} - 4.2\mathbf{j} + 4\mathbf{k}}{5.817} \right) \right] \\ &= 0.6876 T_{DE} \end{aligned}$$

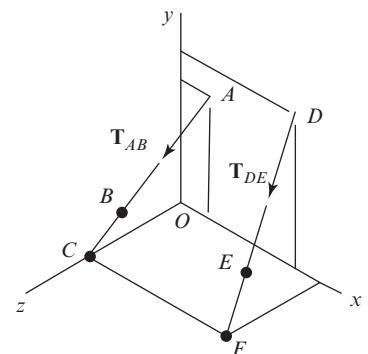
$$y_A = 3.6 \text{ m}$$

$$y_D = 4.2 \text{ m}$$

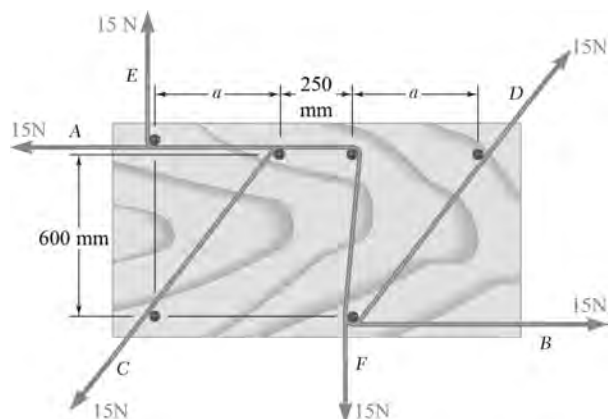
$$M_x = 7092 \text{ N} \cdot \text{m}$$

$$(1135.44 \text{ N})(3.6 \text{ m}) + (0.6876 T_{DE})(4.2 \text{ m}) = 7092 \text{ N} \cdot \text{m}$$

$$\text{and} \quad T_{DE} = 1040.34 \text{ N}$$



$$\text{or } T_{DE} = 1.040 \text{ kN} \quad \blacktriangleleft$$



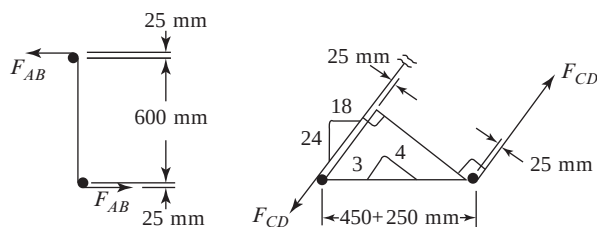
PROBLEM 3.152

A wiring harness is made by routing either two or three wires around 50 mm-diameter pegs mounted on a sheet of plywood. If the force in each wire is 15 N, determine the resultant couple acting on the plywood when $a = 450$ mm and (a) only wires AB and CD are in place, (b) all three wires are in place.

SOLUTION

In general, $M = \Sigma dF$, where d is the perpendicular distance between the lines of action of the two forces acting on a given wire.

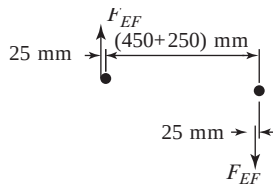
(a)



$$\begin{aligned} \text{We have } M &= d_{AB}F_{AB} + d_{CD}F_{CD} \\ &= (50 + 600) \text{ mm} \times 15 \text{ N} + \left(50 + \frac{4}{5} \times 700 \right) \text{ mm} \times 15 \text{ N} \end{aligned}$$

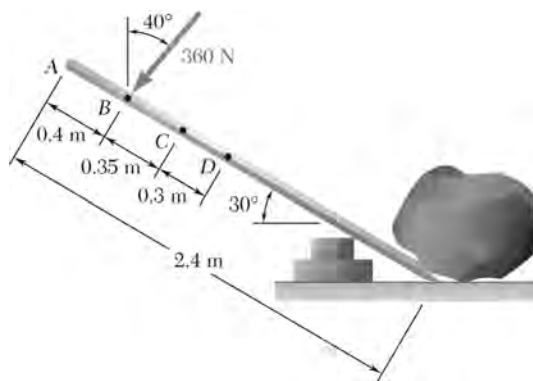
$$\text{or } \mathbf{M = 18900 \text{ N} \cdot \text{mm} \quad \blacktriangleleft}$$

(b)



$$\begin{aligned} \text{We have } M &= [d_{AB}F_{AB} + d_{CD}F_{CD}] + d_{EF}F_{EF} \\ &= 18900 \text{ N} \cdot \text{mm} - 700 \text{ mm} \times 15 \text{ N} \end{aligned}$$

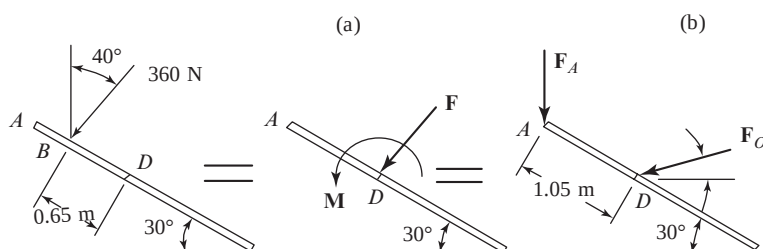
$$\begin{aligned} \text{or } \mathbf{M} &= 8.400 \text{ N} \cdot \text{m} \quad \blacktriangleleft \\ &= 8.40 \text{ N} \cdot \text{m} \end{aligned}$$



PROBLEM 3.153

A worker tries to move a rock by applying a 360-N force to a steel bar as shown. (a) Replace that force with an equivalent force-couple system at D. (b) Two workers attempt to move the same rock by applying a vertical force at A and another force at D. Determine these two forces if they are to be equivalent to the single force of Part a.

SOLUTION



(a) We have $\Sigma \mathbf{F}$: $360 \text{ N}(-\sin 40^\circ \mathbf{i} - \cos 40^\circ \mathbf{j}) = -(231.40 \text{ N})\mathbf{i} - (275.78 \text{ N})\mathbf{j} = \mathbf{F}$

or $\mathbf{F} = 360 \text{ N} \nearrow 50^\circ \blacktriangleleft$

We have $\Sigma \mathbf{M}_D$: $\mathbf{r}_{B/D} \times \mathbf{F} = \mathbf{M}$

where $\mathbf{r}_{B/D} = -[(0.65 \text{ m}) \cos 30^\circ]\mathbf{i} + [(0.65 \text{ m}) \sin 30^\circ]\mathbf{j}$
 $= -(0.56292 \text{ m})\mathbf{i} + (0.32500 \text{ m})\mathbf{j}$

$$\mathbf{M} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -0.56292 & 0.32500 & 0 \\ -231.40 & -275.78 & 0 \end{vmatrix} \text{ N} \cdot \text{m}$$

$$= [155.240 + 75.206] \text{ N} \cdot \text{m} \mathbf{k}$$

$$= (230.45 \text{ N} \cdot \text{m})\mathbf{k}$$

or $\mathbf{M} = 230 \text{ N} \cdot \text{m} \curvearrowright \blacktriangleleft$

(b) We have $\Sigma \mathbf{M}_D$: $\mathbf{M} = \mathbf{r}_{A/D} \times \mathbf{F}_A$

where

$$\mathbf{r}_{A/D} = -[(1.05 \text{ m}) \cos 30^\circ]\mathbf{i} + [(1.05 \text{ m}) \sin 30^\circ]\mathbf{j}$$

$$= -(0.90933 \text{ m})\mathbf{i} + (0.52500 \text{ m})\mathbf{j}$$

$$\mathbf{F}_A = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -0.90933 & 0.52500 & 0 \\ 0 & -1 & 0 \end{vmatrix} \text{ N} \cdot \text{m}$$

$$= [230.45 \text{ N} \cdot \text{m}]\mathbf{k}$$

PROBLEM 3.153 (Continued)

or $(0.90933F_A)\mathbf{k} = 230.45\mathbf{k}$

$$F_A = 253.42 \text{ N}$$

or $\mathbf{F}_A = 253 \text{ N} \downarrow \blacktriangleleft$

We have $\Sigma \mathbf{F}: \mathbf{F} = \mathbf{F}_A + \mathbf{F}_D$

$$-(231.40 \text{ N})\mathbf{i} - (275.78 \text{ N})\mathbf{j} = -(253.42 \text{ N})\mathbf{j} + F_D(-\cos\theta\mathbf{i} - \sin\theta\mathbf{j})$$

From $\mathbf{i}: 231.40 \text{ N} = F_D \cos\theta$ (1)

$\mathbf{j}: 22.36 \text{ N} = F_D \sin\theta$ (2)

Equation (2) divided by Equation (1)

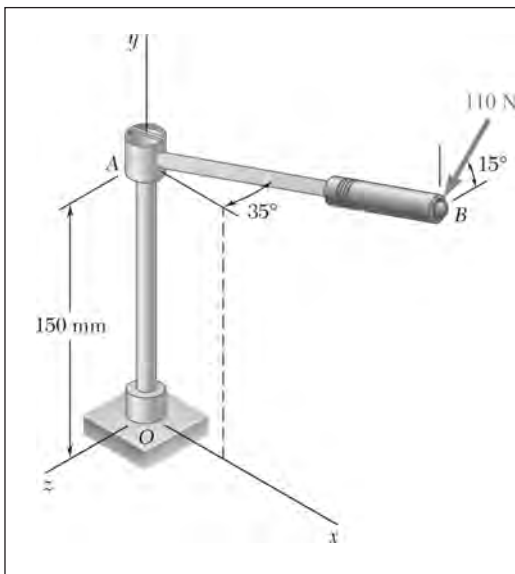
$$\tan\theta = 0.096629$$

$$\theta = 5.5193^\circ \text{ or } \theta = 5.52^\circ$$

Substitution into Equation (1)

$$F_D = \frac{231.40}{\cos 5.5193^\circ} = 232.48 \text{ N}$$

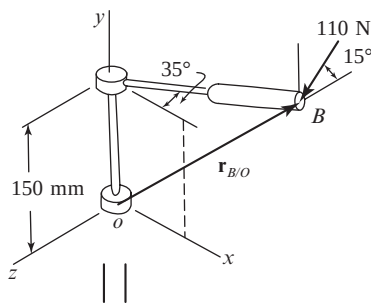
or $\mathbf{F}_D = 232 \text{ N} \nearrow 5.52^\circ \blacktriangleleft$



PROBLEM 3.154

A 110-N force acting in a vertical plane parallel to the yz plane is applied to the 220-mm long horizontal handle AB of a socket wrench. Replace the force with an equivalent force-couple system at the origin O of the coordinate system.

SOLUTION



we have

$$\Sigma \mathbf{F}: \mathbf{P}_B = \mathbf{F}$$

where

$$\begin{aligned} \mathbf{P}_B &= 110 \text{ N} [-(\sin 15^\circ) \mathbf{j} + (\cos 15^\circ) \mathbf{k}] \\ &= -(28.470 \text{ N}) \mathbf{j} + (106.252 \text{ N}) \mathbf{k} \end{aligned}$$

$$\text{or } \mathbf{F} = -(28.5 \text{ N}) \mathbf{j} + (106.3 \text{ N}) \mathbf{k} \quad \blacktriangleleft$$

We have

$$\Sigma \mathbf{M}_O: \mathbf{r}_{B/O} \times \mathbf{P}_B = \mathbf{M}_O$$

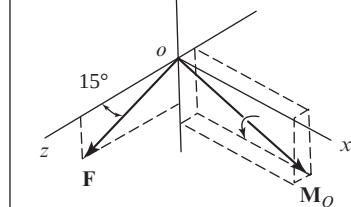
where

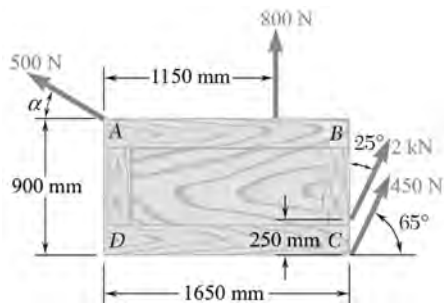
$$\begin{aligned} \mathbf{r}_{B/O} &= [(0.22 \cos 35^\circ) \mathbf{i} + (0.15) \mathbf{j} - (0.22 \sin 35^\circ) \mathbf{k}] \text{ m} \\ &= (0.180213 \text{ m}) \mathbf{i} + (0.15 \text{ m}) \mathbf{j} - (0.126187 \text{ m}) \mathbf{k} \end{aligned}$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.180213 & 0.15 & -0.126187 \\ 0 & -28.5 & 106.3 \end{vmatrix} \text{ N} \cdot \text{m} = \mathbf{M}_O$$

$$\mathbf{M}_O = [(12.3487) \mathbf{i} - (19.1566) \mathbf{j} - (5.1361) \mathbf{k}] \text{ N} \cdot \text{m}$$

$$\text{or } \mathbf{M}_O = (12.35 \text{ N} \cdot \text{m}) \mathbf{i} - (19.16 \text{ N} \cdot \text{m}) \mathbf{j} - (5.13 \text{ N} \cdot \text{m}) \mathbf{k} \quad \blacktriangleleft$$



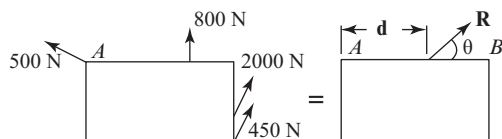


PROBLEM 3.155

Four ropes are attached to a crate and exert the forces shown. If the forces are to be replaced with a single equivalent force applied at a point on line AB , determine (a) the equivalent force and the distance from A to the point of application of the force when $\alpha = 30^\circ$, (b) the value of α so that the single equivalent force is applied at Point B .

SOLUTION

We have



(a) For equivalence $\Sigma F_x: -500 \cos 30^\circ + 2000 \cos 65^\circ + 450 \cos 65^\circ = R_x$

or $R_x = 602.4 \text{ N}$

$$\Sigma F_y: 500 \sin \alpha + 800 + 2000 \sin 65^\circ + 450 \sin 65^\circ = R_y$$

or $R_y = (3020.45 + 500 \sin \alpha) \text{ N}$

(1)

With $\alpha = 30^\circ$ $R_y = 3270.45 \text{ N}$

Then $R = \sqrt{(602.4)^2 + (3270.45)^2}$ $\tan \theta = \frac{3270.45}{602.4}$
 $= 3325.5 \text{ N}$ or $\theta = 79.6^\circ$

Also $\Sigma M_A: (1.150 \text{ m})(800 \text{ N}) + (1.650 \text{ m})(2000 \text{ N}) \sin 65^\circ + (0.650 \text{ m})(2000 \text{ N}) \cos 65^\circ$
 $+ (1.650 \text{ m})(450 \text{ N}) \sin 65^\circ + (0.9 \text{ m})(450 \text{ N}) \cos 65^\circ = d(3270.45)$

or $\Sigma M_A = 53043 \text{ N} \cdot \text{m}$ and $d = 1.622 \text{ m}$ $R = 3330 \text{ N}$ $\nearrow 79.6^\circ$ ◀

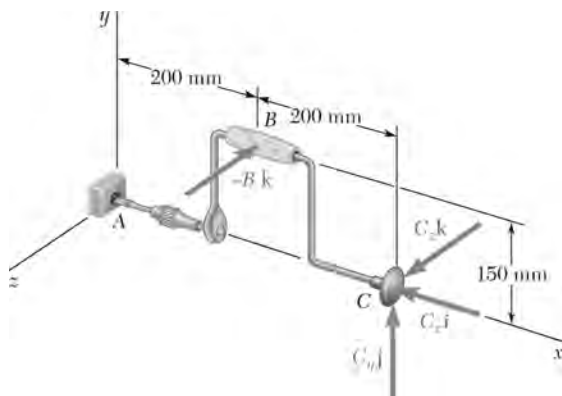
and $\mathbf{R}$ is applied 1.622 m. To the right of A . ◀

(b) We have $d = 1.650 \text{ m}$

Then $\Sigma M_A: 5304.3 \text{ N} \cdot \text{m} = (1.65 \text{ m})(R_y)$

or $R_y = 3214.7 \text{ N}$

Using Eq. (1) $3214.7 = 3020.45 + 500 \sin \alpha$ or $\alpha = 22.9^\circ$ ◀



PROBLEM 3.156

A blade held in a brace is used to tighten a screw at A. (a) Determine the forces exerted at B and C, knowing that these forces are equivalent to a force-couple system at A consisting of $\mathbf{R} = -(30 \text{ N})\mathbf{i} + R_y\mathbf{j} + R_z\mathbf{k}$ and $\mathbf{M}_A^R = -(12 \text{ N} \cdot \text{m})\mathbf{i}$. (b) Find the corresponding values of R_y and R_z . (c) What is the orientation of the slot in the head of the screw for which the blade is least likely to slip when the brace is in the position shown?

SOLUTION

(a) Equivalence requires $\Sigma \mathbf{F}: \mathbf{R} = \mathbf{B} + \mathbf{C}$

$$\text{or} \quad -(30 \text{ N})\mathbf{i} + R_y\mathbf{j} + R_z\mathbf{k} = -B\mathbf{k} + (-C_x\mathbf{i} + C_y\mathbf{j} + C_z\mathbf{k})$$

$$\text{Equating the } \mathbf{i} \text{ coefficients} \quad \mathbf{i}: -30 \text{ N} = -C_x \quad \text{or} \quad C_x = 30 \text{ N}$$

$$\text{Also} \quad \Sigma \mathbf{M}_A: \mathbf{M}_A^R = \mathbf{r}_{B/A} \times \mathbf{B} + \mathbf{r}_{C/A} \times \mathbf{C}$$

$$\text{or} \quad -(12 \text{ N} \cdot \text{m})\mathbf{i} = [(0.2 \text{ m})\mathbf{i} + (0.15 \text{ m})\mathbf{j}] \times (-B)\mathbf{k} + (0.4 \text{ m})\mathbf{i} \times [-(30 \text{ N})\mathbf{i} + C_y\mathbf{j} + C_z\mathbf{k}]$$

$$\text{Equating coefficients} \quad \mathbf{i}: -12 \text{ N} \cdot \text{m} = -(0.15 \text{ m})B \quad \text{or} \quad B = 80 \text{ N}$$

$$\mathbf{k}: 0 = (0.4 \text{ m})C_y \quad \text{or} \quad C_y = 0$$

$$\mathbf{j}: 0 = (0.2 \text{ m})(80 \text{ N}) - (0.4 \text{ m})C_z \quad \text{or} \quad C_z = 40 \text{ N}$$

$$\mathbf{B} = -(80.0 \text{ N})\mathbf{k} \quad \mathbf{C} = -(30.0 \text{ N})\mathbf{i} + (40.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

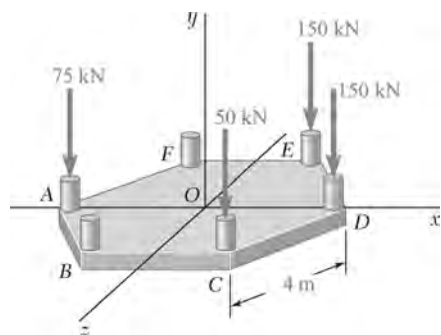
(b) Now we have for the equivalence of forces

$$-(30 \text{ N})\mathbf{i} + R_y\mathbf{j} + R_z\mathbf{k} = -(80 \text{ N})\mathbf{k} + [(-30 \text{ N})\mathbf{i} + (40 \text{ N})\mathbf{k}]$$

$$\text{Equating coefficients} \quad \mathbf{j}: R_y = 0 \quad R_y = 0 \quad \blacktriangleleft$$

$$\mathbf{k}: R_z = -80 + 40 \quad \text{or} \quad R_z = -40.0 \text{ N} \quad \blacktriangleleft$$

(c) First note that $\mathbf{R} = -(30 \text{ N})\mathbf{i} - (40 \text{ N})\mathbf{k}$. Thus, the screw is best able to resist the lateral force R_z when the slot in the head of the screw is vertical. $\blacktriangleleft$



PROBLEM 3.157

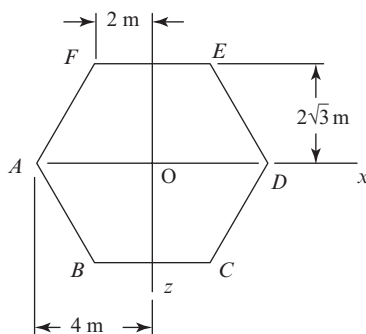
A concrete foundation mat in the shape of a regular hexagon of side 12 m supports four column loads as shown. Determine the magnitudes of the additional loads that must be applied at B and F if the resultant of all six loads is to pass through the center of the mat.

SOLUTION

From the statement of the problem it can be concluded that the six applied loads are equivalent to the resultant $\mathbf{R}$ at O . It then follows that

$$\Sigma \mathbf{M}_O = 0 \quad \text{or} \quad \Sigma M_x = 0 \quad \Sigma M_z = 0$$

For the applied loads.



$$\begin{aligned} \Sigma M_x = 0: & (2\sqrt{3} \text{ m})F_B + (2\sqrt{3} \text{ m})(50 \text{ kN}) - (2\sqrt{3} \text{ m})(100 \text{ kN}) \\ & - (2\sqrt{3} \text{ m})F_F = 0 \end{aligned}$$

$$\text{or} \quad F_B - F_F = 50 \quad (1)$$

$$\begin{aligned} \Sigma M_z = 0: & (4 \text{ m})(75 \text{ kN}) + (2 \text{ m})F_B - (2 \text{ m})(50 \text{ kN}) \\ & - (4 \text{ m})(150 \text{ kN}) - (2 \text{ m})(100 \text{ kN}) + (2 \text{ m})F_F = 0 \end{aligned}$$

$$\text{or} \quad F_B + F_F = 300 \quad (2)$$

$$\text{Then } (1) + (2) \Rightarrow F_B = 175 \text{ kN} \downarrow \blacktriangleleft$$

$$\text{and} \quad F_F = 125 \text{ kN} \downarrow \blacktriangleleft$$